

SECTION-I

Straight Objective Type

6 Q. [3(-1)]

- If $\vec{a} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{j}$, then $\frac{\text{projection of } \vec{a} \text{ on } \vec{b}}{\text{projection of } \vec{b} \text{ on } \vec{a}} =$
 (A) 7/5 (B) 5/7 (C) 4/9 (D) none
- At $t = 1$ sec., a particle is at (1, 0, 0). It moves towards (4, 4, 12) with a constant speed of 65 m/s. The position of the particle is measured in metres and the time in sec. Assuming constant velocity, the position of the particle at $t = 3$ s is :
 (A) $(13\hat{i} - 120\hat{j} + 40\hat{k})$ m (B) $(40\hat{i} + 31\hat{j} - 120\hat{k})$ m
 (C) $(13\hat{i} - 40\hat{j} + 12\hat{k})$ m (D) $(31\hat{i} + 40\hat{j} + 120\hat{k})$ m
- A particle is moving under the influence of force $\vec{F} = [\hat{i} - 2\hat{j}]$ N, is now moved from the point $(x, y, z) = (2, 1, 3)$ m to the point $(x, y, z) = (3, 2, 4)$ m. How much work is done by the force $\vec{F}$ during this time period? ($W = \vec{F} \cdot \vec{s}$)
 (A) 0 J (B) -1 J (C) 1 J (D) None of these
- An object of mass m is attached to a spring. The restoring force of the spring is $F = -\lambda x^3$, where x is the displacement. The oscillation period depends on the mass, λ and oscillation amplitude. Suppose the object is initially at rest. If the initial displacement is D then its period is τ . If the initial displacement is $2D$, find the period. (Hint: use dimension analysis.)
 (A) 8τ (B) 2τ (C) τ (D) $\tau/2$
- According to Maxwell-distribution law, the probability function representing the ratio of molecules at a particular velocity to the total number of molecules is given by

$$f(v) = k_1 \sqrt{\left(\frac{m}{2\pi kT}\right)^3} 4\pi v^2 e^{-\frac{mv^2}{2kT}}$$

Where m is the mass of the molecule,

v is the velocity of the molecule,

T is the temperature

k and k_1 are constants.

The dimensional formulae of k_1 is

- (A) L^2T^{-2} (B) $L^1T^{-1}K^{-3/2}$ (C) $L^1T^{-1}K^{+3/2}$ (D) $L^2T^{-1}K^{+3/2}$
- In a new set of units, 1 unit of force is equal to 100 N, 1 unit of speed is equal to 10^3 m/s, 1 unit of time is equal to 10^{-5} sec, then 1 unit of mass in the new system will be equal to :-
 (A) 1000 kg (B) 10^{-6} kg (C) 10^{-3} kg (D) 10^{+6} kg

Linked Comprehension Type (Single option correct)

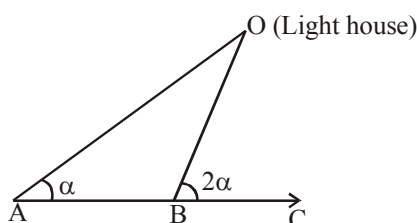
2 Para × 2 Q. [3(-1)]

Paragraph for Question Nos. 7 and 8

Doubling the angle on the Bow



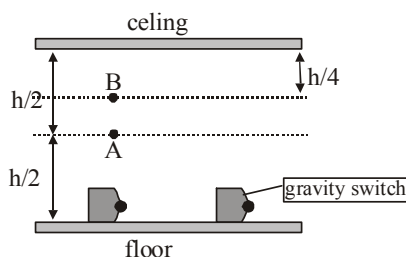
When Captain asks Mr. Gibbs to find the location of his ship “The Black pearl” with respect to Queens light house, Mr. Gibbs uses a technique known as “Doubling the angle on the bow”. Figure shows the straight track (ABC) of ship passing by the lighthouse. At point A, Mr. Gibbs measures the angle α between the line of sight to the lighthouse and the direction of motion of the ship. At point B the measured angle becomes to 2α .



7. Taking light house as the origin, which of the following is NOT equal to the magnitude of displacement of boat when it goes from A to B :-
 (A) $|\vec{OB} - \vec{OA}|$ (B) $|\vec{OB}|$ (C) $|\vec{OA}|$ (D) $|\vec{AB}|$
8. If the ship starts retarding as it passes from point B. What should be the retardation of the ship so that it stops at the point closest to the light house. Velocity of the ship at point B is 10 m/s, distance of the ship from light house at point B is 50 m. (Given $\alpha = 30^\circ$)
 (A) 4 m/s² (B) 2 m/s² (C) 1 m/s² (D) None of these

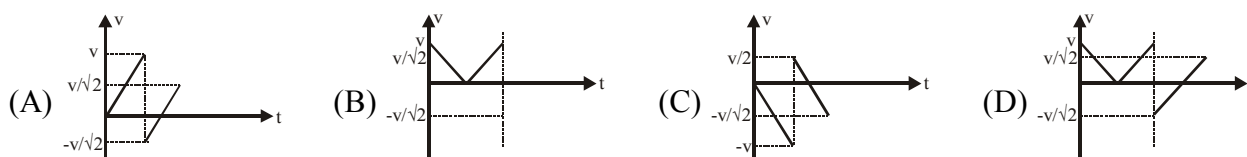
Paragraph for Question 9 and 10

A student tosses a rubber ball vertically upward. When it reaches the top of its trajectory at a point A, another student flips the gravity switch so that acceleration due to gravity now becomes 10 m/s² upwards. The ball bounces back to point B, then to the ceiling and sticks there.



Point A is halfway between the floor and ceiling and point B is 1/4th of the way down from ceiling. Take $y = 0$ at the floor and positive y-axis upward.

9. v-t graph of ball is : (v is initial velocity)



10. If $h = 40$ m, then find the total time taken by ball to stick on ceiling after the flip of gravity switch
 (A) $2 + 2\sqrt{2}$ sec (B) $2 + \sqrt{2}$ sec (C) 2 sec (D) None of these

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

8 Q. [4(-0)]

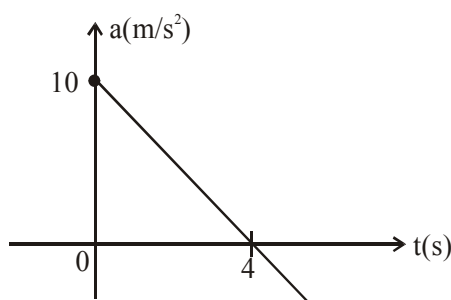
1. A boy A is standing 3 m west and 4 m north to a boy B. A starts moving along a vector $\vec{a} = 1.5\hat{i} + 2\hat{j}$ with a constant speed of 2 m/s for 5 s and stops. Its new position vector with respect to the boy B is $x\hat{i} + y\hat{j}$. Find $\frac{|x|+|y|}{3}$.
2. A sphere is fired downwards into a medium with an initial speed of 27 ms^{-1} . It experiences a deceleration of $a = (-6t) \text{ ms}^{-2}$, where t is in seconds. If the distance (in metre) travelled before it stops is 9α . Then find the value of α .
3. Starting from rest, a particle moving along a straight line is accelerated by the acceleration :

$$a = 4 - t^2 \quad \text{for } 0 \leq t \leq 2 \text{ sec}$$

$$a = \frac{1}{3} \quad \text{for } t > 2 \text{ sec}$$

What is the velocity (in m/s) after 4 sec?

4. The acceleration-time graph of a particle moving along a straight line is shown in figure. At what time (in sec) the particle acquires its initial velocity ?



5. A Diwali rocket moves up with a constant acceleration $a_1 = 20/3 \text{ m/s}^2$. After some time its fuel gets exhausted and then it falls freely. If the maximum height attained is 50 m and if v (in m/s) is speed when the fuel is exhausted. Find the value of $\frac{v}{10}$.
6. Two forces $\vec{F}_1$ & $\vec{F}_2$ of magnitude 12 N & 5N respectively are exerted on a body. $|\vec{F}_1 + \vec{F}_2| = 13 \text{ N}$. If the angle between $\vec{F}_1$ & $\vec{F}_2$ is $15x$ (in degree) then fill the value of x in OMR sheet.
7. Forces acting on a particle have magnitudes of 14, 7 and 7 N act in the direction of vectors $6\hat{i} + 2\hat{j} + 3\hat{k}$, $3\hat{i} - 2\hat{j} + 6\hat{k}$ and $2\hat{i} - 3\hat{j} - 6\hat{k}$ respectively. The forces remain constant while the particle is displaced from point A(2, -1, -3) to B(5, -1, 1). Find the total work done (in joule) on the particle. The coordinates are specified in meters. If your answer is x fill value of $x/15$.
8. A particle starts moving from origin with velocity $u = 2\hat{i} + 4\hat{j}$ under an acceleration of $\vec{a} = 4t\hat{i} - 6\hat{k}$ where t is the time elapsed since it started from origin. Find the value of t at which acceleration is perpendicular to initial velocity.

SECTION-I

Single Correct Answer Type

5 Q. [Marks 3 (–1)]

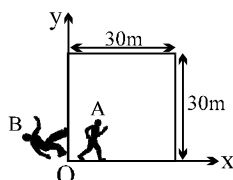
- A ball is thrown vertically upwards with some speed. It reaches two points A and B one after another such that heights of A and B are one fourth and three-fourth of the maximum height attained. If the total time of flight is T, the maximum time taken by the ball to travel from A to B, is :-
 (A) $\left(\frac{\sqrt{3}+1}{4}\right)T$ (B) $\left(\frac{\sqrt{3}-1}{2}\right)T$ (C) $\left(\frac{\sqrt{3}+1}{2}\right)T$ (D) $\frac{T}{\sqrt{2}}$
- Two particles P and Q start from rest and move for equal time on a straight line. Particle P has an acceleration of 2 m/s^2 for the first half of the total time and 4 m/s^2 for the second half. The particle Q has an acceleration of 4 m/s^2 for the first half of the total time and 2 m/s^2 for the second half. Which particle has travelled larger distance?
 (A) both have travelled the same distance
 (B) P has travelled larger distance
 (C) Q has travelled larger distance
 (D) none of these
- A spherical balloon whose initial radius was 3m is expanded such that its surface area is increasing at a constant rate of $72 \pi \text{ m}^2/\text{s}$. What is the speed of a point on the surface in radially outward direction when radius become 9m :-
 (A) 1 m/s (B) 2 m/s (C) 2.5 m/s (D) 3 m/s
- A boy standing at a corner noticed a thief on bike moving with 54 km/hr and after some time a police jeep crosses him with speed 72 km/hr. If jeep catches the thief at a distance of 1 km from the boy then the time gap between bike & jeep will be when they cross the boy-
 (A) 3.33 sec (B) 16.67 sec (C) 20 sec (D) 27.5 sec
- The radius of an air bubble is increasing at the rate of $\frac{1}{2} \text{ cm/s}$. At what rate is the volume of the bubble increasing when the radius is 1 cm?
 (A) $2\pi \text{ cm}^3/\text{s}$ (B) $2\pi \text{ cm}^2/\text{s}$
 (C) $2\pi \text{ m}^2/\text{s}$ (D) $1.5\pi \text{ cm}^2/\text{s}$

Multiple Correct Answer Type

7 Q. [Marks 4 (0)]

- Which of the following statement(s) is/are true for a uniform motion i.e. motion with uniform velocity?
 (A) The motion is always in same direction
 (B) The motion must be along a straight line
 (C) Average velocity is equal to the instantaneous velocity
 (D) Distance covered > |Displacement|

7. Let $\vec{v}$ and $\vec{a}$ denote the velocity and acceleration respectively of a body in one-dimensional motion :-
- $|\vec{v}|$ must decrease when $\vec{a} < 0$
 - Speed must increase when $\vec{a} > 0$
 - Speed will increase when both $\vec{v}$ and $\vec{a}$ are < 0
 - Speed will decrease when $\vec{v} < 0$ and $\vec{a} > 0$
8. Two particles A and B start simultaneously from origin and move in a horizontal plane. A has an initial velocity u_1 due east and acceleration a_1 due north. B has an initial velocity u_2 due north and acceleration a_2 due east :-
- Their paths must intersect at some point
 - They must collide at some point
 - They will collide only if $a_1 u_1 = a_2 u_2$
 - If $u_1 > u_2$ and $a_1 < a_2$, the particles may have the same speed at some point of time
9. Two persons start from the same location O and walk around a square (in horizontal plane) in different directions with constant speeds. The square is 30 m by 30 m. A's speed is 2 m/s and B's speed is 1 m/s.



- The coordinates of the point where A and B will meet for the first time is (10, 30)
 - Their displacement when they meet is $10\sqrt{10}$ m.
 - The average velocity $\vec{v}_A$ of A is $\frac{\hat{i} + 3\hat{j}}{4}$ m/s.
 - The average velocity $\vec{v}_B$ of B is $\frac{\hat{i} + 3\hat{j}}{4}$ m/s.
10. A coin tossed straight up from a point 3 m above the ground. It takes 3 s for the coin to reach the ground from its initial release point. Take $g = 10 \text{ m/s}^2$.
- Its maximum height from the ground is 9.8 m.
 - Its maximum height from the ground is 12.8 m.
 - Its initial speed of projection is 14 m/s.
 - Its initial speed of projection is 11 m/s.
11. A freely falling body travelled 35 m in last second of its journey :-
- The height of the point of release from ground is 80 m
 - The distance travelled in the first second is 5 m
 - The maximum speed attained by the body is 40 m/s
 - The maximum speed attained by the body is 20 m/s
12. Which of the following statement(s) is/are correct?
- A body can have constant speed but varying velocity.
 - A body can have constant velocity but varying speed.
 - A body can have non-zero acceleration with constant velocity.
 - A body can have velocity without having acceleration.

Linked Comprehension Type (Single option correct)
1 Para × 3 Q. [3(−1)]
Paragraph for Question Nos. 13 to 15

A balloon rises up with constant acceleration of 10 m/s^2 from earth's surface at $t = 0$. After 4 seconds a particle drops from the balloon.

13. Displacement of particle in time interval $0 \leq t \leq 4 \text{ s}$ is :-
 (A) 80 m (B) 60 m (C) 40 m (D) 90 m
14. Total air time of particle is :-
 (A) $(2+4\sqrt{2}) \text{ s}$ (B) $(4+2\sqrt{2}) \text{ s}$ (C) $(4+4\sqrt{2}) \text{ s}$ (D) $(4-4\sqrt{2}) \text{ s}$
15. Maximum height attained by the particle is :-
 (A) 160 m (B) 140 m (C) 130 m (D) 165 m

Linked Comprehension Type
(1 Para × 2 Q.) [Marks 4 (−1)]
(Multiple Correct Answer Type)
Paragraph for Question no. 16 and 17

Two motor vehicles left point A simultaneously and reached point B in 2 hrs. The first vehicle travelled half of the distance at a speed of $v_1 = 30 \text{ km/hr}$ and the other half at a speed of $v_2 = 45 \text{ km/hr}$. Second vehicle covered the entire distance with a constant acceleration.

16. At what moment of time speed of both the vehicles be same :
 (A) 0.83 hr after departure (B) 1.25 hr after departure
 (C) 1 hr after departure (D) 2.0 hr after departure
17. During journey ($t > 0$)
 (A) A overtakes B (B) B overtakes A
 (C) First A overtakes, then B overtakes (D) Neither overtake each other

SECTION-II

Numerical Answer Type Question
1 Q. [3(0)]
(upto second decimal place)

1. A stone is thrown upwards with an initial speed of 10 m/s while standing on the edge of a cliff. Find the distance travelled (in m) by the stone till 5 sec.

SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9)
2 Q. [Marks 4(0)]

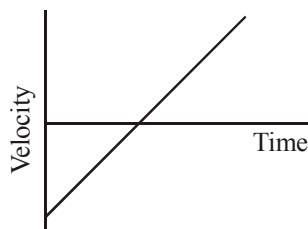
1. A large procession of people is moving along a road of width 10 m. There is a railway track across the road. The number of people present per square meter of the road is 6 (on an average). The average speed at which the procession is moving is 0.15 m/s . Find the number of people crossing the railway track per second.
2. A ball is thrown upwards with a speed of 40 m/s . When the speed becomes half of the initial speed, gravity is switched off for next 2 second. After that gravity is again switched on but magnitude of gravity is doubled. The total distance travelled by the ball from $t = 0$ to the time it reaches the maximum height is 55β . The value of β is

SECTION-I

Straight Objective Type

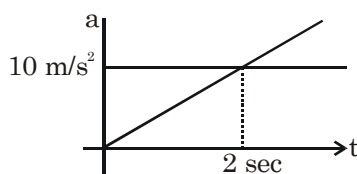
9 [3(-1)]Marks

- A ball is dropped from a height h from the floor. The velocity of the ball is halved and reversed after every impact. The total distance travelled by the ball will be
(A) $\frac{4h}{3}$ (B) $4h$ (C) $\frac{5h}{3}$ (D) infinite
- A man in a balloon rising vertically starting from rest with an acceleration of 4.9 m/s^2 releases a ball 2 seconds after the balloon is let go from the ground. The greatest height above the ground reached by the ball is ($g = 9.8 \text{ m/s}^2$)
(A) 14.7 m (B) 19.6 m (C) 9.8 m (D) 24.5 m
- A passenger who just missed the train stands on the platform, sadly watching the last two boggies of the train. The second last boggy takes time 3 sec. to pass by the passenger, and the last one takes time 2 sec. to pass by. How late is the passenger for the departure of the train? Assume that the train accelerates at constant rate
(A) $3/2 \text{ s}$ (B) $7/2 \text{ s}$ (C) $9/2 \text{ s}$ (D) None of these
- If velocity of a particle moving along a straight line changes with time as $V(\text{m/s}) = 4 \sin(\pi/2)t$, its average velocity over time interval $t = 0$ to $t = 2(2n - 1)$ sec, (n being any +ve integer) is
(A) $\frac{8}{\pi(2n-1)} \text{ m/s}$ (B) $\frac{4}{\pi(2n-1)} \text{ m/s}$ (C) Zero (D) $\frac{16(2n-1)}{\pi} \text{ m/s}$
- A particle is moving in a straight line according to equation $x = \frac{t^3}{3} - \frac{5}{2}t^2 + 6t$. The time interval in which velocity i.e. instantaneous rate of change of position w.r.t. time is negative is
(A) $0 < t < 3$ (B) $0 < t < 2$
(C) $2 < t < 3$ (D) $t > 3$ and $t < 2$
- The graph below shows the velocity with respect to time of an object moving in a straight line. The positive direction is to the right and the negative direction is to the left. Which of the following statements best describes the motion of this object?

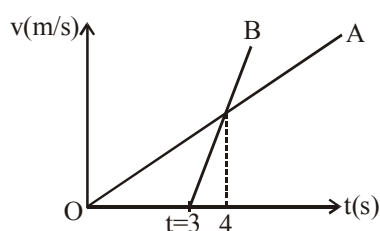


- The object starts at a location to the left of the origin and travels at a constant speed toward the right.
- The object starts at a location to the left of the origin at a slow speed and speeds up as it moves to the right.
- The object slows down as it moves to the left, stops, and starts moving to the right.
- The object slows down as it moves to the right, stops, and continues moving to the right.

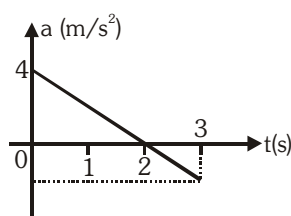
7. Two object starts from rest and their acceleration is shown in figure. The time when their relative velocity is again zero is :-



- (A) 2 sec (B) 4 sec (C) 1 sec (D) 6 sec
8. v-t graph of two boys A and B starting from same point but at different time is given. Time when they cross each other, is :-



- (A) 4 sec (B) 6 sec (C) 8 sec (D) 16 sec
9. The acceleration-time graph of a particle moving in x-axis with initial velocity $\vec{u} = -3\hat{i}$ m/s is given. The velocity of the particle at $t = 3$ s is-



- (A) $-\hat{i}$ m/s (B) $-2\hat{i}$ m/s (C) $\hat{i}$ m/s (D) Zero

Linked Comprehension Type (Single option correct)

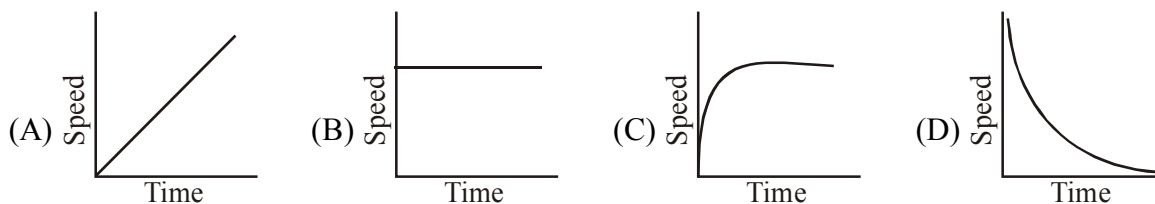
2 Para $\times$ 3 Q. [3(-1)]

Paragraph for Question Nos. 10 to 12

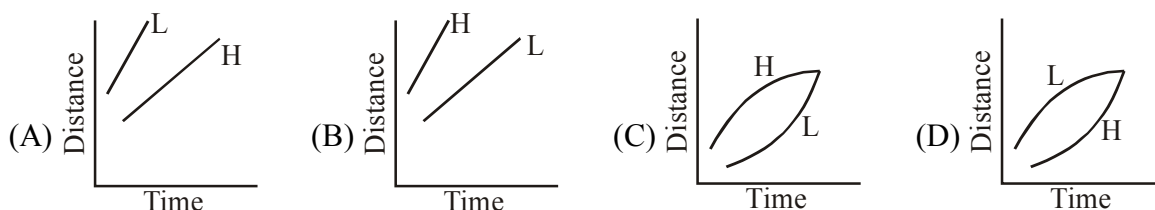
According to Aristotle, when an object is removed from its natural place, it possesses a tendency, to return to its place. Natural motion results from this tendency of an object to seek its natural place. Hence fire (or smoke), being naturally light, would rise, whereas rocks being naturally heavy, would fall. Furthermore, Aristotle asserted that any object, after it is released quickly reaches some final speed which it maintains to the end of its path. When we pick up a stone and release it, the stone strives to return to its natural place, the earth, and quickly gains a speed that it maintains during its entire fall. From the common observation that a heavy stone falls faster than a feather, Aristotle reasoned that mass is a factor that governs the speed of the fall. Consequently, the heavier an object, the greater would be its potential to return to the earth. In turn this stronger tendency would cause a greater speed of fall. Aristotle's description of motion agreed well with common observations of falling leaves, raindrops, and stones. In all cases, the body encounters resistance to its fall from the air.

Galileo attempted to show that Aristotle's belief was logically inconsistent by the following argument. The m and M ($m < M$) together to form a double stone. Then, during fall, m should retard M , since it tends to fall more slowly, as it falls down individually. So the combination would fall faster than m but more slowly than M ; but according to Aristotle the double body ($M + m$) is heavier than M and hence should fall faster than M .

10. Which of the following graph of speed versus time correctly illustrates Aristotle's description of falling body motion ?



11. Which of the following graphs, of distance fallen versus time for a heavy body H and light body L , at the end of path, would an Aristotelian construct ?

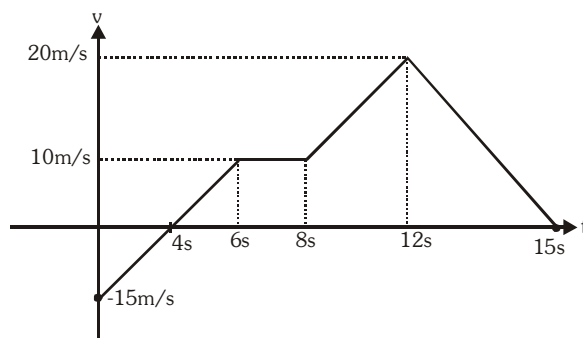


12. Choose correct option :

- (A) According to Aristotle, all the bodies must take equal time in falling equal heights
(B) According to Aristotle, falling of bodies towards earth is because of earth's attraction force
(C) According to Galileo, falling of bodies towards earth is because of earth's attraction force
(D) According to Aristotle, falling of bodies towards earth is a natural tendency of a body

Paragraph for Question 13 to 15

A moving particle is acted upon by three forces at different times to bring it to rest. Its velocity versus time graph is given below



13. The average speed for the first 6 s is

- (A) zero (B) $\frac{5}{3} \text{ ms}^{-1}$ (C) $\frac{10}{3} \text{ ms}^{-1}$ (D) $\frac{20}{3} \text{ ms}^{-1}$

14. The average velocity for the first 12s is
(A) zero (B) 5 m/s (C) $\frac{10}{3}$ m/s (D) 10 m/s
15. The average acceleration from $t = 5$ s to $t = 15$ s is
(A) zero (B) -0.5 m/s^2 (C) $+0.5 \text{ m/s}^2$ (D) 1 m/s^2

SECTION-II**Numerical Answer Type Question****1 Q. [3(0)]****(upto second decimal place)**

1. The position coordinate of a particle that is confined to move along a straight line is given by $x = 2t^3 - 24t + 6$ where x is measured from a convenient origin and t is in seconds. Determine the distance (in m) travelled by the particle during the interval from $t = 1$ sec to $t = 4$ sec.

SECTION-III**Numerical Grid Type (Ranging from 0 to 9)****2 Q. [4(-0)]**

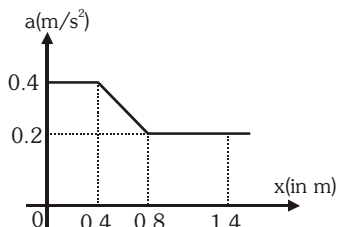
1. Two identical pellet guns are fired simultaneously from the edge of a cliff. These guns impart as initial speed of 35.0 m/s to each pellet. Gun A is fired straight upward, with the pellet going up and falling back down, eventually hitting the ground beneath the cliff. Gun B is fired straight downward. In the absence of air resistance, how long after pellet B hits the ground does pellet A hit the ground? [$g = 10 \text{ m/s}^2$]
2. An engineer works at a plant out-of-town. A car is sent for him from the plant every day that arrives at the railway station at the same time as the train he takes. One day the engineer arrived at the station 10 minutes before his usual time and, without waiting for the car, started walking to work. On his way he met the car and reached his plant 4 minutes before the usual time. How long (in minute) did the engineer walk before he met the car?

SECTION-I

Single Correct Answer Type

4 Q. [Marks 3 (-1)]

1. The acceleration of a particle which moves along the positive x-axis varies with its position as shown. If the velocity of the particle is 0.8 m/s at $x = 0$, the velocity of the particle at $x = 1.4$ is (in m/s) :-



- (A) 1.6 (B) 1.2 (C) 1.4 (D) None of these
2. The maximum possible acceleration of a train moving on a straight track is 10 m/s^2 and maximum possible retardation is 5 m/s^2 . If maximum achievable speed of train is 10 m/s then minimum time in which train can complete a journey of 135 m starting from rest and ending at rest, is :-
- (A) 5s (B) 10s (C) 15s (D) 20s
3. Having gone through a plank of thickness h , a bullet changed its velocity from v_0 to v . Find the time of motion of the bullet in the plank, assuming the resistance force to be proportional to the square of the velocity.

(A) $t = \frac{h(v_0 + v)}{v_0 v \ln(v_0/v)}$ (B) $t = \frac{v_0 v \ln(v_0/v)}{h(v_0 - v)}$ (C) $t = \frac{h(v_0 - v)}{v_0 v \ln(v_0/v)}$ (D) $t = \frac{h(v_0 - v)}{v_0 v \ln(v \times v_0)}$

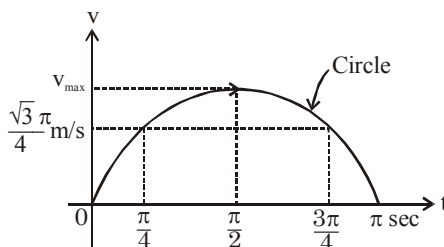
4. If a particle moves along a straight line according to the law $v = 2(x \sin x + \cos x)$ then find its acceleration (i.e. $\frac{dv}{dt}$) at $x = \pi/2$:-

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{2\sqrt{2}}$ (C) $\frac{\pi}{4\sqrt{2}}$ (D) zero

Multiple Correct Answer Type

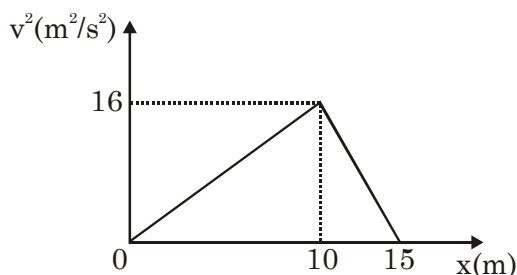
3 Q. [Marks 4 (0)]

5. Velocity time graph of a particle is as shown in figure which have geometry of a circle. At $t = 0$ particle is at origin.

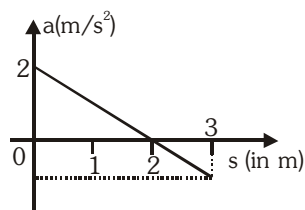


- (A) Displacement of particle from $t = 0$ to $t = \frac{\pi}{4}$ sec is $\frac{\pi^2}{8} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) \text{ m}$
- (B) Displacement of particle from $t = 0$ to $t = \frac{\pi}{4}$ sec is $\frac{\pi^3}{32} \text{ m}$
- (C) Displacement of particle from $t = 0$ to $\frac{\pi}{4}$ sec is $\frac{\sqrt{3}\pi^2}{32} \text{ m}$
- (D) Magnitude of acceleration of particle first decreases then increases.

6. A particle is moving on a straight line along x-direction. A graph between square of its velocity & position is drawn as shown below. Choose the **CORRECT** statement(s) :-



- (A) Acceleration of the particle is 0.8 m/s^2 for $0 < x < 10$.
 (B) Acceleration of the particle is -3.2 m/s^2 for $10 < x < 15$.
 (C) Velocity of particle first increases linearly with position till 10m & then decreases linearly with position till 15 m.
 (D) The particle's velocity increase linearly with time for first five seconds & decreases linearly with time till it stops.
7. The acceleration-displacement graph of a particle moving in x-direction is shown. If the initial speed of the particle is $\vec{u} = -3\hat{i} \text{ m/s}$, the velocity of the particle at $s=3\text{m}$ is-



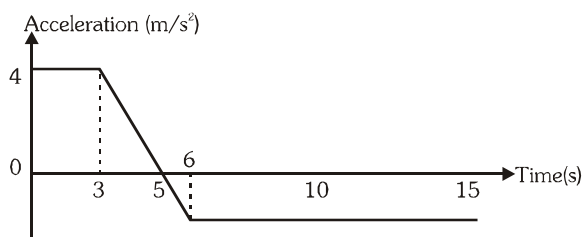
- (A) $\sqrt{13}\hat{i} \text{ m/s}$ (B) $\sqrt{14}\hat{i} \text{ m/s}$ (C) $-2\sqrt{3}\hat{i} \text{ m/s}$ (D) $2\sqrt{3}\hat{i} \text{ m/s}$

Linked Comprehension Type

(3 Para $\times$ 3 Q.) [Marks 3 (-1)]

Paragraph for Questions no. 8 to 10

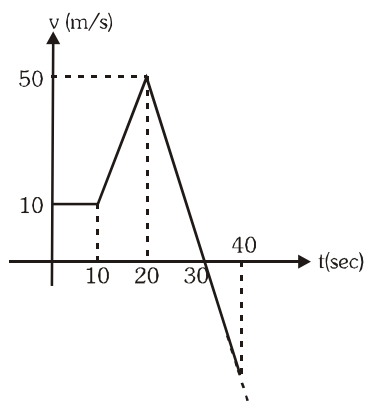
A particle starts from rest from the origin at the instant $t = 0 \text{ s}$ and moves along the x-axis. During first 15s of its motion, it is subjected to acceleration, which varies according to the given graph. After the instant $t = 15 \text{ s}$ acceleration vanishes.



8. Its speed increases during the time interval or intervals
 (A) 0 s to 3 s only (B) 0 s to 5 s only
 (C) 0 s to 3 s and 13.5 s to 15 s (D) 0 s to 5 s and 13.5 s to 15 s
9. Its speed decreases during the time interval
 (A) 3 s to 5 s (B) 3 s to 6 s (C) 5 s to 6 s (D) 5 s to 13.5 s
10. It reverses direction of motion at the instant
 (A) $t = 3 \text{ s}$ (B) $t = 5 \text{ s}$ (C) $t = 6 \text{ s}$ (D) $t = 13.5 \text{ s}$

CLASS TEST

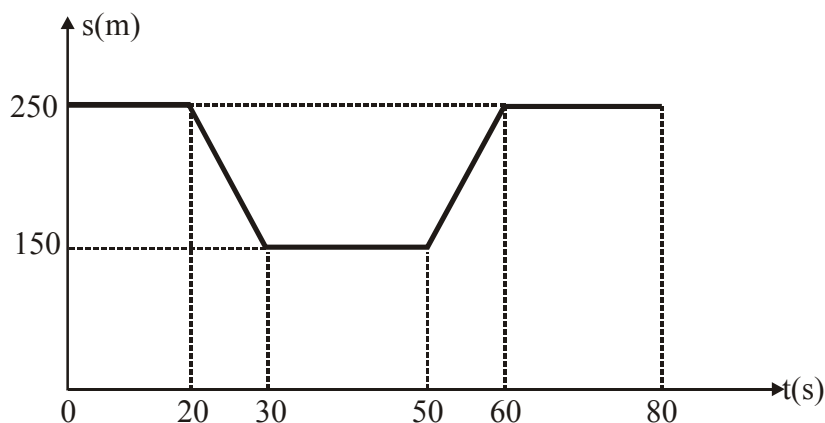
Paragraph for Question 11 to 13



11. What is the displacement (in m) in 40 sec?
(A) 450 (B) 500 (C) 400 (D) 600
12. What is the velocity (in m/s) at $t = 12$ sec?
(A) 19 (B) 18 (C) 20 (D) 21
13. For what time in seconds ($t > 0$) the displacement of the particle is zero?
(A) $2\sqrt{60} + 30$ (B) $2\sqrt{63} + 30$ (C) $2\sqrt{67} + 30$ (D) $2\sqrt{65} + 30$

Paragraph for Question No. 14 to 16

A long straight road connects two towns A and B. There are two small towns C and D between towns A & B. The condition of road between towns C and D is bad. Two cars travel from A to B at constant speed v_1 everywhere except between towns C and D where they travel at speed v_2 . In the following graph, separation between the cars is shown w.r.t. time for a time interval of 80 seconds.



14. The value of v_1 is
(A) 10 m/s (B) 15 m/s (C) 20 m/s (D) 25 m/s
15. The value of v_2 is
(A) 10 m/s (B) 15 m/s (C) 20 m/s (D) 25 m/s
16. The distance between towns C and D is
(A) 300 m (B) 400 m (C) 450 m (D) 500 m

SECTION-II

Numerical Answer Type Question

1 Q. [3(0)]

(upto second decimal place)

1. A big Diwali rocket is projected vertically upward so as to attain a maximum height of 160m. The rocket explodes just as it reaches the top of its trajectory sending out luminous particles in all possible directions all with same speed v . The display, consisting of the luminous particles, spreads out as an expanding, brilliant sphere. The bottom of this sphere just touches the ground when its radius is 80m. With what speed (in m/s) are the luminous particles ejected by the explosion?

SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9)

2 Q. [Marks 4(0)]

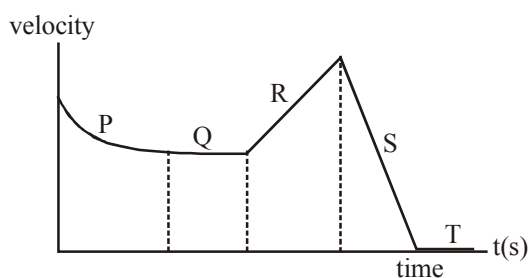
1. The acceleration of a marble in a certain fluid is proportional to the speed of the marble squared and is given (in SI units) by $a = -\frac{v^2}{3}$ for $v > 0$. If the marble enters this fluid with a speed of 1.50 m/s, how long will it take before the marble's speed is reduced to half of its initial value?
2. A particle moves along a straight line. The acceleration of particle as function of time is given by $a = 6t - 12$. Initial velocity of particle is 9 m/s and distance traveled by particle in 5 sec is $7x$ (in m). Find the value of x .

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Study the velocity-time graph pictured below and match each segment of the graph with the description of this motion at that time.



Column-I

- (A) The object is motionless in
- (B) The object is moving with constant velocity in
- (C) The object is having acceleration in
- (D) The object is travelling in north but accelerating in south in

Column-II

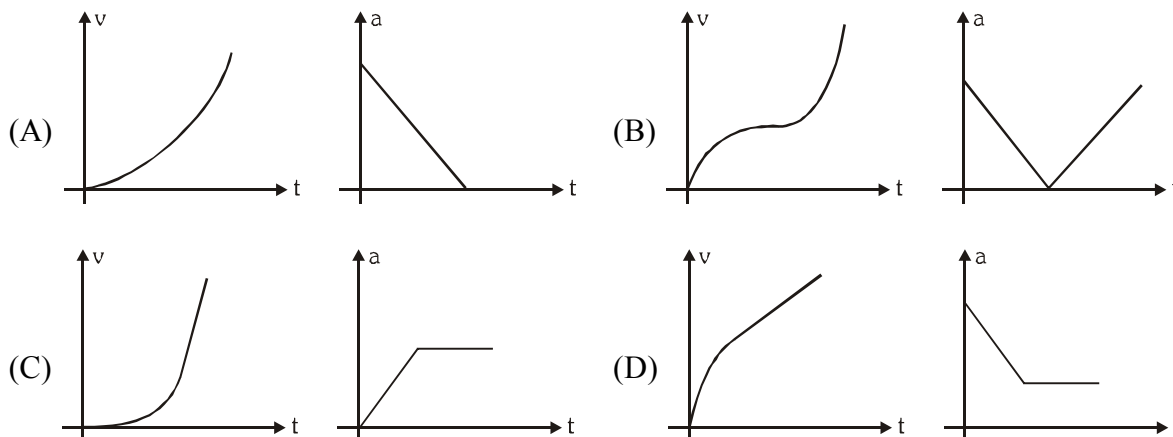
- (P) Segment
- (Q) Segment
- (R) Segment
- (S) Segment
- (T) Segment

SECTION-I

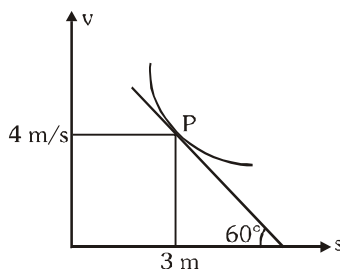
Single Correct Answer Type

7 Q. [Marks 3 (–1)]

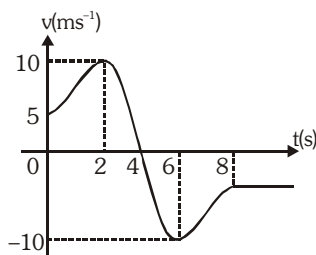
1. Which of the following pairs of graphs does not represent motion of the same particle in the same interval



2. A rectangular farm- house has a 1 km difference between its sides. Two farmers simultaneously leave one vertex of the rectangle for a point at the opposite vertex. One farmer crosses the farmhouse along its diagonal and other walks along the bank. The speed of each farmer is 4km/hr. If one of them arrives half an hour earlier than the other then the size of farmhouse is-
- (A) 12 km $\times$ 13km (B) 4 km $\times$ 5km (C) 3 km $\times$ 4km (D) None of these
3. A particle is moving along a straight line whose velocity-displacement graph is as shown in figure : A tangent is drawn at point P on the graph. At the point P

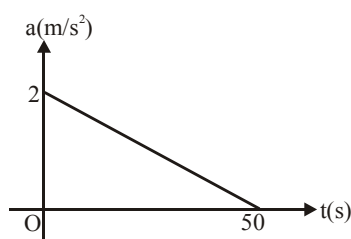


- (A) the particle is speeding up
(B) numerical value of velocity and acceleration of the particle are equal
(C) numerical value of velocity is more than the numerical value of acceleration of the particle
(D) numerical value of acceleration is more than the numerical value of velocity of the particle
4. A particle is moving in a straight line $y=3x$. Its velocity time graph is shown in figure. Its speed is minimum at $t = \dots\dots\dots$



- (A) 2s (B) 4s (C) 6s (D) 8s

5. A man travelling in a car with maximum constant speed of 20 ms^{-1} watches his friend, who starts off at a distance of 100 m ahead on a motor cycle with constant acceleration 'a'. The maximum value of 'a' for which the man in the car can reach his friend will be-
(A) 2 ms^{-2} (B) 1 ms^{-2} (C) 4 ms^{-2} (D) 0.5 ms^{-2}
6. A bus is beginning to move with an acceleration of 1 m/sec^2 . A boy who is 48 m behind the bus starts running at 10 m/sec. The times at which the boy can catch the bus will be-
(A) 8 sec (B) 10 sec (C) 12 sec (D) 14 sec
7. Referring to a-t diagram, find the velocity of the particle at $t = 20 \text{ s}$. Assume that the particle was at the origin at $t = 0$ and was moving with a velocity of 10 m/s in -ve x-direction :-

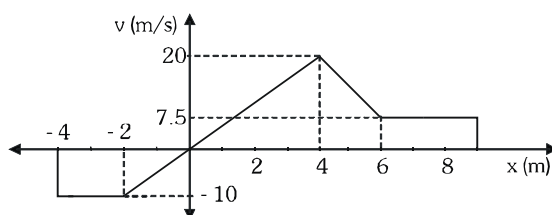


- (A) 20 (B) 22 (C) 18 (D) 15

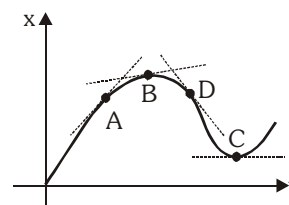
Multiple Correct Answer Type

6 Q. [Marks 4 (0)]

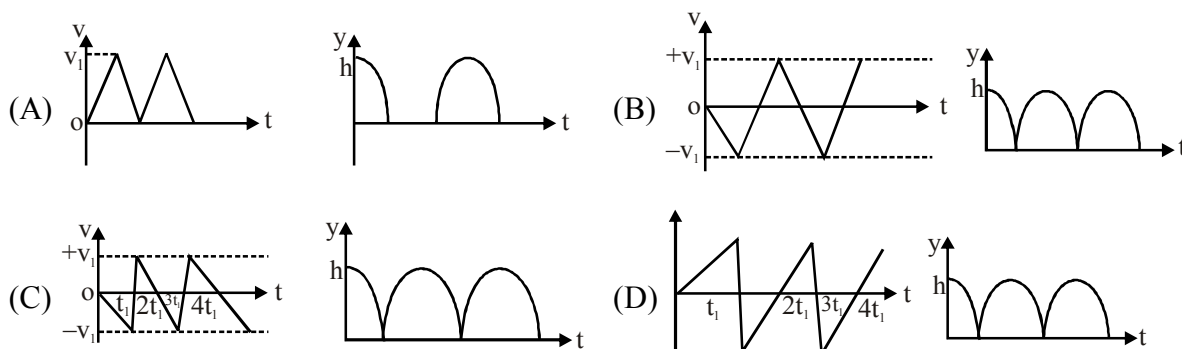
8. Velocity-position graph of a particle moving along the x-axis is shown in the figure. Which of the following function most suits the graph?



- (A) $v = -10 \text{ m/s}$ in position interval $[-4\text{m}, -2\text{m}]$
(B) $v = 5x \text{ m/s}$ in position interval $[-2\text{m}, 4\text{m}]$
(C) $v = 7.5 \text{ m/s}$ in position interval $[6\text{m}, 9\text{m}]$
(D) $v = 7.5 \text{ m/s}$ at $x = 5\text{m}$
9. A particle is moving along a straight line $y - 2 = 0$. Its displacement-time graph is shown in figure. Point A, B, C and D are marked on graph at four different instants. Select the correct alternative (s). [Given velocity $v = dx/dt$]
(A) Magnitude of velocity at A is greater than magnitude of velocity at B.
(B) Magnitude of velocity at B is greater than magnitude of velocity at C.
(C) Magnitude of velocity at C is greater than magnitude of velocity at D.
(D) Magnitude of velocity at D is greater than magnitude of velocity at C.
10. The position of a particle varies according to the expression $x = t(t - 1)(t - 2)$ then
(A) velocity will be zero at $t_2 = 1 - \frac{1}{\sqrt{3}}$ second that $t_2 = 1 + \frac{1}{\sqrt{3}}$ sec
(B) Acceleration changes its direction between $t_1 = 0$ and $t_2 = 2$
(C) Acceleration remains constant in direction between $t_1 = 0$ and $t_2 = 2$
(D) None of these

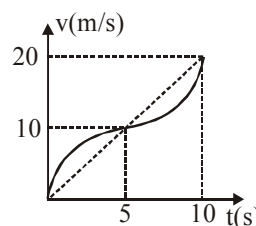


11. Consider a rubber ball freely falling from a height $h = 4.9$ m onto a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic. Then the velocity as a function of time will be

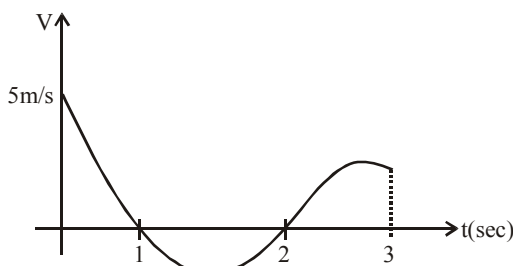


12. A velocity-time graph of a car moving on a straight road is shown below. The correct statement(s) is/are:-

- (A) Acceleration is negative in interval $t = 0$ to $t = 5$ s
(B) Velocity is zero at $t = 5$ s
(C) Car never changes its direction
(D) Displacement in interval $t = 0$ to $t = 5$ is greater than 25 m.



13. A particle is moving along x axis starting from $x = -20$. Figure shows the velocity time graph of a particle. Taking positive direction from origin towards positive x axis, which of the following statement is/are **INCORRECT** :-

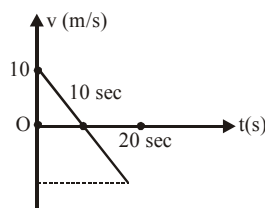


- (A) During 0 to 1 sec particle is moving towards origin
(B) During 1 to 2 sec speed is increasing continuously
(C) During 2 to 3 sec particle is slowing down
(D) During 2 to 3 sec acceleration is positive

Linked Comprehension Type (1 Para $\times$ 3Q.) (1 Para $\times$ 2 Q.) [Marks 3 (-1)]

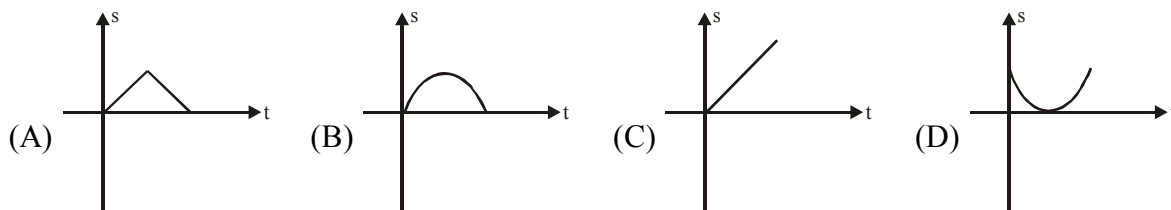
Paragraph for question 14 to 16

A particle is moving along a straight line. Ram observes the motion and draw its v-t graph which is given. But he has to do a lot of home-work in mathematics so he call his brother Shyam and let him draw remaining graphs. Shyam was confused and he made certain conclusions now you have to help Ram. Choose appropriate conclusions.



14. What is the total distance travelled by the particle from 0 to 20 sec
(A) 0 m (B) 50 m (C) 100 m (D) 200 m

15. Displacement v/s time(s-t) graph is

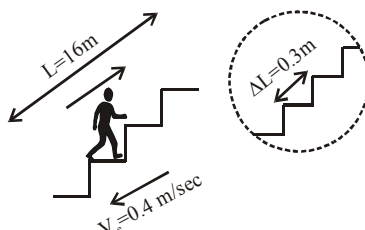


16. Which of the following statements about the particle is correct

- (A) slows down for entire 20 sec
- (B) slows down for first 10 sec and then speed up
- (C) speeds up for first 10 sec and then slows down
- (D) speeds up for entire 20 sec

Paragraph for Question no. 17 and 18

Kids enjoy running up & down escalators in departmental stores, specially in the wrong directions as shown in diagram. Consider an escalator going down an incline at speed $V_e = 0.4$ m/sec with steps that are $\Delta L = 0.3$ m apart. The full length of escalator is $L = 16$ m.



17. Suppose a kid can run down the length of the escalators in $\frac{1}{3}$ of the time that it takes to run up the escalator but his running speed V_r relative to the escalators steps is the same in each case. What is his running speed V_r :-
- (A) 0.8 m/sec (B) 0.6 m/sec (C) 1 m/sec (D) 1.2 m/sec
18. If his running speed relative to the escalator steps is $V_r = 1.2$ m/sec, how many steps do he pass on the way up the escalator ?
- (A) 140 steps (B) 100 steps (C) 120 steps (D) 80 steps

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

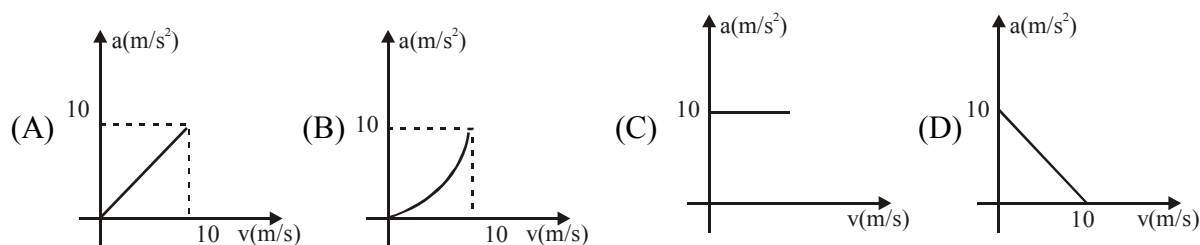
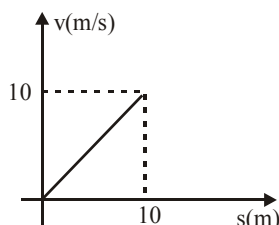
- A bird is flying in a straight line initially at 10 m/s. It uniformly increases its speed to 15 m/s while covering a distance of 25 m. The time taken by the bird in seconds is.
- A car, which can have constant acceleration of 4 m/s^2 and a constant retardation of 8 m/s^2 is travelling on a high way where maximum speed limit is 72 km/h. It starts from rest and comes to rest in minimum time t_1 after travelling 300 meter and if there is no speed limit then minimum time taken is t_2 , to cover the same distance. Then find the value of $\frac{4t_1}{t_2}$.

SECTION-I

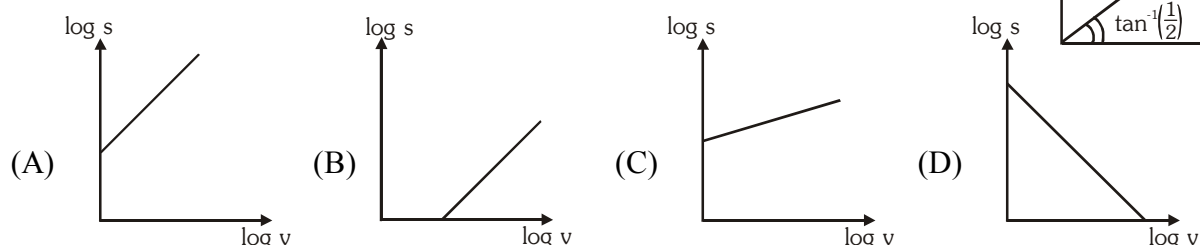
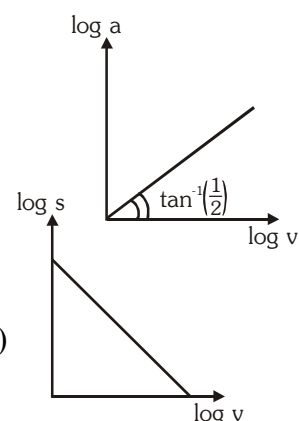
Single Correct Answer Type

5 Q. [3 M (-1)]

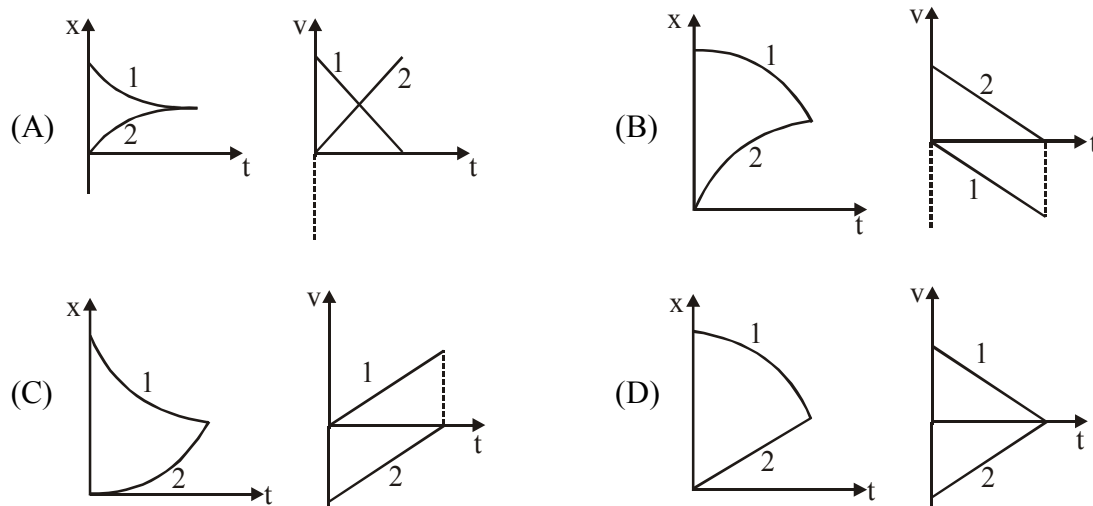
1. Velocity versus displacement graph of a particle moving in a straight line is shown in figure. Corresponding acceleration versus velocity graph will be :-



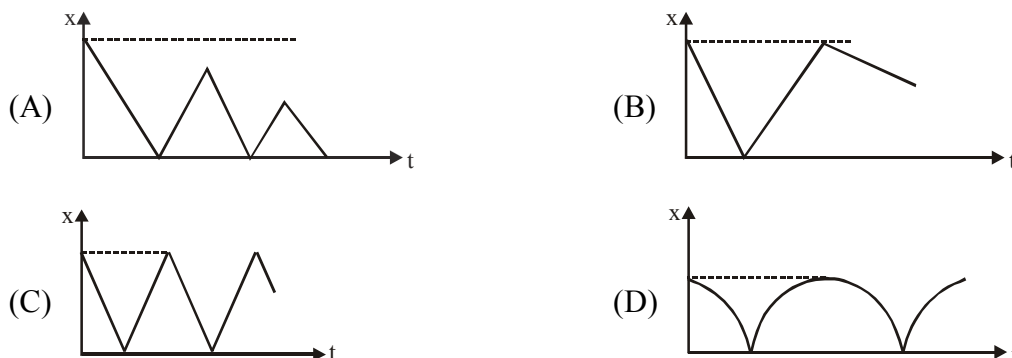
2. Figure shows the graph of acceleration a and velocity v on the log-log scale. Which of the following graphs represent the plot of displacement s and velocity v on log-log scale. (Assume the initial velocity to be zero?)



3. A ball 1 is dropped from top of a tower. Another ball 2 is thrown upwards along the same line. They collide in midair. Which of the following pairs of x - t & v - t graph correctly represents the situation :-



4. A ball is between two walls, where on one of the walls A is the origin. The ball is rolled towards wall A from wall B at a constant speed. The ball after striking each wall starts rolling in opposite direction at a constant speed half of that before striking. The correct x-t (position-time) graph will be :-

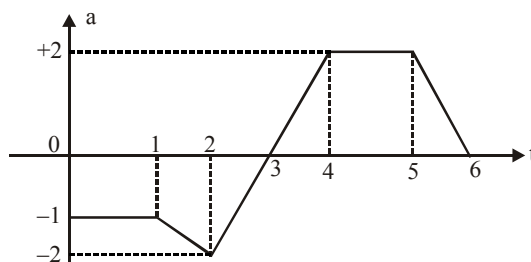


5. Three balloons B_1 , B_2 and B_3 are moving vertically upwards with velocities v , $2v$ and $3v$ respectively, and at some instant they are at the same height. If at this instant, each one of the balloons drops identical bombs, which of the following will happen ?
- (A) Bomb from B_1 will reach the ground first (B) Bomb from B_2 will reach the ground first.
(C) Bomb from B_3 will reach the ground first (D) All the bombs will reach the ground at the same time.

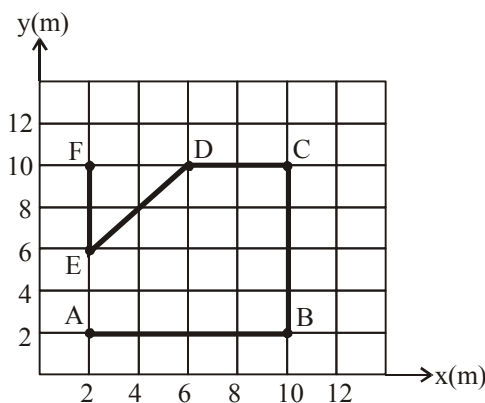
Multiple Correct Answer Type

5 Q. [4 M (-1)]

6. If initial velocity of the particle is 2.5 m/s , the particle will be moving along the positive direction :-



- (A) 0 to 2sec (B) 2 to 4 sec (C) 4 to 6 sec (D) None of these
7. Look at the picture given below. An object starts moving from point A through B, C, D, E and stops at point F.



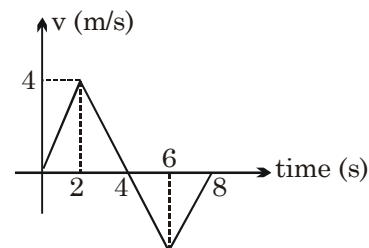
- (A) The magnitude of displacement AC of the object is $8\sqrt{2} \text{ m}$.
(B) The distance traveled by the object from point A to D is 20 m.
(C) The magnitude of average acceleration of the body for total path is zero.
(D) $\vec{BF} = -8\hat{i} + 8\hat{j}$.

8. A body travelling along a straight line with a uniform acceleration has velocities 5 m/s at a point A and 15 m/s at a point B respectively. If M is the mid point of AB, then :
- (A) The ratio of times taken by the body to cover distance MB and AM is $\left[\frac{\sqrt{5}-1}{2} \right]$
- (B) The velocity at M is $5\sqrt{5}$ m/s
- (C) Average velocity over AM is $\frac{5(\sqrt{5}+1)}{2}$ m/s
- (D) The product of the acceleration and the distance AB is $100 \text{ m}^2/\text{s}^2$.
9. A particle is projected vertically upwards from ground with a velocity 50 m/sec & motion is considered till particle reaches the ground again. Find the **CORRECT** statement(s) :
- (A) It's distance covered in 5th sec & 6th sec will be equal.
- (B) For first 4 seconds distance & displacement are same
- (C) Displacement in first & last second of the motion are equal.
- (D) Distance in first & last second of the motion are equal.
10. The engine of a car can produce a maximum acceleration of 5 m/sec^2 . Its brakes can produce a maximum retardation of 10 m/s^2 . The driver of car starts from point A and reaches point B. Whatever be the distance between A & B, driver always takes minimum time to reach from point A to point B. Then choose **CORRECT** statement(s), if maximum speed the car can attain is 50 m/s :-
- (A) If distance between A & B is 375 m then maximum velocity gained by car is 50 m/sec.
- (B) If distance between A & B is 300 m then maximum velocity gained by car is $20\sqrt{5}$ m/sec.
- (C) Time taken by car is 20 sec when distance between A & B is 625 m.
- (D) Time taken by car is 15 sec. when distance between A & B is 625 m.

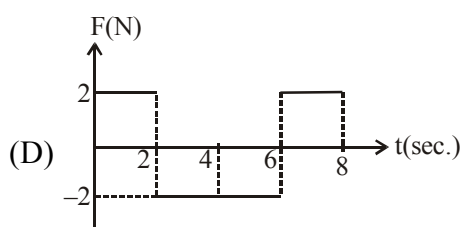
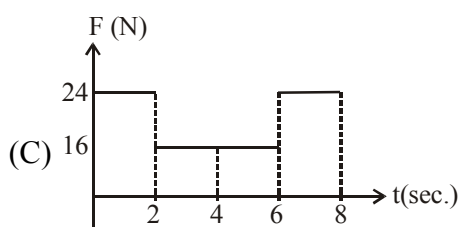
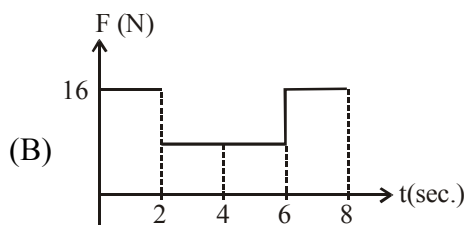
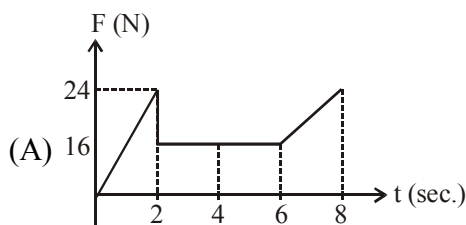
Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question no. 11 and 12

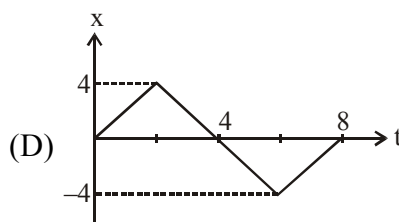
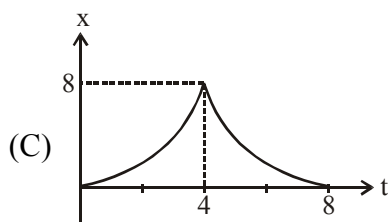
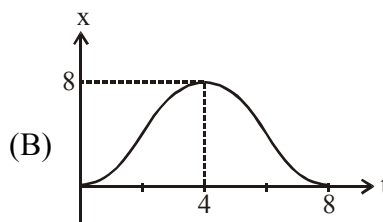
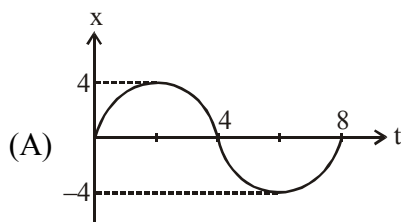
In a certain building, a lift moves with a constant magnitude of acceleration. The v-t graph (upward velocity being taken positive) shows the lift's motion between the ground floor and the first floor. The lift starts at $t = 0$ from ground floor and comes back to the ground again after reaching upto the first floor.



11. A spring balance is attached to the ceiling of the lift. Plot a graph for the reading of the spring balance (in Newtons) with respect to time if a 2 kg mass is hung from it ($g = 10 \text{ m/s}^2$) Ignore any fluctuation in reading during change in acceleration of lift.



12. Draw the x-t graph for the motion taking $x = 0$ at ground level and upward direction as positive.



SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

7 Q. [4 M (0)]

1. A body starts moving on x-axis from rest from $x = -2$ m. Acceleration $\vec{a}$ of the particle varies according

to the equation $\vec{a} = -(\pi \text{ rad/sec})^2 (\vec{x})$ for $0 \leq t \leq \frac{1}{2}$ sec, $\vec{a} = (5\pi \text{ m/s}^2) \left(\frac{\vec{x}}{|\vec{x}|} \right)$ for $t > \frac{1}{2}$ sec,

where $\vec{x}$ is the position vector of the particle at any instant t . Then velocity of the particle at $x = \frac{\pi}{2}$ m is

$A\pi$ m/s then the value of "A" is :-

- An object loses 20% of its speed after running 108 m. If maximum distance travelled by the object before stopping is $100x$ (in meter). Then value of x is.
- A body moving with uniform acceleration passes over distances of 12 and 22 m in the seventh and twelfth seconds respectively. Find its acceleration (in m/s^2).
- Velocity of particle depend on x as $v = 4x^2 - 4$ where x is positive of the particle then find acceleration at $x = 1$ is.
- An engine-driven rocket rises from the ground vertically upwards with a uniform velocity $V = 2$ m/s. The engine stops functioning after some time and rocket falls back on the ground. If the total time of its journey is $t = \frac{5}{2}$ sec, the height (in m) at which the engine stopped functioning. (Assume 'g' to be constant).
- A body covers one-third of the distance with speed v_1 . For the remaining distance, it travels for half of the time with speed v_2 & another half of the time with speed v_3 . Find its average speed. If average speed is $\frac{nv_1(v_2 + v_3)}{4v_1 + v_2 + v_3}$. Fill n in OMR Sheet
- A particle moving with uniform acceleration along a straight line passes three successive points A, B & C where the distances AB : BC is 3 : 5 & the time taken from A to B is 40 sec. If the velocities at A & C are 5 m/s & 15 m/s respectively. Total distance from A to C is $K \times 10^2$ m. Value of K is.

CLASS TEST

SECTION-IV

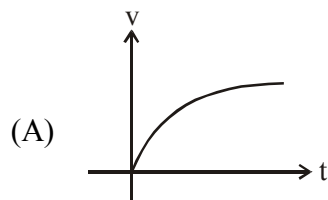
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

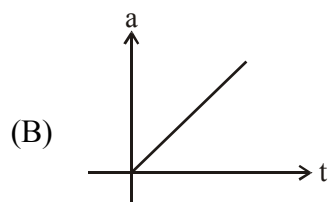
1. Column-I describes motion of a particles along a straight line and Column-II shows about various parameters

Column-I

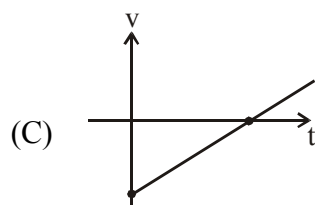
Column-II



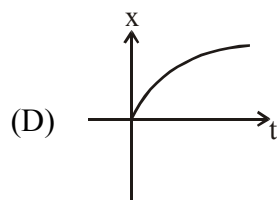
(P) Speed is increasing



(Q) Speed is decreasing



(R) Acceleration is +ive



(S) Acceleration is negative

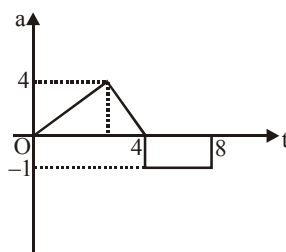
(T) Particle may change its directional motion

SECTION-I

Single Correct Answer Type

2 Q. [3 M (-1)]

1. The acceleration time graph of a particle is shown in the figure. What is the velocity of particle at $t = 8$ s if its initial velocity is 3 m/s ?



- (A) 4 m/s (B) 5 m/s (C) 6 m/s (D) 7 m/s
- 2.. A particle is dropped in uniform gravity ($g = 10 \text{ m/s}^2$) from the top of a tall building. A graph is plotted between velocity and displacement of particle. Magnitude of slope of this graph at $t = 1$ s :-

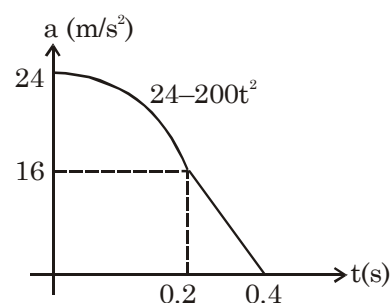
- (A) $\sqrt{3}$ (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{3}}$

Multiple Correct Answer Type

2 Q. [4 M (-1)]

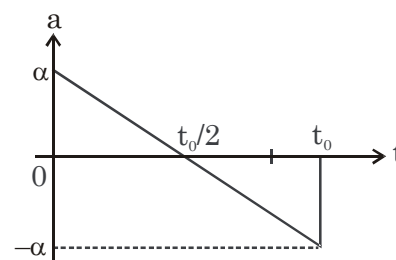
3. An accelerometer record for the motion of the given part of mechanism is approximated by an arc of a parabola for 0.2 sec and an straight line for next 0.2 sec as shown the diagram. Given that $v = 0$ when $t = 0$ and $x = 0.8$ m when $t = 0.4$ sec.

- (A) Acceleration of particle at $t = 0.3$ sec is 8 m/s^2 .
- (B) Velocity of particle at $t = 0.1$ sec is $\left(\frac{7}{3}\right) \text{ m/s}$.
- (C) Velocity of particle at $t = 0.3$ sec is 6 m/s .
- (D) Position of particle at $t = 0.2$ sec is 0.9 m



4. A person stands in an elevator. Starting at rest at $t = 0$ the elevator moves upward, coming to rest again at time $t = t_0$. The acceleration of the elevator during this period is shown graphically below. Which of the following statement(s) is/are **CORRECT**?

- (A) The maximum speed of the elevator is $\frac{\alpha t_0}{4}$
- (B) The total distance traveled by the elevator is $\frac{\alpha t_0^2}{6}$
- (C) Acceleration does not change sign during motion
- (D) Magnitude of acceleration continuously decreases



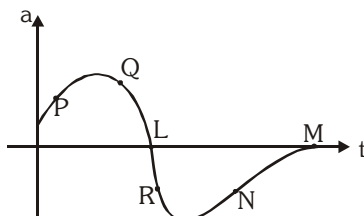
Linked Comprehension Type (Single Correct Answer Type)

(3 Para × 3Q.)

[3 M (-1)]

Paragraph for Question No. 5 to 7

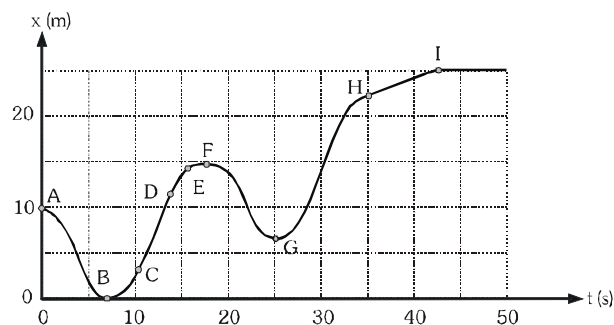
Acceleration–time graph of a moving particle is shown in figure



5. Does particle speeding up just before reaching at L?
Information I : Velocity of particle at L is in positive direction
Information II : Velocity of particle at L is in negative direction
 (A) Question can be solved by using information I only
 (B) Question can be solved by using information II only
 (C) Question can be solved by using either information I or information II
 (D) Question can't be solved by using either information I or information II
6. Does speed of particle at M is less than speed at L?
Information I : Velocity of particle at L is in positive direction
Information II : Velocity of particle at L is in negative direction
 (A) Question can be solved by using information I only
 (B) Question can be solved by using information II only
 (C) Question can be solved by using either information I or information II
 (D) Question can't be solved by using either information I or information II
7. Select the correct alternative
 (A) Particle at L is accelerating
 (B) Particle at M is accelerating
 (C) Particle at R is speeding up
 (D) None of these

Paragraph for Question 8 to 10

An observer records positions of a particle moving on a straight-line path at various instants of time. He starts his stopwatch when the particle is passing the point $x = 10$ m. With the help of these data he prepares the following graph, where position x is shown on the ordinate in meters and time t on the abscissa in seconds.



8. At the instant $t = 0$,
 (A) the particle was moving in the negative x -direction and the observer started his stopwatch.
 (B) the particle was moving in the positive x -direction and the observer started his stopwatch.
 (C) the particle started its motion with a negative velocity and the observer started his stopwatch.
 (D) the particle started its motion with a positive velocity and the observer started his stopwatch.
9. Speed of the particle
 (A) first increases then decreases in time interval between points B and F
 (B) first increases then decreases in time interval between points D and F
 (C) always increases between points H and I.
 (D) always decreases between points F and G.
10. The particle is changing its direction of motion at the instant corresponding to points
 (A) A and E only.
 (B) B, F and G only.
 (C) C, D, E and H only.
 (D) C, D, E, H and I only.

Paragraph for question nos. 11 to 13

A student designed a rocket. The rocket was launched from ground straight up into the air. At $t = 0$, the rocket is at $y = 0$ with $V_y(t = 0) = 0$. The velocity of the rocket is given by : $V_y = (24t - 3t^2)$ m/s for $0 \leq t \leq t_b$ where t_b is the time at which fuel burns out. Vertically upward direction is taken as positive. ($g = 10 \text{ m/s}^2$)

11. The expression for the acceleration $a_y(t)$ valid at all times in the interval $0 < t < t_b$ is
(A) $12t^2 - t^3$ (B) $24 - 6t$ (C) $24t - 6t^2$ (D) $24 - 6t - g$
12. The displacement of the rocket till the fuel burns out ($t = t_b$) is
(A) 128 m (B) 486 m (C) 203 m (D) 242 m
13. The time taken for rocket to reach its maximum height is
(A) 4 sec. (B) 8 sec. (C) 8.8 sec (D) 9.6 sec.

Linked Comprehension Type (1 Para $\times$ 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question no 14 and 15

Tim and Rick both can run at speed v_r and walk at speed v_w , with $v_w < v_r$. They set off together on a journey of distance D . Rick walks half of the distance and runs the second half. Tim walks half of the time and runs the other half.

14. Mark the incorrect statement :

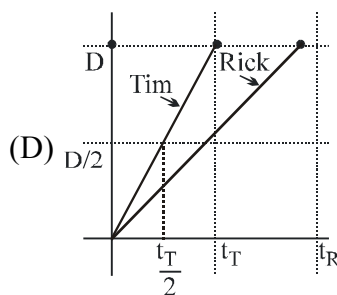
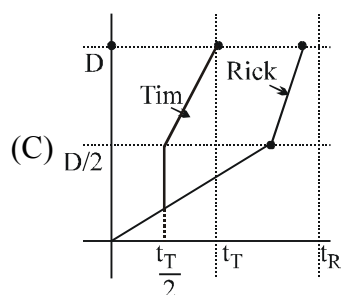
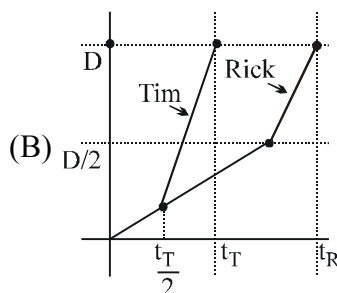
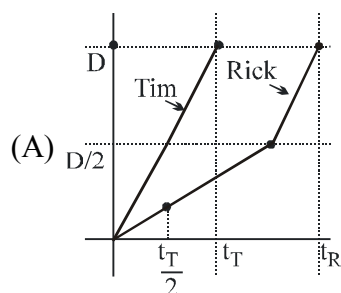
(A) Time taken by Rick to cover the distance D is $t_R = \frac{D}{2} \left(\frac{v_w + v_R}{v_w v_R} \right)$

(B) Time taken by Tim to cover the distance is $t_T = \frac{2D}{V_r + V_w}$

(C) Rick's average speed for covering the distance D is $\frac{2V_r V_w}{V_r + V_w}$

(D) Average speed of Rick for covering distance D is $\frac{V_r + V_w}{2}$

15. Which of the following graphs shows the positions of both Tim and Rick versus time.



Linked Comprehension Type

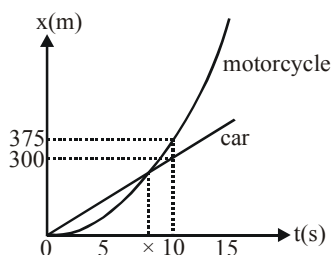
(1 Para × 2 Q.)

[4 M (–1)]

(Multiple Correct Answer Type)

Paragraph for Question No. 16 and 17

A car travelling at constant velocity, passes a stationary motor cycle at a traffic light. As the car overtakes the motorcycle, the motorcycle accelerates uniformly from rest. The following displacement-time graph represents the motions of both vehicles from the traffic light onwards.



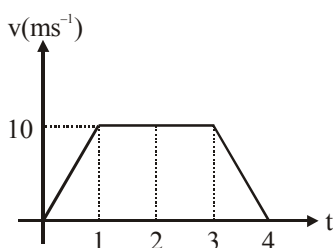
16. In which of following time interval/s speed of the motorcycle may be less than the speed of the car :-
(A) 0 to 2 sec (B) 0 to 4 sec (C) 0 to 8 sec (D) 5 to 10 sec
17. Choose the correct statement(s) :-
(A) The time at which motorcycle will cross the car is 8 sec
(B) Acceleration of the motorcycle is 7.5 m/s^2
(C) Average velocity of the motorcycle from starting point to the point of overtake is greater than that of car.
(D) Speed of the motorcycle at the point of overtake is equal to the speed of car.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. Find the modulus of average acceleration (in m/s^2) of the block from time $t=2$ sec to $t=4$ sec.



2. A person who had to catch a train having 16 coaches each 16 m long, when ran into the platform found that the train has already started moving. He could not board the train but saw the front of 10th coach, from the engine, pass by the position where front of 1st coach was positioned. It takes 8s for the rest of the train to pass. The uniform acceleration of the train is $\left(\frac{\alpha}{4}\right) \text{ m/s}^2$ then the value of α is :

SECTION-I

Single Correct Answer Type

2 Q. [3 M (-1)]

1. A street car moves rectilinearly from rest from station A to the next station B and stops. Its acceleration (f) is varying according to the law $f = a - bx$, where a and b are positive constants and x is the distance from station A. The distance between the two stations & the maximum velocity are :-

(A) $x = \frac{2a}{b}$; $v_{\max} = \frac{a}{\sqrt{b}}$

(B) $x = \frac{b}{2a}$; $v_{\max} = \frac{a}{b}$

(C) $x = \frac{a}{2b}$; $v_{\max} = \frac{b}{\sqrt{a}}$

(D) $x = \frac{a}{b}$; $v_{\max} = \frac{\sqrt{a}}{b}$

2. The velocity of a particle varies with time as $\vec{v} = -t\hat{i} - e^{-t}\hat{j} + \frac{c}{t}\hat{k}$. If the particle accelerates perpendicular to the direction of its motion after one second. The value of c is :

(A) $c = \sqrt{1 - e^2}$

(B) $\sqrt{e^2 - 1}$

(C) $\sqrt{\frac{e^2 + 1}{e^2}}$

(D) $\sqrt{1 - \frac{1}{e^2}}$

Multiple Correct Answer Type

8 Q. [4 M (-1)]

3. The body moves along the x-axis. At time $t = 0$, the velocity of particle is +5.0 m/s. The acceleration of particle in different time interval is given as:

$$a = \begin{cases} -2 \text{ m/s}^2 & 0 \leq t \leq 3 \text{ s} \\ -4 \text{ m/s}^2 & 3 \leq t \leq 6 \text{ s} \\ +6 \text{ m/s}^2 & 6 \leq t \leq 10 \text{ s} \end{cases}$$

- (A) The speed of particle at $t = 2.5$ s is zero. (B) The speed of particle at $t = 6$ s is -13 m/s.
(C) The speed of particle at $t = 7$ s is 7 m/s. (D) The speed of particle at $t = 10$ s is 11 m/s.
4. A particle starts from rest and the acceleration at any time is given by $a = f - kt^2$ where f and k are constants. If s & u represents the displacement and velocity of the particle respectively at the time when its acceleration is zero then :-

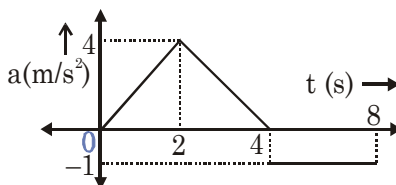
(A) $u = \frac{2}{3}\sqrt{\frac{f^3}{k}}$

(B) $u = \frac{1}{3}\sqrt{\frac{f^3}{k}}$

(C) $s = \frac{5f^2}{12k}$

(D) $s = \frac{f^2}{12k}$

5. $a - t$ graph of a particle is shown. At $t = 0$, $v = 3 \text{ m/sec}$



- (A) $v = 7 \text{ m/s}$ at $t = 8 \text{ s}$ (B) $v = 4 \text{ m/s}$ at $t = 8 \text{ s}$ (C) $v = 0 \text{ m/s}$ at $t = 4 \text{ s}$ (D) $v = 11 \text{ m/s}$ at $t = 4 \text{ s}$

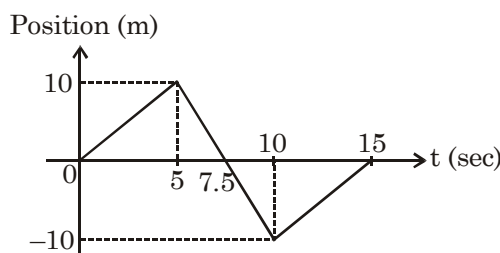
6. Position of a particle is given as

$$x = \frac{t^4}{4} - 2t^3 + \frac{11t^2}{2} - 6t + 5$$

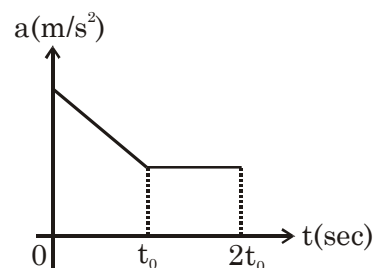
Choose the **CORRECT** option(s) :-

- (A) The particle moves on a straight path.
(B) The particle changes its direction of motion at $t = 1, 2$ & 3 sec .
(C) Minima of acceleration occurs at $t = 2 \text{ s}$.
(D) Distance travelled by particle from 0 to 2 sec is greater than magnitude of displacement from 0 to 2 s.

7. If position v/s time graph of a particle is given as shown in figure :



- (A) Average velocity of particle from $t = 0$ to $t = 15$ sec is zero
 (B) Average velocity of particle from time $t = 0$ to $t = 7.5$ sec is same as from $t = 0$ to $t = 15$ sec.
 (C) Average speed of particle from time $t = 5$ to $t = 10$ sec is 4 m/s
 (D) Maximum speed of the particle is 4 m/s
8. Position of a particle is given by $x = t^3 - 3t^2 - 6t + 5$, where t is the time (in sec.). Then :
 (A) Initially particle is at rest
 (B) Acceleration of particle is constant
 (C) Particle return it's path only once in it's motion.
 (D) Initially speed of particle is increasing.
9. An object is in motion with constant acceleration. Which of the following statement is/are incorrect?
 (A) The object must move in a straight line
 (B) The distance traveled must be proportional to the square of the time
 (C) The speed of the object must increase uniformly with time
 (D) None of these
10. The acceleration-time graph of a particle moving along the x-axis is given below. Initial velocity of particle is along the positive x-axis. Mark the **CORRECT** statement(s) :
 (A) Velocity of particle is decreasing in time interval $t = 0$ to $t = t_0$.
 (B) Velocity of particle is increasing in time interval $t = 0$ to $t = t_0$.
 (C) Velocity of particle is constant in time interval $t = t_0$ to $t = 2t_0$.
 (D) Velocity of particle is increasing in time interval $t = t_0$ to $t = 2t_0$.



Linked Comprehension Type (2 Para × 2Q.)

[3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question Nos. 11 and 12

Raindrops drip from a spout at the edge of a roof and fall to the ground. Assume that the drops drip at a steady rate of n drops per second (where n is large) and that the height of the roof is h .

11. How many drops are in the air at one instant ?

(A) $2n\sqrt{\frac{h}{g}}$ (B) $\frac{n}{2}\sqrt{\frac{h}{g}}$ (C) $n\sqrt{\frac{2h}{g}}$ (D) $n\sqrt{\frac{h}{2g}}$

12. What is the median height of these drops from the ground (i.e., the height above and below which an equal number of drops are found) ?

(A) $\frac{3h}{2}$ (B) $\frac{3h}{4}$ (C) $\frac{h}{4}$ (D) $\frac{h}{2}$

Paragraph for Question no. 13 and 14

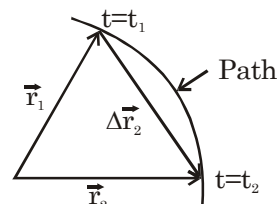
Let the particle 'P' move through an curve between two positions 1 and 2 at time $t = t_1$ and $t = t_2$ respectively. During the time interval $\Delta t (t_2 - t_1)$. The position vector of the particle changes from

$\vec{r}_1$ to $\vec{r}_2$ by $\Delta \vec{r}$ hence, on the average, the position vector changes with time at a rate $\frac{\Delta \vec{r}}{\Delta t}$.

$$\vec{v}_{av} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1}$$

For the instantaneous velocity $\Delta t \rightarrow 0$

$$\therefore \vec{v} = \lim_{\Delta t \rightarrow 0} \vec{v}_{av} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$



13. A particle moves so that its position vector varies with time as $\vec{r} = A \cos \omega t \hat{i} + A \sin \omega t \hat{j}$. The initial velocity of the particle will be :

(A) $A\omega \hat{i}$ (B) $-A\omega \hat{i} + A\omega \hat{j}$ (C) 0 (D) $A\omega \hat{j}$

14. The speed of the particle at any instant and average velocity in time interval from $t = 0$ to $t = \frac{\pi}{\omega}$ sec respectively :

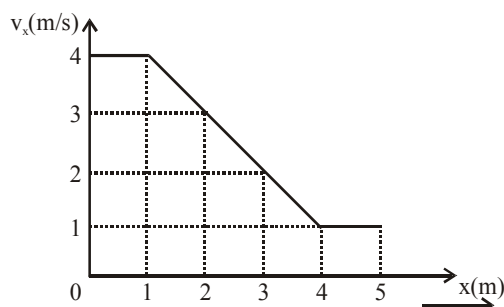
(A) $\frac{A\omega}{\pi}, \frac{A\omega}{\pi} \hat{j}$ (B) $\frac{2A\omega}{\pi}, \frac{-2A\omega}{\pi} \hat{i}$ (C) $A\omega, \frac{-2A\omega}{\pi} \hat{i}$ (D) $2A\omega, \frac{-2A\omega}{\pi} \hat{j}$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

1. A body moves in a straight line for which graph of v_x vs x is shown in the graph. Find the magnitude of acceleration of the body at $x = 3\text{m}$.



2. Earthquake produce several types of shock waves. The most known are the P-waves and the s-waves. In the earth's crust, the P-waves travel at around 6 km/s, while in the S-waves at about 3 km/s. The time delay between the arrival of these two waves at a seismic recording station tells geologists how far away the earthquake occurred. If the time delay is 50s and distance where earthquake occur from the seismic station in $100x$ km, the find x .
3. A ball is dropped from a tower of height 'H' on a planet named 'Gravito' having acceleration due to gravity of 2 m/s^2 . It was found that distance travelled by the ball in first 5 sec is same as travelled in last second of its journey. If the ball is projected downward with velocity 8.8 m/s from the tower of same height on earth, what will be time (sec) required by ball to reach the bottom of tower? Round off to nearest integer.
4. The velocity of a body moving along the x-axis is given by the equation $v = 3t^2$ where t is in sec. & v in m/s. The body is at $x = +36 \text{ m}$ at $t = 0$. The average velocity (in m/s) of the body in the time interval between $t = 0$ and $t = 3 \text{ sec}$.

SECTION-I

Single Correct Answer Type

3 Q. [3 M (-1)]

1. A juggler projects a ball upward in a room of height $H = 20\text{m}$. Ball's projection speed was such that it can just reach the ceiling. Now the ball is projected with two times of the previous velocity. After what time interval in second does this ball return to hand of juggler. Assume collision with roof results in reversal in direction of velocity with no change in magnitude.

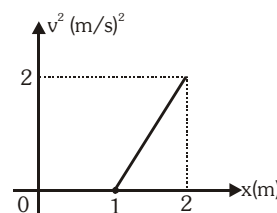
(A) $8 - 4\sqrt{3}$ (B) $8 + 4\sqrt{3}$ (C) $4 - 2\sqrt{3}$ (D) $4 + 2\sqrt{3}$

2. A projectile is thrown with a velocity of $10\sqrt{2} \text{ ms}^{-1}$ at an angle of 45° with vertical. The interval between the moments when speed is $5\sqrt{5} \text{ ms}^{-1}$: (Take $g = 10 \text{ ms}^{-2}$)

(A) 1.5 s (B) 0.5 s (C) 1.0 s (D) 2.0 s

3. The v^2 - x graph is given for a particle undergoing a straight line motion. The acceleration of the particle after $x = 1\text{m}$:

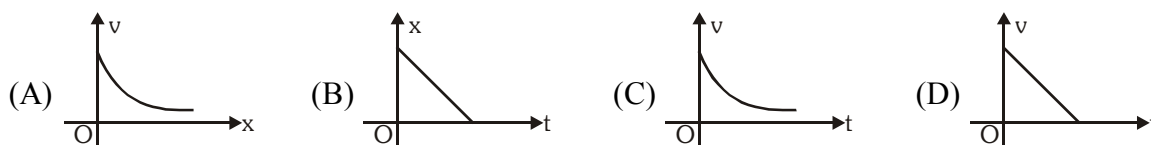
(A) $2\hat{i} \text{ m/s}^2$
(B) $1\hat{i} \text{ m/s}^2$
(C) $-\frac{1}{2}\hat{i} \text{ m/s}^2$
(D) $\frac{3}{2}\hat{i} \text{ m/s}^2$



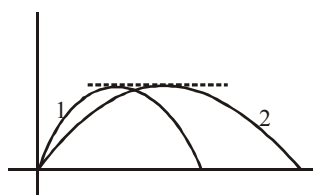
Multiple Correct Answer Type

5 Q. [4 M (-1)]

4. A motor car moves in a straight track with a retardation kv^2 where k is positive constant. Its initial velocity is v_0 at origin. Then, which of the following graphs is/are correct?

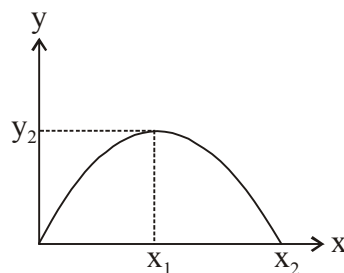
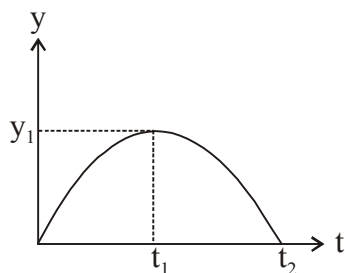


5. Trajectories of two stones projected from level ground is shown. Let T_1, T_2 be their time of flights and u_1, u_2 their speeds of projection then.



(A) $T_2 > T_1$ (B) $u_2 > u_1$ (C) $T_2 = T_1$ (D) $u_2 = u_1$

6. $y-t$ graph of a projectile parabola is drawn in figure 1 and its path is drawn in figure 2. (y vertical up, x horizontal).



- (A) $y_1 = y_2$ (B) $t_1 = 2t_2$ (C) $x_2 = 2x_1$ (D) $\frac{x_1}{x_2} = \frac{t_1}{t_2}$
7. Two particles are projected from a horizontal plane with the same initial velocity v_0 at two different angles of projection θ_1 and θ_2 , such that their ranges are the same. The ratio of their maximum heights reached is
(A) $\tan^2\theta_1$ (B) $\cot^2\theta_2$ (C) $\sin^2\theta_1 \operatorname{cosec}^2\theta_2$ (D) $\sin^2\theta_1 \cos^2\theta_2$
8. A particle is projected from a point O with a velocity u at an angle α (upwards) to the horizontal. At a certain point P it moves at right angles to its initial direction. It follows that :
(A) OP makes an angle $\tan^{-1}(u/2g)$ to the horizontal
(B) the distance of P from O is $u/(2g \sin \alpha)$
(C) the time of flight from O to P is $u/(g \sin \alpha)$
(D) the velocity of the particle at P is $u \cot \alpha$

Linked Comprehension Type (2 Para $\times$ 3Q.)

[3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question No. 9 to 11

Speed of a platform is decreasing uniformly at rate 2 m/s^2 . When its velocity is 4 m/s upwards, a stone is thrown from the platform vertically upwards. The stone falls back on the platform after 4 s . Assume acceleration due to gravity 10 m/s^2 .

9. Velocity of projection of the stone relative to the platform is
(A) 20 m/s (B) 24 m/s (C) 16 m/s (D) 12 m/s
10. What maximum height the stone would reach during its flight (height is measured from the instant when stone was thrown)
(A) 20 m (B) 24 m (C) 16 m (D) 12 m
11. What is the displacement of the stone relative to ground during its flight?
(A) zero (B) 32 m (C) 16 m (D) 20 m

Paragraph for Question No. 12 to 14

Two Aeroplanes, P and Q, move with constant velocities 3 ms^{-1} and 6 ms^{-1} along two mutually perpendicular straight tracks toward the intersection point O. At the moment $t = 0$, the Aeroplanes P and Q were located at distances 10^5 mm and $2 \times 10^4 \text{ cm}$ respectively from the point O.

12. The distance between them at time t is :-
(A) $\sqrt{(200)^2 + (100)^2} \text{ m}$ (B) $\sqrt{(200 - 6t)^2 + (100 - 3t)^2} \text{ m}$
(C) $[(200 - 6t) + (100 - 3t)] \text{ m}$ (D) $\sqrt{(200 - 3t)^2 + (100 - 6t)^2} \text{ m}$
13. At which time the distance between them will be shortest?
(A) $\frac{100}{3} \text{ s}$ (B) $\frac{125}{3} \text{ s}$ (C) $\frac{250}{3} \text{ s}$ (D) 50 s

14. The shortest distance between them is :-

- (A) $\frac{200\sqrt{5}}{3}$ m (B) $\frac{200\sqrt{3}}{5}$ m (C) $\frac{250}{3}$ m (D) None of these

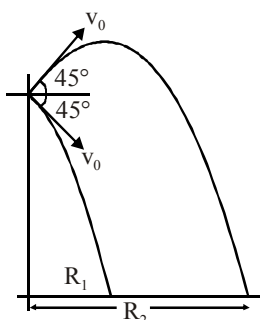
SECTION-II

Numerical Answer Type Question

2Q.[3(0)]

(upto second decimal place)

1. Two balls are thrown from the top of a cliff of unknown height with equal initial speed $v_0 = 10$ m/s one is projected at an angle $\theta = 45^\circ$ above horizontal and second is projected at the same angle below horizontal. What is difference between range (in m) of projectiles.



2. A projectile takes off with an initial velocity of 50 m/s at an angle of 37° with horizontal. It is just able to clear two hurdles of height 25 m each, separated from each other by a distance d (in m). Calculate d .

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. A particle moves in xy plane according to scheme $x = -8 \sin \pi t$ and $y = -2 \cos 2\pi t$ where t is time in s and x and y are in m. Find minimum distance (in m) of particle from origin.
2. A projectile is thrown with some initial velocity u at an angle θ to the horizontal. Its speed when it is at

the highest point is $\sqrt{\frac{2}{5}}$ times the speed v when it is at height half of the maximum height. Then ratio

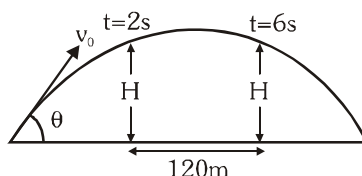
$$\frac{v}{u} = \sqrt{\frac{n}{8}}. \text{ Find the value of } n.$$

SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. A projectile crosses two walls of equal heights symmetrically as shown.



Column-I

- (A) Velocity of projection
(B) Height of each wall
(C) Maximum height of the projectile is
(D) Range of the projectile

Column-II

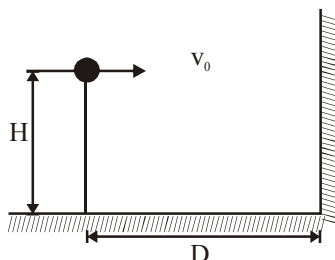
- (P) 60
(Q) 80
(R) 240
(S) 50
(T) 300

SECTION-I

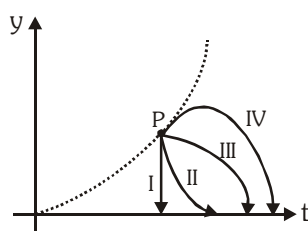
Single Correct Answer Type

5 Q. [3 M (-1)]

- Velocity of an object depends on displacement as $V^{3/2} = K8(y)^{3/4}$, where V is velocity (in m/s), y is displacement (in meter) & K is constant, then acceleration in m/s^2 when $y = 16$:-
(A) $8 K^{2/3}$ (B) 8 (C) $8K^{4/3}$ (D) $32 K^{4/3}$
- A ball is projected horizontally from a table such that it collides with wall then with ground. If after collision component of velocity perpendicular to the surface is reversed in direction without change in magnitude and component of velocity parallel to the surface remains unchanged. At what distance from the wall does the ball collide with the ground. ($H = 500$ m, $v_0 = 20$ m/s, $D = 100$ m)



- (A) 100 m (B) 200 m (C) 50 m (D) 150 m
- Statement -1** : Two projectiles having same range must have the same time of flight.
Statement -2 : Horizontal component of velocity is constant in projectile motion under gravity.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is false, statement-2 is true.
(D) Statement-1 is true, statement-2 is false.
- An elevator is moving upwards with constant acceleration. The dashed curve in the given figure shows the position y of the ceiling of the elevator as a function of time t . At the instant indicated by the point P a bolt breaks loose and drops from the ceiling. Which of the solid curves shown best describes the position of the bolt as function of time ?



- (A) I (B) II (C) III (D) IV
- A boy moves along the y -axis such that his velocity is given by $v = \frac{1}{y}$. If he is initially at $y = 1$ m, then find the time when he reaches $y = 3$ m.
(A) 3s (B) 4s (C) 6 s (D) he can't reach at $y = 3$ m

Multiple Correct Answer Type

2 Q. [4 M (-1)]

- Which of the following statements are true about a ground to ground projectile motion.
(A) Average velocity for time of flight is $u \cos \theta$.
(B) Change in velocity from the time of projection to the time it reaches maximum height has magnitude $u \sin \theta$.
(C) Average acceleration during entire time of flight is zero.
(D) Horizontal component of velocity remains constant.

7. Two particles were projected one by one with the same initial velocity from the same point on level ground. They follow the same parabolic trajectory and are found to be in the same horizontal level, separated by a distance of 1m, 2 seconds after the second particle was projected. Assume that the horizontal component of their velocities is 0.5 m/s. Which of the following statements will be true about their motion ?
- (A) The horizontal range of the parabolic path is 3 m.
 (B) The maximum height for the parabolic path is 45 m.
 (C) The total time of flight in the parabolic path for each particle = 4s
 (D) The horizontal range of the parabolic path is 6 m.

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Question 8 to 10

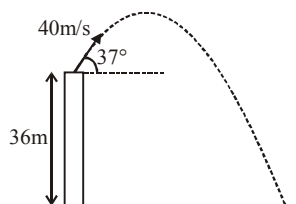
A particle is moving on the sides of an n-sided regular polygon with a constant speed v. Distance of centre from vertex is R.

8. Find the magnitude of average acceleration for the time interval during which the particle while moving from one vertex just crosses the next vertex.
- (A) $\frac{v^2}{R} \sin\left(\frac{\pi}{2n}\right)$ (B) $\frac{v^2}{R}$ (C) $\frac{v^2}{R} \sin\left(\frac{\pi}{n}\right)$ (D) None of these
9. Find direction of average acceleration calculated in previous question.
- (A) directed along one of the sides
 (B) directed towards centre at the vertex of turn
 (C) directed along exterior angle bisector of two adjoining sides
 (D) None of these
10. If $n = 6$, find the magnitude of average velocity of the time interval $t=0$ to $\frac{3R}{v}$.
- (A) $\frac{v}{3}$ (B) $\frac{2v}{3}$ (C) v (D) zero

Matching List Type (4 × 4)

1Q.[3 M (-1)]

11. Consider a situation in which a projectile is fired at 40 m/s from the top of a tower in absence of air resistance at $t = 0$.



List-I

- (P) Time at which particle moves at right angle with the initial direction
 (Q) Time at which kinetic energy is least
 (R) Time at which it strikes the ground
 (S) Time when the $\vec{v}$ make 45° with horizontal

List-II

- (1) 2.4 sec
 (2) $\frac{20}{3}$ sec
 (3) 5.6 sec
 (4) 6 sec

Code :-

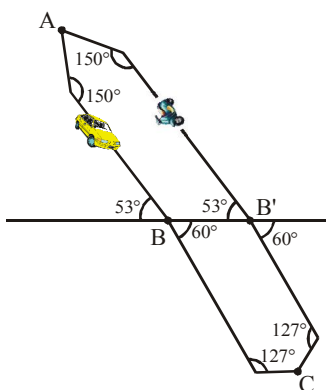
	P	Q	R	S
(A)	3	1	2	4
(B)	2	2	1	3
(C)	2	1	4	3
(D)	1	3	4	2

SECTION-III

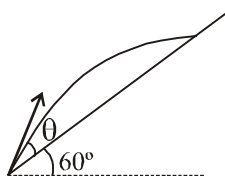
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

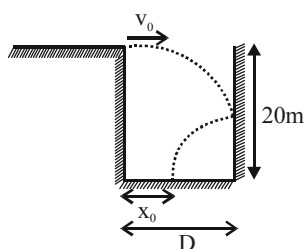
- In a lawn a water sprinkler is sprinkling water in all the possible directions while rotating. The velocity of water from nozzle is $\frac{5}{\sqrt{\pi}}$ m/s and is horizontal. Nozzle is at height 1 m above ground. What is maximum area (in m²) on ground that can be sprinkled?
- A car and a scooter both starts moving from point A. Car follows the path ABC while scooter follows the path AB'C as shown in figure. Both complete their journey in same time. Average speed of car between point A & B is v_1 and between point B & C is v_2 , while average speed of Scooter between point A & B' is v_1 and between point B' & C is v_2 . Then find $50 \left(\frac{v_1 - v_2}{v_2} \right)^2$



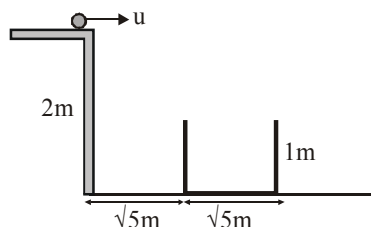
- A projectile is thrown on inclined plane as shown in the figure. Tangent of angle of projection θ (with respect to inclined plane) when range on inclined plane is maximum is x_0 . The value of $x_0 + \sqrt{3}$ is



- A ball leaves a horizontal table with velocity $v_0 = 5$ m/s. The ball bounces elastically from a vertical wall at a horizontal distance $D (= 8$ m) from the table, as shown in figure. The ball then strikes the floor a distance x_0 from the table ($g = 10$ m/s²). The value of x_0 (in meter) is :



5. An open box of height 1 m is placed near a 2m high table as shown. Marble rolling on the table, leave the table and are collected in the box. $u_{\max}$ and $u_{\min}$ are the maximum and minimum values of velocities of marbles to fall into the box. Find the value of $u_{\max} - u_{\min}$ (in m/s).



SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. A shell fired from horizontal ground hits a target on the ground. Equation of its trajectory is

$$y = 7x - \frac{x^2}{80}. \text{ Here the } x \text{ and } y\text{-axes represent the horizontal and vertically upward directions. Assume}$$

no air resistance and acceleration due to gravity 10 m/s^2 .

Column-I	Column-II
(A) Speed of projection	(P) 28 SI Units
(B) Air-time	(Q) $100\sqrt{2}$ SI Units
(C) Maximum height	(R) 560 SI Units
(D) Horizontal Range	(S) 980 SI Units
	(T) 1000 SI units

SECTION-I

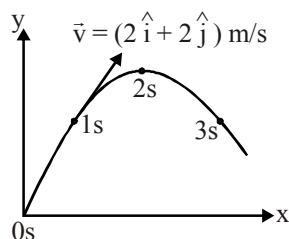
Multiple Correct Answer Type

7 Q. [Marks 4 (0)]

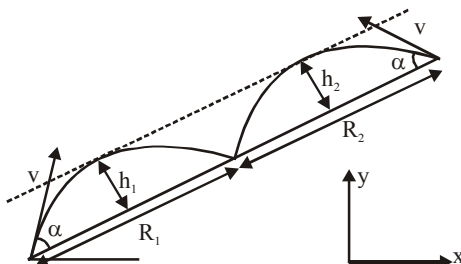
1. A particle is projected from the ground with an initial speed of u at an angle θ with horizontal. The average velocity of the particle between its point of projection and highest point of trajectory is :-

(A) $\frac{u}{2}\sqrt{1+2\cos^2\theta}$ (B) $\frac{u}{2}\sqrt{4-3\sin^2\theta}$ (C) $\frac{u}{2}\sqrt{1+3\cos^2\theta}$ (D) $u\cos\theta$

2. A physics student on Planet Exidor throws a ball, and it follows the parabolic trajectory shown in figure. The ball's position is shown at 1s intervals until $t=3$ s. At $t=1$ s, the ball's velocity is $\vec{v} = (2.0\hat{i} + 2.0\hat{j})$ m/s. At 2 sec ball is at its highest point



- (A) The acceleration of ball is 2 m/s^2 (B) Speed of ball at highest point is 3 m/s
(C) Speed of ball at $t=3$ sec is $2\sqrt{2} \text{ m/s}$ (D) The acceleration of ball is 9.8 m/s^2
3. The displacement x of a particle moving along a straight line varies with time according to the relation $x = x_0(1 - e^{-at})$, where x_0 and a are positive constants then
(A) Maximum displacement of the particle is x_0 (B) Maximum velocity of the particle is ax_0
(C) Acceleration of the particle is negative (D) Particle's speed decreases with time
4. An experimental rocket starts vertically from rest with a constant upward acceleration during flight a small piece detaches at 1s from start and rocket engine shuts at 5s from start. The detached small piece falls to ground 2s after it is separated from rocket. Identify the correct statements :-
(A) Upward acceleration of rocket is 8 m/s^2
(B) Height of rocket is 100 m above starting position when rocket engine shuts.
(C) Maximum height reached by rocket is 180 m.
(D) Total time of flight for rocket to fall down is 15 sec.
5. Two projectiles A and B are projected with equal speeds v as shown in figure. Trajectory shown is representative only (not actual). In standard notation :-



(A) $T_1 = T_2$ (B) $R_2 > R_1$ (C) $h_1 > h_2$ (D) $(y_1)_{\max} > (y_2)_{\max}$

6. To give trainee astronauts experience at feeling weightless, NASA has the insides of large planes removed. How can a plane be flown so it has an acceleration of 9.8 ms^{-2} vertically down for some time?
 (A) vertically up with decreasing speed.
 (B) on a parabolic path with a constant speed
 (C) on a parabolic path with increasing speed while moving upwards and decreasing speed when moving downwards
 (D) on a parabolic path with decreasing speed while moving upwards and increasing speed when moving downwards
7. A particle is projected under gravity. Its equation of trajectory is $2y = 4x - x^2$ then select the CORRECT statement(s) :- (Assuming horizontal as x and vertically upward as y-axis)
 (A) Maximum height of particle is 4 unit
 (B) Range of particle is 4 unit
 (C) At $x = 1$ particle is moving at an angle of 45° with horizontal
 (D) Maximum height of particle is 2 unit

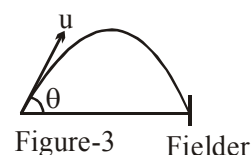
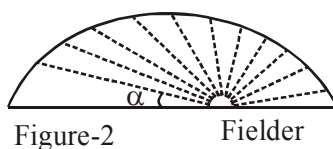
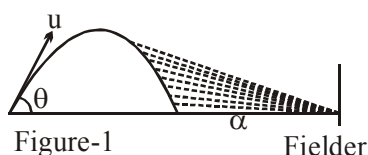
Linked Comprehension Type

(2 Para $\times$ 3 Q.) [Marks 3 (–1)]

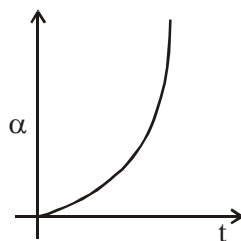
Paragraph for questions no. 8 to 10

When a cricket ball is hit for the boundary, how does the player in the area know where to be to catch it? Are there clues hidden in ball's motion that can guide the fielder. Lets just imagine ourselves in the cricket field. Let's assume for simplicity that we are in the plane of projection of the ball. All we need to decide is whether we should move towards the ball, away from it, or stand still to catch.

If the ball falls short of you, the path of ball is as shown in figure-1. In such a case, the line joining ball and you makes increasing angle α with horizontal and then after a certain maximum, the angle α with the horizontal starts decreasing. If the ball goes right over your head as in figure-2, the line joining ball and you makes continuously increasing angle with the horizontal. Then you should move back. If you are at the right position, the angle α made by the line joining ball and you makes increasing angle with horizontal and is maximum when it reaches you (figure-3). when it is about to reach you, angle becomes almost constant.



8. The angle α made by the line joining you and ball has a graph as shown below. To catch the ball, you should

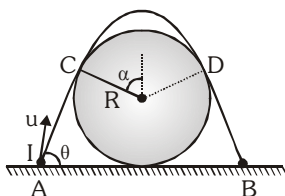


- (A) run back (B) stand still (C) run towards the ball (D) can't be said
9. In figure-1, when angle of line joining you and ball with horizontal is maximum,
 (A) The ball is moving perpendicular to its initial velocity.
 (B) The ball is moving at an angle θ below horizontal.
 (C) the ball is moving horizontally.
 (D) The velocity of the balls is towards you.
10. The constant in last line of paragraph refers to
 (A) θ (B) $90^\circ - \theta$ (C) 0° (D) 2θ

Paragraph for Questions no. 11 to 13

An insect I crosses a cylinder of radius R with minimum jumping speed. Path of the insect is shown in figure. When insect will be just touching the cylinder at C & D its velocity will be tangent to the cylinder at that point. At point C radius of the cylinder is making an angle α from the vertical. Minimum velocity

of insect is u, at an angle θ from the horizontal $\left[\text{Given : } R = \frac{2(\sqrt{2}-1)}{10} \text{ m \& } g = 10 \text{ m/s}^2 \right]$



11. Value of α is

- (A) $\sin^{-1} \frac{1}{(2)^{1/4}}$ (B) $\cos^{-1} \frac{1}{\sqrt{3}}$ (C) 45° (D) 30°

12. Value of minimum velocity u in m/s is

- (A) $\sqrt{2(2-\sqrt{2})}$ (B) $\sqrt{2\sqrt{2}}$ (C) 2 (D) 4

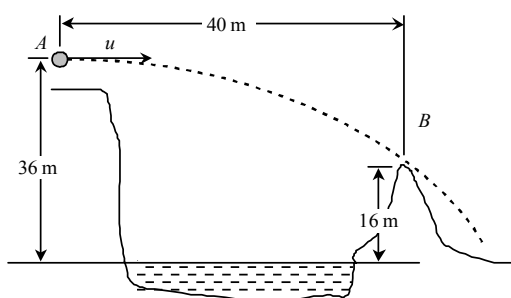
13. Value of θ is

- (A) $\sin^{-1} \frac{1}{(2)^{1/4}}$ (B) $\cos^{-1} \sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$ (C) 45° (D) 30°

SECTION-II

Numerical Grid Type (Single digit Ranging from 0 to 999) 3 Q. [Marks 4(0)]

1. With what minimum horizontal velocity 'u', (in m/s) can a boy throw a rock at A so that it just clear the obstruction at B



2. A fountain jet situated at a height of 11.25m above the ground projects water droplets in all directions with velocity of 20 ms^{-1} . Calculate the maximum distance (in m) on ground reached by any water droplet.

3. A helicopter is moving vertically upwards with a velocity 10 m/s. When the helicopter is at a height of 15 m from ground, a stone is thrown with a velocity $(6\hat{i} + 8\hat{j}) \text{ m/s}$ from the helicopter w.r.t. the man in it. Considering the point on ground vertically below the helicopter as the origin of coordinates, and the ground below as xy plane, if distance (in m) between the helicopter & the stone, at the instant the stone strikes the ground is $d\sqrt{3.25}$, find the value of d. (Assume helicopter moves upwards with constant velocity).

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. An eagle is flying horizontally at 30 ms^{-1} , 100 m above the ground. It was carrying a mouse in its grasp, which is released at a certain instant. The eagle continues on its path at the same speed for 2 s before attempting to retrieve its prey. To catch the mouse it dives in a straight line at constant speed and recaptures the mouse 20 m above the ground. Neglect air resistance ($g = 10 \text{ ms}^{-2}$)

Column-I

- (A) The diving speed of the eagle in ms^{-1}
- (B) Diving distance of the eagle in m
- (C) Mouse's free fall time in second
- (D) Angle made by eagle with vertical during descent in degrees

Column-II

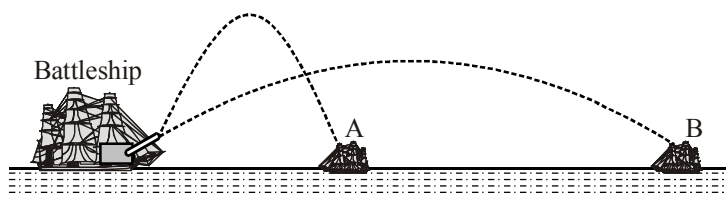
- (P) 4
- (Q) 100
- (R) 37
- (S) 50
- (T) 53

SECTION-I

Single Correct Answer Type

2 Q. [Marks 3 (-1)]

1. A gun is mounted in a stationary battleship which is used to hit two different boats A and B. The initial speed of both the shots is the same. Choose the correct statement.



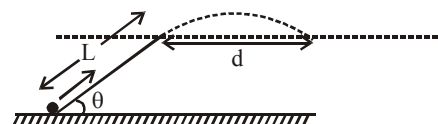
- (A) Both shots have same time of flight and same final speed
(B) Both shots have same time of flight but different final speeds
(C) Both shots have different time of flight but same final speed
(D) Both shots have different time of flight and different final speed
2. A particle is projected under gravity with velocity $\sqrt{2ag}$ from a point at a height h above the level plane at an angle θ to it. The maximum range R on the ground is

- (A) $\sqrt{(a^2 + 1)h}$ (B) $\sqrt{a^2 h}$ (C) $\sqrt{ah}$ (D) $2\sqrt{a(a+h)}$

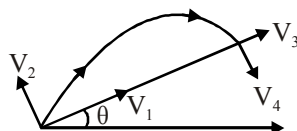
Multiple Correct Answer Type

3 Q. [Marks 4 (0)]

3. A cannon, when aimed vertically, is observed to fire a ball to a maximum height of L . Another ball is then fired with this same speed, but with the cannon aimed up along a plane of length L , inclined at an angle θ , as shown in figure.
- (A) Distance d is maximum at $\theta = 45^\circ$.
(B) Time of flight will be maximum for $\theta = 90^\circ$.
(C) Value of d will be zero if θ is increased to 90° keeping L fixed.
(D) Minimum length of plane inclined at an angle θ for which the ball will not leave the plane is given by $\frac{L}{\sin \theta}$.



4. A projectile is projected on the inclined plane as shown. V_1 & V_2 are components of its initial velocity along the incline and perpendicular to incline. and V_3 & V_4 are components of its final velocity along the incline and perpendicular to incline. {Here we are comparing the magnitudes only}



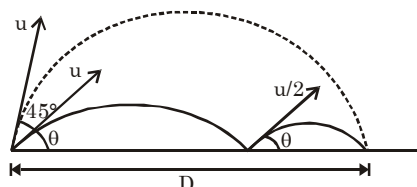
- (A) $V_1 > V_3$ (B) $V_1 = V_3$ (C) $V_2 = V_4$ (D) $V_2 > V_4$
5. An object is projected horizontally from a 80 m high tower with initial velocity of 10 m/s. Due to air flow, the ball gets a horizontal acceleration which depends on the vertical velocity given by $a_x = 2v_y$. Choose the **CORRECT** statement(s) :
- (A) Time of flight of the object is 4s.
(B) Speed just before it strike the ground is 40 m/s
(C) Horizontal range of the projectile may be $\frac{760}{3}$ m.
(D) Path of the object will be parabolic.

Linked Comprehension Type

(1 Para × 2 Q.) [Marks 3 (–1)]

Paragraph for Question 6 and 7

When baseball players throw the ball in from the out-field they usually allow it to take one bounce before it reaches the in field, on theory that the ball arrives sooner that way. Suppose the angle at which a bounced ball leaves the ground is same as the angle at which the outfielder throw it as shown in figure.



Ball speed after the bounce is one-half of what it was before bounce. Assume that ball is always thrown with same initially speed.

6. At what angle θ should the fielder throw the ball to make it go the same distance D with one bounce as a ball thrown upward at 45° with no bounce.
(A) 26.5° (B) 53° (C) 37° (D) 18.5°
7. Find out ratio of time interval for one-bounce throw to the flight time for no bounce throw ?
(A) $\sqrt{\frac{9}{10}}$ (B) $\sqrt{\frac{10}{9}}$ (C) $\sqrt{\frac{10}{3}}$ (D) $\sqrt{\frac{10}{12}}$

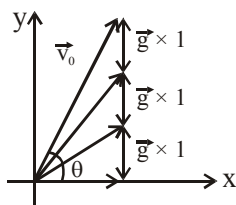
Linked Comprehension Type

(1 Para × 2 Q.) [Marks 4 (–1)]

(Multiple Correct Answer Type)

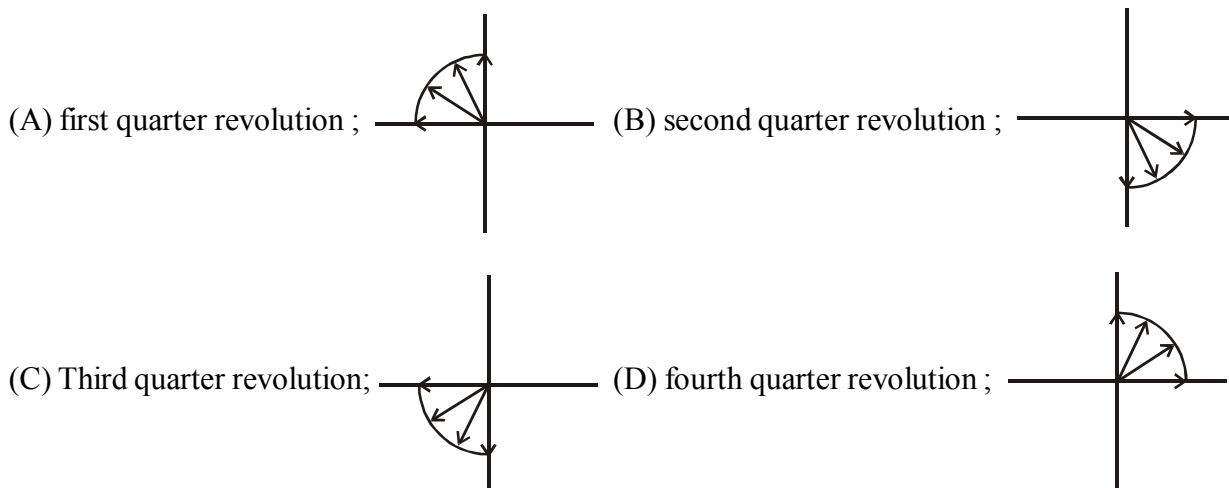
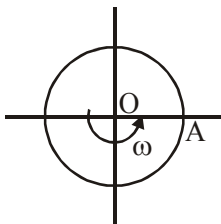
Paragraph for question no. 8 and 9

The hodograph is a plot of velocity of the body on the x-y plane. The velocity vector of the body is drawn to scale such that its starting point is the origin. To illustrate the definition of hodograph, we take the case of a projectile projected under influence of gravity. Initially, the particle is moving at an angle θ to the horizontal with a velocity $\vec{v}_0$ as shown. As time passes, its velocity changes under the influence of constant acceleration $\vec{g}$. By equation of motion, we can say $\vec{v} = \vec{v}_0 + \vec{g}t$. Note that $\vec{g}t$ is always vertically downward. Here this velocity vector is drawn at 3 successive seconds. At the end of 3 seconds the velocity vector in this case becomes horizontal. This means that the body has reached the highest point. The trajectory is more difficult to work with because it is parabolic in shape.



8. The area of the triangle formed till any time is :-
(A) $\frac{1}{2} \times \text{horizontal velocity} \times \text{change in vertical velocity}$
(B) $\frac{1}{2} g \times \text{horizontal displacement}$
(C) Distance travelled by projectile $\times g$
(D) $\frac{g^2 \times \text{Horizontal range} \times \text{time}}{4 \times \text{Initial vertical velocity}}$

9. A particle is moving in a circular path with constant angular velocity ω , starting from A. ω is anticlockwise. Its hodographs for subsequent 4 quarter revolutions are :-

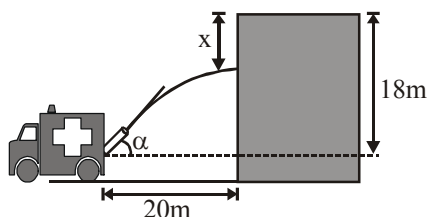


SECTION-II

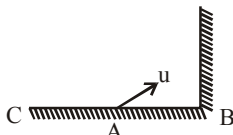
Numerical Grid Type (Single digit Ranging from 0 to 9)

6 Q. [Marks 4(0)]

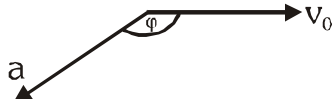
1. A fire nozzle situated at a distance 20 m from the building discharges water with an initial velocity of 20 ms^{-1} as shown in figure. If the height of the building is 18 meter (above nozzle), find the minimum possible height x (in m) of that portion of the building that escapes the water jets.



2. A ball projected as shown collides with the wall elastically. If it falls at point C such that $AB = AC = R/x$. Then value of x (where R is range,)



3. The figure shows the velocity and the acceleration of a point like body at the initial moment of its motion. The direction and the absolute value of the acceleration remain constant. Find the time in seconds when the speed becomes minimum. (Given : $a = 4 \text{ m/s}^2$, $v_0 = 40 \text{ m/s}$, $\phi = 143^\circ$)



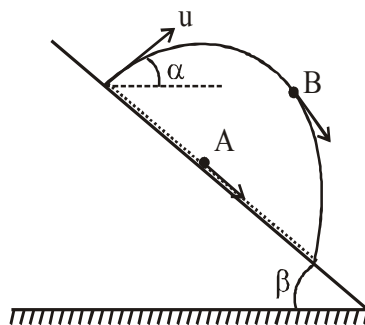
4. A particle moves along the parabolic path $x = y^2 + 2y + 2$ in such a way that y -component of velocity 2 m/s is constant during the complete motion, Find the magnitude of acceleration (in m/s^2).

5. Displacement X of a particle at time t moving along a straight line path is given by $X^2 = at^2 + 2bt + c$

where a , b and c are constant, acceleration of particle varies as $\frac{a - v^n}{X^m}$. Where ' v ' is velocity of particle at time t . Then find $n + m$.

6. A particle A is thrown with initial velocity 5 m/s along a frictionless inclined plane of inclination β . Another particle B is thrown with initial velocity 10 m/s at an angle α with the horizontal at the same moment when

A is thrown. Both the particle meet again on the inclined plane. The value of $\left(\frac{\pi}{\alpha + \beta}\right)$ is

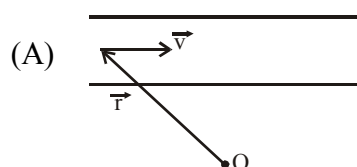


SECTION-IV

Matrix Match Type (4 × 5)

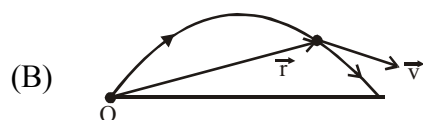
3 Q. [8 M (for each entry +2(0))]

1. Column-I shows motion of a particle ; where $\vec{r}$ = position vector of particle, $\vec{v}$ = velocity vector of particle and O represents origin.



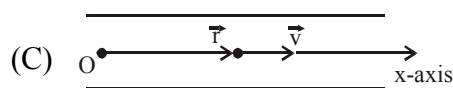
A car moving along a straight road with constant speed

(P) $\frac{d\vec{r}}{dt} \neq 0$



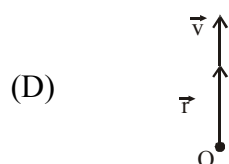
A projectile in gravitational field

(Q) $\frac{d\vec{v}}{dt} \neq 0$



A car moving along a straight road with constant speed

(R) $\frac{d|\vec{v}|}{dt} \neq 0$



A ball projected vertically up under gravity

(S) $\left|\frac{d\vec{v}}{dt}\right| = 0$

(T) $\frac{d|\vec{r}|}{dt} \neq 0$

CLASS TEST

2. Column-I shows different situations in which two projectile are projected. In column-II we have to compare the given situation for two projectiles. (Neglect air resistance)

Column-I

- (A) Particles are projected at same speed & different direction from same point on level ground such that their range is same.
- (B) For ground to ground projection, particles are projected with different velocity from same point on ground such that maximum height reached is same.
- (C) Particles are projected horizontally from same point at a height with different initial velocities.
- (D) Particle are projected from the same point from a height with same initial speed, direction of velocity makes equal angle with horizontal one below and the other above horizontal.

Column-II

- (P) Time of flight will be same
- (Q) Speed just before reaching ground will be same
- (R) Vertical component of velocity just before reaching ground will be same.
- (S) Minimum kinetic energy during the flight will be equal
- (T) Velocity vector makes same angle with horizontal direction at the instant of impact with ground

3. For a particle moving in x-y plane initial velocity of particle is $\vec{u} = u_x \hat{i} + u_y \hat{j}$ and acceleration of particle is always $\vec{a} = a_x \hat{i} + a_y \hat{j}$ where u_x, u_y, a_x, a_y are constants. Some parameters of motion are given in column-I, match the corresponding path given in column-II

Column - I

- (A) If $u_x \neq 0, u_y = 0, a_x \neq 0, a_y \neq 0$
- (B) If $u_x = 0, u_y \neq 0, a_x \neq 0, a_y \neq 0$
- (C) If $u_x = 0, u_y = 0, a_x \neq 0, a_y \neq 0$
- (D) If $u_x \neq 0, u_y \neq 0, a_x \neq 0, a_y \neq 0$

Column - II

- (P) Path of particle must be parabolic
- (Q) Path of particle must be straight line
- (R) Path of particle may be parabolic
- (S) Path of particle may be straight line
- (T) Particle has variable velocity

SECTION-I

Multiple Correct Answer Type

5 Q. [Marks 4 (0)]

- Equation of path of a particle moving in x-y plane is given by $y = 10x - 2x^2$ for the given particle select correct statement. Assume acceleration of gravity as 10 m/s^2 in negative y-direction :-
 (A) Maximum height of the particle is 12.5 m.
 (B) Range of the particle shown is $x = 5$.
 (C) At $x = 2.25$ particle is moving at an angle of 45° with x-axis.
 (D) Initially particle is projected at an angle of 53° .
- A particle is moving in x-y plane and equation of its path is given by $y = x^3 + 2x^2 + 5x + 3$. At $t = 0$ particle is located at $x = 0$. Consider the case for $x \geq 0, y \geq 0$. Choose the **CORRECT** option(s):-
 (A) Angle made by velocity vector of the particle at $t = 0$ is $\tan^{-1}(5)$ with x-axis.
 (B) Particle will never cross the origin.
 (C) Angle made by its velocity at $x = 1\text{m}$ is $\tan^{-1}(4)$ with x-axis.
 (D) If velocity of particle makes an angle θ with x-axis at point of projection, it will never make the same angle with x-axis during its motion.
- A stone is projected with speed 'u' on an inclined plane inclined at an angle α with horizontal. It was found the stone strikes the incline perpendicularly. Initial velocity of projection was at an angle β with incline. Choose the **CORRECT** statement(s) :
 (A) If u is doubled, angle at which stone strikes the incline remains same.
 (B) The stone was projected up the incline.
 (C) If angle of projection is changed keeping angle of incline same, still the stone can strike the incline perpendicularly by adjusting speed of projection
 (D) Range of stone on the incline is $\frac{2u^2 \sin \beta}{g \cos^2 \alpha} \cos(\alpha + \beta)$.
- A particle is projected with a speed of 50 m/s at an angle of 37° with the horizontal from the top of a tower. Then select the **CORRECT** statement(s) :-
 (A) Equation of trajectory of a particle is $y = \frac{3}{4}x - \frac{x^2}{320}$, if origin is taken at the point of projection.
 (B) Particle moves perpendicular to initial direction at $\frac{25}{3} \text{ sec}$
 (C) Minimum speed of particle in subsequent motion is zero
 (D) Maximum height attained by the particle from the point of projection is 45 m
- Two shells are fired by guns facing each other from points A and B simultaneously in the same vertical plane with velocities 60 m/s at 30° above the horizontal and 50 m/s respectively. The points A and B are 100 m apart on horizontal ground. If both the shells hit each other, find the angle of projection at point B and time when they collide.

- (A) 37° (B) 53° (C) $\left(\frac{20}{2-3\sqrt{3}}\right)\text{s}$ (D) $\left(\frac{10}{4+3\sqrt{3}}\right)\text{s}$

Linked Comprehension Type (1 Para × 5Q.) (2 Para × 3 Q.) [Marks 3 (–1)]

Paragraph for Question No. 6 to 10

A projectile is observed moving with velocity $\vec{v} = (30\hat{i} - 10\hat{j})$ m/s after 5 s of its projection from the ground. On the basis of above information find the following.

6. The velocity of projection is
 (A) $(30\hat{i} - 40\hat{j})$ m/s (B) $(30\hat{i} + 40\hat{j})$ m/s (C) $(30\hat{i} + 10\hat{j})$ m/s (D) None of these
7. The time of flight of the projectile is
 (A) 8 sec (B) 4 sec (C) 2 sec (D) 1 sec
8. The maximum height attained by the projectile is
 (A) 40 m (B) 100 m (C) 20 m (D) 80 m
9. The horizontal range of the projectile is
 (A) 100 m (B) 180 m (C) 120 (D) 240m
10. The equation of trajectory of the projectile is
 (A) $y = \frac{4}{8}x\left(1 - \frac{x}{240}\right)$ (B) $y = \frac{4}{3}x\left(1 - \frac{x}{240}\right)$ (C) $y = \frac{4}{3}x\left(1 - \frac{x^2}{120}\right)$ (D) $y = \frac{16}{3}x\left(1 - \frac{x^2}{120}\right)$

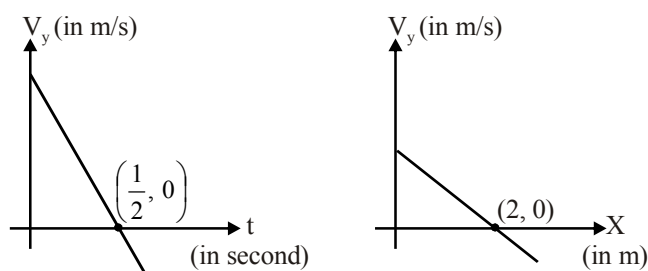
Paragraph for Question No. 11 to 13

Two projectiles following the same trajectory are found to be 8 m apart at the same horizontal level 2 s after the second was projected. If both projectiles were projected with the same initial horizontal velocity of 4 m/s from the same point.

11. The time of flight will be :
 (A) 5 sec (B) 4 sec (C) 6 sec (D) 8 sec
12. What will be the range of the path
 (A) 24 m (B) 10 m (C) 12 m (D) 45 m
13. The maximum height will be
 (A) 80m (B) 24 m (C) 45 m (D) 10 m

Paragraph for Question No. 14 to 16

Two graphs of the same projectile motion (in the xy-plane) projected from origin are shown. x-axis is along horizontal direction and y-axis is vertically upwards. Take $g = 10 \text{ m/s}^2$.



14. The projection speed is :
 (A) $\sqrt{37}$ m/sec (B) $\sqrt{41}$ m/sec (C) $\sqrt{14}$ m/sec (D) $\sqrt{40}$ m/sec
15. Projection angle with the horizontal is :
 (A) $\tan^{-1}\left(\frac{4}{5}\right)$ (B) $\tan^{-1}\left(\frac{2}{3}\right)$ (C) $\tan^{-1}\left(\frac{5}{4}\right)$ (D) $\tan^{-1}\left(\frac{1}{2}\right)$
16. Maximum height attained from point of projection is :
 (A) 1.25 m (B) 12.5 m (C) 2.25 m (D) None of these

CLASS TEST

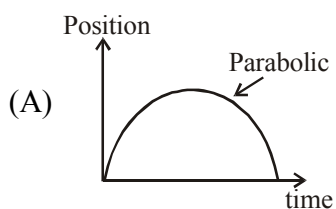
SECTION-IV

Matrix Match Type (4×5)

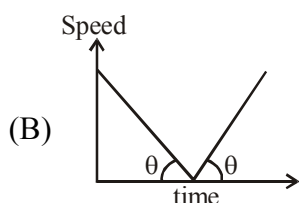
2 Q. [8 M (for each entry +2(0))]

1. Column-I

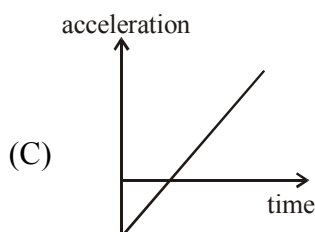
Column-II



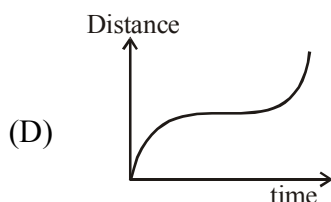
(P) Particle must change its direction during the subsequent motion



(Q) Particle may return to its initial position during subsequent motion



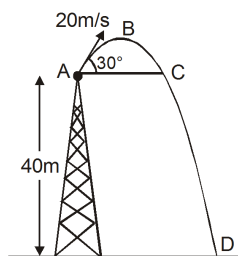
(R) Acceleration vector of a particle must be constant throughout the motion



(S) Particle come to rest at least once during its motion

(T) Initial velocity of particle is zero.

2. A projectile is fired from top of a 40 m high tower with velocity 20 m/s at an angle of 30° with the horizontal (see figure). $g = 10 \text{ m/s}^2$.



Column I

- (A) Ratio of time taken from A to D with time taken from A to C is equal to
(B) Ratio of vertical distance travelled from A to D with the maximum height from ground is less than.
(C) Ratio of final speed at D with the initial speed at A is less than
(D) Ratio of horizontal displacement from A to D with height of tower is greater than

Column II

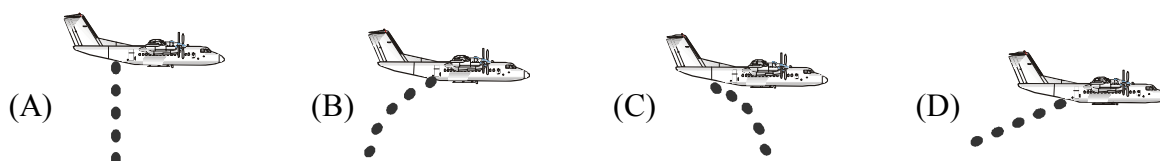
- (P) 1
(Q) 2
(R) 3
(S) 4
(T) 5

SECTION-I

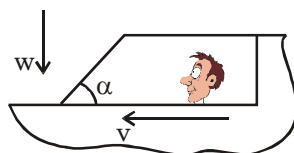
Single Correct Answer Type

8 Q. [Marks 3 (–1)]

1. Figure shows a still photograph from a war movie. Bombs have been dropped from the plane B-52 at regular intervals. Air plane moves with constant speed in horizontal direction. Which figure may be true still photograph. [Assume that there is no wind.]



2. A ferry boat is sailing at 12 km/h 30° W of N with respect to a river that is flowing at 6.0 km/h E. As observed from the shore, the ferry boat is sailing :
(A) 30° E of N (B) due N (C) 30° W of N (D) 45° E of N
3. A boy is standing on a flatcar that is moving with uniform acceleration on a level track. At the instant when flatcar is moving with velocity 10 m/s due east, the boy throws a ball with velocity 20 m/s in direction 45° above the east with respect to the flatcar. What should be the acceleration of the flatcar so that the boy can catch the ball without moving anywhere on the flatcar? Assume acceleration due to gravity 10 m/s^2 .
(A) 10 m/s^2 eastwards (B) 10 m/s^2 westwards (C) 20 m/s^2 eastwards (D) 20 m/s^2 westwards
4. An aircraft is flying horizontally with velocity v . It moves within a band of rain that falls uniformly vertically at speed w . The cockpit has two transparent surfaces: horizontal top and front screen inclined at an angle α to the horizontal (Fig.). Each of the surfaces has an area of S . Find the ratio of the number of drops of water falling per unit of time on the front glass, to the number of drops falling per unit time on the top glass.

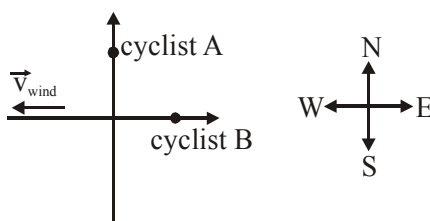


- (A) $\sin \alpha + \frac{v}{w} \cos \alpha$ (B) $\cos \alpha + \frac{w}{v} \sin \alpha$ (C) $\cos \alpha - \frac{v}{w} \sin \alpha$ (D) $\cos \alpha + \frac{v}{w} \sin \alpha$
5. A particle is dropped from the top of a tower and at the same time another particle is thrown horizontally from the same point. Their relative velocity with time in air is
(A) vertical
(B) constant, in a direction different from horizontal as well as vertical
(C) horizontal
(D) variable

6. Some persons hire a boat for 4 hours. The river flows at a speed of 5 km/hr and the boat moves with speed of 15 km/hr relative to the water. How far along the flow direction, they can go if they have to return in 4 hrs.

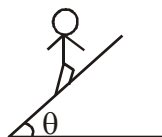
(A) 5 km (B) 7.5 km (C) $\frac{80}{3}$ km (D) 15 km

7. Figure shows two cyclists A and B moving towards North and East respectively. Wind is also blowing towards West. Possible direction of fluttering of flag on cycle A as seen by cyclist B is :-



(A)  (B)  (C)  (D) 

8. A man is standing on incline which makes θ with horizontal and rain is falling vertically with respect to man. Now man starts walking along the incline. Mark the correct option.

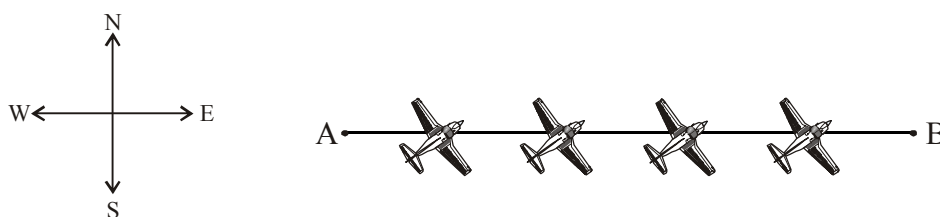


- (A) If man walks up the incline, rain may appear to come horizontally
(B) If man walks down the incline, rain may appear to come horizontally
(C) No matter whether man walks up or down, rain can never appear to come horizontally.
(D) If man walks (up or down) along incline rain may appear to come along the incline.

Multiple Correct Answer Type

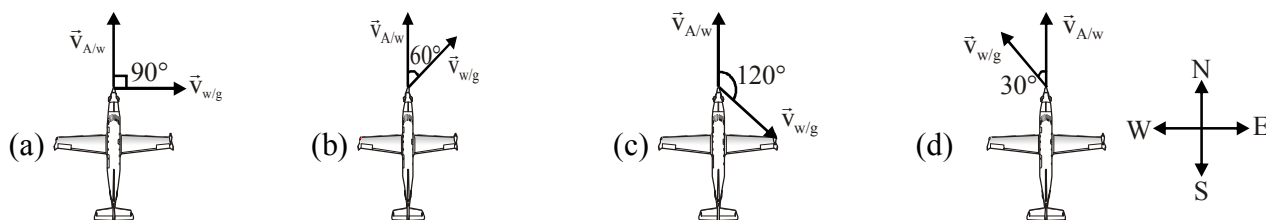
4 Q. [Marks 4 (0)]

9. Figure shows top view of an airplane moving in horizontal plane. Path of air plane as seen from ground is a straight line AB where A is initial point. Through out the journey orientation of airplane with respect to line AB is fixed. Direction of wind is nearly towards.

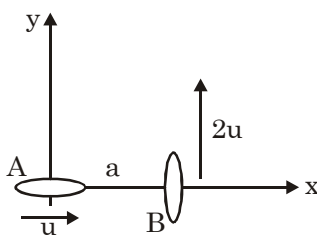


(A) South east (B) North east (C) North west (D) South west

10. Figure shows top view of an airplane blown off course by wind in various directions. Assume the magnitude of velocity of airplane relative to wind and magnitude of velocity of wind to be same in each case. $\vec{v}_{A/w}$ = velocity of airplane relative to wind, $\vec{v}_{w/g}$ = velocity of wind in ground frame.



- (A) Airplane travels fastest across ground in case (d)
(B) Airplane travels slowest across ground in case (c)
(C) Airplane experiences maximum lateral displacement in case (a)
(D) In none of the cases velocity of wind with respect to air planes can be directed along south-west
11. A boat is traveling due east at 12 m/s. A flag on the boat flaps at 53° N of W. Another flag on the shore flaps due north.
(A) Speed of wind with respect to ground is 16 m/s
(B) Speed of wind with respect to ground is 20 m/s
(C) Speed of wind with respect to boat is 20 m/s
(D) Speed of wind with respect to boat is 16 m/s
12. Consider two ships on the sea as shown in the figure. Ship A moves with velocity u directed to East. Ship B moves with velocity $2u$ directed to North. At time $t = 0$, ship B crosses the path of ship A at distance a in front of ship A.



- (A) The shortest distance between the ships $\frac{2a}{\sqrt{3}}$
(B) The time at which they reaches position of minimum distance is $\frac{a}{5u}$
(C) Suppose at $t = 0$, the speed of ship A remains u , but its direction can be adjusted such that the shortest distance between the two ships is minimum which is given by $\frac{a}{\sqrt{2}}$
(D) Suppose at $t = 0$, the speed of ship A remains u , but its direction can be adjusted such that the shortest distance between the two ships is minimum which is given by $\frac{\sqrt{3}a}{2}$

Linked Comprehension Type

(1 Para × 4 Q.) [Marks 3 (–1)]

Paragraph for Question 13 to 16 (4 Question)

A ball is thrown from the top of a 75 m high tower with velocity 20 m/s at 30° dip angle. Then

13. Find its time of flight

- (A) 5 sec (B) 3 sec (C) 4 sec (D) 10 sec

14. Find maximum height above the ground :-

- (A) 100 m (B) 150 m (C) 75 m (D) 50 m

15. Its horizontal range is :-

- (A) $30\sqrt{3}$ m (B) $\frac{30}{\sqrt{3}}$ m (C) 60 m (D) $10\sqrt{3}$ m

16. What will be the velocity with which it hits the ground :-

- (A) 10 m/s (B) $10\sqrt{19}$ m/s (C) 20 m/s (D) 150 m/s

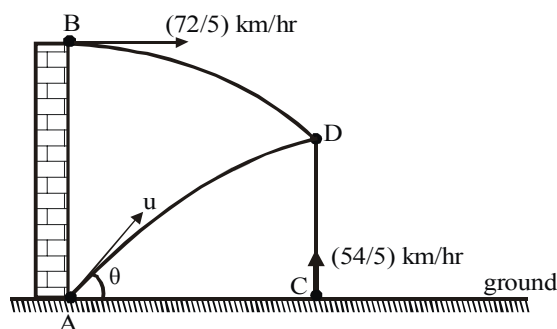
SECTION-IV

Numerical Grid Type (Single digit Ranging from 0 to 9)

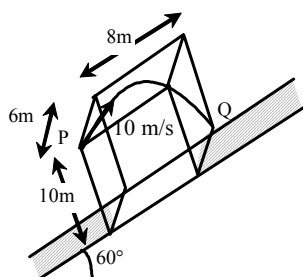
2 Q. [Marks 4(0)]

1. In the given figure points A and C are on the horizontal ground & A and B are in same vertical plane. Simultaneously bullets are fired from A, B and C and they collide at D. The bullet at B is fired horizontally

with speed of $\frac{72}{5}$ km/hr and the bullet at C is projected vertically upward at velocity of $\frac{54}{5}$ km/hr. Find velocity of the bullet projected from A in m/s.



2. A box of dimension $10\text{ m} \times 6\text{ m} \times 8\text{ m}$ is kept on a frictionless inclined plane as shown in the figure. A stone is thrown from one corner P with speed 10 m/s parallel to the ceiling of the box such that it hits the floor of the box at point Q as shown. At the same instant, the box is released to slide down the plane. Calculate the time of flight in second. [$g = 10\text{ m/s}^2$]

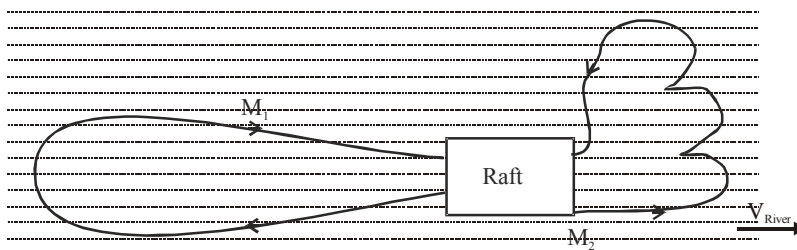


SECTION-I

Single Correct Answer Type

13 Q. [Marks 3 (-1)]

1. Two motorboats M_1 & M_2 start at time $t = 0$ from raft in river and after time t_0 they come back to the raft. Speed of motorboats in still water is equal. Paths of M_1 and M_2 as seen by observer on raft are shown in figure then (River is flowing uniformly with non-zero velocity) :-



- (A) w.r.t. Raft during the time interval $t = 0$ to $t = \frac{t_0}{2}$ both travel same distance.
- (B) w.r.t. ground during the time interval $t = 0$ to $t = \frac{t_0}{2}$ both travel same distance.
- (C) w.r.t. Raft during the time interval $t = 0$ to $t = \frac{t_0}{2}$ magnitude of displacement is same for both.
- (D) w.r.t. ground during the time interval $t = 0$ to $t = \frac{t_0}{2}$ magnitude of displacement is same for both.
2. A particle is released from rest from top of a fixed wedge of inclination angle α with horizontal. Another particle is projected at an angle θ with horizontal such that its horizontal range is equal to the base length of the wedge. Also time taken by first particle to reach bottom of wedge & time of flight of second particle is same then angle θ is

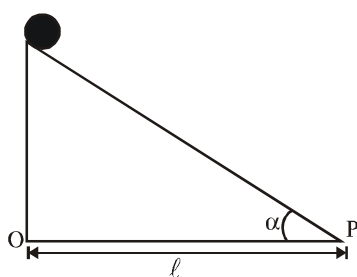


Fig. (i)

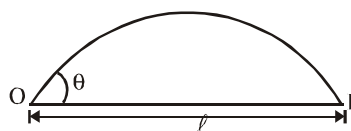
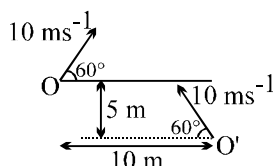


Fig. (ii)

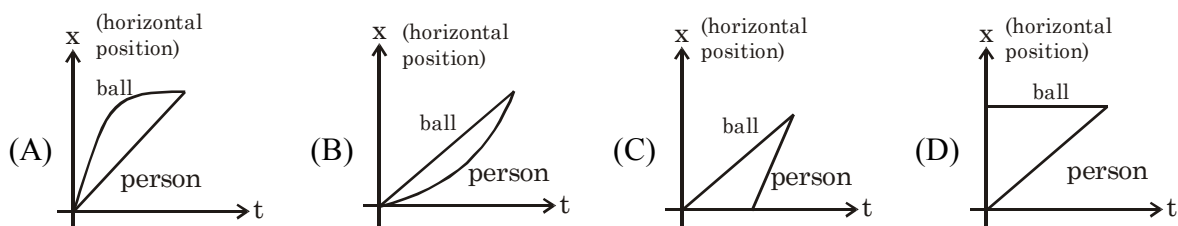
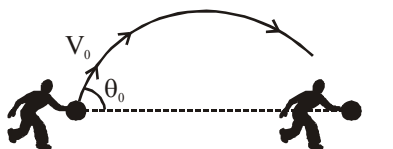
- (A) $\tan^{-1} (2 \operatorname{cosec} \alpha)$ (B) $\tan^{-1} [2 \sin 2\alpha]$ (C) $\tan^{-1} [2 \operatorname{cosec} 2\alpha]$ (D) $\tan^{-1} [3 \operatorname{cosec} 2\alpha]$
3. Two particles are projected simultaneously from two points O and O' such that 10 m is the horizontal and 5 m is the vertical distance between them as shown in the figure. They are projected at the same inclination 60° to the horizontal with the same velocity 10 ms^{-1} . The time after which their separation becomes minimum is



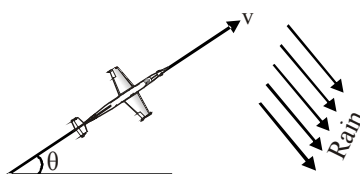
- (A) 2.5 sec (B) 1 sec (C) 5 sec (D) 10 sec

CLASS TEST

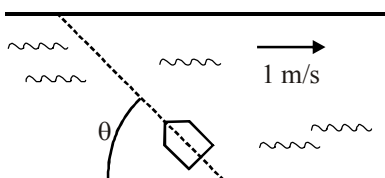
4. A person initially at rest throws a ball upward at an angle θ_0 with horizontal with an initial speed v_0 . He tries to catch the ball by accelerating with a constant acceleration. He catches the ball at exactly the same height he threw the ball. Let g be the gravitational acceleration. Which of the following graphs represents the above situation?



5. Rain is falling with speed $12\sqrt{2}$ m/s at an angle of 45° with vertical line. A man in a glider going at a speed of v at angle of 37° with horizontal with respect to ground. Find the speed (in m/s) of glider so that rain appears to him falling vertically. Consider motion of glider and rain drops in same vertical plane.



- (A) 15 (B) 60 (C) 25 (D) none of these
6. A boat moves with full power downstream in a river. At $t = 0$ it was right in front of a point A on the shore. At $t = 1$ hr it was right in front of a point B where it turns back and starts moving upstream with full power. A plank that is floating in river was met at $t = 0$ by the boat and was again met at C while going upstream. If $AB = 6$ km and $AC = 4$ km, the value of t when boat returns back in front of A is :-
(A) 2 hr (B) 3 hr (C) 4 hr (D) 5 hr
7. A boat is traveling upstream at 14 km/h with respect to a river that is flowing at 6 km/h (with respect to the ground). A man runs directly across the boat, from one side to the other, at 6 km/h (with respect to the boat). The speed of the man with respect to the ground is
(A) 10 km/h (B) 14 km/h (C) 18.5 km/h (D) 21 km/h
8. A boy wishes to row across a river in the shortest possible time. He can row at 2 m/s in still water and the river is flowing at 1 m/s. At what angle θ should he point the bow (front) of his boat ?



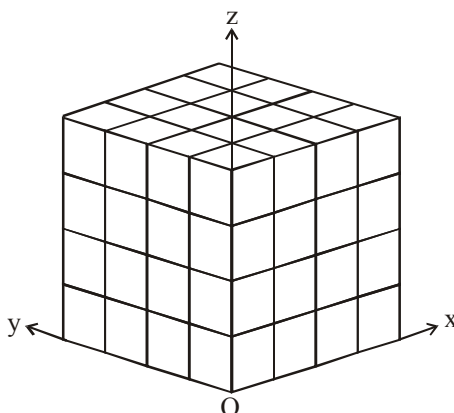
- (A) 30° (B) 45° (C) 60° (D) 90°
9. A projectile of mass 1.50 kg leaves the ground with a kinetic energy of 200 J. Only gravity acts on the projectile after it leaves the ground. At the highest point in its trajectory, its kinetic energy is 100 J. To what vertical height, relative to its launch point, did it rise?
(A) 12.2 m (B) 5.10 m (C) 9.14 m (D) 6.67 m
10. A particle moves in XY plane such that its position, velocity and acceleration are given by

$$\vec{r} = x\hat{i} + y\hat{j}; \vec{v} = v_x\hat{i} + v_y\hat{j}; \vec{a} = a_x\hat{i} + a_y\hat{j}$$

Which of the following condition is correct if the particle is speeding down?

- (A) $xv_x + yv_y < 0$ (B) $xv_x + yv_y > 0$ (C) $a_xv_x + a_yv_y < 0$ (D) $a_xv_x + a_yv_y > 0$

11. A child starts at one corner of a cubical jungle gym in a playground and climbs up to the diagonally opposite corner. The original corner (O) is the coordinate origin, and the x-, y- and z-axes are oriented along the jungle gym edges. The length of each side is 2 m. The child's displacement is:



- (A) $2\hat{i} + 2\hat{j} + 2\hat{k}$ (B) $2.8\hat{i} + 2.8\hat{j} + 2\hat{k}$ (C) $2\hat{i} + 2\hat{j} + 2.8\hat{k}$ (D) $2\hat{i} + 2\hat{j} + 3.5\hat{k}$
12. The time of flight of a bullet over a horizontal range R is T second. The angle of projection will be equal to :-
- (A) $\tan^{-1}\left(\frac{gT^2}{2R}\right)$ (B) $\tan^{-1}\left(\frac{2R}{gT^2}\right)$ (C) $\tan^{-1}\left(\frac{T}{Rg}\right)$ (D) $\tan^{-1}\left(\frac{Rg}{T}\right)$
13. A projectile is projected with speed u at an angle of 60° with horizontal from the foot of an inclined plane. If the projectile hits the inclined plane horizontally, the range on inclined plane will be :-

- (A) $\frac{u^2 \sqrt{21}}{2g}$ (B) $\frac{3u^2}{4g}$ (C) $\frac{u^2}{8g}$ (D) $\frac{\sqrt{21} u^2}{8g}$

Linked Comprehension Type

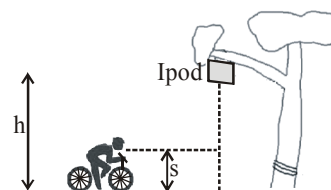
(1 Para $\times$ 2 Q.) [Marks 4 (-1)]

(Multiple Correct Answer Type)

Paragraph for Question no. 14 and 15

A bicycle rider is traveling at a constant speed along a straight road and then gradually applies the brakes during a time interval $0 < t < t_f$ until the bicycle comes to a stop. The combined mass of the cyclist and bicycle is m . Assume that the magnitude of the braking acceleration increases linearly in time according to $|\vec{a}| = bt/m$, $0 < t < t_f$, where $b > 0$ is a constant.

At the instant the person applies the brakes, a horizontal distance from the rider, the wind blows and snaps an iPod off the branch of the tree with a zero initial speed. The iPod was initially at a height h above the ground. The cyclist catches the iPod at the instant the cyclist has come to a stop. You may assume that the cyclist catches it at a height ' s ' above the ground.



14. Which of the following is/are **CORRECT** ?

- (A) The time at which the cyclist catches the iPod is $\sqrt{\frac{2(h-s)}{g}}$
- (B) The time at which the cyclist catches the iPod is $\sqrt{\frac{(h-s)}{g}}$
- (C) The initial speed of the cyclist is $\frac{b(h-s)}{mg}$
- (D) The initial speed of the cyclist is $\frac{2b(h-s)}{mg}$

15. Which of the following is/are **CORRECT** ?

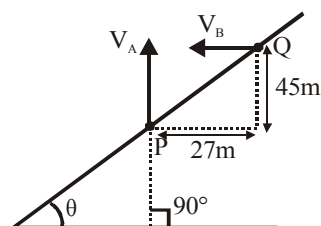
- (A) The retardation of cyclist at the moment he catches the iPod is $\frac{b}{m} \sqrt{\frac{2(h-s)}{g}}$
- (B) The retardation of cyclist at the moment he catches the iPod is $\frac{b}{m} \sqrt{\frac{(h-s)}{g}}$
- (C) The distance travelled by cyclist till the instant he catches the iPod is $\frac{b}{3m} \left[\frac{2(h-s)}{g} \right]^{3/2}$
- (D) The distance travelled by cyclist till the instant he catches the iPod is $\frac{b}{3m} \left[\frac{2h-s}{g} \right]^{3/2}$

SECTION-IV

Numerical Grid Type (Single digit Ranging from 0 to 9)

3 Q. [Marks 4(-2)]

- On a day rain is falling vertically. A boy cycling on the road finds rain falling at an angle 30° with vertical relative to him when moves with speed v_1 . On increasing speed to v_2 rain falls at an angle 60° with vertical relative to him. Find $\frac{v_2}{v_1}$.
- A particle is moving in a plane with velocity given by $\vec{v} = v_0 \hat{i} + a \omega \cos \omega t \hat{j}$. If particle is at origin at $t=0$ then its trajectory is given by equation $y = a^m \sin \left(\frac{\omega x^n}{v_0} \right)$. Find the value of $(m+2n)^2$.
- Two particles A and B are projected simultaneously from points P and Q respectively as shown in figure. Particle A was projected vertically while particle B was projected horizontally. If horizontal and vertical component of PQ are 27 m and 45 m respectively then find the minimum value of V_B in m/s for which a collision is possible between A and B.



Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. Match the following question.

Column-I

- (A) Positive acceleration
(B) Constant negative acceleration
(C) Distance does not change with time
(D) Acceleration is perpendicular to velocity

Column-II

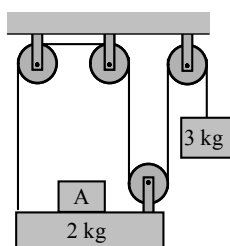
- (P) Speed may increase uniformly
(Q) Speed may decrease uniformly
(R) Speed is zero
(S) Speed is constant
(T) Speed may increase non uniformly

SECTION-I

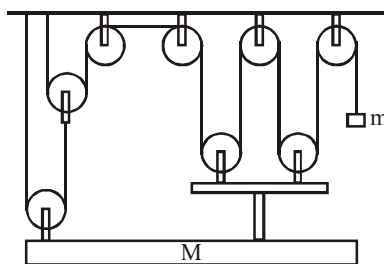
Single Correct Answer Type

12 Q. [Marks 3 (–1)]

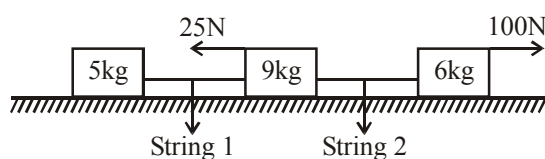
1. The arrangement shown in the figure is in equilibrium. The contact force between the block A and 2 kg block is closest to :



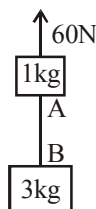
- (A) 10 N (B) 20 N (C) 30 N (D) 70 N
2. The minimum value of mass m required to lift the load M shown in figure is :-



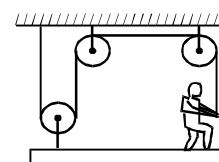
- (A) $\frac{M}{8}$ (B) $\frac{M}{4}$ (C) $\frac{M}{16}$ (D) M
3. In the given figure find tension at the mid-point of string-2, if mass of string-1 is 6 kg and mass of string-2 is 4 kg.



- (A) 70 N (B) 60 N (C) 80 N (D) 50 N
4. Tension in string AB is (Given AB is massless string and $g = 10 \text{ m/s}^2$).

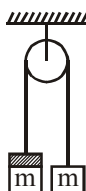


- (A) 60 N (B) 30 N (C) 45 N (D) Zero
5. A 50 kg person stands on a 25 kg platform. He pulls on the rope which is attached to the platform via the frictionless pulleys as shown in the figure. The platform moves upwards at a steady rate if the force with which the person pulls the rope is :-
- (A) 500 N (B) 250 N (C) 25 N (D) None

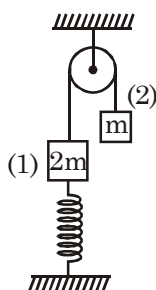
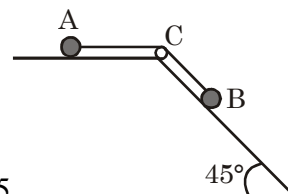


CLASS TEST

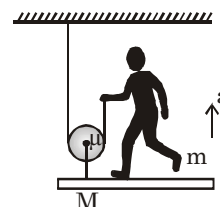
6. Two objects of mass m are connected by a thread that passes over a light, frictionless pulley as shown below. The objects are at rest. If a third object with mass m' is added on top of one of the objects, & objects are released, the tension in the string becomes $\frac{4}{3}$ times of the initial tension. Find the value of m'



- (A) $\frac{m}{2}$ (B) m (C) $2m$ (D) $3m$
7. In the shown diagram particles A and B are of mass 10 kg and 8 kg respectively and rest on planes. They are connected by a light inextensible string passing over a smooth fixed pulley at C. (Assume all surfaces are frictionless) The acceleration (in m/s^2) of the system is p . The value of $\frac{9}{10\sqrt{2}}p$ is :-
- (A) 2 (B) 4 (C) 6.5 (D) 0.5
8. In the figure shown string is massless and inextensible. Pulley and spring are massless. When the string is cut, ratio of acceleration of block 2 to acceleration of block 1 is :-

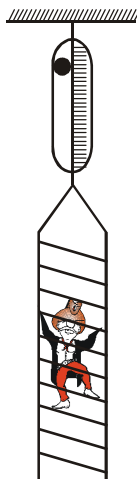


- (A) 1 (B) 2 (C) ∞ (D) 0
9. A person stands on a platform and pulley system as shown in the figure. The masses of platform, person and pulley are M , m and μ respectively. The rope is massless. Let the person pull up on the rope so that he has acceleration ' a ' upwards. Tension in the rod connecting the pulley to the platform is :
- (A) $(M + m + \mu)(g + a)$ (B) $(M + 2m + \mu)(g + a)$
(C) $(2M + 2m + \mu)(g + a)$ (D) $(2M + m + 2\mu)(g + a)$
10. An elevator accelerates upwards at a constant rate. A uniform string of length L and mass m hangs from the ceiling of the elevator and supports a small block of mass M . The tension at a distance ℓ from the ceiling is T . The acceleration of elevator is :-

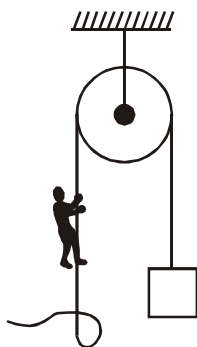


- (A) $\frac{T}{M + m - \frac{m\ell}{L}} - g$ (B) $\frac{T}{2M + m - \frac{m\ell}{L}} + g$ (C) $\frac{T}{M + \frac{m\ell}{L}} - g$ (D) $\frac{T}{2M - m + \frac{m\ell}{L}} - g$

11. Figure shows a 5 kg ladder hanging from a string that is connected with a ceiling and is having a spring balance connected in between. A man of mass 25 kg is climbing up the ladder at acceleration 1 m/s^2 . Assuming the spring balance and the string to be massless, the reading of the spring balance is :-



- (A) 30 kg (B) 32.5 kg (C) 35 kg (D) 37.5 kg
12. A person holds the rope (assumed massless) that passes over a pulley. The person is balanced by a block of mass m hanging at the other end of the rope. Initially both the person and block are motionless. The person then starts climbing the rope by pulling on it with a constant force in order to reach the block. The person moves a distance L relative to the rope. Friction is absent between rope & pulley. By how much distance does the block move ?



- (A) L (B) $2L$ (C) $L/2$ (D) $L/4$

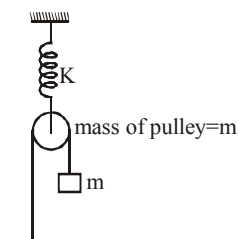
Multiple Correct Answer Type

13. Figure shows a spring and a pulley, block system in equilibrium
(A) Net force on pulley is $3mg$.

(B) Elongation of spring will be $\frac{3mg}{K}$

(C) If block m is detached, pulley will have acceleration $2g \uparrow$

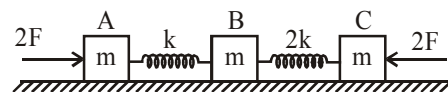
(D) If block m is detached, pulley will have acceleration $g \downarrow$



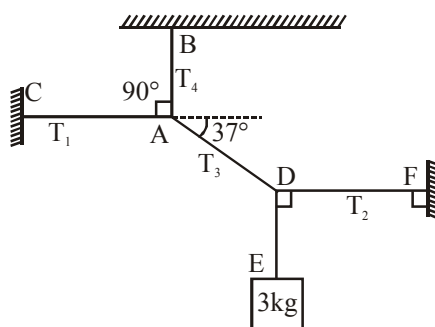
3 Q. [4 M (-2)]

14. In the figure shown blocks are placed on smooth surface and springs are ideal at the instant shown all the blocks are in equilibrium. Suddenly forces at both ends are removed. Mark the correct statement(s):

- (A) Instantaneous acceleration of block A is $\frac{2F}{m}$ towards left
(B) Instantaneous acceleration of block B is zero.
(C) Instantaneous acceleration of block C is $\frac{2F}{m}$ towards right
(D) Ratio of compressions in springs initially is $\frac{x_{AB}}{x_{BC}} = 2$.



15. In the figure shown string CA & DF are horizontal & T_1 , T_2 , T_3 and T_4 are tension in the string CA, DF, AD and AB. then



- (A) Value of $T_1 = 60$ N, $T_2 = 80$ N
(B) Value of $T_1 = T_2 = 40$ N
(C) Value of $T_3 = 50$ N
(D) Value of $T_4 = 40$ N

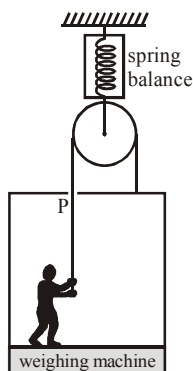
Linked Comprehension Type

(Single Correct Answer Type)

(1 Para $\times$ 2 Q.) [Marks 3 (0)]

Paragraph for Question No. 16 and 17

A man & lift having mass 50 kg & 30 kg respectively as shown in the figure. Man manages to keep the system in rest. There is a small hole at P so that string is not in the contact with lift at point P.



16. Find reading of weighing balance in lift :-

- (A) 500 N (B) - 100 N (C) 0 N (D) 100 N

17. Find reading of spring balance :-

- (A) 500 N (B) 300 N (C) 400 N (D) 800 N

SECTION-IV

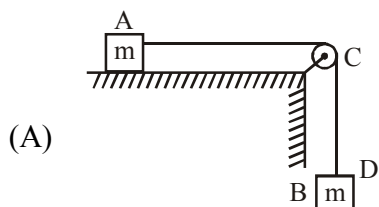
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

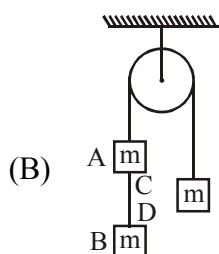
1. In the column-I some arrangement with ideal string & frictionless & light pulley are shown. In string CD tension may be written as $T = \eta mg$. Then match the column-II with column-I.

Column-I

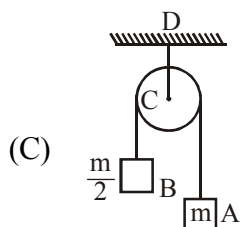
Column-II



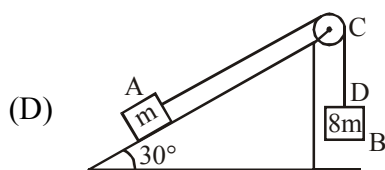
(P) $\eta = \frac{1}{2}$



(Q) $\eta = \frac{2}{3}$



(R) $\eta = \frac{4}{3}$



(S) Acceleration of block A and B will be same

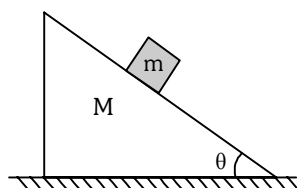
(T) Acceleration of block A and B will be different.

SECTION-I

Single Correct Answer Type

7 Q. [Marks 3 (–1)]

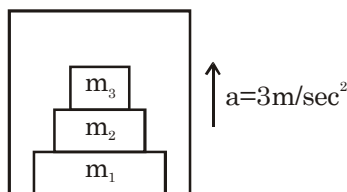
1. A block of mass m slides down on a wedge of mass M as shown in figure. Let $\vec{a}_1$ be the acceleration of the wedge and $\vec{a}_2$ the acceleration of block w.r.t. ground. N_1 is the normal reaction between block and wedge and N_2 the normal reaction between wedge and ground. Friction is absent everywhere. Select the incorrect alternative



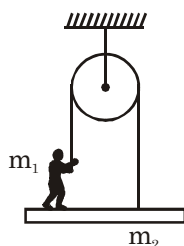
- (A) $N_2 < (M + m)g$ (B) $N_1 = m(g \cos\theta - |\vec{a}_1| \sin\theta)$
(C) $N_1 \sin\theta = M|\vec{a}_1|$ (D) $m\vec{a}_2 = -M\vec{a}_1$
2. Three blocks are connected by strings and pulled by a force $F = 60$ N as shown in figure. If $m_A = 10$ kg, $m_B = 20$ kg and $m_C = 30$ kg, then



- (A) Acceleration of the system is 2 m/s^2 (B) Acceleration of the system is 5 m/s^2
(C) $T_1 = 10 \text{ N}$ & $T_2 = 30 \text{ N}$ (D) $T_1 = 20 \text{ N}$ & $T_2 = 40 \text{ N}$
3. A lift is moving with acceleration 3 m/sec^2 . Three masses are put as shown in figure. Find the normal contact force exerted by floor of lift on mass m_1 . ($m_1 = 5\text{kg}$, $m_2 = 3\text{kg}$ and $m_3 = 1 \text{ kg}$)

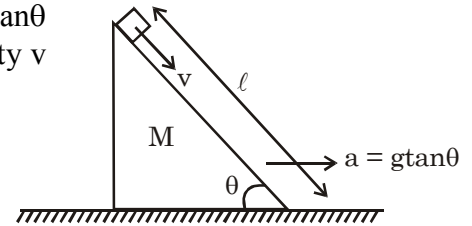


- (A) 116 N (B) 117 N (C) 52 N (D) 13 N
4. A man of mass m_1 holding one end of a massless inextensible string that passes over the smooth pulley, is accelerating up while standing on a platform of mass m_2 . Find the force applied by the man on the platform, if it moves up with an upward acceleration 'a'.

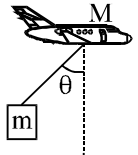


- (A) $m_1 g$ (B) $m_1(g + a)$ (C) $\frac{(m_1 - m_2)}{2} g$ (D) $\frac{m_1 - m_2}{2} (g + a)$

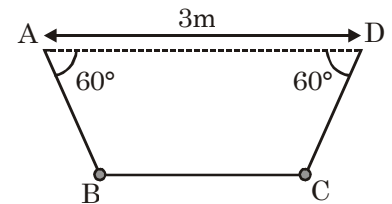
5. A smooth wedge of mass M is pushed with an acceleration $a = g \tan \theta$ and a block of mass m is projected down the slant with a velocity v relative to the wedge. The most **CORRECT** statement is :



- (A) The time taken by the block to reach the ground is $\frac{l}{v}$.
(B) The normal reaction between wedge and block is $mg \sec \theta$.
(C) The normal reaction offered by ground to the wedge is $(M + m)g$.
(D) All of the above.
6. A helicopter of mass M is carrying a box of mass m at the end of a rope and is moving horizontally with constant acceleration 'a'. The acceleration due to gravity is 'g'. Neglect air resistance. The rope is stretched out from the helicopter at a constant angle θ to the vertical. What is this angle?



- (A) $\sin \theta = a/g$ (B) $\cos \theta = a/g$
(C) $\tan \theta = a/g$ (D) $\sin \theta = ma/(Mg)$
7. ABCD is an elastic string of natural length 3 m and particles of equal mass are attached to the unstretched string at points B and C where $AB = BC = CD$. The ends A and D are then attached to two points on the same horizontal level and 3 m apart. The particles hang in equilibrium so that the string sections AB and CD are each at 60° to the horizontal. Force constant of the string is 50 N/m.

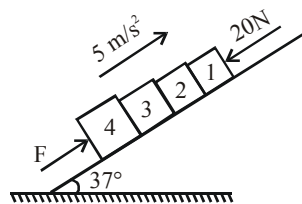


- (A) Extension in AB is $\frac{2}{3}m$
(B) Extension in BC is $\frac{2}{3\sqrt{3}}m$
(C) Extension in AB is equal to extension in BC.
(D) Mass of each particle is 5kg.

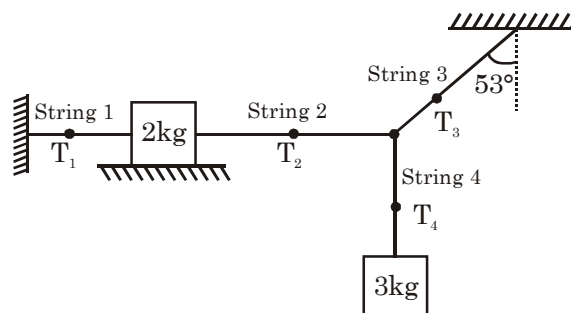
Multiple Correct Answer Type

5 Q. [4 M (-2)]

8. Four block system having masses 1kg, 2kg, 3kg & 4 kg respectively are placed on smooth inclined plane and all blocks are moving upward with acceleration 5 m/s^2 . If net force action on 1kg, 2kg, 3kg & 4 kg blocks are F_1 , F_2 , F_3 & F_4 respectively then :-



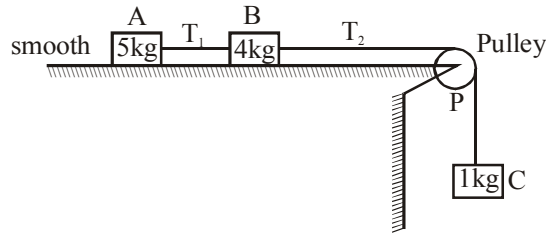
- (A) $F_1 = 5 \text{ N}$ (B) $F_2 = 10 \text{ N}$ (C) $F_3 = 15 \text{ N}$ (D) $F_4 = 20 \text{ N}$
9. A system of two blocks are connected by the strings as shown in figure. All strings are inextensible and massless and surface is smooth.



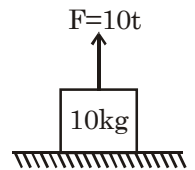
- (A) $T_1 = 40 \text{ N}$ (B) $T_2 = 40 \text{ N}$ (C) $T_3 = 50 \text{ N}$ (D) $T_4 = 30 \text{ N}$

CLASS TEST

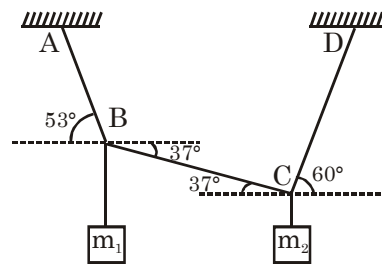
10. Shown system is released at $t = 0$. Select correct statement(s) about subsequent motion of blocks. Here strings are massless and pulley is frictionless & massless. [$g = 10 \text{ m/s}^2$]



- (A) Acceleration of A is equal to 1 m/s^2 (B) Tension $T_1 = 5\text{N}$
(C) Tension $T_2 = 9\text{N}$ (D) Clamp force on pulley is 9N .
11. A block of $m = 10 \text{ kg}$ is kept on ground. A vertically upward force $F = (10t) \text{ N}$, where t is the time in seconds starts acting on it at $t = 0$ as shown in figure.
- (A) Time at which block loose contact from the surface is 10 sec .
(B) Velocity of block at $t = 20 \text{ sec}$ is 50 m/s .
(C) Velocity of block at $t = 20 \text{ sec}$ is 200 m/s .



- (D) Displacement of block in 20 sec is $\frac{500}{3} \text{ m}$.
12. For the diagram shown below, $m_1 = 10 \text{ kg}$ then :



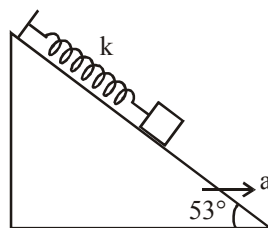
- (A) Tension in string AB is $\frac{2000}{7} \text{ N}$
(B) Tension in string BC is 1500 N
(C) Tension in string CD is $\frac{2400}{7} \text{ N}$
(D) Tension in string CD is 4800 N

Linked Comprehension Type (Single Correct Answer Type)

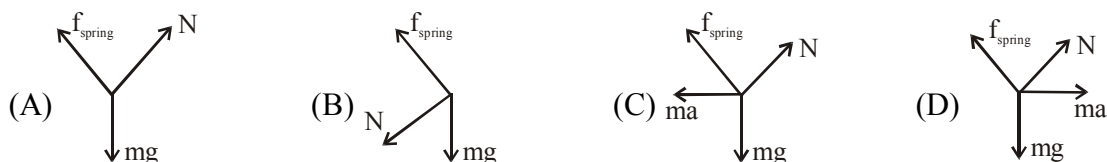
(1 Para $\times$ 2 Q.) [Marks 3 (0)]

Paragraph for Question No 13 and 14

A block of mass m connected to a spring on a wedge as shown in figure. Wedge is accelerated towards right.



13. Which of the following free body diagram is **CORRECT** w.r.t wedge :-



14. Which of the following statement is **CORRECT**

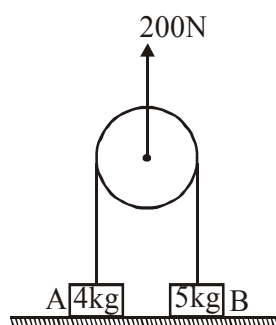
- (A) If $a = g$ spring will be compressed.
(B) If $a = \frac{4g}{3}$ spring will be in natural length
(C) Normal reaction between block and wedge reduces as acceleration of wedge is increased.
(D) None

SECTION-III

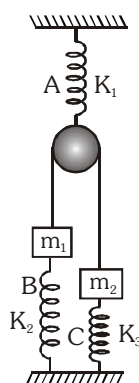
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

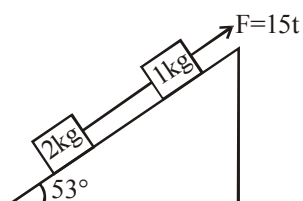
1. Two blocks of masses 4 kg and 5 kg are at rest on the ground. The masses are connected by a massless string passing over a smooth and light pulley. A 200 N force is applied on the pulley. Find the difference in accelerations of blocks in m/s^2 .



2. The system shown in figure is in equilibrium. Masses m_1 and m_2 are 2 kg and 2.5 kg respectively. Spring constants K_1 , K_2 and K_3 are 200 Nm^{-1} , 1000 Nm^{-1} & 500 Nm^{-1} respectively. If the elongation in spring A is 20 cm and the compression in spring C is 1 cm then find out the elongation in spring B in cm.



3. In the given figure the maximum value of tension in the string connecting blocks of 1 kg and 2 kg is 50 N. A force varying with time $F = 15t$ is applied on the 1 kg block as shown. The time (in sec) at which the string breaks is :

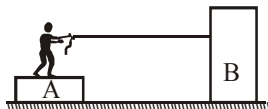


SECTION-I

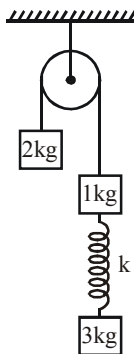
Single Correct Answer Type

10 Q. [Marks 3 (–1)]

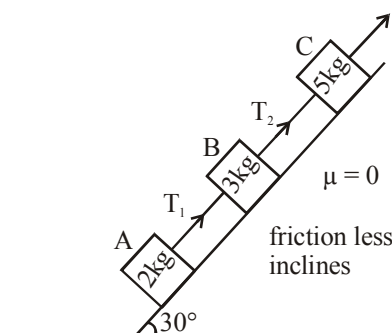
1. Figure shows a boy on a horizontal platform A on a smooth horizontal surface, holding a rope attached to a box B. Boy pulls the rope with a constant force of 50 N. (boy does not slip over the platform). The combined mass of platform A and boy is 250 kg and that of box B is 500 kg. The velocity of A relative to the box B, 5 s after the boy on A begins to pull the rope, will be



- (A) 1 m/s (B) 1.5 m/s (C) 2 m/s (D) 0.5 m/s
2. From the fixed pulley, masses 2kg, 1kg and 3kg are suspended as shown in the figure. Find the extension in the spring if $k = 100 \text{ N/m}$. (Neglect oscillations due to spring) :-

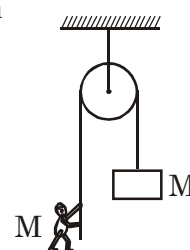


- (A) 0.1 m (B) 0.2 m (C) 0.3 m (D) 0
3. Three blocks A, B, C of mass 2kg, 3kg, 5kg are placed on a frictionless inclined as shown in figure. Force F of magnitude 150 N is applied on block 'C' as shown. Find tension in string 1 (T_1) :-



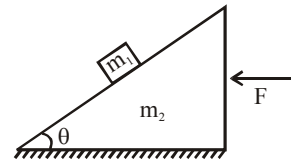
- (A) 20 N (B) 30 N (C) 10 N (D) 40 N
4. A block is hanging at one end of the massless rope passing over a fixed smooth pulley. From the other end of the rope a man of same mass climbs at an acceleration x relative to the rope. Find the value of x if acceleration of rope is ' a ' :-

- (A) a (B) $\frac{a}{2}$
(C) $2a$ (D) $\frac{5a}{2}$

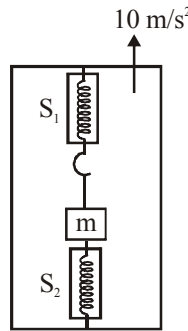


5. A block of mass m_1 is placed on wedge (mass m_2) shown in fig. Find out magnitude of force 'F' such that block m_1 remain stationary w.r.t wedge. (All surfaces are frictionless.)

(A) $(m_1 + m_2)g \sin \theta$ (B) $(m_1)g \tan \theta$
(C) $(m_2)g \tan \theta$ (D) $(m_1 + m_2)g \tan \theta$



6. Reading shown in two spring balances, S_1 and S_2 is 60 kg and 30 kg respectively. Spring S_1 is in elongated state & spring S_2 is in compressed state. Lift is accelerating upward with acceleration 10 m/s^2 . The mass is stationary with respect to lift. Then the mass of the block will be :-

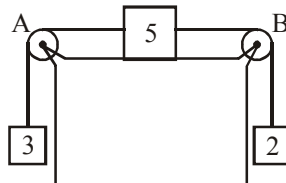


(A) 45 kg (B) 30 kg (C) 60 kg (D) 15 kg

7. A weighing machine is kept upside down on head of a man of mass 50 kg, mass of weighing machine is 3 kg. What will be reading of weighing machine ?

(A) 0 kg (B) 50 kg (C) 3 kg (D) 53 kg

8. Three blocks are arranged by the two string as shown in the diagram. Then the force exerted by the pullyon the clamp is



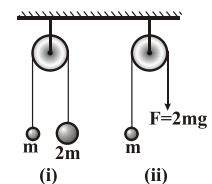
(A) $22\sqrt{2}$ (B) $27\sqrt{2}$ (C) $20\sqrt{2}$ (D) Zero

9. The rear side of a truck is open and smooth. A box of mass 20 kg is placed on the truck 4 m away from the open end, and $g = 10 \text{ m/s}^2$. The truck starts from rest with an acceleration of 2 m/s^2 on a straight road. The distance moved by the truck when box falls from the truck is:-

(A) 4 m (B) 8 m (C) 16 m (D) 32 m

10. The pulley arrangements shown in figure are identical, the mass of the rope being negligible. In case I, the mass m is lifted by attaching a mass $2m$ to the other end of the rope. In case II, the mass m is lifted by pulling the other end of the rope with a constant downward force $F = 2mg$, where g is acceleration due to gravity. The acceleration of mass m in case I is:

(A) zero (B) more than that in case II
(C) less than that in case II (D) equal to that in case II



Multiple Correct Answer Type

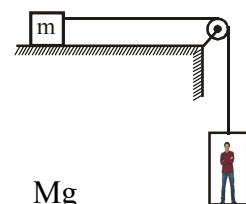
11. In the figure, a man of true mass M is standing on a weighing machine placed in a cabin. The cabin is joined by a string with a body of mass m . Assuming no friction, and negligible mass of cabin and weighing machine, then choose the correct statement(s) (normal force between the man and the machine is proportional to the mass) :-

(A) Measured mass of man is $\frac{Mm}{(M+m)}$

(B) Acceleration of man is $\frac{Mg}{(M+m)}$

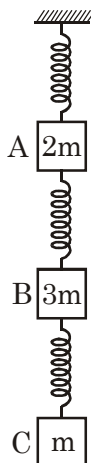
(C) Acceleration of man is $\frac{mg}{(M+m)}$

(D) Measured mass of man is M .

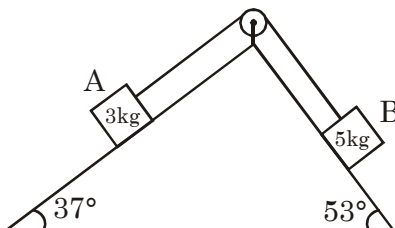


3 Q. [4 M (-2)]

12. Initially system is in equilibrium. If the spring between A & B is cut then just after cutting. (Assume all springs are massless and identical)



- (A) Acceleration of block A is $2g$ upwards (B) Acceleration of block A is $4g$ upwards
(C) Acceleration of block B is $4g/3$ downwards (D) Acceleration of block B is $4g$ downwards
13. A system of two block A & B having mass 3kg and 5kg . If system is released from rest as shown in figure. If surface between inclined & both blocks is smooth than choose **INCORRECT** statement (pulley & string are ideal) :-



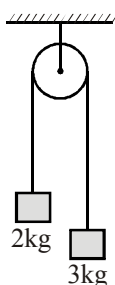
- (A) Magnitude of acceleration of 3kg block along the inclined plane is $\frac{3}{2} \text{ m/sec}^2$.
(B) Acceleration of both blocks along the inclined plane will be same.
(C) Tension in string is $\frac{105}{4} \text{ N}$.
(D) Acceleration of B w.r.t. A is $\frac{11}{2} \text{ N}$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

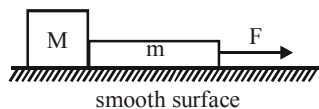
4 Q. [4 M (0)]

1. The atwood machine system shown in the figure is released from rest. At what time (in sec) will the speed of the block of mass 3 kg be 6 m/s ?

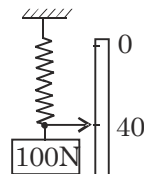


CLASS TEST

2. A block of mass M is pulled along horizontal frictionless surface by a rope of mass m . If tension at the middle point of rope & force applied by string on the block is in ratio 3 : 2 then find the value of M/m .



3. A vertical rod of mass 4 kg is hanging on a rope and 3kg cat is grasping the bottom end of the rod. The rope is cut off and the frightened cat beings to run up on the rod. While the rod is falling vertically, the cat remains at the same height, with respect to the ground. If the acceleration of the rod is $ng/4 \text{ m/s}^2$ then what is value of n .
4. An ideal spring, with a pointer attached to its end, hangs next to a scale. With a 100 N weight attached and at rest, the pointer indicates '40' cm on the scale as shown. Using a 200 N weight instead results in 60 cm on the scale. Using an unknown weight 'X' instead results in '30' cm on the scale. Fill value of $\frac{X}{10}$ (in newton).



SECTION-IV

Matrix Match Type (4×5)

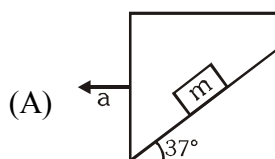
1 Q. [8 M (for each entry +2(0))]

1. A block of mass $m=1\text{kg}$, is placed on a wedge. The wedge can be accelerated with acceleration ($a=g$) in four manners as shown. The normal reaction between block and incline is N .

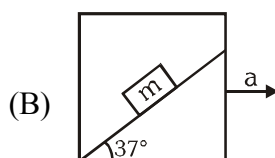
Column I

Column II

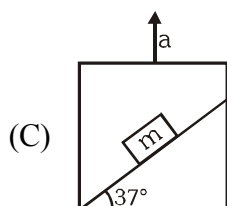
(Normal reaction and acceleration of block w.r.t. incline)



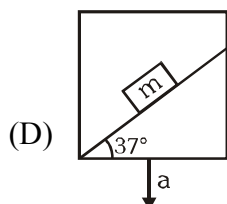
(P) $\frac{7g}{5}$



(Q) $\frac{g}{5}$



(R) $\frac{8g}{5}$



(S) Zero

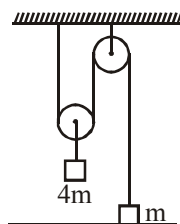
(T) $\frac{6g}{5}$

SECTION-I

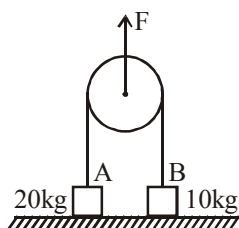
Single Correct Answer Type

6 Q. [Marks 3 (–1)]

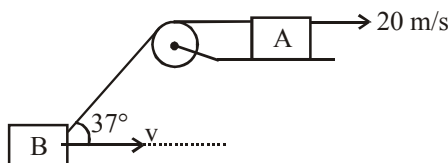
1. The mass of the body which is hanging on the rope attached to the movable pulley is four times as much as the mass of the body which is fixed to the ground. At a given instant the heavier body is released. What is the initial acceleration of block m (in m/s^2) ? (The mass of the pulleys and the ropes are negligible.)



- (A) 5 m/s^2 (B) 10 m/s^2 (C) 20 m/s^2 (D) 2.5 m/s^2
2. In the figure shown blocks are initially at rest on the floor. Mark the correct statement(s) :



- (A) When applied force on pulley is 200 N the normal reaction on block A is zero.
(B) When applied force is 400 N acceleration of B is 10 m/s^2
(C) When applied force is 240 N acceleration of pulley is 2 m/s^2
(D) For a force $F = 100 \text{ N}$, normal reaction on B is zero.
3. Two block A & B are connected by an ideal string as shown in figure. If block B has velocity v & block A has velocity 20 m/s as shown find v :-

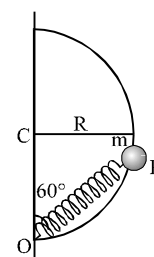


- (A) 25 m/s (B) 16 m/s (C) 20 m/s (D) 10 m/s
4. A smooth semicircular wire track of radius R is fixed in a vertical plane as shown. One end of a light spring of natural length $\frac{3R}{4}$ is attached to the lower point O of the wire track. A small bead of mass m which can slide on the track, is attached to the other end of the spring.

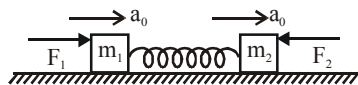
The bead is held stationary at point P such that spring makes an angle of 60° with the vertical.

The spring constant is $k = \frac{mg}{R}$. The restoring force in the spring at that instant is :-

- (A) mg (B) $\frac{mg}{4}$ (C) $\frac{mg}{8}$ (D) $\frac{mg}{2}$



5. Two blocks of masses m_1 and m_2 being connected with a light spring of stiffness k are driven with forces F_1 and F_2 on a smooth horizontal plane. If the blocks move with same acceleration a_0 , the value of a_0 is:-



- (A) $\frac{F_1 - F_2}{m_1 + m_2}$ (B) $\frac{F_1 m_2 - m_1 F_2}{(m_1 + m_2)^2}$
- (C) $\frac{1}{2} \left(\frac{F_1}{m_1} + \frac{F_2}{m_2} \right)$ (D) decided by the stiffness of the spring
6. Figure represents a painter in a crate which hangs alongside a building. When the painter of mass 100 kg pulls the rope, the force exerted by him on the floor of the crate is 450 N. If the weight of the crate is 25 kg, the acceleration of the painter will be : (Take $g = 10 \text{ m/s}^2$)

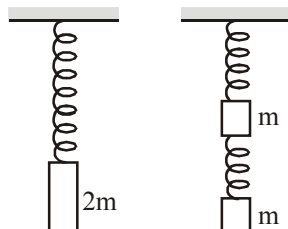


- (A) 1 m/s^2 (B) 2 m/s^2 (C) 3 m/s^2 (D) 4 m/s^2

Multiple Correct Answer Type

6 Q. [4 M (-2)]

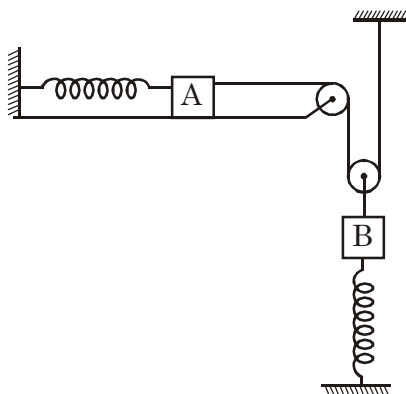
7. Two masses of an Atwood machine are at rest on a tabletop. A small upward force F_p on the pulley keeps the string taut. Suppose that $m_2 > m_1$. Assume that the pulley itself has negligible mass. Mark the **CORRECT** statement(s) :-
- (A) if $|F_p| < 2m_1 |g|$, the system remains at rest
- (B) if $2m_1 |g| < |F_p| < 2m_2 |g|$, the pulley and m_1 move upward but m_2 stays on the table
- (C) if $|F_p| > 2m_2 |g|$, the whole system leaves the table.
- (D) if $m_1 |g| < |F_p| < m_2 |g|$, the pulley and m_1 move upward but m_2 stays on the table
8. A mass $2m$ suspended from a given spring causes it to stretched relative to its relaxed length. The mass and the spring are then each cut into two identical pieces and connected as shown in figure.



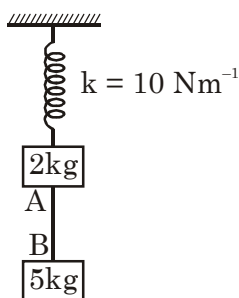
- (A) Bottom of the lower mass is higher than bottom of the original mass
- (B) Bottom of the lower mass is lower than bottom of the original mass
- (C) Bottom of the lower mass is at the same level as the bottom of the original mass
- (D) Spring constant of new springs obtained after cutting is double than the spring constant of original spring.

CLASS TEST

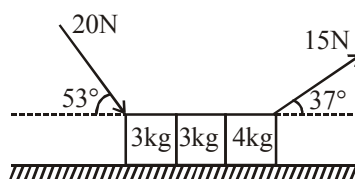
9. The system is released from rest with both the springs in unstretched position. Mass of each block is 1 kg and force constant of each spring is 10 N/m. Assume pulleys and strings are massless and all contacts are smooth. ($g = 10 \text{ m/s}^2$). In equilibrium choose the incorrect options :-



- (A) Compression of vertical spring is $\frac{2}{5} \text{ m}$ and extension of horizontal spring is $\frac{4}{5} \text{ m}$
 (B) Compression of vertical spring and extension of horizontal spring is $\frac{2}{5} \text{ m}$
 (C) Extension of horizontal spring is $\frac{1}{5} \text{ m}$ and compression in vertical spring is $\frac{2}{5} \text{ m}$
 (D) Compression of vertical spring is $\frac{1}{5} \text{ m}$ and extension of horizontal spring is $\frac{2}{5} \text{ m}$
10. An ideal spring of spring constant $K = 10 \text{ N/m}$ is hung vertically by a rigid support. On the other end a system of masses 2 kg & 5 kg connected by a string AB is attached as shown in figure. The system is initially at rest. Now string AB is cut at $t = 0$:-



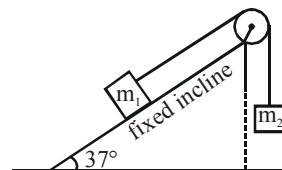
- (A) Acceleration of 5 kg block = 10 m/sec^2 (B) Acceleration of 5 kg block = 0 m/sec^2
 (C) Acceleration of 2 kg block = 25 m/sec^2 (D) Acceleration of 2 kg block = 0 m/sec^2
11. For the system shown below which of the statements is/are correct ?



- (A) Net force on all the blocks is same
 (B) Net force on 4 kg block is more than 3 kg block.
 (C) Net force on both 3 kg blocks is same
 (D) Normal force between 3 kg blocks is equal to normal force between 3 kg and 4 kg blocks.

12. All the surfaces are smooth and system released from rest then :-

- (A) If $m_2 = \frac{m_1}{2}$, then m_1 will move down the plane
(B) If $m_2 = \frac{3}{4}m_1$, then m_1 will move up the plane
(C) if $m_1 = m_2$ then m_2 will move downwards
(D) for the system to be to rest $m_2 = \frac{3}{5}m_1$



Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 3Q.) [Marks 3 (0)]

Paragraph for Question no 13 to 15

An inclined plane of angle of elevation 30° is placed in a lift which accelerates in vertical downward direction with 6 m/s^2 and an object of mass 4 kg is placed at the top of inclined plane. If object can slide on inclined plane without friction and released from rest with respect to lift then :

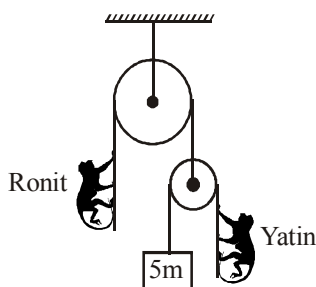
13. The trajectory of sliding object as seen from ground :-
(A) Must be parabola
(B) Must be straight line
(C) Either straight line or parabola depending upon initial velocity of lift
(D) May be circle
14. What is the force exerted by object on inclined plane:-
(A) 13.16 N (B) 14.25 N (C) 16.92 N (D) 14.5 N
15. How much time does object take to arrive at the bottom of inclined plane if length of plane is 1.5 m :-
(A) 1.23 s (B) 1.85 s (C) 2s (D) 1s

SECTION-II

Numerical Answer Type Question
(upto second decimal place)

1Q.[3(0)]

1. Two monkeys Ronit and Yatin of equal mass ' m ' can climb strings of a pulley arrangement as shown in figure. Find magnitude of acceleration (in m/s^2) of Ronit with respect to rope so that block remains stationary. It is given that Yatin is just holding the string. Assume pulley is frictionless and string is massless and inextensible.

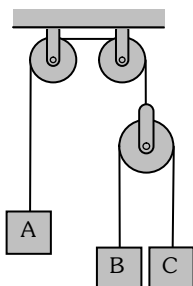


SECTION-III

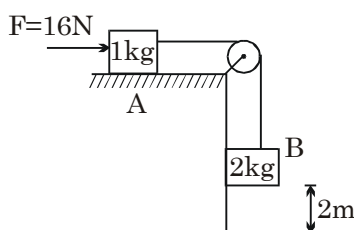
Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. In the system shown, block A is of mass 4.0 kg and blocks B and C are of equal mass each of 3 kg. Find the acceleration in m/s^2 of block C, if the system is set free. [$g = 10 \text{ m/s}^2$]



2. The system shown in the diagram is released from rest. Treat string segments as long. Neglect friction everywhere. Pulley and string are massless. If the time in sec after which block B reaches ground is t . Fill the value of $10t^2$.



SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. A man of mass 50 kg is hanging on a ladder of mass 10 kg. The ladder is hanging from the ceiling with help of a light rope. In first column of the following table are mentioned some situations and in the second column are mentioned values of tension in the rope in newtons. Suggest suitable matches. [$g = 10 \text{ m/s}^2$]

Column-I

- (A) The man is hanging standstill on the ladder.
(B) The man is climbing the ladder with uniform speed.
(C) The man is climbing the ladder with acceleration 1 m/s^2 .
(D) The man is descending the ladder with acceleration 1 m/s^2 .

Column-II

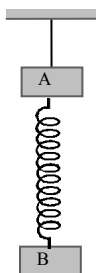
- (P) 540
(Q) 550
(R) 600
(S) 650
(T) 660

SECTION-I

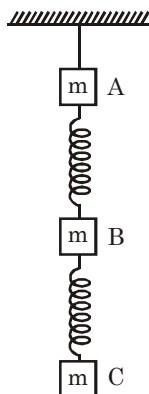
Single Correct Answer Type

7 Q. [Marks 3 (–1)]

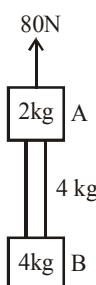
1. In the given figure both the blocks have equal mass. When the thread is cut, acceleration of block A & B are ?



- (A) $0, g\uparrow$ (B) $2g\downarrow, 0$ (C) $g\downarrow, g\downarrow$ (D) $g\downarrow, g\uparrow$
2. Initially system is in equilibrium. If the lower spring will be cut, then acceleration of blocks A, B and C will be (in m/s^2) respectively :-

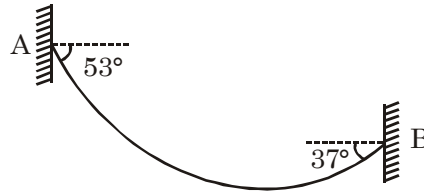


- (A) $0, g\uparrow, g\downarrow$ (B) $0, 0, g\downarrow$ (C) $\frac{g}{2}\uparrow, \frac{g}{2}\uparrow, 0$ (D) $\frac{g}{2}\uparrow, \frac{g}{2}\uparrow, g\downarrow$
3. Two objects A and B of masses 2 kg and 4 kg are connected by a uniform rope of mass 4 kg as shown in the diagram. A force of magnitude 80 N acts on A in vertically upward direction. Tension at mid point of the rope is ($g = 10 m/s^2$) :-

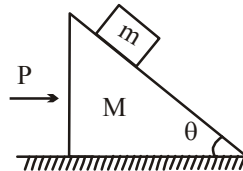


- (A) 32 N (B) 40 N (C) 42 N (D) 48 N

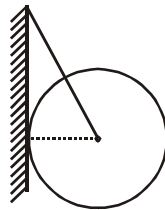
4. A heavy rope is hanging between points A and B. If mass of the rope is 90 kg :



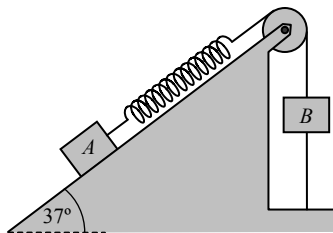
- (A) Tension at point A is 900 N
(B) Tension at point B is 540 N
(C) The horizontal component of tension force is different at each point of the rope.
(D) The vertical component of tension force is same at all the points of the rope.
5. Two wooden blocks are moving on a smooth horizontal surface such that the mass m remains stationary with respect to block of mass M as shown in the figure. The magnitude of force P is:



- (A) $(M + m)g \tan \theta$ (B) $g \tan \theta$ (C) $mg \cos \theta$ (D) $(M + m)g \operatorname{cosec} \theta$
6. A sphere of radius R and mass m is connected to a wall by a string of length $2R$. The normal reaction of wall on sphere is :-



- (A) mg (B) $\frac{mg}{2}$ (C) $\frac{mg}{\sqrt{3}}$ (D) $2mg$
7. The setup shown is in equilibrium. Mass of block A and B are $5m$ and $2m$ respectively. The slope is frictionless. String connecting the block B with the ground is cut. Accelerations a_A and a_B of these blocks immediately after cutting the string are

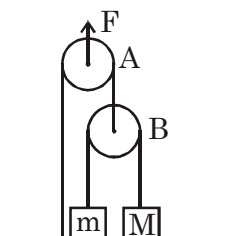


- (A) $a_A = \frac{1}{3}g$ down the plane and $a_B = 0$ (B) $a_A = \frac{1}{3}g$ up the plane and $a_B = 0$
(C) $a_A = 0$ and $a_B = \frac{1}{2}g$ downwards (D) $a_A = 0$ and $a_B = \frac{1}{2}g$ upwards

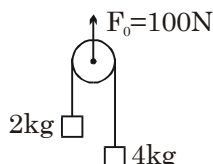
Multiple Correct Answer Type

4 Q. [4 M (–2)]

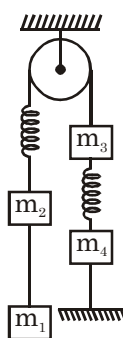
8. Two blocks of mass $m = 5\text{ kg}$ & $M = 10\text{ kg}$ are connected by a string passing over a pulley B as shown. Another string connects the centre of pulley B to the floor and passes over another pulley A as shown. An upward force $F = 300\text{ N}$ is applied at the centre of A. [Both pulley are ideal & massless & $g = 10\text{ m/sec}^2$] :-



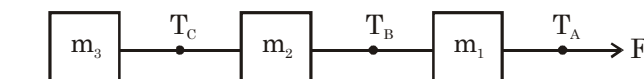
- (A) $a_m = 5\text{ m/sec}^2$ (B) $a_m = 0\text{ m/sec}^2$ (C) $a_M = 0\text{ m/sec}^2$ (D) $a_M = 5\text{ m/sec}^2$
9. Two blocks of masses $m_1 = 2\text{ kg}$ and $m_2 = 4\text{ kg}$ hang over a massless pulley as shown in the figure. A force $F_0 = 100\text{ N}$ acting at the axis of the pulley accelerates the system upwards. Then :



- (A) acceleration of 2 kg mass is 15 m/s^2 (B) acceleration of 4 kg mass is 2.5 m/s^2
- (C) acceleration of pulley is $\frac{35}{4}\text{ m/s}^2$ (D) acceleration of both the masses is upward
10. For the system shown in figure. $m_1 > m_2 > m_3 > m_4$. Initially the system is at rest in equilibrium condition. If the string joining m_4 and ground is cut, then just after the string is cut :



- (A) Acceleration of m_1 & m_2 will be zero. (B) Acceleration of m_4 will be g downward
- (C) Acceleration of m_3 will be zero. (D) Acceleration of m_3 will be non-zero.
11. A child's toy consists of three blocks as shown in figure. The blocks have masses m_1 , m_2 and m_3 ($m_1 < m_2 < m_3$). If they are pulled to the right with a horizontal force 'F', mark the correct statement (T = Tension in the string.)



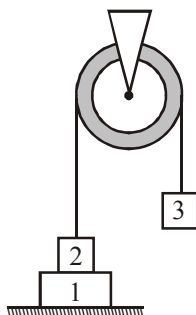
- (A) $T_A > T_B > T_C$ (B) $T_A < 2T_B$ (C) $T_B < 2T_C$ (D) $T_A > 3T_C$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Questions 12 and 13

The drawing shows box 1 resting on a table, with box 2 resting on top of box 1. A massless rope passes over a massless, frictionless pulley. One end of the rope is connected to box 2 and the other end is connected to box 3. The weights of the three boxes are $W_1 = 55 \text{ N}$, $W_2 = 35 \text{ N}$, and $W_3 = 28 \text{ N}$.



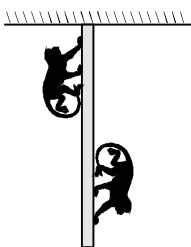
12. The magnitude of the normal force that the table exerts on box 1, is :-
(A) 55 N (B) 62 N (C) 48 N (D) 90 N
13. If the pulley is pulled upward with an acceleration that increases with time as $a = t/4$ where t is the time in seconds, what is the time when the box 2 is lifted off?
(A) 2.5 sec (B) 5 sec (C) 1.25 sec (D) 3.75 sec

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1Q.[3(0)]

1. Two monkeys of masses 10 and 8 kg are moving along a vertical rope, the former climbing up with an acceleration of 2m/s^2 while the latter coming down with a uniform velocity of 2m/s . Find the tension (in N) in the rope at the fixed support.

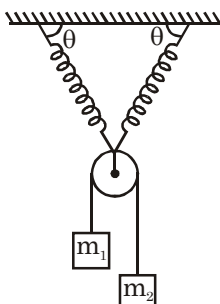


SECTION-III

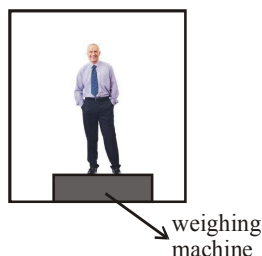
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

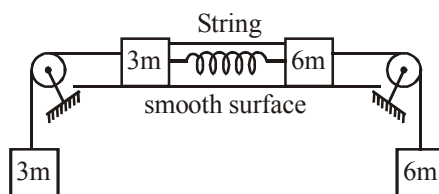
1. Two identical ideal springs of spring constant 1000 N/m are connected by an ideal pulley as shown and system is arranged in vertical plane. At equilibrium θ is 60° and masses m_1 and m_2 are 2 kg and 3 kg respectively. The elongation in each spring when θ is 60° is $p \text{ cm}$. The value of $\frac{\sqrt{3}p}{1.6}$ is :



2. Spring constant of weighing machine reduces due to severe use of it. If a man on weighing machine gets correct reading if elevator goes with 2 m/s^2 if % change in spring constant is P , then fill $(P - 16)$ if lift goes upward or $(P - 17)$ if lift goes downward.



3. Consider the arrangement shown. The system is released from rest and the string (connecting two blocks) shown is simultaneously burnt. Maximum extension in the spring (initially relaxed) during the subsequent motion is nmg/k then find n .



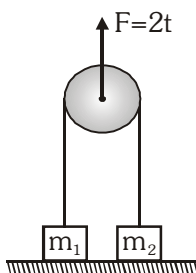
4. A monkey pulls (along the ground) the mid point of a 10 m long light inextensible string connecting two identical objects A & B each of mass 0.3 kg continuously along the perpendicular bisector of line joining the masses. The masses are found to approach each other at a relative acceleration of 5 m/s^2 when they are 6 m apart. The constant force applied by monkey is:

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. In shown figure $m_1 = 2 \text{ kg}$ and $m_2 = 4 \text{ kg}$. The pulley is movable. At $t=0$, both masses touch the ground and the string is taut. A vertically upward, time dependent force $F=2t$ (F is in newton, t is in second) is applied to the pulley. (Take $g=10 \text{ ms}^{-2}$)



Column I

- (A) The time in seconds when m_1 is lifted off the ground
(B) The time in seconds when m_2 is lifted off the ground
(C) Acceleration in m/s^2 of m_1 at $t = 30 \text{ s}$
(D) Acceleration in m/s^2 of m_2 at $t = 60 \text{ s}$

Column II

- (P) 10
(Q) 5
(R) 20
(S) 40
(T) 30

SECTION-I

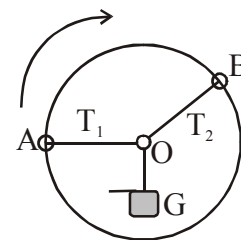
Single Correct Answer Type

5 Q. [Marks 3 (–1)]

1. A rigid hoop can rotate in a vertical plane about the center. Two massless strings are attached to the hoop, one at A, the other at B. These strings are tied together at the center of the hoop at O, and a weight G is suspended from that point. The strings have a fixed length, regardless of the tension, and the weight G is only supported by the strings. Originally OA is horizontal.

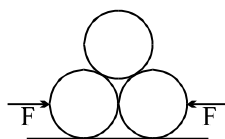
Now, the outer hoop will start to slowly rotate 90° clockwise until OA will become vertical, while keeping the angle between the strings constant and keeping the object static. Which of the following statements about the tensions.

T_1 and T_2 in the two strings is correct?



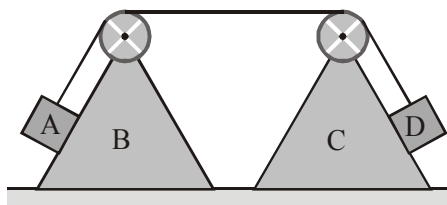
- (A) T_1 always decreases.
(B) T_1 always increases.
(C) T_2 always increases.
(D) T_2 will become zero at the end of the rotation.

2. Two smooth cylindrical bars weighing W each lie next to each other in contact. A similar third bar is placed over the two bars as shown in figure. Neglecting friction, the minimum horizontal force on each lower bar necessary to keep them together is :-



- (A) $W/2$ (B) W (C) $W/\sqrt{3}$ (D) $W/(2\sqrt{3})$

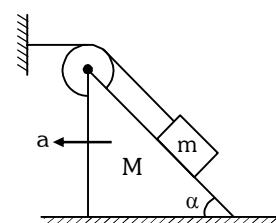
3. The figure shows two wedges and two blocks. Let acceleration of A and D are $\vec{a}_1$ and $\vec{a}_4$ wrt B and C respectively, and acceleration of B and C are $\vec{a}_2$ and $\vec{a}_3$ wrt ground respectively. The correct relation between these four vectors is :-



- (A) $\vec{a}_1 + \vec{a}_4 = \vec{a}_2 + \vec{a}_3$ (B) $\vec{a}_1 + \vec{a}_4 = -(\vec{a}_2 + \vec{a}_3)$
(C) $\vec{a}_1 = \vec{a}_4$ (D) None of the above

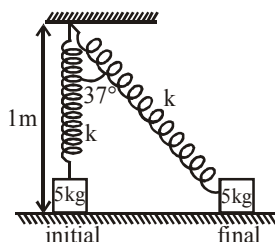
4. In the adjoining figure if acceleration of M with respect to ground is a , then:

- (A) acceleration of m with respect to M is $2a$
(B) acceleration of m with respect to ground is $2a \sin(\alpha/2)$
(C) acceleration of m with respect to ground is a
(D) acceleration of m with respect to ground is $a \tan \alpha$



5. In the figure shown, initially spring is relaxed. Spring constant of spring is $k = 100 \text{ N/m}$. Find minimum

value of μ (coefficient of friction) between surface & block so that block remains in equilibrium in final stage :-



(A) $\mu = 0.1$

(B) $\mu = 0.2$

(C) $\mu = 0.4$

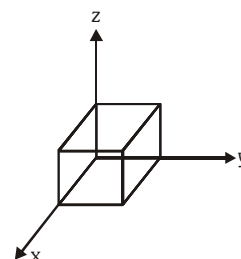
(D) $\mu = 0.5$

Multiple Correct Answer Type

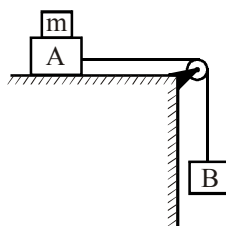
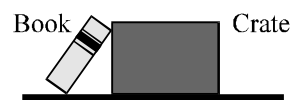
6 Q. [4 M (-2)]

6. A solid cube of mass 5 kg is placed on a rough horizontal surface, in xy-plane as shown. The friction coefficient between the surface and the cube is 0.4. An external force $\vec{F} = 6\hat{i} + 8\hat{j} + 20\hat{k}$ N is applied on the cube. (use $g=10\text{m/s}^2$)

- (A) The block starts slipping over the surface
(B) The friction force on the cube by the surface is 10 N.
(C) The friction force acts in xy-plane at angle 127° with the positive x-axis in anticlockwise direction.
(D) The contact force exerted by the surface on the cube is $10\sqrt{10}$ N.



7. A book leans against a crate on a table. Neither is moving. Which of the following statements concerning this situation is/are **INCORRECT**?
- (A) The force of the book on the crate is less than that of crate on the book.
(B) Although there is no friction acting on the crate, there must be friction acting on the book or else it will fall.
(C) The net force acting on the book is zero.
(D) The direction of the frictional force acting on the book is in the same direction as the frictional force acting on the crate.
8. Two blocks of mass 6 kg (A) and 3 kg (B) are connected with a string passing over a pulley as shown. The block 6 kg lies on a horizontal rough surface. What can be the values of additional mass m placed on the block A so that the system does not accelerate? (The coefficient of friction between all surface is 0.3)



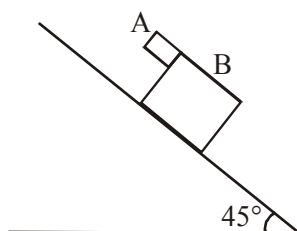
(A) 3 kg

(B) 6 kg

(C) 9 kg

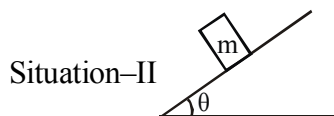
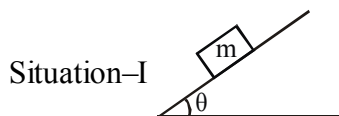
(D) 12 kg

9. Two cubical blocks A & B of mass 1 kg and 5 kg are released from the given position as shown in figure. All the surfaces are smooth :-

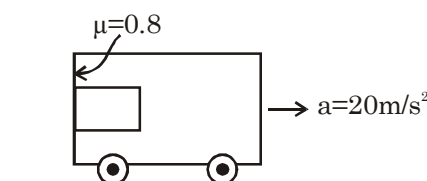


- (A) Acceleration of A relative to B is $5\sqrt{2} \text{ m/s}^2$ (B) Acceleration of A relative to ground 10 m/s^2
(C) Force on B by A is $5\sqrt{2} \text{ N}$ (D) Net force on A is 10 N

10. A block is first placed on its long side and then on its short side on the same inclined plane (see figure). The block slides down in situation II but remains at rest in situation I. A possible explanation is



- (A) The normal contact force is less in situation-II.
(B) The frictional force is less in situation-II because the contact area is less.
(C) The shorter side is smoother.
(D) In situation-I, frictional force is more.
11. A truck is accelerating on horizontal road with an acceleration 20 m/s^2 . A block placed against the rear wall of the truck as shown has a mass of 100 kg and coefficient of friction between it and the wall is 0.8 .



- (A) Acceleration of the box is 20 m/s^2
(B) The friction acting on the block is 1600 N
(C) The friction acting on the block is 1000 N
(D) The contact force between the wall and block is 1000 N .

Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 3Q.) [Marks 3 (0)]

Paragraph for Question no. 12 to 14

A shot putter with a mass of 80 kg pushes the iron ball of mass of 6 kg from a standing position, accelerating it uniformly from rest at an angle of 45° with the horizontal during a time interval of 0.1 seconds. The ball leaves his hand when it is 2 m high above the level ground and hits the ground 2 seconds later.

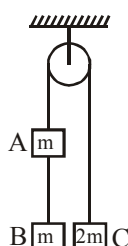
12. The acceleration of the ball in shot putter's hand
(A) $11\sqrt{2} \text{ m/s}^2$ (B) $100\sqrt{2} \text{ m/s}^2$ (C) $90\sqrt{2} \text{ m/s}^2$ (D) $9\sqrt{2} \text{ m/s}^2$
13. The horizontal distance between the point of release and the point where the ball hits the ground
(A) 16 m (B) 18 m (C) 20 m (D) 22 m
14. The minimum value of the static coefficient of friction if the shot putter does not slip during the shot is closest to
(A) 0.28 (B) 0.38 (C) 0.48 (D) 0.58

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

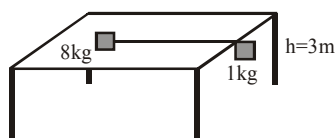
2 Q. [4 M (0)]

1. Two objects A & B are connected by a light string and these blocks are further connected to block C by thread that passes over a light frictionless pulley as shown. The objects are at rest. At $t = 0$, block B is given a velocity 40 m/s in vertical upward direction. Find the time (in sec) after which string between A & B again becomes tight (string is large, blocks do not collide with each other or with pulley)



CLASS TEST

2. On a horizontal table with a height 3m, there is a block of mass 8 kg at rest. The block is connected by a long (length $\gg$ height of table = 3m) massless string to a second block of mass 1 kg which hangs from the edge of table as shown in the figure. The blocks are then released. Find distance (in meter) between the points where the two blocks hit ground. Neglect friction every where, and assume the blocks do not bounce back after collision with ground. ($g = 10 \text{ m/s}^2$).

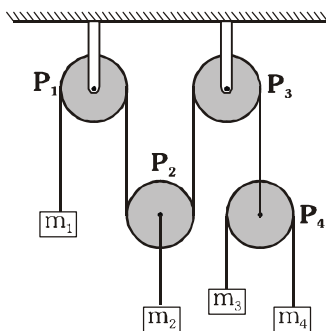


SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

1. Match the entries of column I with column II. Given that the pulleys are massless and frictionless and four masses are of $m_1 = 2\text{kg}$, $m_2 = 3\text{kg}$ and $m_3 = m_4 = 1\text{kg}$ connected with ideal strings as shown in figure.



Column I

- (A) Acceleration of pulley P_4
(B) Acceleration of mass m_3
(C) Tension in string attached to block of mass m_2
(D) Acceleration of mass m_2

Column II

- (P) $\frac{2g}{7}$ vertical up
(Q) $\frac{12g}{7}$ newton
(R) $\frac{g}{7}$ vertical downward
(S) $\frac{g}{7}$ vertical upward
(T) $\frac{24}{7}g$ newton

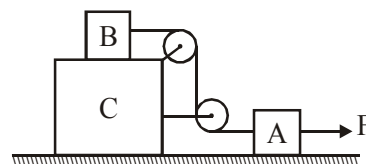
2. In figure, A, B and C have masses 2 kg, 5 kg and 10 kg respectively. A horizontal force F is applied to A. Strings are either horizontal or vertical. In column I is given information reading F and in column II information about accelerations of A, B and C and tension in the string T . [No friction anywhere]

Column I

- (A) $F = 35 \text{ N}$
(B) $F = 21 \text{ N}$
(C) $F = 49 \text{ N}$
(D) $F = 20 \text{ N}$

Column II

- (P) $a_A = 5 \text{ m/s}^2$
(Q) $a_C = 0 \text{ m/s}^2$
(R) $a_B = 3 \text{ m/s}^2$
(S) $T = 5 \text{ N}$
(T) $T = 35 \text{ N}$

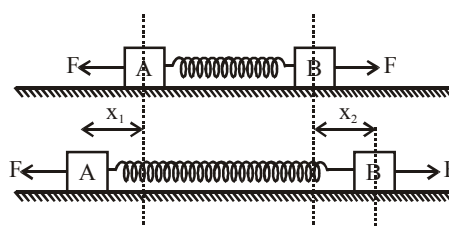


SECTION-I

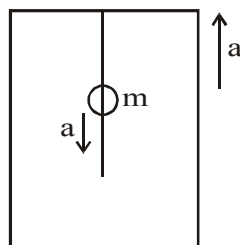
Single Correct Answer Type

8 Q. [Marks 3 (–1)]

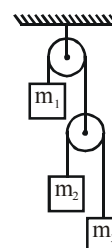
1. A 60 kg boy notices a nice fruit on a tree. Even with stretched arms he cannot reach it and is 0.5 m below the fruit. The boy crouches down - his centre of mass descends 0.4m and then he leaps up. Thus, he just reaches the fruit. Assume that he exerts a constant force on the ground while he is in contact with the ground. What is this force at least? ($g = 10 \text{ m/s}^2$)
(A) 1000 N (B) 1350 N (C) 1800 N (D) 500 N
2. Initially spring is relaxed having spring constant 50 N/m. If blocks A and B are displaced by x_1 and x_2 such that $x_1 = 30 \text{ cm}$ & $x_2 = 10 \text{ cm}$. If force applied by spring on A is F_1 & on B is F_2 then:-



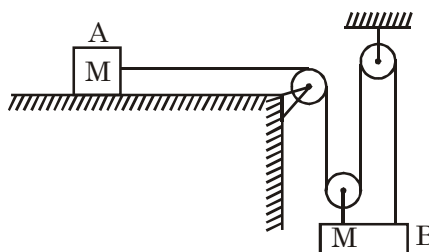
- (A) $F_1 = 15 \text{ N}$; $F_2 = 5 \text{ N}$ (B) $F_1 = 5 \text{ N}$; $F_2 = 15 \text{ N}$
(C) $F_1 = 20 \text{ N}$; $F_2 = 20 \text{ N}$ (D) $F_1 = 15$; $F_2 = 15$
3. A cabin is moving vertically upwards with an acceleration a . A massless string is attached to the ceiling of the cabin on which a bead of mass m is sliding downwards w.r.t. string with an acceleration a . The tension in the string is



- (A) $m(g + 2a)$ (B) $m(g + a)$ (C) mg (D) $m(g - a)$
4. The block of mass m_1 remains stationary if :-
(A) $\frac{1}{m_1} = \frac{4}{m_2} + \frac{1}{m_3}$ (B) $\frac{1}{m_1} = \frac{1}{m_2} + \frac{4}{m_3}$
(C) $\frac{1}{m_1} = \frac{2}{m_2} + \frac{2}{m_3}$ (D) $\frac{4}{m_1} = \frac{1}{m_2} + \frac{1}{m_3}$

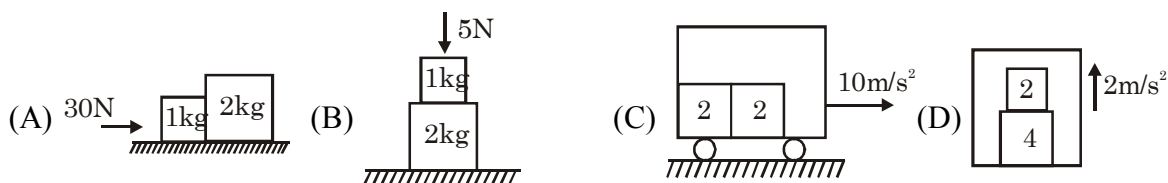


5. In the given arrangement all contact surfaces shown, are smooth. Acceleration of B block will be :

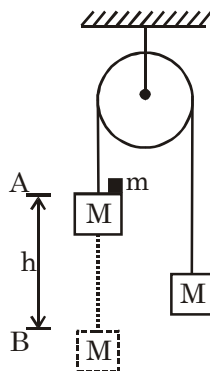
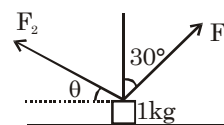


- (A) 1 m/s^2 (B) 2 m/s^2 (C) 3 m/s^2 (D) None

6. In which of the following figures contact force between blocks is minimum :-



7. If the acceleration of particle is $g \text{ m/s}^2$ in vertically upwards direction. For the minimum value of F_2 , what will be the value of F_1 , F_2 and θ will be respectively :-
 (A) 10N , $10\sqrt{3}\text{N}$ and $\theta = 60^\circ$ (B) $10\sqrt{3}\text{N}$, 10N and $\theta = 30^\circ$
 (C) $10\sqrt{3}\text{N}$, 10N and $\theta = 60^\circ$ (D) 20N , 10N and $\theta = 60^\circ$
8. In the figure, pulley and cord have negligible mass and friction. The system is balanced with equal masses M on each side as shown (solid line), and then a small mass m is gently placed to one block at position A. The combined masses accelerate through a certain distance h , then mass m is gently removed at position B and two equal masses then move on with constant speed, v . Find the value of v . (That corresponds to the values $m = 250 \text{ gm}$, $M = 1 \text{ kg}$, $h = 16.2\text{cm}$, $g = 10 \text{ m/s}^2$)



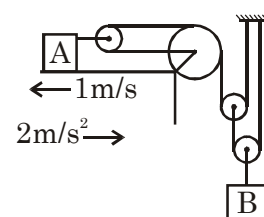
- (A) 0.6 m/s (B) 4 m/s (C) 5 m/s (D) 6 m/s

Multiple Correct Answer Type

5 Q. [4 M (-2)]

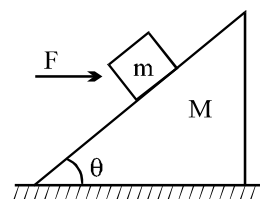
9. In the given figure find the velocity and acceleration of B, if velocity and acceleration of A are as shown. Choose the incorrect options :-

- (A) $v_B = 1 \text{ m/s} \uparrow$ & $a_B = 1 \text{ m/s}^2 \downarrow$
 (B) $v_B = 0.5 \text{ m/s} \uparrow$ & $a_B = 0.5 \text{ m/s}^2 \downarrow$
 (C) $v_B = 1 \text{ m/s} \uparrow$ & $a_B = 0.5 \text{ m/s}^2 \downarrow$
 (D) $v_B = 0.5 \text{ m/s} \uparrow$ & $a_B = 1 \text{ m/s}^2 \downarrow$

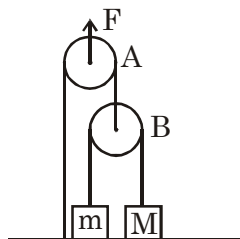


10. A small cubical block of mass m is placed on a triangular block M so that they touch each other along a smooth inclined plane as shown. The inclined surface makes an angle θ with the horizontal. A horizontal force F is to be applied on the block m so that the two bodies move without slipping against each other. Assuming the floor to be smooth also, which of the following statement(s) is/are **CORRECT**?

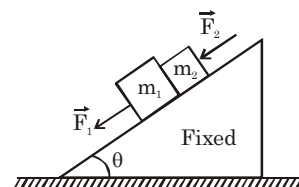
- (A) Normal force with which m and M press against each other is $\frac{mg}{\cos \theta}$.
 (B) Normal force with which m and M press against each other is $mg \cos \theta$
 (C) The magnitude of external force F is $\frac{(M+m)mg \tan \theta}{M}$
 (D) The magnitude of external force F is $\frac{(M+m)mg \cot \theta}{M}$



11. Two blocks of mass $m = 5\text{ kg}$ & $M = 10\text{ kg}$ are connected by a string passing over a pulley B as shown. Another string connects the centre of pulley B to the floor and passes over another pulley A as shown. An upward force $F = 300\text{ N}$ is applied at the centre of A. [Both pulley are ideal & massless & $g = 10\text{ m/sec}^2$] :-



- (A) $a_m = 5\text{ m/sec}^2$ (B) $a_m = 0\text{ m/sec}^2$ (C) $a_M = 0\text{ m/sec}^2$ (D) $a_M = 5\text{ m/sec}^2$
12. Two blocks of masses m_1 and m_2 ($m_1 > m_2$) are placed on the fixed inclined plane. Forces of $\vec{F}_1$ and $\vec{F}_2$ are applied on blocks m_1 and m_2 respectively along the inclined plane in downward direction as shown in figure. Here 'N' is the normal contact force between the blocks. The surface between blocks and inclined plane is smooth. Select the **CORRECT** alternative.
- (A) $N = 0$ if $\vec{F}_1 = \vec{F}_2 = 0$ (B) $N = 0$ if $\vec{F}_1 = \vec{F}_2 \neq 0$
- (C) $N \neq 0$ if $\vec{F}_1 = \vec{F}_2 \neq 0$ (D) $N = 0$ if $\frac{\vec{F}_1}{m_1} = \frac{\vec{F}_2}{m_2}$



13. Each of the system shown below is initially at rest. Pulleys are massless and frictionless.

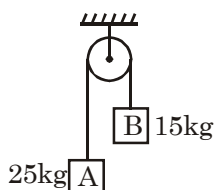


Figure (1)

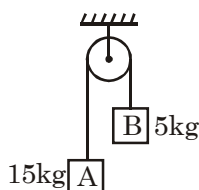


Figure (2)

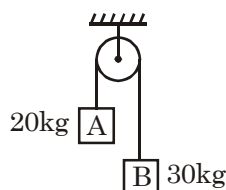


Figure (3)

- (A) Magnitude of acceleration of block A in figure 1 is same as that in figure 2.
- (B) Magnitude of acceleration of block A in figure 1 is less than that in figure 2 and greater than that in figure 3.
- (C) Tension in string connected to ceiling is maximum in figure (3)
- (D) The ratio of magnitude of velocity of block A in figure 1 to that in fig. 2 after it has moved through 1 m is $1/\sqrt{2}$.

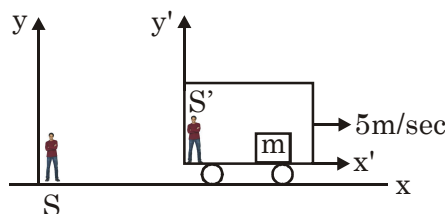
Linked Comprehension Type

(1 Para $\times$ 2 Q.) [4 M (-2)]

(Multiple Correct Answer Type)

Paragraph for Question no. 14 and 15

There are two frames of reference S and S' first is one being fixed to ground and second one is fixed to the cart moving with constant velocity 5 m/sec w.r.t ground as shown in figure. A block having mass $m = 2\text{ kg}$, initially at rest w.r.t. S' is acted upon by a 10 N force at $t = 0\text{ sec.}$ in positive X' direction. (all surfaces are friction less).



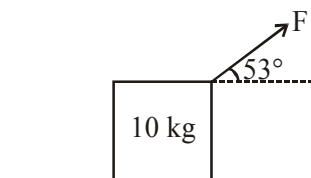
14. If force applied on block of mass m for 2 sec than :-
 (A) Change in velocity of block w.r.t. S' is 10 m/s
 (B) Change in velocity of block w.r.t. S' is 5 m/s
 (C) Change in velocity of block w.r.t. S is 10 m/s
 (D) Change in velocity block w.r.t. S is 5 m/s
15. Choose the **CORRECT** statement :-
 (A) S' frame is inertial frame of reference.
 (B) S frame is inertial frame of reference.
 (C) S' is non-inertial frame of reference because we apply force on mass m kg.
 (D) From the given data we cannot decide the frame is inertial or not.

SECTION-II

Numerical Answer Type Question (upto second decimal place)

8Q.[3(0)]

1. A block of mass 10 kg is kept on a smooth horizontal surface as shown. A force F is acted at an angle of 53° with horizontal. Find the value of F (in N) such that block moves with maximum acceleration on smooth horizontal surface.

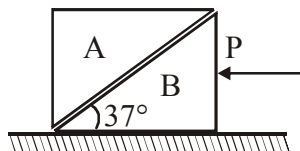


SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. In given figure blocks A and B each have same mass $m = 0.4$ kg. A horizontal force P is applied to B so that A will not slip up on B. What is force (in newton) exerted by B on A. Neglect any friction.

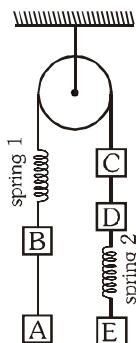


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. The system shown below is initially in equilibrium $m_A = m_B = 3$ kg, $m_C = m_D = m_E = 2$ kg



Column-I

- (A) Just after the spring 2 is cut the block D
 (B) Just after the spring 2 is cut the block B
 (C) Just after the spring 2 is cut the block A
 (D) Just after the spring 1 is cut the block D

Column-II

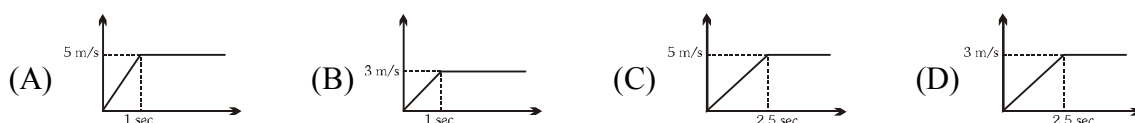
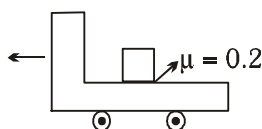
- (P) Accelerates up
 (Q) Accelerates down
 (R) Acceleration is zero
 (S) Acceleration is $3g/2$
 (T) Acceleration is $g/2$

SECTION-I

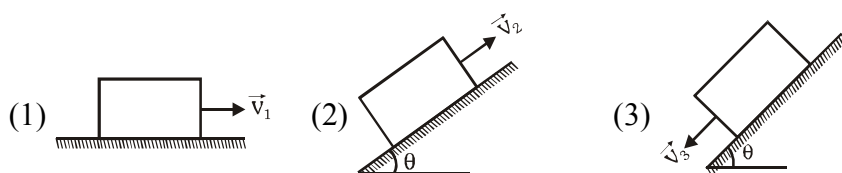
Single Correct Answer Type

5 Q. [3 M (-1)]

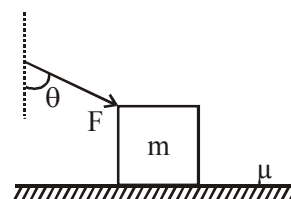
1. A truck starting from rest moves with an acceleration of 5 m/s^2 for 1 sec and then moves with constant velocity. The velocity w.r.t ground v/s time graph for block in truck is (Assume that block does not fall off the truck)



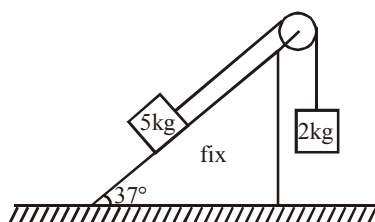
2. Three identical blocks move either on a horizontal surface, up a plane, or down a plane, as shown below. They start with different speeds and continue to move until brought to rest by friction. They all move the same distance. Rank the three situations according to the initial speeds, least to greatest. [Friction is same for each surface]



- (A) The same for all cases
(B) 1, 2, 3
(C) 3, 1, 2
(D) 2, 1, 3
3. If μ is the coefficient of friction between the road and the tyre of a car, the minimum time in which the car can cover a distance 's', starting from rest is
(A) directly proportional to μ
(B) inversely proportional to μ
(C) inversely proportional to $\sqrt{\mu}$
(D) directly proportional to $\sqrt{\mu}$
4. A block of mass $m = 1 \text{ kg}$ is placed on a horizontal surface. It is being pushed by a force $F = 25 \text{ N}$ making an angle $\theta = 37^\circ$ with the vertical. If the friction coefficient is $\mu = 0.8$, then the magnitude of acceleration of the block will be :-
(A) 25 m/s^2
(B) 17 m/s^2
(C) 9 m/s^2
(D) zero



5. In the arrangement shown in figure coefficient of friction between 5kg block and incline plane is $\mu = 0.5$. Friction force acting on the 5kg block is :-



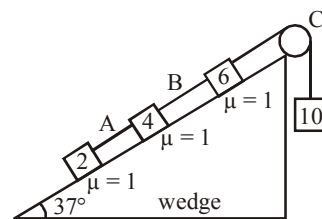
- (A) 20 N
(B) 15 N
(C) 10 N
(D) 5 N

Multiple Correct Answer Type

5 Q. [4 M (-1)]

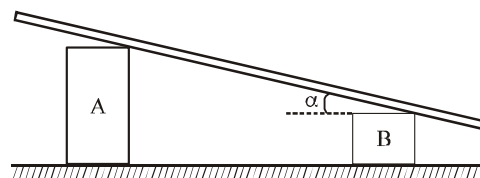
6. Figure shows a system consisting of some blocks and wedge as shown in figure. Wedge is fixed and all block can move on the surface and mass of pulley and string are negligible. Choose the **CORRECT** statement(s).

- (A) Acceleration of block of mass 10kg is zero
(B) Thrust on the pulley is $80\sqrt{5}$ N
(C) The maximum mass that can be hung in place of 10kg block so that it does not move is 16 kg
(D) If the friction suddenly vanishes, the 10 kg block will move downward



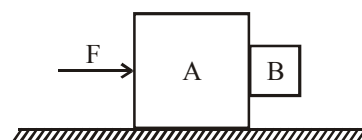
7. A plank inclined at an angle of α to the horizontal lies on two supports A and B (figure), over which it can slip without friction under the action of its own weight Mg . With what acceleration and in what direction should a man of mass m move along the plank so that it should not slip ?

- (A) the man should move down the plank
(B) the acceleration of the man should be $g \sin \alpha \left(1 + \frac{M}{m}\right)$
(C) the acceleration of the man should be $g \sin \alpha \left(1 - \frac{M}{m}\right)$
(D) the man should move up the plank

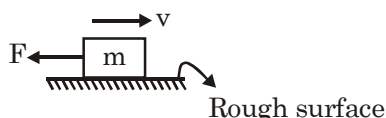


8. A constant horizontal force F is acting on block A, as shown in the figure. Friction between A & horizontal surface is negligible. But friction is present between A & B. Contact surface of A & B is vertical. Due to friction there is no slipping between A & B. Mass of A is $2m$ and that of B is m & value of F is $6mg$. Then :

- (A) Friction force on B must be greater than mg
(B) Coefficient of friction between A & B must be greater than or equal to 0.5
(C) Coefficient of friction between A & B may be 0.8
(D) Friction force on block B may be greater than mg .



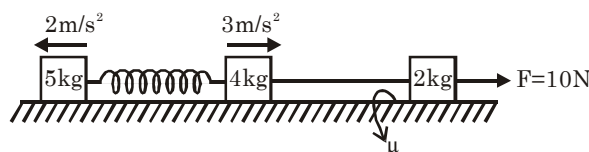
9. An object of mass ' m ' is moving to the right on a rough horizontal surface. At $t = 0$ a force F (large than the limiting friction force) is applied to the left as shown in figure. Select the **CORRECT** alternative :-



- (A) Initially, force $\vec{F}$ and friction force will act in same direction.
(B) Object comes to rest for a moment and after that its motion is accelerating in direction of $\vec{F}$.
(C) Object slow down and remains at rest.
(D) Magnitude of rate of change of speed for block when it is moving to right is same as when it is moving to left.

10. System is shown at an instant. There is friction only between 2kg and ground and rest all surfaces are smooth. Then

- (A) Acceleration of 2kg block at this instant is 2 m/s^2 .
(B) Acceleration of 2kg block at this instant is 3 m/s^2 .
(C) Coefficient of friction between 2kg and ground is $\mu = 0.2$.
(D) Coefficient of friction between 2kg and ground is $\mu = 0.1$.

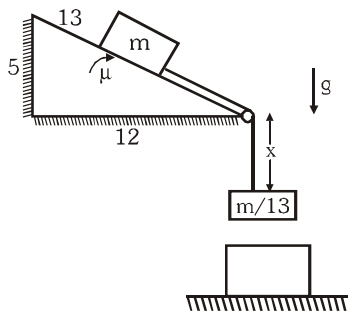


Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Question 11 to 13

A block of mass m slides on an inclined plane of sufficient large size with a slope of $5/12$ (i.e. the slope of the hypotenuse of a $5:12:13$ triangle). A massless rope, guided by a massless pulley, connects the block to a second block of mass $m/13$, which is hanging freely above a lower table.



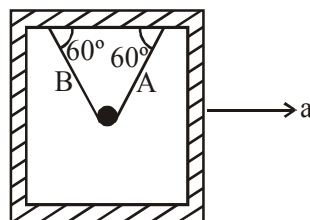
- The two block system is observed to be moving with a constant velocity v_0 . What is the coefficient of sliding friction μ between block and plane?
(A) $1/3$ (B) $1/4$ (C) $1/2$ (D) $1/5$
- After the hanging block hits the table, what is the distance s along the surface of the plane which the top block continues to slide before stopping? (Assume that the plane is long enough that the block does not fall off)
(A) $15v_0^2/4g$ (B) $13v_0^2/2g$ (C) $13v_0^2/3g$ (D) $15v_0^2/2g$
- What will be the acceleration of blocks as a function of length x ($x < L$ and hanging from pulley towards lower block) of uniform string (mass of string in this case is m and total length of string is L). (No friction between string & incline)
(A) $(5L+9x)g/27L$ (B) $(5L+8x)g/27L$ (C) $(5L+3x)g/27L$ (D) Independent of x

SECTION-III

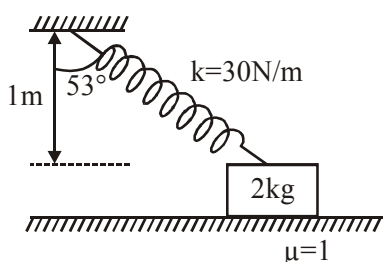
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

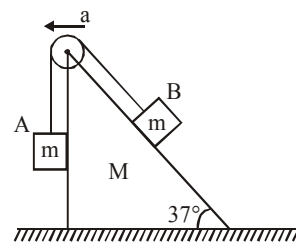
- The steel ball is suspended in an accelerating frame by two cords A and B. If the acceleration of the frame is $a = \frac{g}{3\sqrt{3}}$, then the tension in A is x times the tension in B. Find x .



- The spring shown in the figure has a natural length of 1m . What is the initial acceleration (in m/sec^2) of the block when released?



3. In the given figure a wedge of mass M is kept on a horizontal smooth surface. Two blocks of equal mass m are arranged as shown in figure. All surfaces are smooth. Find the value of acceleration a (in m/s^2), so that blocks A and B do not slip over the wedge.



SECTION-IV

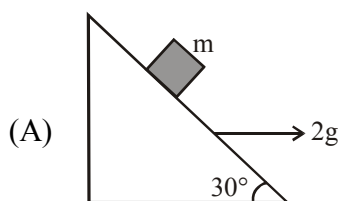
Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

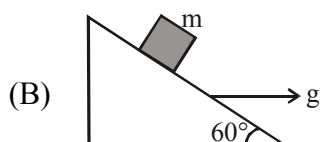
1. In the following arrangements if $\mu = \tan \theta$ then match the situation in column-I with effect in column-II.

Column-I

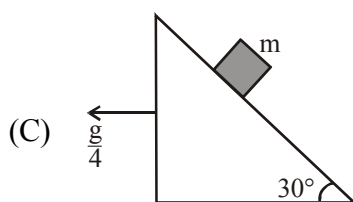
Column-II



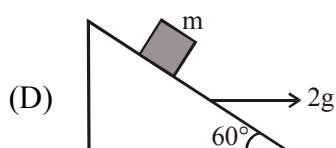
(P) Block will not slide with respect to wedge



(Q) Block will slide upward with respect to wedge



(R) Block will slide down ward with respect to wedge



(S) Friction will act upward

(T) Friction will be equal to μN

2. In the following arrangement string is light and inextensible. All surfaces are smooth. Treating:

a = acceleration of wedge A

b = acceleration of block B

T = tension in the string

N = normal reaction on block B applied by wedge A

Column-I

Column-II

(A) $\frac{4a}{g}$

(P) $\sqrt{3}$

(B) $\frac{8T}{mg}$

(Q) $3\sqrt{3}$

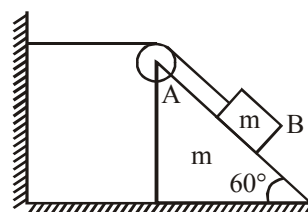
(C) $\frac{8N}{mg}$

(R) 1

(D) $\frac{3b}{a}$

(S) 3

(T) $\frac{1}{\sqrt{3}}$

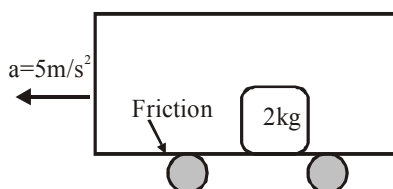


SECTION-I

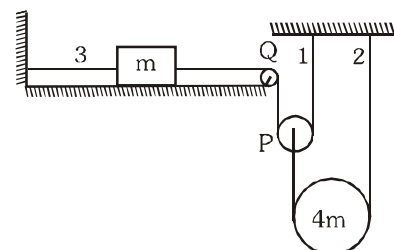
Single Correct Answer Type

6 Q. [3 M (–1)]

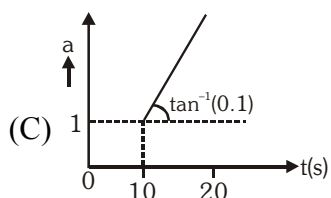
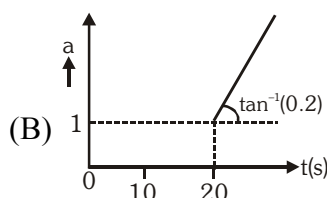
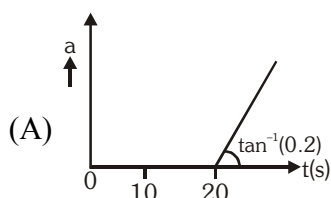
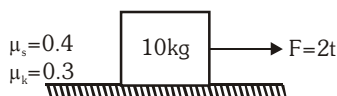
1. A block is placed on a surface with vertical cross-section given by the equation $y = \frac{x^2}{20}$. If the coefficient of friction is 0.5, the maximum height above the ground at which a block can be placed without slipping is
(A) 1.00 m (B) 1.25 m (C) 1.50 m (D) 1.90 m
2. A block of mass 2kg, is kept on a horizontal surface of a trolley which accelerates towards left with an acceleration 5m/s^2 as shown in figure. If friction coefficient between block and trolley is 0.4, find the acceleration of block for a ground observer.



3. Shown in the diagram is a system of two bodies, a block of mass m and a disc of mass $4m$, held in equilibrium. If the string 3 is burnt, find the acceleration of the disc. Neglect the masses of the pulleys P and Q. The co-efficient of friction between the block and horizontal surface is 0.5 and friction between disc and string is zero.



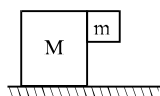
4. For shown situation the acceleration of block is represented by. (Here t is time in second) ($g = 10\text{ms}^{-2}$)



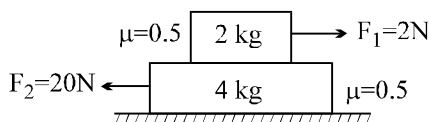
(D) None of these

CLASS TEST

5. With what minimum acceleration mass M must be moved on frictionless surface so that m remains stick to it as shown. The coefficient of friction between M & m is μ .



- (A) μg (B) $\frac{g}{\mu}$ (C) $\frac{\mu mg}{M+m}$ (D) $\frac{\mu mg}{M}$
6. The friction force acting between surfaces in contact in the adjoining figure is best represented by



- (A)
- (B)
- (C)
- (D)

Multiple Correct Answer Type

7. The block shown in figure remains at rest undisturbed. A man wishes to start the block by the rope. Given the ramp angle α and the coefficient of static friction μ_s ,

(A) the value of the angle β that minimizes the force T is given

by $\beta = \tan^{-1} \mu_s$

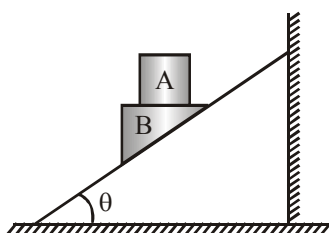
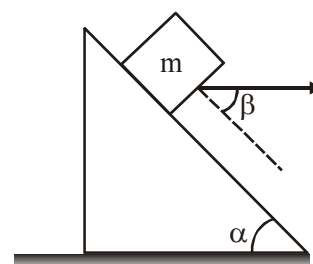
(B) the value of the angle β that minimizes the force T is given

by $\beta = \cot^{-1} \mu_s$

(C) the minimum value of T in terms of α , μ_s , m and g is given by $T = mg \frac{(\mu_s \cos \alpha - \sin \alpha)}{\sqrt{1 + \mu_s^2}}$

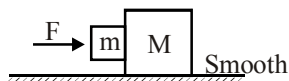
(D) the minimum value of T in terms of α , μ_s , m and g is given by $T = mg \frac{(\mu_s \sin \alpha + \cos \alpha)}{\sqrt{1 + \mu_s^2}}$

8. The coefficient of friction between the block A of mass m and block B of mass $2m$ is $\mu = \sqrt{3}$. There is no friction between block B and the inclined plane. If the system of blocks A and B is released from rest and there is no slipping between A and B, then the possible value(s) of θ can be:

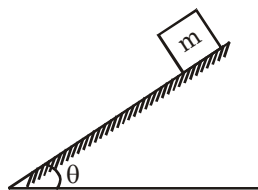
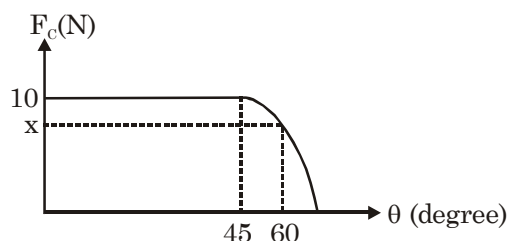


- (A) 30° (B) 45° (C) 53° (D) 70°

9. Figure shows a block of mass m in contact with wedge of mass M . Force F is such that block is just stationary relative to wedge. Contact surface between block and wedge is rough whereas ground is smooth. Coefficient of friction is μ . Mark the **CORRECT** statement.



- (A) Acceleration of block and wedge is $\frac{F}{M+m}$
- (B) Normal reaction between block and wedge has magnitude $\frac{MF}{m+M}$
- (C) Minimum force for which block and wedge move in combination is $F = \frac{mg}{\mu} \left[1 + \frac{m}{M} \right]$
- (D) Minimum force for which block and wedge move in combination is $F = \frac{mg}{\mu}$
10. An object of mass m is kept on a rough inclined surface ($\mu_s = \mu_k$). If angle of inclination is gradually increased, the net contact force F_c offered by surface on the object vary with incline angle θ , as shown in the graph. ($g = 10 \text{ m/s}^2$)



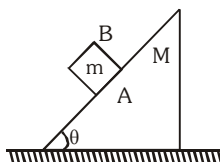
- (A) Mass of object is 1 kg
- (B) Coefficient of friction is $\sqrt{3}$
- (C) Coefficient of friction is 1
- (D) Value of x is $5\sqrt{5} \text{ N}$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Question 11 to 13

A block B of mass m is kept over a wedge A of mass M as shown in the figure. Wedge is kept over a frictionless horizontal surface.



11. The acceleration of wedge so that block does not slip over the wedge is (assume friction between block & wedge negligible)–
- (A) $g \cot \theta$ (B) $g \tan \theta$ (C) $\frac{m}{M+m}g$ (D) $\frac{M}{m+M}g$
12. If coefficient of friction between wedge & block is μ , the maximum acceleration of the wedge so that block does not slip over wedge is ($\tan \theta > \mu$)
- (A) $\left(\frac{\mu + \tan \theta}{1 - \mu \tan \theta} \right) g$ (B) $\left(\frac{\tan \theta - \mu}{1 + \mu \tan \theta} \right) g$ (C) $\left(\frac{m \tan \theta + M}{1 - \mu \tan \theta} \right) g$ (D) $\left(\frac{M - m \tan \theta}{1 + \mu \tan \theta} \right) g$

13. For the situation in question (12), minimum value of acceleration is ($\tan \theta > \mu$)

- (A) $\left[\frac{M - m \tan \theta}{1 + \mu \tan \theta} \right] g$ (B) $\left[\frac{M + m \tan \theta}{1 - \mu \tan \theta} \right] g$ (C) $\left[\frac{\tan \theta - \mu}{1 + \mu \tan \theta} \right] g$ (D) $\left[\frac{\mu + \tan \theta}{1 - \mu \tan \theta} \right] g$

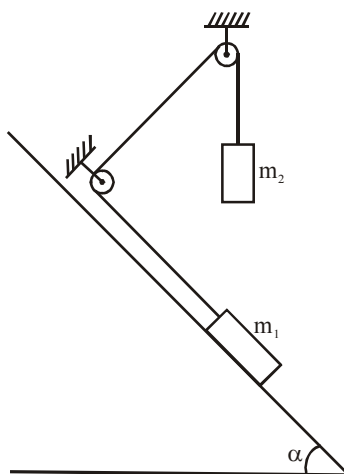
Linked Comprehension Type

(1 Para $\times$ 2 Q.) [4 M (-1)]

(Multiple Correct Answer Type)

Paragraph for Question no. 14 and 15

Figure shows two masses $m_1 = 10$ kg and m_2 connected by string passing over smooth fixed pulleys. Mass m_1 lies on a fixed rough surface having a coefficient of friction $\mu = 0.8$. The angle of incline $\alpha = 37^\circ$.



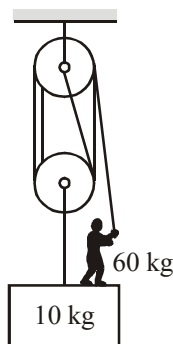
14. The possible value(s) of m_2 so that it remains at rest is :-
(A) 1 kg (B) 5 kg (C) 10 kg (D) 15 kg
15. For $m_2 = 12$ kg and system is released from rest at $t = 1$ s. Choose the **correct** statement(s) :-
(A) Speed of m_1 at $t = 1$ s is zero.
(B) Speed of m_1 at $t = 2$ s is zero.
(C) Speed of m_1 as seen by m_2 at $t = 1$ s is zero.
(D) Speed of m_1 as seen by m_2 at $t = 2$ s is zero.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. A person of mass 60 kg stands on block of mass 10 kg as shown in figure. If the tension in rope that the person must apply so that the system remains in equilibrium is 20α newton. Find α . (radii of pulley is negligible)

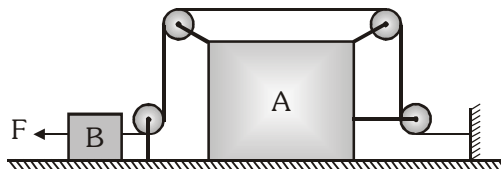


SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

1. In the shown diagram A is of mass 50 kg, B is of mass 5 kg. Coefficient of friction between A and ground is 0.2 and friction is absent between B and ground. A horizontal force F is applied on B. f_1 is friction force between A and ground, a_1 is acceleration of A, a_2 is acceleration of B.



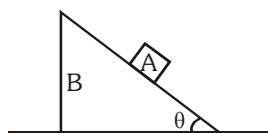
Column I

- (A) $F = 100 \text{ N}$
(B) $F = 125 \text{ N}$
(C) $F = 192.5 \text{ N}$
(D) $F = 80 \text{ N}$

Column II

- (P) $f_1 = 80 \text{ N}$
(Q) $a_1 = 0$
(R) $f_1 = 125 \text{ N}$
(S) $a_1 = 1 \text{ m/s}^2$
(T) $a_2 = 1 \text{ m/s}^2$

2. A block A is placed on wedge B, which is placed on horizontal surface. All the contact surfaces are rough but friction is not sufficient to prevent sliding at any surface. Match Column I and II. Column II indicates possible direction(s) of the physical quantities mentioned under Column I. X and Y axes are along the incline and perpendicular to the incline.



Column-I

- (A) Acceleration of A
(B) Net force applied by A on B
(C) Acceleration of A relative to B
(D) Net force applied by ground on B

Column-II

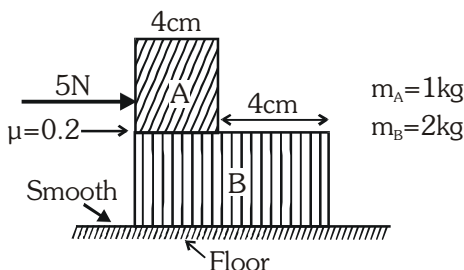
- (P)

SECTION-I

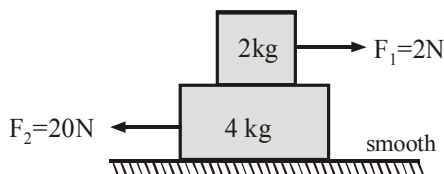
Single Correct Answer Type

7 Q. [3 M (-1)]

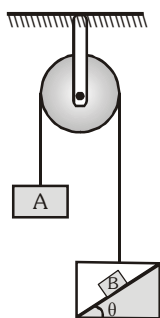
1. For shown situation in figure (Assume : $g = 10 \text{ ms}^{-2}$) –



- (A) The acceleration of the block A is 1 m/s^2
 (B) The acceleration of the block B is 3 m/s^2
 (C) The time taken for the front face of A lining up with the front face of B is 0.2 sec
 (D) The time taken for the front face of A lining up with the front face of B is 0.50 sec
2. In the arrangement shown in figure, coefficient of friction between the two blocks is $\mu = 1/2$. The force of friction acting between the two blocks is

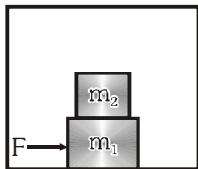


- (A) 8 N (B) 10 N (C) 6 N (D) 4 N
3. String is massless and pulley is smooth in the adjoining figure. Total mass on left hand side of the pulley is m_1 and on right hand side is m_2 . Friction coefficient between block B and the wedge is $\mu = \frac{1}{2}$ and $\theta = 30^\circ$. Select the wrong option

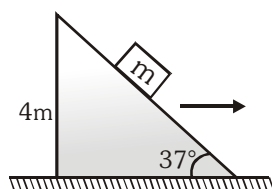


- (A) block B will slide down if $m_1 = m_2$
 (B) block B may remain stationary with respect to wedge for suitable values of m_1 and m_2 with $m_2 > m_1$
 (C) block B can not remains stationary with respect to wedge in any case
 (D) block B will slide down if $m_1 > m_2$

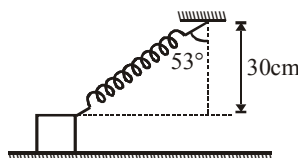
4. A plank of mass $m_1 = 8\text{kg}$ with a bar of mass $m_2 = 2\text{kg}$ placed on its rough surface, lie on a smooth floor of elevator ascending with an acceleration $g/4$. The coefficient of friction is $\mu = 1/5$ between m_1 and m_2 . A horizontal force $F = 30\text{N}$ is applied to the plank. Then the acceleration of bar and the plank in the reference frame of elevator are:-



- (A) $3.5\text{ m/s}^2, 5\text{ m/s}^2$ (B) $5\text{ m/s}^2, \frac{50}{8}\text{ m/s}^2$ (C) $2.5\text{ m/s}^2, \frac{25}{8}\text{ m/s}^2$ (D) $4.5\text{ m/s}^2, 4.5\text{ m/s}^2$
5. If the coefficient of friction between block & wedge is μ , then the maximum horizontal acceleration of the wedge for which block will remain at rest w.r.t. the wedge is-



- (A) $\left(\frac{3-4\mu}{4+3\mu}\right)g$ (B) $\left(\frac{1+\mu}{1-\mu}\right)g$ (C) $\left(\frac{1-\mu}{1+\mu}\right)g$ (D) $\left(\frac{3+4\mu}{4-3\mu}\right)g$
6. A smooth block is released at rest on a 45° incline and then slides a distance d . The time taken to slide on rough incline is n times as much to slide on a smooth incline. The coefficient of friction is-
- (A) $\mu_k = 1 - \frac{1}{n^2}$ (B) $\mu_k = \sqrt{1 - \frac{1}{n^2}}$ (C) $\mu_s = 1 - \frac{1}{n^2}$ (D) $\mu_s = \sqrt{1 - \frac{1}{n^2}}$
7. A block of mass 2 kg is connected with a spring of natural length 40 cm of force constant $K = 200\text{ N/m}$. The coefficient of friction is $\mu = 0.5$. When released from the given position, acceleration of block will be

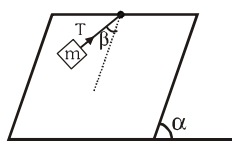


- (A) zero (B) 2 m/s^2 (C) 6 m/s^2 (D) 8 m/s^2

Multiple Correct Answer Type

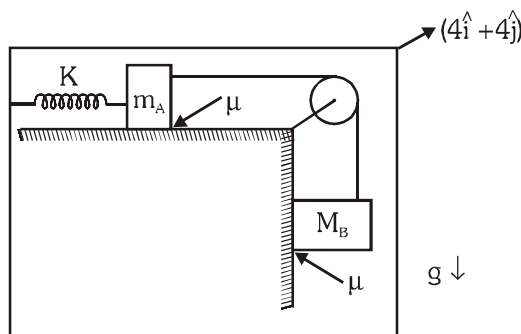
6 Q. [4 M (-1)]

8. A block of mass m is hanged at an angle β with line of greatest slope on an incline of angle α as shown. Minimum value of μ for no slipping is, T is tension at minimum μ .

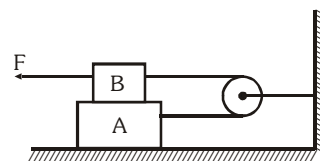


- (A) $\mu_{\min} = \frac{\sin \alpha \sin \beta}{\cos \alpha}$ (B) $\mu_{\min} = \frac{\sin \alpha \cos \beta}{\cos \alpha}$ (C) $T = mg \sin \alpha \cos \beta$ (D) $T = mg \cos \alpha \sin \beta$

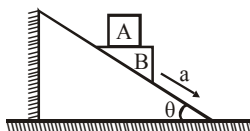
9. The arrangement shown in the diagram is moving with acceleration $\vec{a} = 4(\hat{i} + \hat{j}) \text{ m/s}^2$. An ideal spring of natural length ℓ_0 having spring constant $K = 50 \text{ N/m}$ is connected to block A. Blocks A & B are connected by an ideal string passing through frictionless pulley. Mass of each block A & B is 2 kg . If the friction coefficient between all the surfaces is $5/9$ then just after releasing from rest, (initially spring is in its natural length) :-



- (A) spring force is equal to tension in the string. (B) force exerted by the spring will be zero
(C) the value of tension is close to 23.6 N (D) if $\mu = 2/3$ then spring force is zero.
10. The force F_1 that is necessary to move a body up an inclined plane is double the force F_2 that is necessary to just prevent it from sliding down, then (Where ϕ = angle of friction, θ = angle of inclined plane, w = weight of the body)
(A) $F_2 = w \sin(\theta - \phi) \sec \phi$ (B) $F_1 = w \sin(\theta - \phi) \sec \phi$
(C) $\tan \phi = 3 \tan \theta$ (D) $\tan \theta = 3 \tan \phi$
11. Two blocks each of mass 1 kg are placed as shown. They are connected by a string which passes over a smooth (massless) pulley. There is no friction between A and the ground and the coefficient of friction between A and B is 0.2 . A force F is applied to B. Which of the following statements is/are correct :-
(A) The system will be in equilibrium if $F < 4 \text{ N}$
(B) If $F > 4 \text{ N}$ the tension in the string will be 4 N
(C) if $F > 4 \text{ N}$ the frictional force between the block will be 2 N
(D) if $F = 6 \text{ N}$ the tension in the string will be 3 N

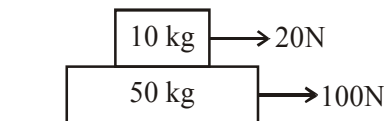


12. In the given figure, a block A rests on a smooth triangular block B and the block B is moved at an acceleration of $a = 2 \text{ m/s}^2$ along the plane :-



- (A) normal force on block A due to block B is $m(g - a)$
(B) Acceleration of block A relative to block B is $a \cos \theta$
(C) If friction is present between block A and B, the coefficient should be greater than $\frac{a}{g} \cos \theta$, for no relative motion between A and B
(D) If friction is present between block and A and B, the coefficient of friction should be greater than $\frac{a \cos \theta}{g - a \sin \theta}$, for no relative motion between A and B

13. Two blocks of mass 10 kg & 50 kg are placed on smooth horizontal surface. If surface between blocks is rough then choose the **CORRECT** statement(s).



- (A) Friction acts on upper block towards right & static
(B) Friction acts on lower block towards right & kinetic
(C) There is not friction between blocks
(D) Both blocks move together

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Question 14 to 16

In diagram, the friction coefficient between the block of mass 1 kg and the plank of mass 2 kg is 0.4 while that between the plank and floor is 0.1. A constant force 'F' starts acting horizontally on the upper 1 kg block.



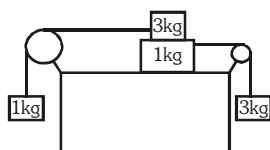
14. The acceleration of plank if $F = 10 \text{ N}$ is
(A) 2.5 m/s^2 (B) 1.5 m/s^2 (C) 0.5 m/s^2 (D) 1.0 m/s^2
15. The friction force between plank and block if $F = 2 \text{ N}$, is
(A) 3 N (B) 4 N (C) 2.5 N (D) 2 N
16. For what value of F will the block move with double the acceleration of that of the plank?
(A) 6 N (B) 10 N (C) 5 N (D) 12.5 N

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. The value of the friction coefficient acting between the blocks of 1kg and 3kg, so that all the blocks move with same acceleration is μ . Find the value of 6μ . Consider table as smooth.



SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

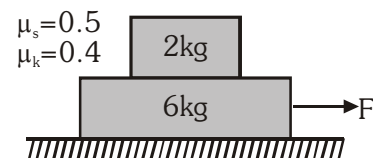
1. All surfaces in contact have same value of friction coefficients. Let frictional force between 2kg & 6kg be f_1 and between 6kg and ground be f_2 . ($g = 10 \text{ ms}^{-2}$)

Column-I

- (A) $F = 36\text{N}$
(B) $F = 48\text{N}$
(C) $F = 64\text{N}$
(D) $F = 96\text{N}$

Column-II

- (P) $f_1 = 4\text{N}, f_2 = 32\text{N}$
(Q) $f_1 = 8\text{N}, f_2 = 32\text{N}$
(R) $f_1 = 10\text{N}, f_2 = 32\text{N}$
(S) $f_1 = 0, f_2 = 36\text{N}$

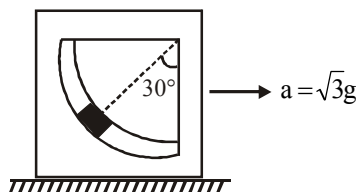


SECTION-I

Single Correct Answer Type

8 Q. [3 M (-1)]

1. A wedge with a rough groove in the shape of a quarter of a circle is kept on a smooth table (see figure). A disc is placed in the groove with a small clearance. Friction exists between groove and disc. The wedge is moved with an acceleration $\sqrt{3}g$. If disc is to remain stationary relative to groove, the coefficient of friction required can be.



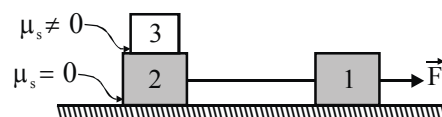
- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{5}$ (D) $\frac{9}{10}$
2. Three blocks of equal mass M are initially at rest on smooth floor. A force $\vec{F}$ is applied to the system so that the three blocks are to move together. Mark the correct option.

(A) The minimum coefficient of friction required is $\frac{F}{3Mg}$.

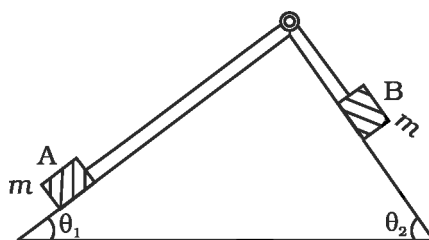
(B) The minimum coefficient of friction required is $\frac{F}{Mg}$.

(C) The minimum coefficient of friction required is $\frac{F}{2Mg}$.

(D) The minimum coefficient of friction required is $\frac{3F}{2Mg}$.

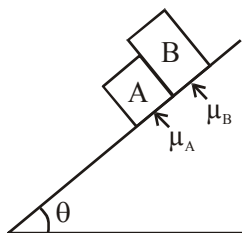


3. In figure, a body A of mass m slides on plane inclined at angle θ_1 to the horizontal and μ_1 is the coefficient of friction between A and the plane. A is connected by a light string passing over a frictionless pulley to another body B, also of mass m , sliding on a frictionless plane inclined at angle θ_2 to the horizontal. Which of the following statements is **INCORRECT** ?

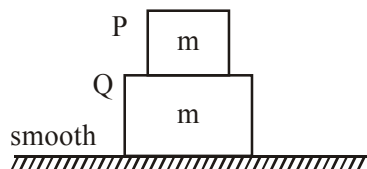


- (A) B will never move up the plane.
- (B) A will just start moving up the plane when $\mu_1 = \frac{\sin \theta_2 - \sin \theta_1}{\cos \theta_1}$
- (C) For A to move up the plane, θ_2 must always be greater than θ_1
- (D) B will always slide down with constant speed.

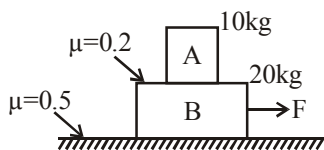
4. For given situation select the **INCORRECT** statement



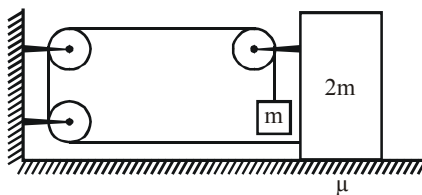
- (A) If $\mu_A > \mu_B$ then contact force depends on mass as well as angle of inclination of inclined plane.
 (B) In every condition contact force depends on mass and the inclination of the inclined plane
 (C) If $\mu_A = \mu_B$ then contact force between the blocks is zero
 (D) If $\mu_A = \mu_B = 0$ then contact force between them will be zero
5. Two blocks P and Q of same mass 'm' are placed over one another, on a smooth horizontal surface. The lower block Q is held fixed and force is applied to P. The minimum force required to slide P on Q is 12N. Now if Q is free to move and force is applied to Q, then the minimum force required to slide P on Q is:



- (A) 12 N (B) 18 N (C) 24 N (D) 30 N
6. A block of mass 2.5 kg is kept on a rough horizontal surface. It is found that the block does not slide if a horizontal force less than 15 N is applied to it. Also it is found that it takes 5 seconds to slide through the first 10 m if a horizontal force of 15 N is applied and the block is gently pushed to start the motion. Taking $g = 10 \text{ m/s}^2$, calculate the ratio of coefficients of static and kinetic friction between the block and the surface :-
- (A) $\frac{3}{5}$ (B) $\frac{13}{15}$ (C) $\frac{15}{13}$ (D) $\frac{4}{5}$
7. Two blocks A and B are as shown in figure. The minimum horizontal force F applied to the block B for which slipping begins between B and ground is :



- (A) 20 N (B) 120 N (C) 150 N (D) 170 N
8. In the system shown in figure the friction coefficient between ground and bigger block is μ . There is no friction between both the blocks. The string connecting both the block is light, all three pulley are light and frictionless. Then the minimum limiting value of μ so that the system remains in equilibrium is :-

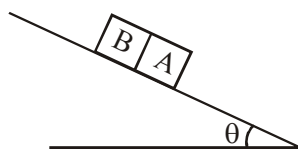


- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

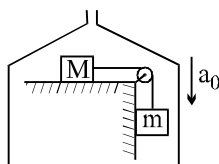
Multiple Correct Answer Type

6 Q. [4 M (–1)]

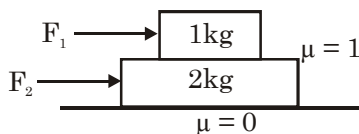
9. Two blocks having same mass are placed on rough incline plane and the coefficient of friction between A and incline is $\mu_1 = 1.0$ and between block B and incline is $\mu_2 = \frac{3}{4}$. As the inclination of the plane 'θ' with respect to horizontal increases, choose the correct answer (s).



- (A) There is no contact force between block A and B for $\theta \leq 37^\circ$
 (B) There is no contact force between block A and B for $\theta \leq 45^\circ$
 (C) As $\mu_1 > \mu_2$ both the blocks move together
 (D) They start moving at an angle $\theta = \tan^{-1} \frac{7}{8}$
10. Imagine a situation in which the given arrangement is placed inside an elevator that can move only in the vertical direction and compare the situation with the case when it is placed on the ground. When the elevator accelerates downward with $a_0 (< g)$. Coefficient of friction between M and surface in contact is μ while m is smooth, then (pulley and string are ideal)

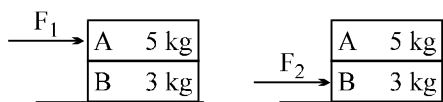


- (A) the limiting friction force between the block M and the surface decreases
 (B) the system can accelerate with respect to the elevator even when $m < \mu M$
 (C) the system does not accelerate with respect to the elevator unless $m > \mu M$
 (D) the tension in the string decreases
11. Two blocks of masses 1 kg and 2 kg are placed one over another on a smooth surface as shown. Coefficient of friction between the blocks is $\mu = 1$. Horizontal forces F_1 and F_2 are acting on 1 kg and 2kg respectively :-

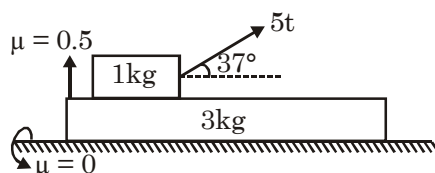


- (A) If $F_1 = 9\text{ N}$ and $F_2 = 0$, then both the blocks move together with common acceleration 3 m/s^2
 (B) If $F_1 = 9\text{ N}$ and $F_2 = 0$, then frictional force on 1 kg will be along the direction of force.
 (C) If $F_1 = 0$ and $F_2 = 36\text{ N}$, both the blocks move together with common acceleration 12 m/s^2 .
 (D) If $F_1 = 1\text{ N}$ and $F_2 = 2\text{ N}$, frictional force will not be acting on both the blocks.

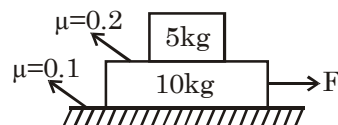
12. A block A (5 kg) rests over another block B (3 kg) placed over a smooth horizontal surface. There is friction between A and B. A horizontal force F_1 gradually increasing from zero to a maximum is applied to A so that the blocks move together without having motion relative to each other. Instead of this, another horizontal force F_2 gradually increasing from zero to a maximum is applied to B so that the blocks move together without relative motion. The magnitudes of friction between the blocks in the two cases are f_1 & f_2 respectively during the variation of F_1 and F_2 respectively. Then



- (A) $f_{1\max} > f_{2\max}$ (B) $F_{1\max} : F_{2\max} = 3 : 5$
(C) $F_{1\max} : F_{2\max} = 5 : 3$ (D) $f_1 < f_2$
13. A block of mass 1kg is placed over a long plank of mass 3kg. The friction coefficient between block and plank is 0.5. The system is placed over a smooth horizontal surface. A time varying force $F = 5t$ Newton starts acting on the block as shown in figure. Select the **CORRECT** alternative:-



- (A) Block loose contact from the plank before relative motion starts between two blocks.
(B) Block loose contact from the plank after the starts of relative motion.
(C) Relative motion between block and plank starts at $t = \frac{10}{9}$ sec.
(D) Relative motion between block and plank starts at $t = \frac{30}{9}$ sec.
14. Mass of the blocks and coefficient of friction between each surfaces are shown in the figure. Choose **CORRECT** statements :-
- (A) Minimum force required to move the system is $F = 10$ N.
(B) For $F = 30$ N, both the blocks move together.
(C) For $15\text{N} < F \leq 45\text{N}$, both blocks moves with same acceleration.
(D) For $F \leq 50\text{N}$, there is no relative slipping between blocks.

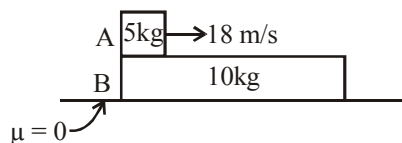


Linked Comprehension Type (Single Correct Answer Type)

(2 Para $\times$ 2Q.) [3 M (-1)]

Paragraph for Question No 15 and 16

Block A of mass 5kg is on a plank of mass 10kg. There is no friction between plank and ground but coefficient of friction between block and plank is 0.4. Block is given velocity 18 m/s as shown in the figure at $t = 0$.

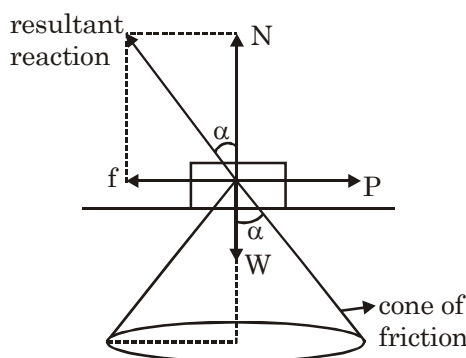


15. After how much time relative motion stops between block and plank :-
(A) 4.5 sec (B) 3 sec (C) 6 sec (D) 5 sec
16. Minimum length of plank so that block does not fall off the plane :-
(A) 27 m (B) 36 m (C) 18 m (D) 24 m

Paragraph for Questions 17 and 18

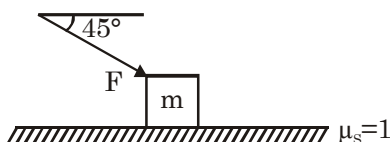
When a body is about to move on a surface in the direction of applied force, the resultant reaction force makes a limiting angle α with normal. If direction of applied force is gradually changed through an angle 360° , keeping the force in the same plane the resultant reaction of the surface, generates a right circular cone with semicentral angle α . This inverted cone with semicentral angle α is called cone of friction. If the resultant reaction force of the surface is

- within this cone the body is stationary
- on the surface of cone, the motion of body is impending.
- outside the cone of friction, the body is in motion.



It can be inferred easily that if direction of net force [force of gravity + applied force] exerted on object placed on a surface is such that its line of action is inside the frictional cone, then object will not move regardless of magnitude of force.

- An object is kept on a rough horizontal surface having coefficient of static friction $\mu_s = \sqrt{3}$. The semicentral angle of frictional cone will be :-
(A) 30° (B) 45° (C) 60° (D) 90°
- An object is kept on rough horizontal surface having coefficient static friction $\mu_s = 1$, as shown. A force F is applied on the block at an angle 45° .



- Object may move if F is sufficiently large
- Object must move for all values of F
- Object will not move, irrespective of value of force F
- More information is needed.

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

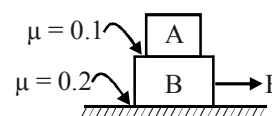
- In the figure shown, the coefficient of friction between the two blocks is 0.1 and coefficient of friction between the block B and ground is 0.2, masses of A and B are 20 kg and 40 kg respectively then match the following, where a_A = acceleration of A and a_B = acceleration of B.

Column-I

- $F = 10 \text{ N}$
- $F = 50 \text{ N}$
- $F = 150 \text{ N}$
- $F = 300 \text{ N}$

Column-II

- $a_A = 0 \text{ m/s}^2$
- $a_A = 0.5 \text{ m/s}^2$
- $a_B = 0.5 \text{ m/s}^2$
- $a_A = 1 \text{ m/s}^2$



SECTION-I

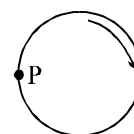
Single Correct Answer Type

4 Q. [3 M (-1)]

1. In a rotor, a hollow vertical cylindrical structure rotates about its axis and person rest against the inner wall. At a particular speed of the rotor, the floor below the person is removed and the person hangs resting against the wall without any floor. If the radius of the rotor is 2m and coefficient of static friction between the wall and person is 0.2, then the minimum speed at which the floor may be removed is ($g = 10 \text{ m/sec}^2$)

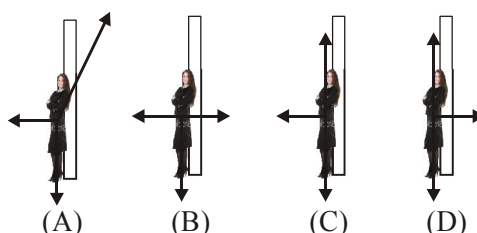
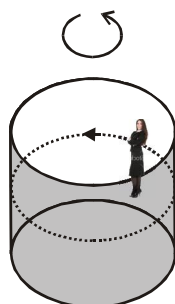
(A) 10ms^{-1} (B) 1 ms^{-1} (C) 100ms^{-1} (D) 5ms^{-1}

2. The diagram shows a CD rotating clockwise (as seen from above) in the CD-player. After turning it off, the CD slows down. Assuming it has not come to a stop yet, the direction of the acceleration of point P at this instance is



(A) (B) (C) (D)

3. A girl finds herself stuck with her back to the wall of a cylinder rotating about its axis. Which diagram correctly shows the forces acting on her?



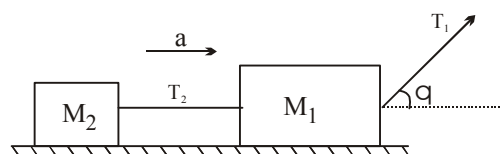
4. A man tows his daughter on a sled on level ice, and she in turn tows behind her a toy sled. The girl and her sled have a combined mass of M_1 and the toy sled a mass M_2 . Dad's tow rope is inclined at an angle θ to the horizontal. The coefficient of kinetic friction between the sled runners and the ice is μ , and the mass of the ropes is negligible. If the sleds accelerate with acceleration a , derive an expression for the tension in each rope.

(A) $T_1 = \frac{M_2(a + \mu g) + M_1(a + \mu g)}{\cos \theta + \mu \sin \theta}$; $T_2 = 0$

(B) $T_1 = M_2(a + \mu g)$; $T_2 = 0$

(C) $T_1 = \frac{M_2(a + \mu g) + M_1(a + \mu g)}{\cos \theta + \mu \sin \theta}$; $T_2 = M_2(a + \mu g)$

(D) $T_1 = 0$; $T_2 = M_2(a + \mu g)$



Multiple Correct Answer Type

1 Q. [4 M (-1)]

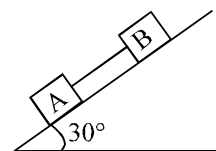
5. Two blocks A and B are placed on an inclined plane of inclination 30° as shown in figure. If A and B each have mass m and $\mu_A = 0$ and $\mu_B = \sqrt{\frac{2}{3}}$, where μ_A and μ_B are the coefficients of friction between plane and the bodies A & B respectively,

(A) The acceleration of the system is $\frac{g}{2} \left(1 - \frac{1}{\sqrt{2}} \right)$

(B) Tension in the string is $\frac{mg}{2\sqrt{2}}$

(C) The acceleration of the system is $\frac{g}{2} \left(1 + \frac{1}{\sqrt{2}} \right)$

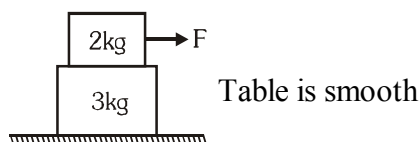
(D) Tension in the string is zero



Linked Comprehension Type (Single Correct Answer Type)

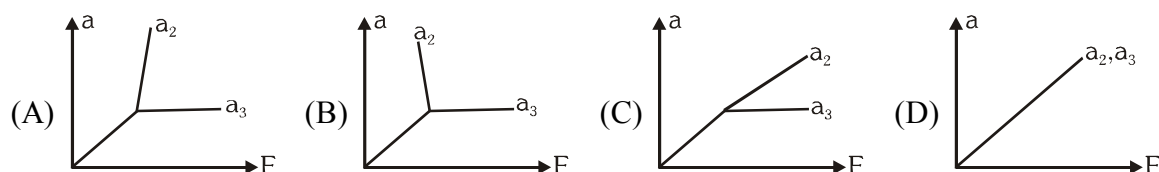
(2 Para × 3Q.) [3 M (-1)]

Paragraph for Question 6 to 8



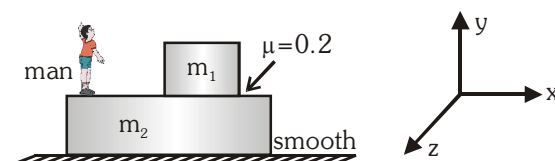
The slipping starts between the blocks when the force of 25 N acts on 2 kg block.

6. Acceleration of both the blocks when 40 N force acts on 2 kg block is
 (A) $a_2 = 7.5 \text{ m/s}^2$, $a_3 = 5 \text{ m/s}^2$ (B) $a_2 = 12.5 \text{ m/s}^2$, $a_3 = 5 \text{ m/s}^2$
 (C) $a_2 = 7.5 \text{ m/s}^2$, $a_3 = 25/3 \text{ m/s}^2$ (D) $a_2 = 12.5 \text{ m/s}^2$, $a_3 = 25/3 \text{ m/s}^2$
7. The force applied on 3 kg block so that slipping start between 2 kg & 3 kg blocks
 (A) 25 N (B) 37.5 N (C) 50 N (D) 62.5 N
8. Select the correct graph of acceleration of blocks with force F is



Paragraph for Question 9 to 11

The system shown in the figure is initially at origin and is moving with a velocity $(5\text{ms}^{-1})\hat{k}$. A force $(120t \text{ newton})\hat{i}$ acts on mass m_2 , where t is time in seconds. The man throws a ball of very small mass, at the instant when m_1 starts slipping on m_2 , with a velocity 10 ms^{-1} vertically up w.r.t himself. Taking the masses of blocks and man as 60kg each and assuming that the man never slips on m_2 . Take $g=10\text{ms}^{-2}$ and neglect the dimensions of the system.



9. The time at which man throws the ball-

(A) 1s (B) 3s (C) 6s (D) $4/3 \text{ s}$

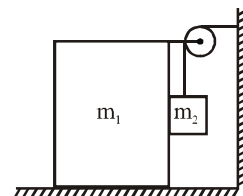
10. Projection velocity of ball w.r.t. ground is
(A) $3\hat{i} + 10\hat{j} + 5\hat{k}$ (B) $1.5\hat{i} + 10\hat{j} + 5\hat{k}$ (C) $0.5\hat{i} + 10\hat{j} + 5\hat{k}$ (D) $10\hat{j} + 5\hat{k}$
11. z-coordinate of the point where ball lands, is-
(A) 10m (B) 15m (C) 25m (D) 33m

SECTION-III

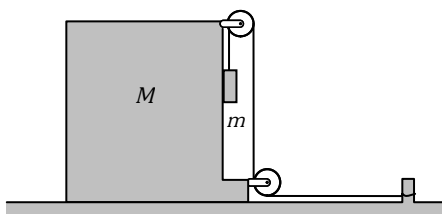
Numerical Grid Type (Ranging from 0 to 9)

6 Q. [4 M (0)]

1. In the arrangement shown friction between masses m_1 & m_2 ($m_1 = 4m_2$) is negligible, but friction is present between m_1 & horizontal surface. If acceleration of m_1 is found $\frac{g}{10}$ horizontally, coefficient of friction between m_1 & horizontal surface is $x/49$, then what is the value of x ?

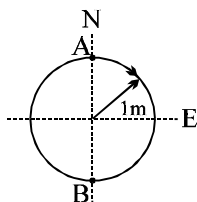


2. In the system shown, coefficient of static friction between all the surfaces in contact is 0.2.

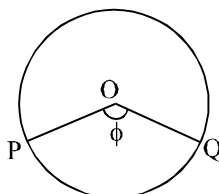


For which minimum ratio of values of masses M to m , there is no slipping anywhere?

3. A projectile is projected from horizontal ground with velocity 18 km/hr at an angle of 60° from horizontal. Find angular speed (in rad/s) as observed from the point of projection at the time of landing. [$g = 10 \text{ m/s}^2$]
4. A particle starts from rest and move on a circular path of radius 36.0 m. Its speed increases uniformly by 0.25 m/s in each meter of the distance traveled. Calculate the magnitude of its acceleration in SI units when it has traveled a distance of 48.0 m.
5. In one second a particle moves with constant speed from point A to point B along the circular track of radius 1.0 m as shown in the figure. The average acceleration of the particle (in m/s^2) during this motion is found to be $n\pi$ where n is an integer. Find the value of n . [Particle is moving from A to B in clockwise direction.]



6. A car starts at point P from rest, but with a constant tangential acceleration of 2 m/s^2 . The car is constrained to travel on a circular track of unknown radius. When the car reaches point Q, the angle between the acceleration and velocity vectors is 37° . The centre of the track is point O. What is the angle (in radians) POQ? If your answer is in the form of α/β where α & β are least integer values then find the value of $\beta - \alpha$.

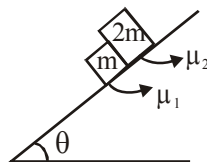


SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

1. Two blocks of mass m and $2m$ are slowly just placed in contact with each other on a rough fixed inclined plane as shown. Initially both the blocks are at rest on inclined plane. The coefficients of friction between the blocks and inclined surface are shown. There is no friction between the blocks. Column I gives four situations. Column II gives conditions under which statements in column I are true. Match the statement in column I with corresponding conditions in column II.



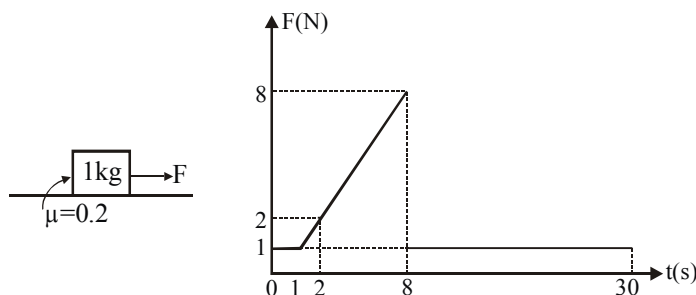
Column I

- (A) The magnitude of acceleration of both blocks are same, if
(B) The normal reaction between both the blocks is zero, if
(C) The net reaction exerted by inclined surface on each block make same angle with inclined surface, if
(D) The net reaction exerted by inclined surface on block of mass $2m$ is double that of net reaction exerted by inclined surface on block of mass m , if

Column II

- (P) $\mu_1 = \mu_2 > \tan\theta$
(Q) $\mu_1 > \mu_2 > \tan\theta$
(R) $\mu_1 < \mu_2 < \tan\theta$
(S) $\mu_1 = \mu_2 < \tan\theta$
(T) $\mu_1 < \tan\theta; \mu_2 > \tan\theta$

2. A block of mass 1 kg is placed on a rough horizontal surface of coefficient friction $\mu = 0.2$. A force is applied on the block horizontally whose variation with time is shown in the figure. Match column I with column II.



Column I

- (A) Velocity of particle is zero at
(B) Acceleration of particle is non-zero at
(C) Friction force = 1 N
(D) Friction force $\leq F$

Column II

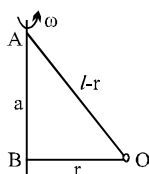
- (P) $t = 1\text{ s}$
(Q) $t = 4\text{ s}$
(R) $t = 20\text{ s}$
(S) $t = 25\text{ s}$
(T) $t = 30\text{ s}$

SECTION-I

Single Correct Answer Type

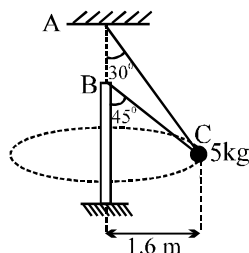
2 Q. [3 M (-1)]

1. A string of length l has its ends fastened to two points A & B in the same vertical line at a distance 'a' apart. If a smooth bead O on the string rotates uniformly about B so that BO is always horizontal, then angular velocity is :



- (A) $\ell \sqrt{\frac{2g}{a(\ell^2 - a^2)}}$ (B) $\frac{\ell}{2} \sqrt{\frac{2g}{a(\ell^2 - a^2)}}$ (C) $\frac{\ell}{2} \sqrt{\frac{g}{a(\ell^2 - a^2)}}$ (D) $\ell \sqrt{\frac{g}{a(\ell^2 - a^2)}}$

2. Two light wires AC and BC are tied at C to a sphere which revolves at constant speed v in the horizontal circle shown in the figure. Determine the range of values of v for which both wires remain tight.



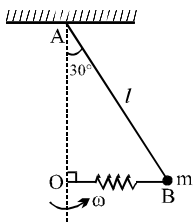
- (A) $\frac{2}{5^{1/4}} < V < 2$ (B) $2 < V < 4$ (C) $\frac{4}{3^{1/4}} < V < 4$ (D) can't determine

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q. & 1 Para × 4Q.) [3 M (-1)]

Paragraph for Question No. 3 to 5 (3 Questions)

The end A of an inextensible light string of length l is attached to a fixed point, and an object of mass m is attached to the other end B. A light spring of natural length $\frac{1}{4}l$ and stiffness $\frac{\sqrt{3}mg}{l}$ is attached to B and to a smooth pivot at the point O. The system rotates with angular speed ω about the vertical line OA. The angle $OAB = 30^\circ$ and OB is horizontal, as shown in the figure.



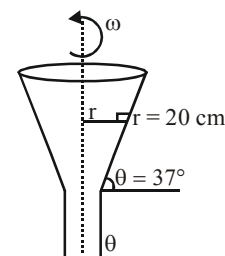
3. Evaluate the tension in the string "AB" in terms of 'm' and 'g'

- (A) mg (B) $\frac{2}{\sqrt{3}} mg$ (C) $\frac{4}{\sqrt{3}} mg$ (D) $2mg$

4. Evaluate the tension in the spring OB in terms of 'm' and 'g'.
- (A) $3mg$ (B) $\frac{\sqrt{3}}{4}mg$ (C) $\frac{4}{\sqrt{3}}mg$ (D) 0
5. Find ' ω ' in terms of 'g' and ' ℓ '.
- (A) $\left[\frac{7\sqrt{3}g}{6\ell}\right]^{\frac{1}{2}}$ (B) $\left[\frac{3\sqrt{3}g}{4\ell}\right]^{\frac{1}{2}}$ (C) $\left[\frac{3\sqrt{2}g}{\ell}\right]^{\frac{1}{2}}$ (D) 0

Paragraph for Question No. 6 to 9 (4 Questions)

A very small cube of mass 2kg is placed on the surface of a funnel as shown in figure. The funnel is rotating about its vertical axis of symmetry with angular velocity ' ω '. The wall of funnel makes an angle 37° with horizontal. The distance of cube from the axis of rotation is 20 cm and frictional coefficient is μ . (Take $g = 10 \text{ m/s}^2$)



6. The frictional force acting between the block and surface (if $\mu = 0.3$) of funnel at $\omega = 5 \text{ rad/s}$ is :-
(A) 6.6 N (B) 4N (C) 2.2 N (D) Zero
7. For what value of ω there would be no frictional force acting between the surfaces ?
(A) 5 rad/sec (B) $\sqrt{\frac{75}{2}}$ rad/sec (C) $\sqrt{6}$ rad/sec (D) $\sqrt{40}$ rad/sec
8. The maximum value of angular velocity for which no relative slipping occurs and also direction of frictional force is : (take $\mu = 2/3$) :-
(A) $\sqrt{\frac{425}{3}}$ rad/sec, down the surface of funnel (B) $\sqrt{\frac{425}{3}}$ rad/sec, up the surface of funnel
(C) $\sqrt{\frac{25}{9}}$ rad/sec, up the surface of funnel (D) $\sqrt{\frac{25}{9}}$ rad/sec, down the surface of funnel
9. The minimum value of angular velocity for which no relative slipping occurs is : ($\mu = 2/3$) :-
(A) $\sqrt{\frac{25}{9}}$ (B) $\sqrt{\frac{25}{3}}$ (C) $\sqrt{\frac{9}{25}}$ (D) $\sqrt{\frac{3}{25}}$

SECTION-II

Numerical Answer Type Question (upto second decimal place)

2 Q. [3(0)]

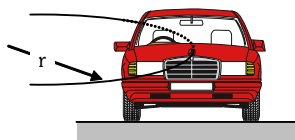
1. A disc revolves with a speed $33\frac{1}{3}$ rev./min and has a radius of 15 cm. Two coins are placed at 4 cm and 14 cm. away from the centre of the disc. If the coefficient of friction between the coins and the disc is 0.15. Coin placed at x cm will not revolve with record. Find x :-
2. A toy car of mass 10 kg is moving with constant speed on a circular track of radius $10\sqrt{3}$ m bank at angle 30° . Coefficient of friction between wheels of the car and track is $\frac{1}{\sqrt{3}}$. When the car moves on this track with maximum safe constant speed, the normal reaction (in N) between the car and the track is N. Find numerical value of $N\sqrt{3}$.

SECTION-III

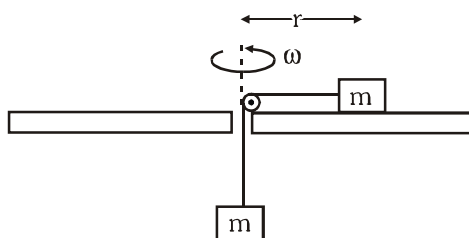
Numerical Grid Type (Ranging from 0 to 9)

9 Q. [4 M (0)]

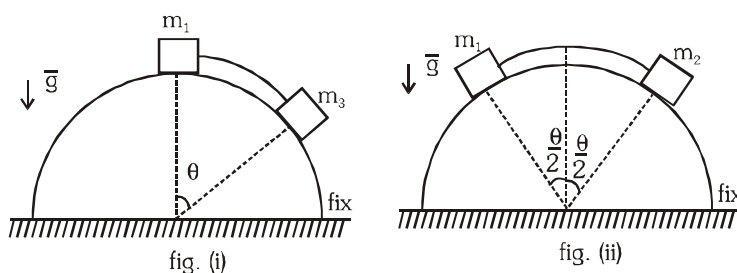
1. Two-wheeler can tilt while turning, but four-wheeler cannot. Estimate maximum safe speed of a four-wheeler on turn of radius 10 m assuming coefficient of static friction between tires and the flat road to be 0.64 and acceleration due to gravity to be 10 m/s^2 .



2. Two equal masses m are attached by a string. One mass lies at radial distance r from the center of a horizontal turntable which rotates with constant angular velocity ω , while the second hangs from the strings inside the turntable's hollow spindle (see figure). The coefficient of static friction between the turntable and the mass lying on it is $\mu_s = 0.5$. The maximum and minimum values $r_{\max}$, $r_{\min}$ of r such that the mass lying on the turntable does not slide. Find the ratio of $r_{\max} / r_{\min}$?

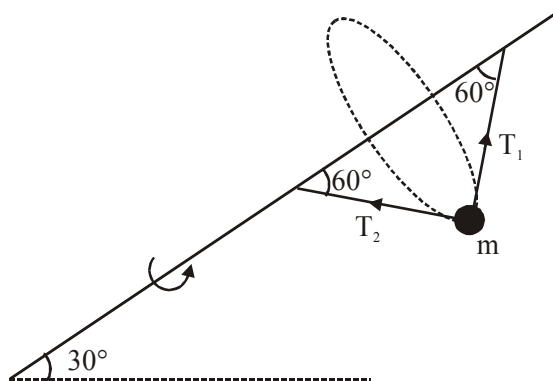


3. There are three blocks of masses $m_1 = m$, $m_2 = 4m$ and $m_3 = m$. Two block system is kept over a smooth hemisphere with two different arrangement one by one as shown in figure (i) & (ii) respectively and are connected with massless & inextensible string. In arrangement figure (i), after releasing the system from rest acceleration of blocks is found to be $\left(a = \frac{5\sqrt{3}}{2}\right) \text{ m/s}^2$. In second arrangement block m_1 & m_2 are kept symmetrically. Find the acceleration of block (in m/s^2) after releasing the system from rest according to figure (ii).

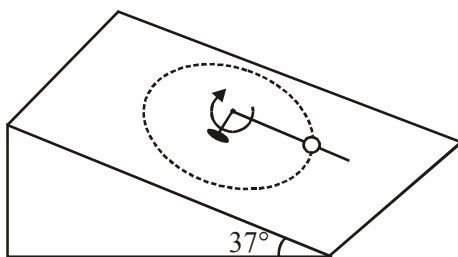


4. A car is accelerating on a rough circular track, according to the equation $\theta = \left(\frac{t^2}{2} + 2\right) \text{ rad}$ where θ is its angular displacement and t is time in sec. The angle between tangential friction and total friction is β then find the value of $\tan\beta$ at $t = 3 \text{ s}$.
5. The corner of a motor racing track is banked at an angle of 30° . If the coefficient of friction between the tyre and track is $\sqrt{3}/2$, the ratio of the maximum speed at which car can take the turn with the corresponding speed of car if there is no friction is $\sqrt{\alpha} : \beta$. where α and β are smallest integers. Find the value of $\alpha + \beta$.

6. Figure shows a rod arranged at an angle of 30° from the horizontal attached to the two strings is the mass m as shown. The rod is rotated maintaining its direction in space, so that m travels in a circular path. The strings are of equal length and make angle of 60° with the rod as shown. Calculate the minimum value of the tangential speed (m/sec) of the mass such that the string with tension T_2 does not become slack when the mass is directly above rod. Take length of string as $\ell = 2.4\text{m}$



7. An airplane is moving in a vertical circle with constant speed in a such a way that a passenger of mass 70 kg inside it experiences weightlessness at the top most point. The total contact force exerted by airplane seat when the velocity of plane is making 60° angle downwards with horizontal is given by $100x$ (in N) then x is.
8. A rod is rotating with constant angular velocity about an end, parallel to the inclined plane as shown in the figure. A bead of mass m is moving with rod on a circular path of radius 2 m. The bead does not touch the inclined plane. The friction coefficient between rod and bead is 1. Find the maximum possible ω (in rad/s) at which the bead will not slip on the rod :-



9. A particle moves in a circle with a uniform speed, when it goes from a point A to a diametrically opposite point B, the momentum of the particle changes by $\vec{P}_A - \vec{P}_B = 2\text{kgm/s}(\hat{j})$ and the centripetal force acting on it changes by $\vec{F}_A - \vec{F}_B = 8\text{N}(-\hat{i})$ where $\hat{i}$, $\hat{j}$ are unit vectors. Then the angular velocity of the particle (in rad/s) is.

SECTION-I

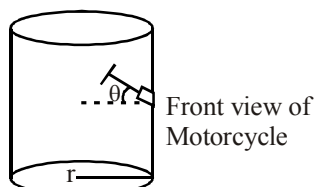
Single Correct Answer Type

3 Q. [3 M (–1)]

1. A railway track is banked for a speed v , by making the height of the outer rail h higher than that of the inner rail. The distance between the rails is d . The radius of curvature of the track is r . Then choose the correct relation

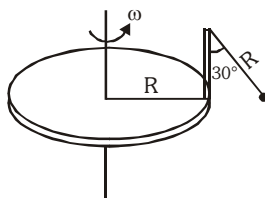
(A) $\frac{h}{d} = \frac{v^2}{rg}$ (B) $\tan\left(\sin^{-1} \frac{h}{d}\right) = \frac{v^2}{rg}$ (C) $\tan^{-1}\left(\frac{h}{d}\right) = \frac{v^2}{rg}$ (D) $\frac{h}{r} = \frac{v^2}{dg}$

2. A trick cyclist rides his bike around a well of death in the form of a vertical cylinder (see figure). The maximum frictional force parallel to the surface of the cylinder is equal to a fraction μ of the normal force exerted by the wall. At what minimum constant speed must the cyclist go to avoid slipping down? (Assume that size of motorcycle is very small as compared to cylinder.)



(A) $\left[\frac{2gr \cos \theta}{\mu}\right]^{1/2}$ (B) $\left[\frac{gr \sin \theta}{2\mu}\right]^{1/2}$ (C) $\left(\frac{gr}{\mu}\right)^{1/2}$ (D) $\left(\frac{2gr}{\mu}\right)^{1/2}$

3. A disc of radius R has a light pole fixed perpendicular to the disc at the circumference which in turn has a pendulum of length R attached to its other end as shown in figure. The disc is rotated with a constant angular velocity ω . The string is making an angle 30° with the rod. Then the angular velocity ω of disc is :-

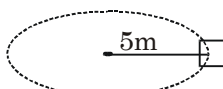


(A) $\left(\frac{\sqrt{3}g}{R}\right)^{1/2}$ (B) $\left(\frac{\sqrt{3}g}{2R}\right)^{1/2}$ (C) $\left(\frac{g}{\sqrt{3}R}\right)^{1/2}$ (D) $\left(\frac{2g}{3\sqrt{3}R}\right)^{1/2}$

Multiple Correct Answer Type

1 Q. [4 M (–1)]

4. A block of mass 25 kg rests on a horizontal floor ($\mu = 0.2$). It is attached by a 5m long horizontal rope to a peg fixed on floor. The block is pushed along the ground with an initial velocity of 10 m/s so that it moves in a circle around the peg. Choose the correct statement(s) :-



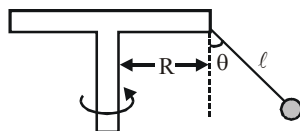
- (A) Magnitude of tangential acceleration of the block is 2 m/s^2
(B) Speed of the block at time $t = 1 \text{ sec}$ is 8 m/s
(C) Time when tension in rope becomes zero is 5 sec
(D) Friction force acts towards centre of the circle

Linked Comprehension Type (Single Correct Answer Type)

(3 Para × 3Q. & 1 Para × 2Q.) [3 M (-1)]

Paragraph for Questions 5 to 7

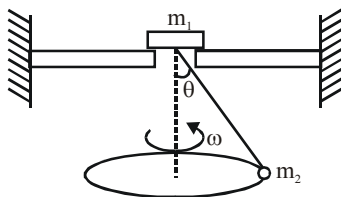
A pendulum bob of mass m hangs from one end of a horizontal rod which is rotating about vertical axis with an unknown angular speed ω , such that the light inextensible string of length ℓ of the pendulum makes a constant angle θ with vertical.



5. The angular speed ω is
- (A) $\sqrt{\frac{2g \tan \theta}{R + \ell \sin \theta}}$ (B) $\sqrt{\frac{g \tan \theta}{R + \ell \sin \theta}}$ (C) $\sqrt{\frac{g \tan \theta}{2R + \ell \sin \theta}}$ (D) $\frac{g \tan \theta}{R + \ell \sin \theta}$
6. The tension in the string is,
- (A) $2mg \sec \theta$ (B) $mg \cos \theta$ (C) $3mg \sec \theta$ (D) $mg \sec \theta$
7. The acceleration of the bob is
- (A) $g \tan \theta$ (B) $2g \tan \theta$ (C) $g \cot \theta$ (D) $2g \cot \theta$

Paragraph for Questions 8 to 10

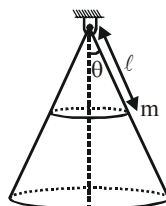
A pendulum bob of mass m_2 hangs from a fixed block of mass m_1 by a light inextensible string of length ℓ . The coefficient of static friction between the block and the fixed horizontal surface is μ .



8. Find the angle θ made by the string with vertical, if the bob swings in a horizontal plane.
- (A) $\theta \leq \tan^{-1} \left(\frac{\mu(m_1 + m_2)}{2m_2} \right)$ (B) $\theta \leq \tan^{-1} \left(\frac{\mu m_2}{(m_1 + m_2)} \right)$
- (C) $\theta \leq \tan^{-1} \left(\frac{\mu(m_1 + m_2)}{m_2} \right)$ (D) $\theta \leq \tan^{-1} \left(\frac{2\mu(m_1 + m_2)}{m_2} \right)$
9. Referring to previous Question, what will be the range of possible values of tension in the string.
- (A) $T \leq g\sqrt{m_1^2 + \mu^2(m_1 + m_2)^2}$ (B) $T \leq g\sqrt{m_2^2 + \mu^2(m_1 + m_2)^2}$
- (C) $T \leq \frac{m_2 g}{m_1} \sqrt{m_1^2 + \mu^2(m_1 + m_2)^2}$ (D) $T \leq \frac{m_1 g}{m_2} \sqrt{m_2^2 + \mu^2(m_1 + m_2)^2}$
10. Referring to previous Question,, find the ranges of values of the angular speed of revolution ω relative to vertical axis.
- (A) $\omega \leq \sqrt{\frac{g\sqrt{m_2^2 + \mu^2(m_1 + m_2)^2}}{m_2 \ell}}$ (B) $\omega \leq \sqrt{\frac{g\sqrt{m_1^2 + \mu^2(m_1 + m_2)^2}}{m_1 \ell}}$
- (C) $\omega \leq \sqrt{\frac{g\sqrt{m_2^2 + \mu^2(m_1 + m_2)^2}}{m_1 \ell}}$ (D) $\omega \leq \sqrt{\frac{g\sqrt{m_1^2 + \mu^2(m_1 + m_2)^2}}{m_2 \ell}}$

Paragraph for Questions 11 to 13

A block of mass m is resting on a smooth fixed conical surface by an inextensible light string of length ℓ tied at the apex of the cone. If the block moves with a constant angular speed ω about vertical.



11. Find the normal reaction on the block

- (A) $2mg \sin \theta \left(1 - \frac{\ell \omega^2 \cos \theta}{g} \right)$ (B) $mg \sin \theta \left(1 - \frac{\ell \omega^2 \cos \theta}{g} \right)$
(C) $mg \sin \theta \left(1 - \frac{\ell \omega^2 \cos \theta}{2g} \right)$ (D) $mg \sin \theta \left(1 - \frac{\ell \omega^2 \sin \theta}{g} \right)$

12. Tension in the string is

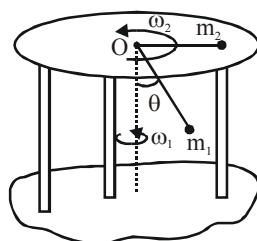
- (A) $m(g \cos \theta + \omega^2 \ell \sin^2 \theta)$ (B) $2m(g \cos \theta + \omega^2 \ell \sin^2 \theta)$
(C) Zero (D) $m(g \sin \theta + \omega^2 \ell \cos^2 \theta)$

13. If the coefficient of kinetic friction between the block and conical surface is μ , assuming $\omega = \sqrt{\frac{g}{\ell}}$, find the acceleration of the block

- (A) $2g \sin \theta \sqrt{1 + \mu^2 (1 - \cos \theta)^2}$ (B) $g \sin \theta \sqrt{1 + 2\mu^2 (1 - \cos \theta)^2}$
(C) $g \sin \theta \sqrt{2 + \mu^2 (1 - \cos \theta)^2}$ (D) $g \sin \theta \sqrt{1 + \mu^2 (1 - \cos \theta)^2}$

Paragraph for Questions 14 and 15

Two bobs of masses m_1 and m_2 connected by a light inextensible string of length ℓ are revolving about a vertical axis passing through a hole made at the point O of the smooth table. The angular speeds of m_1 and m_2 relative to the axis of their revolution are ω_1 and ω_2 respectively. If m_2 swings in a horizontal circle while the string is making a constant angle θ with vertical,



14. The value of θ is

- (A) $\cos^{-1} \left(\frac{3m_1 g}{2\ell} \left(\frac{m_1 \omega_1^2 + m_2 \omega_2^2}{m_1 m_2 \omega_1^2 \omega_2^2} \right) \right)$ (B) $\cos^{-1} \left(\frac{m_1 g}{\ell} \left(\frac{2m_1 \omega_1^2 + m_2 \omega_2^2}{m_1 m_2 \omega_1^2 \omega_2^2} \right) \right)$
(C) $\cos^{-1} \left(\frac{m_1 g}{\ell} \left(\frac{m_1 \omega_1^2 + m_2 \omega_2^2}{m_1 m_2 \omega_1^2 \omega_2^2} \right) \right)$ (D) $\sin^{-1} \left(\frac{m_1 g}{\ell} \left(\frac{m_1 \omega_1^2 + m_2 \omega_2^2}{m_1 m_2 \omega_1^2 \omega_2^2} \right) \right)$

15. The tension in the string is

- (A) $\frac{m_1 m_2 \ell \omega_1^2 \omega_2^2}{m_2 \omega_2^2}$ (B) $\frac{m_1 m_2 \ell \omega_1^2 \omega_2^2}{m_1 \omega_1^2 + m_2 \omega_2^2}$ (C) $\frac{2m_1 m_2 \ell \omega_1^2 \omega_2^2}{m_1 \omega_1^2 + m_2 \omega_2^2}$ (D) Zero

SECTION-II

Numerical Answer Type Question

1 Q. [3(0)]

(upto second decimal place)

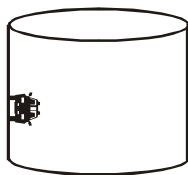
1. A block of mass $m = 15$ kg is kept at a distance $R = 1$ m from central axis of rotation of a round turn table (A table whose surface can rotate about central axis). Table starts from rest and rotates with constant angular acceleration, $\alpha = 2$ rad/sec². The friction coefficient between block and table is $\mu = 0.8$. At time t sec from starting of motion (i.e. $t = 0$ sec) the block is just about to slip. Find the value of $8t^4$.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. You are shown a photo of a car driven on a vertical inside wall of a huge cylinder with a radius of 50 m. The coefficient of static friction between the car tires and the cylinder is $\mu_s = 0.8$. The minimum speed in (m/s), at which the car can be driven like that is $5x$. Find x .



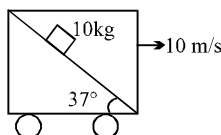
2. An aircraft of 300 kg flies at a speed $v = 360$ km/hr and makes a turn of radius $R = 750$ m. Determine the magnitude of the lift force (in kN) needed to perform the turn in a horizontal plane. The lift force is always directed perpendicularly to the plane of the aircraft wings.

SECTION-I

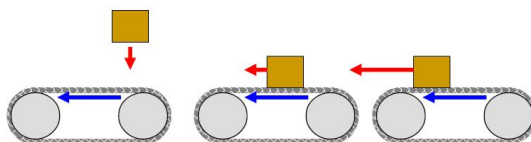
Single Correct Answer Type

13 Q. [3 M (-1)]

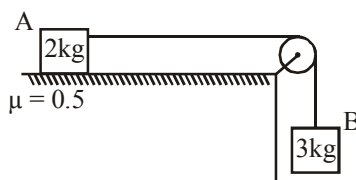
1. A block of mass 10 kg is released on a fixed wedge inside a cart which is moved with constant velocity 10 m/s towards right. Take initial velocity of block with respect to cart to be zero. Then work done by normal reaction on block in two second in ground frame will be ($g = 10 \text{ m/s}^2$)



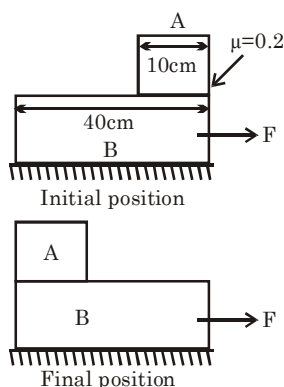
- (A) zero (B) 960 J (C) 1200 J (D) none of these
2. A 300 kg crate is dropped vertically onto a conveyor belt that is moving at 1.20 m/s. A motor maintains the belt's constant speed. The belt initially slides under the crate, with a coefficient of friction of 0.400. After a short time, the crate is moving at the speed of the belt. During the period in which the crate is being accelerated, find the work done by the motor which drives the belt :-



- (A) 432 J (B) 216 J (C) 108 J (D) 54 J
3. The system shown is released from rest. Speed of block A, after block B has descended by 2 cm, is :- (take $g = 10 \text{ m/s}^2$)

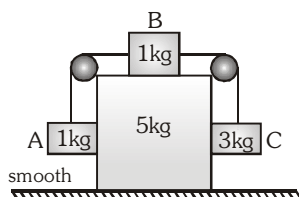


- (A) 0.4 m/s (B) 0.5 m/s (C) 0.6 m/s (D) 0.8 m/s
4. A block 'A' of mass 45 kg is placed on a block 'B' of mass 123 kg. Now block 'B' is displaced by external agent by 50 cm horizontally towards right. During the same time block 'A' just reaches to the left end of block B. Initial & final position are shown in figure. Refer to the figure & find the workdone by frictional force on block A in ground frame during above time :-



- (A) - 18 Nm (B) 18 Nm (C) 36 Nm (D) - 36 Nm

5. A force $\vec{F} = (3t\hat{i} + 5\hat{j})$ N acts on a body due to which its displacement varies as $\vec{s} = (2t^2\hat{i} - 5\hat{j})$. Work done by this force in $t = 0$ to 2 sec is :
 (A) 23 J (B) 32 J (C) zero (D) can't be obtained.
6. A force of magnitude 50 N acting along $\hat{i} + \hat{j} + \hat{k}$, displaces a point mass from (5, 9, 7) to (4, 8, 6). The work done by this force during this displacement is :-
 (A) 150 J (B) $50\sqrt{3}$ J (C) $-50\sqrt{3}$ J (D) $50/\sqrt{3}$ J
7. In the given diagram calculate the work done by the normal force exerted on each block by the bigger block in 1s

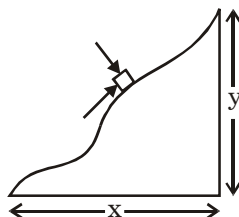


- (A) $W_A \neq 0, W_B \neq 0, W_C \neq 0$ (B) $W_A = 0, W_B = 0, W_C = 0$
 (C) $W_A \neq 0, W_B = 0, W_C = 0$ (D) $W_A = 0, W_B = 0, W_C \neq 0$
8. A conveyor belt is moving with a constant velocity 2 m/s. Block is projected with speed of 4 m/s in opposite direction of velocity of belt. if mass of block is '4 kg' and coefficient of friction is 0.2. Find work done against friction on conveyor belt up to the instant when slipping between block and belt ceases :
 (A) 24 Joule (B) 48 Joule (C) 12 Joule (D) 36 Joule
9. Force acting on a particle moving in a straight line varies with the velocity of the particle as $F = \frac{K}{v}$. Here K is a constant. The work done by this force in time t is :
 (A) $\frac{K}{v^2} \cdot t$ (B) 2Kt (C) Kt (D) $\frac{2Kt}{v^2}$
10. During displacement of a body two forces are acting on it simultaneously. Forces are $\vec{F}_1 = 2\hat{i} - 2\hat{j}$ and $\vec{F}_2 = \hat{i} + 5\hat{j}$. Work done by individual forces on body are (-8 joule) and (2 joule) respectively then displacement of body will be :-
 (A) $-3\hat{i} + \hat{j}$ (B) $3\hat{i} - \hat{j}$ (C) $3\hat{i} + \hat{j}$ (D) $-3\hat{i} - \hat{j}$
11. A body of mass m is lifted from ground by a force which varies with height 'h' as $F = 2(1 - bh)mg$, here 'b' is constant. Work done by this force over the first half distance of the ascent of body :-
 (A) $\frac{5mg}{2b}$ (B) $\frac{3mg}{8b}$ (C) $\frac{3mg}{4b}$ (D) 10 mgb
12. A deliveryman moves 10 cartons from the sidewalk, along a 10-meter ramp to a loading dock, which is 1.5 meters above the sidewalk. If each carton has a mass of 25 kg, what is the total work done by the deliveryman on the cartons to move them to the loading dock? ($g = 10 \text{ m/s}^2$)



- (A) 2500 J (B) 3750 J (C) 10000 J (D) 25000 J

13. A block is placed on rough hill. Block is slowly moved over the surface. One of the force of magnitude 'F' is applied perpendicular to the surface and other is always parallel to surface. Select correct statement :

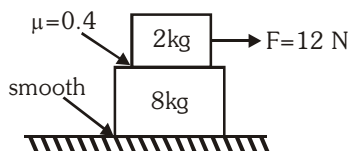


- (A) Work done by the friction on the block is μmgx .
 (B) Work done by friction on the block is $-\mu mgs$ where s is path length of block
 (C) Work done by friction on the block is $-\mu(mgx + Fs)$, where s is path length of block
 (D) Work done by friction in the round trip on the block is zero.

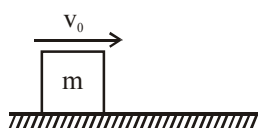
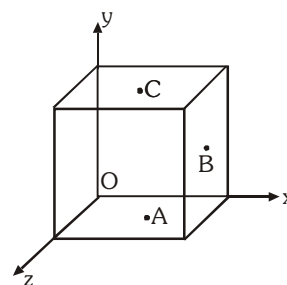
Multiple Correct Answer Type

4 Q. [4 M (-1)]

14. The friction coefficient between the two blocks is 0.4 and floor is smooth. If we assume three reference frames S_1 , S_2 and S_3 attached rigidly to 2kg block, 8kg block and ground, then the work done by the force F in the three frame be W_1 , W_2 and W_3 then-

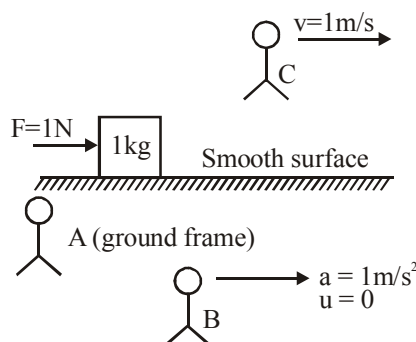


- (A) $W_1 = W_2 = W_3$
 (B) $W_3 > W_2 > W_1$
 (C) $W_3 > W_2 = W_1$
 (D) If $F = 8\text{ N}$ Option (C) is correct
15. A particle is shifted from A to B and then from B to C where A, B and C are the midpoints of the corresponding faces of a cube of side 2m. If a force $\vec{F} = (3\hat{i} + 4\hat{j} - 5\hat{k})\text{ N}$ is continuously acting on the particle, then select correct alternative is
- (A) work done from A to B is 7 J
 (B) work done from B to C is 1 J
 (C) work done A to C is 8 J
 (D) force $\vec{F}$, is conservative force
16. A block of mass m is projected with velocity v_0 over a rough surface as shown in the figure, at $t = 0$. After time t_0 block stops. Consider a reference frame S , which is moving with constant velocity v_0 in the direction of motion of block, then work done by friction from $t = 0$ to $t = t_0$



- (A) On block is $-\frac{1}{2}mv_0^2$ w.r.t. ground
 (B) On block is $\frac{1}{2}mv_0^2$ w.r.t. S
 (C) On ground is zero w.r.t. ground
 (D) On ground is $\frac{1}{2}mv_0^2$ w.r.t. S

17. An observer A is at rest in ground frame, observer B is moving with constant acceleration of 1 m/s^2 and another observer C is moving with constant velocity 1 m/s . Comment on work on block of mass 1 kg .



- (A) work done by all force (both real and pseudo) in 1 sec in frame of A is 0.5 J .
 (B) work done by all force (both real and pseudo) in 1 sec in frame of B is 0 J .
 (C) work done by all force (both real and pseudo) in 1 sec in frame of C is -0.5 J .
 (D) change in K.E. in frame of B is zero.

Linked Comprehension Type

(1 Para $\times$ 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question no. 18 to 20

In a second ODI match between England and India Bhuvnesh kumar bowled his medium fast bowling. He starts his run up from the distance of 50 m from the stumps (bowling end) and at the time of bowling a ball, his speed is 36 km/hr and the speed of ball is 144 km/hr . If his mass is 70 kg and mass of the ball is $\frac{1}{4} \text{ kg}$ and the heat produce in his body is one tenth of the total work done by him, then

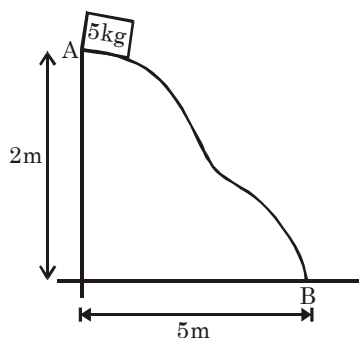
18. Select the incorrect statement :
 (A) Work done by the friction on him is negative
 (B) Work done by the ball on him is negative
 (C) Work done by ground on him is zero
 (D) Work done by the man on the ball is positive
19. Energy spent by bowler in throwing one ball is :-
 (A) 3500 J (B) 200 J (C) 4070 J (D) 3700 J
20. If he accelerates uniformly to the stumps find the average friction force acting on him :-
 (A) 750 N (B) 70.25 N (C) 90.25 N (D) 70 N

SECTION-I

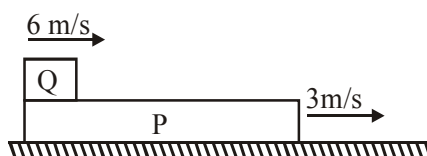
Single Correct Answer Type

9 Q. [3 M (-1)]

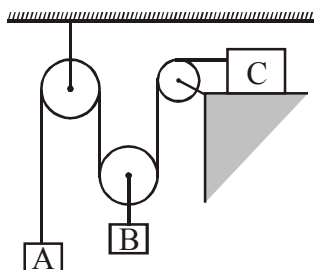
1. A particle of mass 5kg is taken from point A to point B slowly with help of external force on a rough inclined surface as shown in figure. If coefficient of friction force between block & inclined plane is $\mu = 0.5$. Then find the work done by external force on block :-



- (A) 25 J (B) -225J (C) +225 J (D) -25 J
2. A plank P and block Q are arranged as shown on a smooth table top. They are given velocities 3 m/s and 6 m/s respectively. The length of plank is 1m and block is of negligible size. After some time when the block has reached the other end of plank it stops slipping on plank. The velocity of plank then is (coefficient of friction between plank and block is 0.3 and mass of plank is double of block).



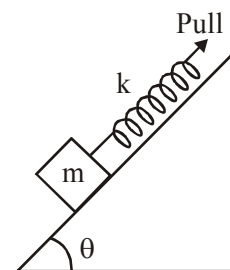
- (A) 4 m/s (B) 5 m/s (C) 4.5 m/s (D) zero
3. In the arrangement shown block A and C are of mass 2 kg and B is of unknown mass. When released, A moves downward by 0.5 m and C moves towards pulley by 0.5 m. Velocity of A after the displacement given is (friction is absent)



- (A) $\sqrt{2}$ m/s (B) $\sqrt{3}$ m/s (C) $\sqrt{4}$ m/s (D) $\sqrt{5}$ m/s
4. Two bars of masses m_1 and m_2 connected by a non-deformed light spring rest on a horizontal plane. The coefficient of friction between the bars and the surface is equal to μ . What minimum constant force has to be applied in the horizontal direction to the bar of mass m_1 in order to shift the other bar?

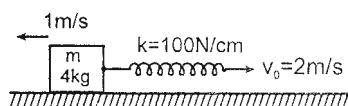
- (A) $\left(m_1 + \frac{m_2}{2}\right)\mu g$ (B) $\left(m_2 + \frac{m_1}{2}\right)\mu g$ (C) $\left(m_1 - \frac{m_2}{2}\right)\mu g$ (D) $\left(2m_2 + \frac{m_1}{2}\right)\mu g$

5. A block of mass m , connected to a spring of spring constant k , rests on a rough incline; the angle of incline is θ . The coefficients of friction are μ_s and μ_k respectively. The spring is slowly pulled up the incline until the block starts to move. What is the value of μ_k such that the block comes to rest when the spring is neither extended or compressed.

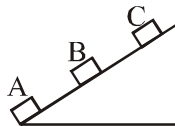


- (A) $\mu_k = \frac{1}{2}(\mu_s - \tan \theta)$ (B) $\mu_k = (\mu_s - \tan \theta)$
(C) $\mu_k = \frac{3}{2}(\mu_s - \tan \theta)$ (D) $\mu_k = \frac{1}{2}(\mu_s - 2 \tan \theta)$

6. The spring block system lies on a smooth horizontal surface. The free end of the light spring is being pulled towards right with constant speed $v_0 = 2\text{ m/s}$. At $t = 0$ sec, the spring of spring constant $k = 100\text{ N/cm}$ is unstretched and the block has a speed 1 m/s to left. The maximum elongation in the spring is :-



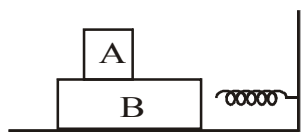
- (A) 2 cm (B) 4 cm (C) 6 cm (D) 8 cm
7. A block has an initial kinetic energy of 128 J. It slides up from point A at the bottom of the inclined plane with uniform deceleration. When it passes point B, its kinetic energy is reduced by 80 J, and its mechanical energy is reduced by 35 J. Calculate the work done against friction when the block moves from A to the highest point C on the inclined plane.



- (A) 42 J (B) 48 J (C) 56 J (D) 72 J
8. A box of mass m is initially at rest on a horizontal surface. A constant horizontal force of $\frac{mg}{2}$ is applied to the box, directed to the right. The coefficient of friction changes with the distance pushed as $\mu = \mu_0 x$, where x is the distance from initial location. For what distance is the box pushed until it comes to rest again.

- (A) $\frac{2}{\mu_0}$ (B) $\frac{1}{\mu_0}$ (C) $\frac{1}{2\mu_0}$ (D) $\frac{1}{4\mu_0}$

9. A block A is placed over block B having mass m & $2m$ respectively. Block B is resting on a frictionless surface and there is friction between block A and B. The system of blocks is pushed towards a spring with a velocity v_0 such that A doesn't slip on B by the time the system comes to momentary rest. The correct statement is :-

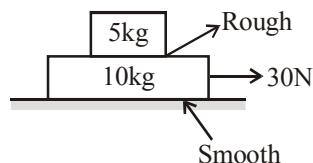


- (A) Work done by friction on A is zero (B) Work done by friction on B is $-\frac{1}{2}mv_0^2$
(C) Work done by spring on B is $-\frac{3}{2}mv_0^2$ (D) None of these

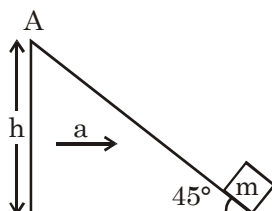
Multiple Correct Answer Type

4 Q. [4 M (-1)]

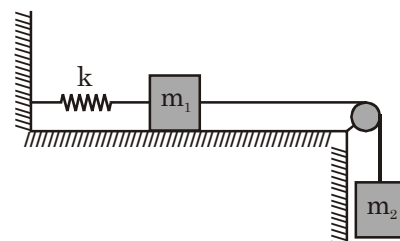
10. In the figure shown ground is smooth, whereas surface between 10 kg block and 5 kg block is rough. A force of magnitude 30 N is applied on 10 kg block, horizontally. If net work done by friction on the system (on 10 kg and 5 kg blocks) in any time frame is zero, then coefficient of friction between 10 kg and 5 kg can be :



- (A) 0.1 (B) 0.3 (C) 0.4 (D) 0.6
11. An ideal spring with spring-constant k is hung from the ceiling and a block of mass M is attached to its lower end. The mass is released with the spring initially unstretched. Then :-
 (A) The maximum extension in the spring is Mg/k
 (B) The velocity of block is maximum when it has moved by distance of Mg/k
 (C) The block returns to initial position with zero velocity
 (D) The velocity of block is maximum when it has moved by a distance of $\frac{2Mg}{k}$
12. A block of mass m is kept on a smooth wedge of height h at rest. At $t = 0$, wedge starts moving with constant acceleration $a = \frac{3}{2}g$ as shown in figure. V represents the relative velocity of block w.r.t. wedge at A & H is the maximum height achieved by the block measured from ground.



- (A) $V = \sqrt{2gh}$ (B) $V = \sqrt{gh}$ (C) $H = \frac{5h}{4}$ (D) $H = \frac{3h}{2}$
13. In the system in figure, the blocks have masses m_1 and m_2 the spring constant is k ; coefficient of friction between the block 1 and the surface is μ . The system is released with zero initial speed from the position where the spring is not stretched.



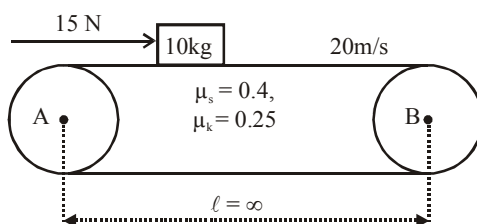
- (A) the maximum possible speed of the blocks is $\frac{g(m_2 - \mu m_1)}{\sqrt{k(m_1 + m_2)}}$
- (B) the maximum possible speed of the blocks is $\frac{g(m_1 - \mu m_2)}{\sqrt{k(m_1 + m_2)}}$
- (C) the maximum possible speed of the blocks if friction is absent is $\frac{gm_2}{\sqrt{k(m_1 + m_2)}}$
- (D) the maximum possible speed of the blocks if friction is absent is $\frac{gm_1}{\sqrt{k(m_1 + m_2)}}$

Linked Comprehension Type (Single Correct Answer Type)

(2 Para × 2Q. 1 Para × 3Q.) [3 M (-1)]

Paragraph for Question Nos. 14 & 15

A block of mass 10 kg is put gently on a belt-conveyor system of infinite length at $t = 0$ sec, which is moving with constant speed 20 m/sec rightward at all time, irrespective of any situation by means of a motor-system as shown in the figure. A constant force of magnitude 15 N is applied on the block continuously during its motion.



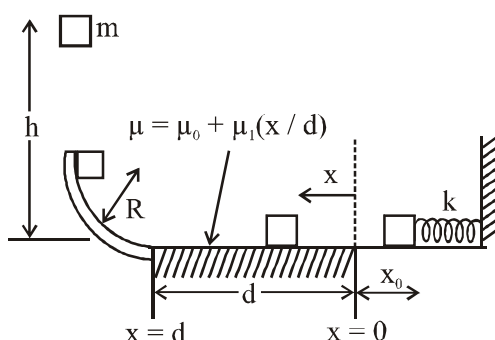
14. Work done by the kinetic friction on the block of mass 10 kg is :-
(A) 1250 Joule (B) 2500 Joule (C) -1250 Joule (D) Zero
15. The magnitude of acceleration of the block of mass 10 kg at $t = 6$ sec is :-
(A) 4 m/s^2 (B) 3 m/s^2 (C) 2 m/s^2 (D) Zero

Paragraph for Question no. 16 and 17

An object of mass m is released from an initial state of rest from a spring of constant k that has been compressed a distance x_0 . After leaving the spring (at the position $x = 0$ when the spring is relaxed) the object travels a distance d along a horizontal track that has a coefficient of friction that varies with position as

$$\mu = \mu_0 + \mu_1(x/d)$$

Following the horizontal track, the object enters a quarter turn of a frictionless loop whose radius is R . Finally, after exiting the quarter turn of the loop the object travels vertically upward to a maximum height, h , (as measured from the horizontal surface).

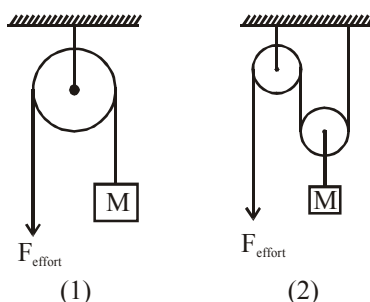


16. Net work done by the friction force on block from $x = 0$ to $x = d$ is :-
(A) $-mgd(\mu_0 + \mu_1)$ (B) $-mgd(\mu_0 + \mu_1/2)$ (C) $-2mgd(\mu_0 + \mu_1)$ (D) $-\frac{mgd}{2}(\mu_0 + \frac{\mu_1}{2})$
17. Maximum height (h) attained by the object is :
(A) $\frac{kx_0^2}{2mg} - d(\mu_0 + \mu_1/2)$ (B) $\frac{kx_0^2}{2mg} - d(\mu_0 + \mu_1)$
(C) $\frac{kx_0^2}{mg} - d(\mu_0 + \mu_1/2)$ (D) $\frac{kx_0^2}{2mg} - 2d(\mu_0 + \mu_1/2)$

CLASS TEST

Paragraph for Question no. 18 to 20

With almost four and half months are over in your present syllabus, it is foolish to ask "have you heard of pulleys". Of course you all know about pulleys, but have you really thought about utility of a pulley. The most important use of a pulley lies in the fact that a pulley not only can change direction of your effort (force), but can also make you lift a bigger weight with a smaller force. Let us consider figure 1 & 2. In both cases we have to just lift the block of mass M .

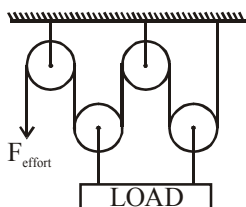


It is obvious that in case (2), F_{effort} will be half of F_{effort} in case (1) if we neglect mass of pulley and rope. This ability of pulley to lift greater weight by applying smaller force can be measured by quantity called "Mechanical Advantage". Mechanical advantage (M.A) of a pulley system can be defined as

$$\text{M.A} = \frac{\text{Weight of the load to be lifted}}{\text{Applied force to just lift the given load}}$$

So M.A. in figure (1) is 1 and in figure (2) is 2.

18. Mechanical advantage of system shown is (Assume pulleys and strings are massless and frictionless) :



- (A) 2 (B) 3 (C) 4 (D) 6

19. A pulley system has M.A. = 3. If load moves up by 2cm, then distance moved by point at which effort force is applied is :

- (A) 6 cm (B) 4 cm (C) 2 cm (D) 1 cm

20. Mark the **CORRECT** statement :

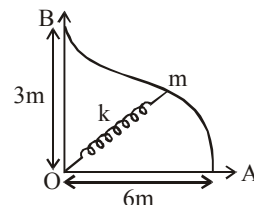
- (A) A pulley system with higher MA indicates that we can obtain more work by doing less work
(B) A pulley system with higher MA indicates that we can obtain less work by doing more work
(C) A pulley system with lower MA indicates that we can obtain more work by doing less work
(D) None of the above options

SECTION-I

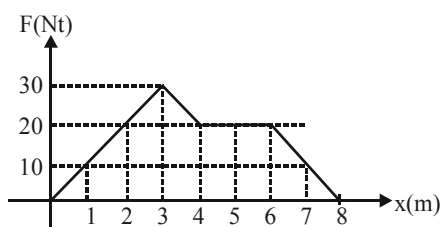
Single Correct Answer Type

5 Q. [3 M (-1)]

1. A small object with mass $m = 2 \text{ kg}$ is attached to the free end of an ideal spring with $k = 10 \text{ Nm}^{-1}$. The other end of the spring is connected to a fixed frictionless pivot located at the origin O as shown in the Figure. The relaxed length of the spring is 1 m . An unspecified force F carries the object at rest from point A to point B. At point B, the object has the speed 5 ms^{-1} . The work done by the force F is (Neglect gravity) :-



- (A) 130 J (B) 80 J (C) -105 J (D) -80 J
2. A body of 4 kg mass placed on a smooth horizontal surface experiences a force varying with displacement of block as shown in figure. Find the speed of the body when force ceases to act on it :-



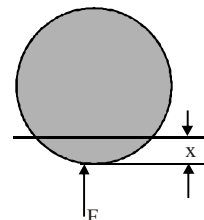
- (A) $\sqrt{65} \text{ m/s}$ (B) $\sqrt{85} \text{ m/s}$ (C) $\sqrt{45} \text{ m/s}$ (D) None of these
3. Experimentally it has been found that the force F needed to compress elastically a ball through a distance x (as shown in the figure) follows the formula $F(x) = ax + bx^2 + cx^3$ where a , b and c are constants. The small ball of mass m at rest is dropped from a great height h . It bounces elastically off the floor and is compressed a maximum distance ' d ' during the bounce. The height h is :- (where ' g ' is the acceleration due to gravity)

(A) $\frac{1}{mg} \left(\frac{1}{2} ad^2 + \frac{1}{3} bd^3 + \frac{1}{4} cd^4 \right)$

(B) $\frac{1}{mg} (ad + bd^2 + cd^3)$

(C) $\frac{1}{mg} (ad^2 + bd^3 + cd^4)$

(D) $\frac{1}{mg} (ad^2 + 2bd^3 + 3cd^4)$



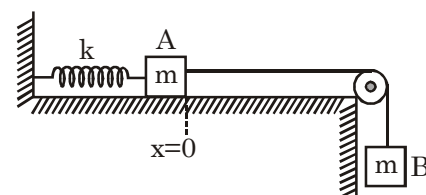
4. The system is released from rest when spring is at its natural length. Spring constant is 100 N/m and mass of each block is 10 kg . The velocity ' v ' of block A as a function of position x is given as $v^2 = ax - bx^2$. The value of a/b is equal to.

(A) 1

(B) 2

(C) 3

(D) 4



5. A bungee jumper is jumping with help of elastic ideal rope (Force constant k). Jumper steps off the bridge and falls from the rest towards the river below. He does not hit the water. The mass of jumper is m , natural length of rope is l . Gravity is g , assume every thing ideal. then, choose the incorrect option :

(A) Jumper comes to rest first time after falling a distance $S = \frac{(kl + mg) + \sqrt{2mgkl + m^2g^2}}{k}$

(B) Maximum speed attained is $v = \sqrt{2gl + \frac{mg^2}{k}}$

(C) time of free fall from rest $= \sqrt{2l/g}$

(D) None of the above options is correct

Multiple Correct Answer Type

4 Q. [4 M (-1)]

6. A block of mass 1kg kept on a rough horizontal surface ($\mu = 0.4$) is attached to a light spring (force constant = 200 N/m) whose other end is attached to a vertical wall. The block is pushed to compress the spring by a distance d and released. Find the value(s) of ' d ' for which (spring + block) system loses its entire mechanical energy in form of heat.

(A) 4cm (B) 6cm (C) 8cm (D) 10 cm

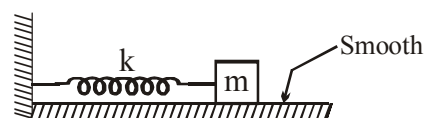
7. Figure shows an ideal spring block system, force constant of spring is k which has been compressed by an amount x_0 . If x is instantaneous deflection of spring from its natural length, mark the correct option(s).

(A) Instantaneous power developed by spring is $P = kx \sqrt{\frac{k}{2m}(x_0^2 - x^2)}$

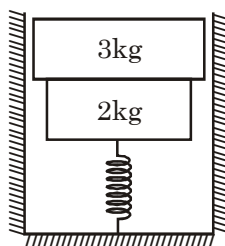
(B) Maximum power of spring is $\frac{k}{2} \sqrt{\frac{k}{m}} x_0^2$

(C) Maximum power occurs at $x = \frac{x_0}{\sqrt{2}}$

(D) Maximum power occurs at $x = \frac{x_0}{2}$



8. A 3 kg block rests on a 2 kg block on a spring of spring constant 50 N/m. The upper block is suddenly removed.



- (A) If 2 kg block is not attached to spring, maximum velocity of the block is 3 m/s.
(B) If 2 kg block is not attached to the spring, maximum height of the block above the natural length of the spring is 0.25 m.
(C) If 2 kg block is attached to the spring, maximum velocity of the block is 3 m/s.
(D) If 2 kg block is attached to the spring, maximum height of the block above the natural length of the spring is 0.5 m

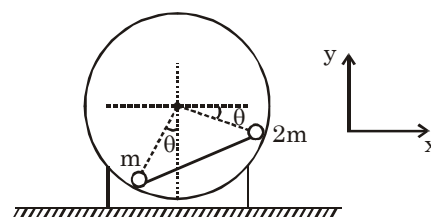
9. The two particles of mass m and $2m$, respectively are connected by a light rod of negligible mass and slide with negligible friction on a circular path of radius r inside a fixed vertical circular ring. If the system is released from rest at $\theta = 0^\circ$ and θ is taken from positive x -axis in clockwise direction.

(A) The speed of the particles when the rod passes the horizontal position is $\sqrt{\frac{2gr}{3} \left(\frac{3}{\sqrt{2}} - 1 \right)}$

(B) The maximum speed of the particles is $\sqrt{\frac{2}{3}gr(\sqrt{5} - 1)}$

(C) The maximum speed of the particles is at $\theta = \tan^{-1} \left(\frac{1}{2} \right)$

(D) The maximum value of θ is $2\tan^{-1}(2)$.

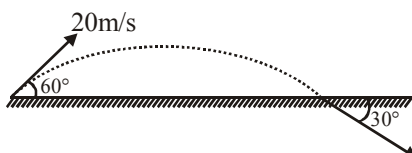


Linked Comprehension Type (Single Correct Answer Type)

(3 Para × 2Q.) [3 M (-1)]

Paragraph for Question No. 10 and 11

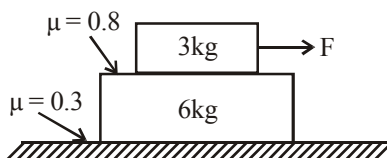
A particle of mass 10 kg is projected from a horizontal surface with a velocity of 20 m/s at an angle of 60° with the horizontal. When it hits the horizontal surface, its velocity makes an angle of 30° with the horizontal. Apart from gravity, a horizontal force acts on the particle during the motion ($g = 10 \text{ m/s}^2$)



10. The speed of the particle when it hits the ground is :-
 (A) 20 m/s (B) $20\sqrt{3}$ m/s (C) 10 m/s (D) $10\sqrt{3}$ m/s
11. The work done by the horizontal force during the complete motion is :-
 (A) 2000 J (B) 6000 J (C) 5000 J (D) 4000 J

Paragraph for Question no. 12 and 13

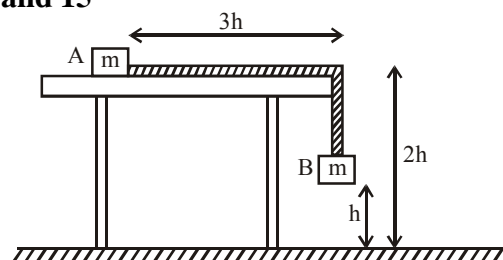
Two Blocks 3kg and 6kg are placed on a rough surface ($\mu = 0.3$) as shown in the figure. The friction coefficient between two blocks is $\mu = 0.8$. If a force F is applied on the upper block then



12. Choose the **CORRECT** statement.
 (A) work done by friction on 6kg block may be negative from ground frame
 (B) For any value of F , 6kg block will not move
 (C) 3kg block will not move for any value of $F < 27 \text{ N}$
 (D) Friction force on 6kg block can have greater value between block and ground than friction between blocks.
13. If F is applied on 6kg block instead of 3kg block then choose the **INCORRECT** statement
 (A) Work done by friction on 3kg block will be non negative from ground frame.
 (B) If acceleration of 6kg block is 10 m/s^2 then acceleration of 3kg block is 8 m/s^2
 (C) If acceleration of 6kg block is 1 m/s^2 then acceleration of 3kg block is 1 m/s^2
 (D) Work done by friction on 6kg block may be positive from ground frame.

Paragraph for Question no. 14 and 15

Two small blocks, each of mass m are connected by a string of constant length $4h$ and negligible mass. Block A is placed on a smooth table top as shown, and block B hangs over the edge of the table. The table top is at a distance $2h$ above the floor. Block B is then released from rest at a distance h above the floor at time $t = 0$. At time t_0 block B strikes the floor and does not bounce



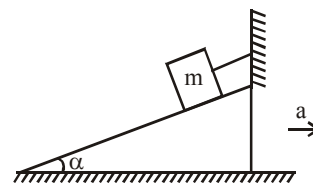
14. Determine the velocity of block A, when it leaves the table
 (A) $\sqrt{\frac{gh}{2}}$ (B) $\sqrt{gh}$ (C) $\sqrt{2gh}$ (D) $2\sqrt{gh}$
15. Determine the distance between the landing points of the two blocks
 (A) $4h$ (B) $2h$ (C) $\frac{9h}{8}$ (D) $\frac{7h}{6}$

Linked Comprehension Type (Multiple Correct Answer Type)

(1 Para × 2 Q.) [4 M (–1)]

Paragraph for Question no. 16 and 17

A smooth wedge moves to the right on the horizontal plane with an acceleration a . A mass m is attached to the wedge through a thread as shown in the figure. The thread is parallel to the incline.



16. Which among the following equations is/are **correct** :-
 (A) The tension in the thread is $T = m(g \sin \alpha + a \cos \alpha)$
 (B) The tension in the thread is $T = m(g \sin \alpha - a \cos \alpha)$
 (C) The Normal acting on mass 'm' is $N = m(g \cos \alpha + a \sin \alpha)$
 (D) The Normal acting on mass 'm' is $N = m(g \cos \alpha - a \sin \alpha)$
17. Which among the following equations is/are **correct** :-
 (A) Tension does non zero work in ground frame during any time 't'.
 (B) Tension does non zero work in wedge frame during any time 't'.
 (C) Pseudo force does non zero work in ground frame during any time 't'.
 (D) Pseudo force does non zero work in wedge frame during any time 't'.

SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. Column-I shows some situations with specified observers. Surface-1 refers to block and surface-2 refers to lower contacting surface. In all the cases velocity of block is in ground frame.

Column-I

- (A) Block projected on rough surface, observer on ground.
- (B) Block projected on rough surface observer moving with constant velocity $\vec{v}_0$.
- (C) Block projected with velocity $\vec{v}_0$ on rough belt moving with constant velocity $2\vec{v}_0$ observer on belt
- (D) Block projected on rough belt moving with initial velocity $\vec{v}_0$ and constant acceleration $\vec{a}_0$ initially block is at rest observer is on ground.

Column-II

- (P) work done by friction on surface-2 is negative
- (Q) Till block stops in given reference frame friction on surface-1 is in +ve x direction
- (R) Till block stops in given reference frame friction on surface-1 is in –ve x direction
- (S) Net work done by kinetic friction is $-\frac{1}{2}mv_0^2$
- (T) When block stops in given reference frame friction is static in nature.

SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

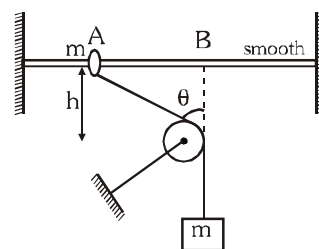
1. A 0.20 kg particle moves along the x-axis only due to its interaction with a stationary object. The potential energy is given by $U(x) = (8.0 x^2 + 2.0 x^4)$ J, where x in meters is position coordinate of the particle. If the particle has a speed of 5.0 m/s when it is at $x = 1.0$ m, what is its speed when it is at the origin?

(A) $2\sqrt{5}$ m/s (B) $3\sqrt{5}$ m/s (C) $4\sqrt{5}$ m/s (D) $5\sqrt{5}$ m/s

2. A ring of mass m slides from rest on the smooth rod as shown in the figure, due to the block of mass m. Pulley and string are massless. Then find the speed of ring when the string becomes straight.

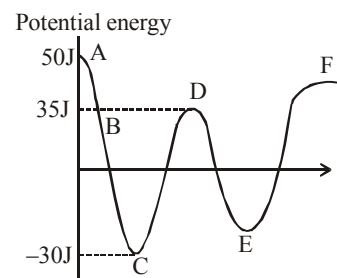
(Given $\theta = 60^\circ$)

(A) $4\sqrt{gh}$ (B) $\sqrt{2gh}$
(C) $2\sqrt{2gh}$ (D) $8\sqrt{gh}$

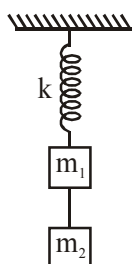


3. An object is in the region where potential energy is changing as shown in the figure. Its total mechanical energy is 30 J. If object is initially at C. Then select the correct statement.

(A) Object will be able to reach at E.
(B) At point C kinetic energy of object is 50 Joules
(C) Object will be oscillating in between B and D.
(D) Object can reach at D.

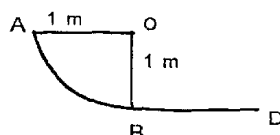


4. System is initially at equilibrium. Find potential energy stored in the springs ($m_1 > m_2$) :-



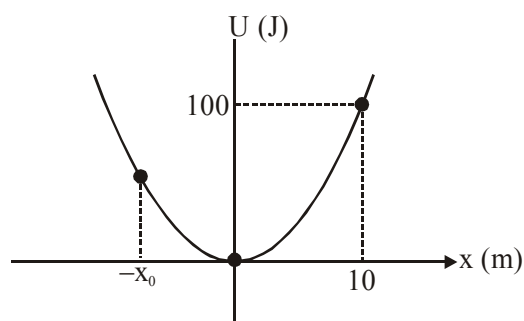
(A) $\frac{(m_1 g)^2}{2k}$ (B) $\frac{(m_2 g)^2}{2k}$ (C) $\frac{(m_1 + m_2)^2 g^2}{2k}$ (D) $\frac{(m_1 - m_2)^2 g^2}{2k}$

5. In the track shown in figure, section AB is a quadrant of a circle of 1 metre radius in vertical plane. A block is released at A and slides without friction until it reaches at B. After B it moves on a rough horizontal floor and comes to rest at D, 3 metres from B. The coefficient of friction between floor and the body will be



(A) 1/3 (B) 2/3 (C) 1/4 (D) 3/8

6. Potential energy vs position graph of a particle in a particular system is given and here P.E. is $U = ax^2$. There is a non conservative force also acting on a particle in such a manner that mechanical energy loss due to its work is equal to four times the distance travelled. Initially particle is at $x = 10$, has kinetic energy 36 J and moving towards origin. When it reaches origin second time then kinetic energy of particle is :-
- (A) 100 J
(B) Particle will never reach origin second time
(C) 32 J
(D) 96 J



Multiple Correct Answer Type

5 Q. [4 M (-1)]

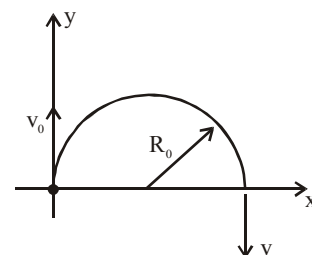
7. A bead slides on a fixed frictionless wire bent into a horizontal semicircle of radius R_0 as shown in figure. In addition to any normal forces exerted by the wire, the bead is subjected to an external force that points directly away from origin and depends on distance r from the origin according to the formula

$$\vec{F} = F_0 \left(\frac{r}{R_0} \right)^2 \hat{r}$$

- (A) Given force is a central force.
(B) Given force is a conservative force.

- (C) Work done by external force as bead leaves the track is $\frac{8F_0R_0}{3}$.

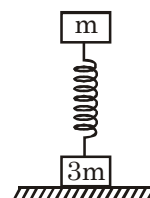
- (D) Speed of bead as it leaves the wire is $\sqrt{v_0^2 + \frac{18F_0R_0}{3m}}$.



8. In figure shown, the spring constant is K . The mass of the upper block is m and that of the lower block is $3m$. The upper block is depressed down from its equilibrium position by a distance δ and released at $t = 0$.

- (A) The minimum value of δ for which the lower block loses contact with the ground is $4mg/K$
(B) The value of δ for which the minimum normal reaction on $3m$ from ground is mg is $3mg/K$

- (C) The minimum value of δ for which the lower block loses contact with the ground is $6mg/K$
(D) The value of δ for which the minimum normal reaction on $3m$ from ground is $4mg/K$



9. The potential energy of a particle of mass 1kg is given by $U = 4x^3 - 5x^2$, where x is in meters.
- (A) The particle is at stable equilibrium at $x = 0$.
(B) The particle is at unstable equilibrium at $x = 5/6$ m.
(C) If kinetic energy of particle at $x = 5/6$ m is 1J, the particle oscillates about its equilibrium position.
(D) The magnitude of acceleration of the particle at $x = 1$ m is 2 m/s^2 .
10. The potential energy of a particle of mass 4kg is given as $U = -4x^2 + 8x^4$, where x is the position of the particle along positive x -axis from origin. The particle moves along x -axis. Then :
- (A) Potential energy of the particle is maximum at $x = 0$
(B) Potential energy of the particle is minimum at $x = \frac{1}{2}$
(C) Potential energy of the particle is maximum at $x = 1$
(D) Potential energy of the particle is minimum at $x = 2$

11. Choose the **CORRECT** statement(s) :

- (A) On a system work done by all external forces equal to change in kinetic energy of system
- (B) On a system work done by all external and non-conservative forces equal to change in kinetic energy of system
- (C) On a system work done by all external, conservative and non-conservative forces equal to change in kinetic energy of system
- (D) On a system work done by all external force, non-conservative force and negative of change in potential energy is equal to change in kinetic energy

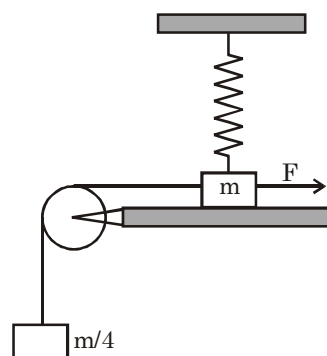
Linked Comprehension Type

(2 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question no. 12 and 13

In the figure shown, a constant horizontal force $F = \frac{mg}{2}$ starts acting on the block of mass 'm', from the position shown. The spring is undeformed in the position shown and has a natural length 'L', while the blocks are initially stationary. The spring constant is k. The floor is frictionless. The mass of the hanging block is $\frac{m}{4}$, while the pulley is massless and frictionless. Assume that the block never leaves contact with surface.



12. What is the initial acceleration of the block of mass m ?

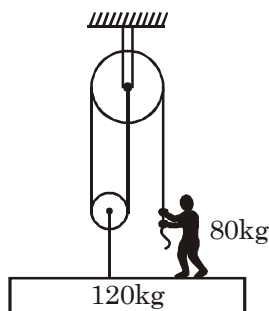
- (A) $\frac{g}{5}$
- (B) $\frac{g}{3}$
- (C) $\frac{g}{7}$
- (D) $\frac{g}{4}$

13. The maximum displacement of the heavier block is found to be $\sqrt{3} L$. Based on this information, find the spring constant :-

- (A) $\left(\frac{\sqrt{3}}{4}\right) \frac{mg}{L}$
- (B) $\left(\frac{\sqrt{3}}{2}\right) \frac{mg}{L}$
- (C) $\left(\frac{3\sqrt{3}}{2}\right) \frac{mg}{L}$
- (D) $\left(\frac{3\sqrt{3}}{4}\right) \frac{mg}{L}$

Paragraph for Question no. 14 and 15

A gentleman sits on a platform and raises himself by a rope and pulley arrangement as shown in figure. Assuming that pulley and ropes are ideal and friction is absent every where. By pulling rope system goes up. Mass of man is 80 kg and platform is 120 kg and system goes up with constant velocity ($a_{\text{system}} = 0$).



14. If man goes up by height of 3m then potential energy of (man + platform) system increased by.

- (A) 6 kJ
- (B) 2.4 kJ
- (C) 3.6 kJ
- (D) 8 kJ

15. What is the force exerted by man on rope ?

- (A) 2000/3 N
- (B) 200 N
- (C) 800 N
- (D) 1200 N

Matching List Type (4×4)

1 Q. [3 M (-1)]

16. List-I give work done by the forces and list-II represent its effect on the system. Match the correct order.

List-I

- (P) Work done by all the forces is positive
(Q) Work done by internal conservative force is negative. No information about work of other forces
(R) Work done by external forces is negative (assume there is no non-conservative force)
(S) Work done by all the force is negative

List-II

- (1) KE must decreases
(2) Mechanical energy of system decreases
(3) K.E must increases
(4) P.E. of system increases

Codes :

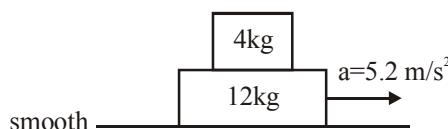
	P	Q	R	S
(A)	4	3	1	2
(B)	1	3	2	4
(C)	3	4	2	1
(D)	1	4	2	3

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

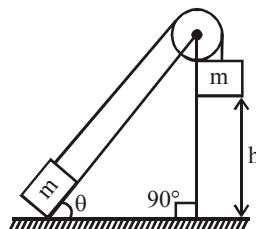
3 Q. [4 M (0)]

1. A 4 kg block is kept on to a long 12 kg block, which is accelerating along a smooth horizontal table at $a = 5.2 \text{ m/s}^2$ under application of an external constant force. Coefficient of friction between the two blocks is half the minimum value which will prevent the relative slipping between 4 kg and 12 kg blocks. If the amount of heat generated due to sliding between the two blocks during the time in which 12 kg block moves 10 m starting from rest is $\alpha \text{ J}$. Find the value of $\frac{\alpha}{13}$.



2. Two blocks of same mass starts their motion as shown. After t time their separation becomes minimum.

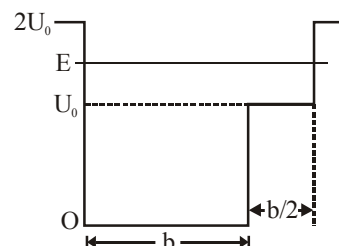
If $t = \sqrt{\frac{kh}{g \sin \theta (1 - \sin \theta)}}$ the value of k is



3. The potential energy of a particle of mass m as a function of its position along the x axis is as shown. (The discontinuous jumps in the value of U are not physically realistic but may be assumed to approximate a real situation.) The particle has a total mechanical energy E equal to $3U_0/2$. The period of

motion of particle from origin O to $x = \frac{3b}{2}$ is $\frac{(2 + \sqrt{x})}{2} \times \left(\frac{mb^2}{yU_0} \right)^{1/2}$. Then

find the value of $x + y$. (Given x & y are integers)

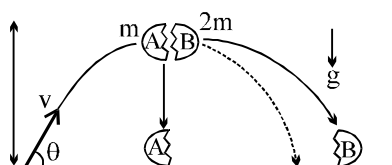


SECTION-I

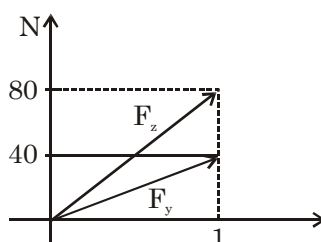
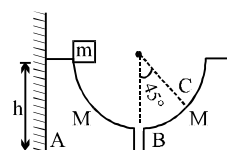
Single Correct Answer Type

8 Q. [3 M (-1)]

- A string passes over a fixed pulley. Two boys P and Q of the same mass hang at the same height at each end. Both start to climb upwards at the same time to reach the pulley. The velocity of P relative to the string is v and that of Q is $3v$, then the time taken by P to reach the pulley is equal to
(A) $1/3$ rd of the time taken by Q
(B) 3 times the time taken by Q
(C) the time taken by Q
(D) twice the time taken by Q
- A projectile is launched from the origin with speed v at an angle θ from the horizontal. At the highest point in the trajectory, the projectile breaks into two pieces, A and B, of masses m and $2m$, respectively. Immediately after the breakup piece A is at rest relative to the ground. Neglect air resistance. Which of the following sentences most accurately describes what happens next?

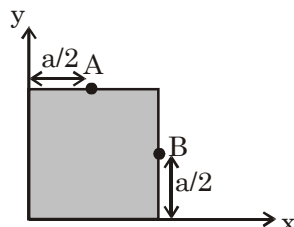


- Piece B will hit the ground first, since it is more massive.
 - Both pieces have zero vertical velocity immediately after the breakup, and therefore they hit the ground at the same time.
 - Piece A will hit the ground first, because it will have a downward velocity immediately after the breakup.
 - There is no way of knowing which piece will hit the ground first, because not enough information is given about the breakup.
- Two identical blocks having mass M each are smoothly conjugated and placed on a smooth horizontal floor as shown in figure. On the left of block A there is a wall. A small block of mass m is released from the position. Then velocity of block B is maximum
(A) when m is at highest position on B
(B) when m is at lowest position and moving towards left w.r.t B
(C) when m is at point C
(D) is equal to $\frac{m\sqrt{2gh}}{m+M}$
 - A particle is acted upon by a force for 1 sec. Whose x component remains constant at $F_x = 30\text{N}$ but y and z components vary with time as shown. The magnitude of change in momentum (in kg m/s) of particle in 1 sec.



- (A) 90 (B) 100 (C) $29\sqrt{10}$ (D) $10\sqrt{29}$

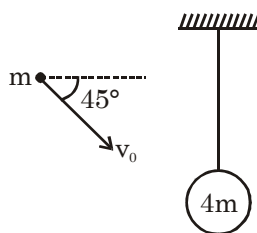
5. A square plate of variable mass density has mass '4m' and side length 'a'. It is kept as shown in figure. A point particle is kept on the plate. The y-coordinate of centre of mass of plate and particle system comes at same point if either a 'm' mass particle is kept at point A or a '4m' mass particle is kept at point B. Then y-coordinate of centre of mass of plate is :-



- (A) $\frac{a}{6}$ (B) $\frac{a}{4}$ (C) $\frac{a}{3}$ (D) $\frac{2a}{3}$
6. A smooth semicircular tube AB of radius R is fixed in a vertical plane and contains a heavy flexible chain of length πR . Find the velocity v with which it will emerge from the open end 'B' of tube, when slightly displaced.



- (A) $\sqrt{2gR(2\pi + 2/\pi)}$ (B) $\sqrt{\frac{gR}{2}\left(\frac{\pi}{4} + 4\pi\right)}$ (C) $\sqrt{2gR\left(\frac{2}{\pi} + \frac{\pi}{2}\right)}$ (D) $\sqrt{gR\left(\pi + \frac{1}{\pi}\right)}$
7. A particle of mass $m = 1$ kg moves in a circle of radius $R = 2$ m with uniform speed $v = 3\pi$ m/s. The magnitude of impulse given by centripetal force to the particle in one second is :-
- (A) $\sqrt{2}\pi$ Ns (B) $\sqrt{3}\pi$ Ns (C) $2\sqrt{3}\pi$ Ns (D) $3\sqrt{2}\pi$ Ns
8. A point mass m collides with a block of mass 4m connected with string of length ℓ as shown in the diagram. The collision is elastic in nature and the block deflects to an angle 60° after the collision. The collision is along the line joining the center of the two objects. Mark the **CORRECT** option(s) :-

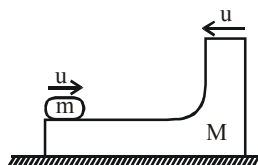


- (A) The speed v_0 of the point mass before collision is $\frac{9}{4}\sqrt{\frac{g\ell}{2}}$
- (B) The impulse due to tension on the block of mass 4m during the collision is $4m\sqrt{g\ell}$
- (C) The impulse due to normal force between point mass m and block of mass 4m is $2m\sqrt{2g\ell}$
- (D) The speed with which the ball rebounds is $\frac{v_0}{\sqrt{2}}$

Multiple Correct Answer Type

1 Q. [4 M (-1)]

9. A small block of mass m is placed on a smooth body of mass M ; the combination is placed on a smooth horizontal surface and are given velocities u towards each other as shown in the diagram. The maximum height to which the small block of mass m rises after breaking off the vertical section of the body of mass M is H , relative to its initial level. Assume $M > m$.



- (A) $u < \sqrt{\frac{gH}{2}}$
- (B) The work done in raising the two bodies through a height H equals 4 times the initial kinetic energy of the body of mass M
- (C) The final velocity of the small block at its height point is $\left(\frac{M-m}{M+m}\right)u$
- (D) The kinetic energy of the system when the smaller block is at its highest point is $\frac{1}{4}(M-m)^2 \frac{gH}{M}$

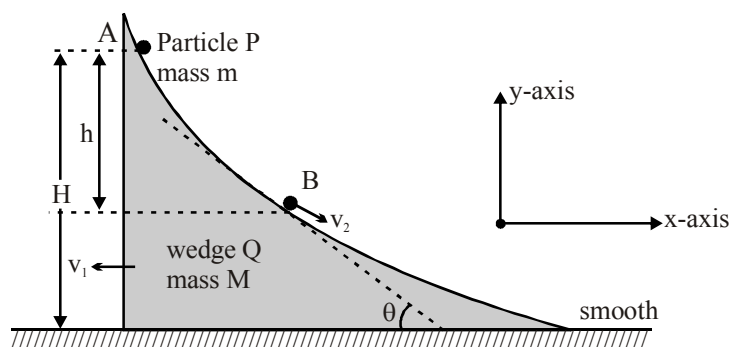
Linked Comprehension Type

(1 Para $\times$ 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for question nos. 10 to 12

A particle (P) of mass m is released from rest at point A on the curved surface of a wedge (Q) of mass M , free to move on a smooth horizontal surface. There is no information regarding the nature of surface of wedge. The speed of P w.r.t. Q when it reaches B is v_2 and the speed of Q at this instant is v_1 .



10. The centre of mass of system comprising of P and Q will be moving in which direction.
- (A) $\leftarrow$ (B) $\downarrow$ (C) $\searrow$ (D) $\rightarrow$
11. Consider the system comprising of P and Q. Linear momentum can not be conserved in vertical direction because, there is non zero vertical net force of gravity and normal from ground. The conservation of linear momentum in horizontal direction for this system gives

$$Mv_1 + m(\quad) = 0.$$

The blank is to be filled by :

- (A) $-v_2 \cos \theta$ (B) $-v_2 + v_1$
- (C) $-v_2 \cos \theta + v_1$ (D) $v_1^2 + v_2^2 - 2v_1v_2 \cos \theta$

12. Now, let the surface of wedge be smooth so that there is no loss of energy. For the system of P and Q, the conservation of mechanical energy yields.

$$\frac{1}{2} M v_1^2 + \frac{1}{2} m (\quad) = mgh$$

The blank is to be filled by :

(A) $v_1^2 + v_2^2 - 2v_1 v_2 \cos \theta$

(B) $v_1^2 + v_2^2$

(C) $v_1^2 + v_2^2 + 2v_1 v_2 \cos \theta$

(D) $v_1^2 + v_2^2 - 2v_1 v_2 \sin \theta$

Linked Comprehension Type

(1 Para × 2 Q.) [4 M (-1)]

(Multiple Correct Answer Type)

Paragraph for Question No. 13 and 14

Consider the potential energy given by

$$U(x) = \frac{1}{4}x^4 - \frac{1}{3}x^3 - 3x^2,$$

where the coefficients of the various powers of x have units such that U has units of joules.

13. Find the equilibrium points and classify them.

(A) $x = 0$ stable

(B) $x = 3$ unstable

(C) $x = -2$ stable

(D) $x = 2$ unstable

14. If total energy is -6 J,

(A) Turning points will be only on +ve side of x-axis

(B) Turning points will be only on -ve side of x-axis

(C) The body can be at $x = 0$.

(D) The body can be at $x = 3$.

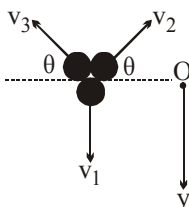
SECTION-II

Numerical Answer Type Question

2 Q. [3(0)]

(upto second decimal place)

1. A rocket is projected straight up and explodes into three equally massive fragments just as it reaches the top of its flight (refer figure). One of the fragments is observed to come straight down in 2 sec, while the other two take 4 sec to come to ground, after the burst. Find the height h (in m) at which the fragmentation occurred.



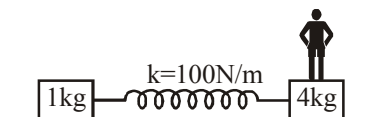
2. A cannon of mass 10×10^3 kg is rigidly bolted to the earth so it can recoil only by a negligible amount. The cannon fires a 2.1×10^3 kg shell horizontally with an initial velocity of 550 m/s. Suppose the cannon is then unbolted from the earth, and no external force hinders its recoil. What would be the velocity (in m/s) of a shell fired horizontally by this loose cannon? (Assume that the burning gunpowder imparts the same kinetic energy to the system.)

SECTION-III

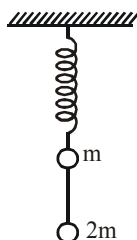
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

1. A boy of mass 10 kg is standing on a 4 kg block as shown. The whole system is on smooth level ground. The boy jumps to the right with a velocity of 7 m/s relative to the 4 kg block. Find the maximum velocity 'v' of 1 kg block (in m/sec) in the subsequent motion.

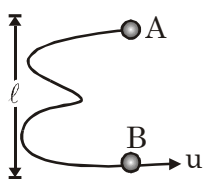


2. A ball of mass $m = 1$ kg is hung from a spring of constant 400 N/m and another ball of mass $2m$ is hanging from it as shown with the help of a thread. The system is in equilibrium. Now the thread is burned. As a result, ball of mass m starts moving upwards. Eventually it will stop under the influence of gravity and spring. In between, there is a point at which its speed is maximum. What is that maximum speed (in m/s) ?

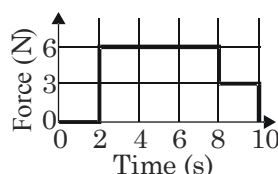
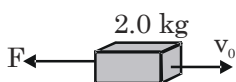


3. Two particles A and B of equal mass m each are attached by a string of length 2ℓ and initially placed over a smooth horizontal table in the position shown in figure. Particle B is projected across the table with speed u perpendicular to AB as shown in figure. The ratio $\frac{v_B}{v_A}$ after the string becomes taut is

$\sqrt{\frac{\alpha}{\beta}}$. Find the least integral value of $(\alpha + \beta)/2$. (where v_B and v_A are speed of B & A respectively after the string become taut.)



4. A block of mass 2 kg is sliding on a smooth surface. At $t = 0$, its speed is $v_0 = 2$ m/s. At $t = 0$, a time-varying force starts acting on the block in the direction opposite to v_0 . If the speed of object at $t = 10$ s is V m/s then fill the value of $\frac{V+8}{3}$.

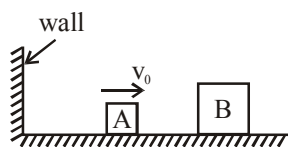


SECTION-I

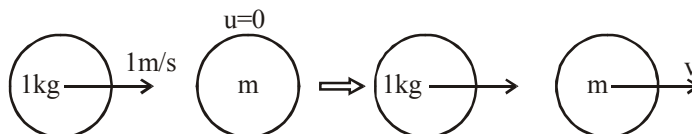
Single Correct Answer Type

10 Q. [3 M (-1)]

- A 40 g ball dropped from a certain height bounces back from the horizontal ground without losing mechanical energy. If its speed is 10 m/s just before making contact with the ground, and the average value of the force of the ground on the ball is equal to 16 N while ball and wall are in contact, how long were they in contact?
(A) 25 ms (B) 50 ms (C) 75 ms (D) 100 ms
- Block A of mass m is moving with a speed v_0 on a frictionless horizontal surface as shown in figure. It collides with a massive block B of mass $M = \alpha m$ ($\alpha > 1$) initially at rest. All collisions are elastic. Condition on α for which the two blocks will undergo only one collision is :

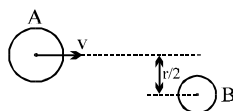


- (A) $\alpha < 3$ (B) $\alpha < 4$ (C) $\alpha < 5$ (D) $\alpha < 6$
- A particle of mass 1 kg moving with velocity 1 m/s, collides elastically with another particle of mass m . In the collision particle of mass 1 kg loses $\frac{3}{4}$ of its K.E. The value of m is :



- (A) $\frac{1}{2}$ kg (B) $\frac{1}{3}$ kg (C) $\frac{1}{4}$ kg (D) $\frac{1}{5}$ kg
- A body of mass m_1 strikes a stationary body of mass m_2 . If the collision is elastic and head-on, the fraction of kinetic energy transmitted by the first body to the second body is :
(A) $\frac{m_1 m_2}{m_1 + m_2}$ (B) $\frac{2m_1 m_2}{m_1 + m_2}$ (C) $\frac{4m_1 m_2}{(m_1 + m_2)^2}$ (D) $\frac{2m_1 m_2}{(m_1 + m_2)^2}$
- The body having mass m_1 moving with a speed of v , collides elastically with a motionless body and after impact jumps aside from it at an angle 90° to the initial direction of motion with speed of $v/2$. find the mass of motionless body.
(A) $\frac{m_1}{6}$ (B) $\frac{5}{3} m_1$ (C) $\frac{4}{3} m_1$ (D) $\frac{3}{5} m_1$
- A ball, moving with a speed of 9 m/s, strikes an identical stationary ball such that after the collision, the direction of each ball makes an angle of 30° with the original line of motion. The speeds of the two balls after the collision are, respectively,
(A) $3\sqrt{3}$ m/s, 3 m/s (B) 3 m/s, $3\sqrt{3}$ m/s (C) 3 m/s, 3 m/s (D) $3\sqrt{3}$ m/s, $3\sqrt{3}$ m/s
- A particle is thrown vertically upward with a speed u from the top of a tower of height h from ground level. If after first impact with ground it just reaches to height h from ground the coefficient of restitution for the collision is :-
(A) 1 (B) u^2 (C) $\sqrt{\frac{2gh}{u^2 + 2gh}}$ (D) $\frac{\sqrt{2gh}}{u}$

8. A disk A of radius r moving on perfectly smooth surface at a speed v undergoes an elastic collision with an identical stationary disk B. Find the velocity of the disk B after collision if the impact parameter is $r/2$ as shown in the figure



- (A) $\frac{v\sqrt{15}}{4}$ (B) $\frac{v}{4}$ (C) $\frac{v}{2}$ (D) $\frac{\sqrt{3}v}{2}$
9. A ball is suspended from the top of a cart by a string of length 1.0 m. The cart and the ball are initially moving to the right at constant speed V , as shown in figure (i). The cart comes to rest after colliding and sticking to a fixed bumper, as in figure (ii). The suspended ball swings through a maximum angle 60° . The initial speed v is

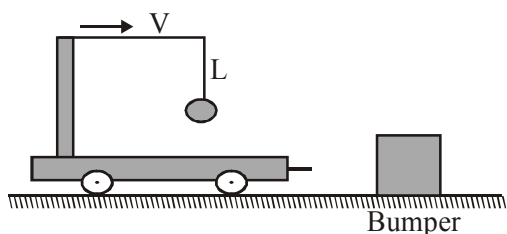


Fig. (i)

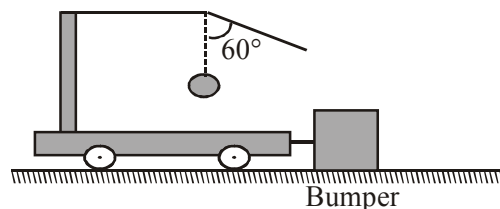
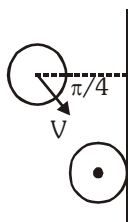


Fig. (ii)

- (A) $\sqrt{10}$ m/s (B) $2\sqrt{5}$ m/s (C) $5\sqrt{2}$ m/s (D) 4 m/s
10. In a carom-board game the striker and the coins are identical and of mass m . In a particular hit the coin is hit when it is placed close to the edge of the board as shown in figure such that the coin travels parallel to the edge. If the striker is moving with speed V before the strike, then the net impulse on the striker during collision if it moves perpendicular to the edge is (assume all collisions to be perfectly elastic)

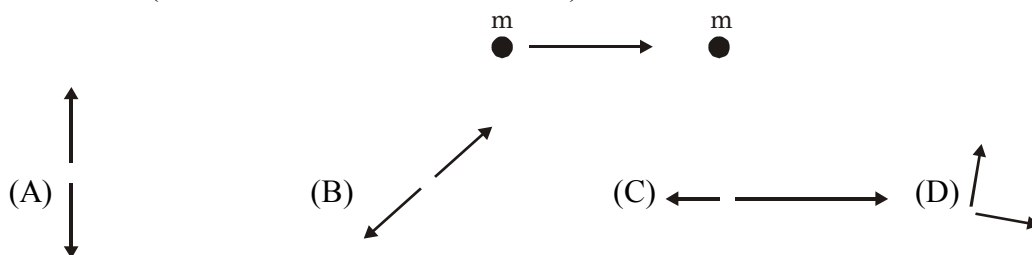


- (A) $mV\sqrt{\frac{5}{2}}$ (B) $2mV$ (C) $\frac{mV\sqrt{3}}{2}$ (D) mV

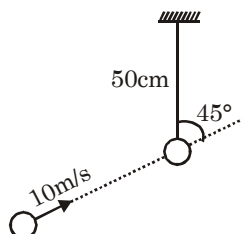
Multiple Correct Answer Type

2 Q. [4 M (–1)]

11. A mass m moves with a given speed and collides (not necessarily head-on) elastically with another mass m that is initially at rest, as shown in Fig. Which of the figures shows an impossible outcome for the two velocities? (The velocities are drawn to scale)



12. On a smooth horizontal surface, a ball of mass 1 kg is placed, tied to a nail O on the same horizontal surface with a string of length 50 cm. Another ball of same mass travelling on the horizontal surface with a speed of 10 m/s, its velocity making an angle of 45° with string as shown, collides with the first ball. The balls stick to each other after collision :-



- (A) Momentum will not be conserved during collision.
(B) Tension in the string just after collision is zero.
(C) Time for which the string remains slack after collision is 0.141 sec.
(D) Velocity of the balls just after the string is taut again is $\frac{5}{\sqrt{2}}$ m/s

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para $\times$ 2Q. & 1 Para $\times$ 3Q.) [3 M (-1)]

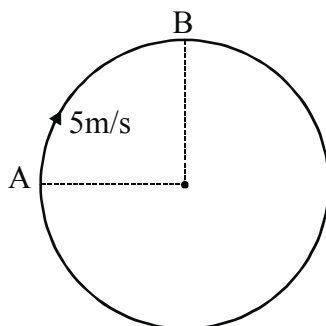
Paragraph for question no. 13 and 14

Two identical particles each of mass 4 kg are placed together at the edge of a tall cliff. They are joined by an inextensible string of negligible mass. One of the particles is projected with a speed of $10\sqrt{3}$ m/s at 60° above the horizontal. The string becomes taut after 2 sec. ($g = 10 \text{ m/s}^2$)

13. The length of the string is :
(A) $5\sqrt{3}$ m (B) 10 m (C) 20 m (D) $10\sqrt{3}$ m
14. Tension in the string when the second particle starts moving is :
(A) $10\sqrt{3}$ N (B) 7.5 N (C) 15 N (D) 10 N

Paragraph for Question no. 15 to 17

Two friends A and B (each weighing 40 kg) are sitting in a frictionless well having vertical circular smooth wall. They subtend angle 90° at the centre of well. 'A' rolls a ball of mass 40 kg in clockwise direction on the platform base towards B along the wall of the well which B catches. Then similarly B rolls the ball towards 'A' anticlockwise and A catches it. The ball has a fixed speed of 5 m/s. (Neglect catching and throwing time).



15. Find the total impulse by the wall of well on whole system by the time ball travels from A to B
(A) 200 kg-m/s (B) 400 kg-m/s (C) zero (D) None of these
16. Find the angular displacement of A till he meets B
(A) 150° (B) 60° (C) 90° (D) 180°

17. When A meets B they catch each other and now they move together i.e. perfectly inelastic collision. What will be the velocity when they (A+B) throw the ball after catching it

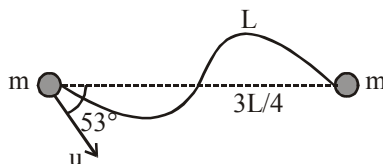
(A) 2.5 m/s (B) $\frac{5}{3}$ m/s (C) $\frac{50}{11}$ m/s (D) None of these

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. In the figure shown, each tiny ball has mass m , and the string has length L . One of the balls is imparted a velocity u , in the position shown, in which the initial distance between the balls is $\frac{3L}{4}$. If the impulse of the tension in the string when it becomes taut is $m\alpha \text{ kg}\cdot\text{ms}^{-1}$, then find the value of α .
(Given : $m = 2\text{kg}$, $u = 15 \text{ m/s}$)



SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. A body of mass m_1 collides one dimensionally with another stationary body of mass m_2 . The coefficient of restitution is e . Find the correct combination of options of column II to satisfy the condition of column I.

Column I

- (A) After the collision, velocity of the second body is maximum when
(B) After the collision, momentum of the second body is maximum when
(C) After the collision, kinetic energy of the second body is maximum when
(D) After the collision, kinetic energy of system is minimum when

Column II

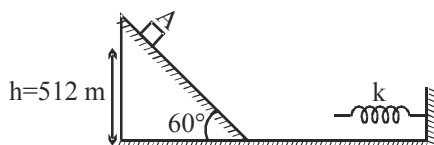
- (P) $e = 1$
(Q) $m_1 \gg m_2$
(R) $e = 0$
(S) $m_1 \ll m_2$
(T) $m_2 = m_1$

SECTION-I

Single Correct Answer Type

7 Q. [3 M (-1)]

1. A small Block A is released from the top of an incline plane at height of $h = 512$ m. All surfaces are frictionless. The block collides with the ground inelastically but does not stick to it. After collision with the spring, it returns back and collides with the incline inelastically but again does not stick to it. Find the maximum height (in m) from the ground attained by block A when it climbs up the incline.



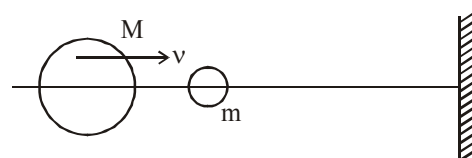
- (A) 16 (B) 32 (C) 6 (D) 8
2. Two beads of masses M and m can move without friction on a horizontal wire as shown in the diagram. Initially M is moving with a speed v and m is stationary. During subsequent collisions speed of m will increase. If all collisions are perfectly elastic then maximum speed attained by m is. (Here : $M \gg m$)

(A) v

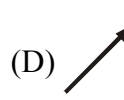
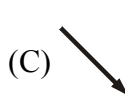
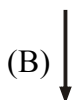
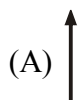
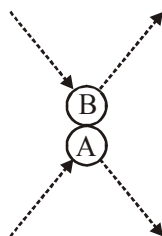
(B) $\sqrt{\frac{m}{M}}v$

(C) $\sqrt{\frac{M}{m}}v$

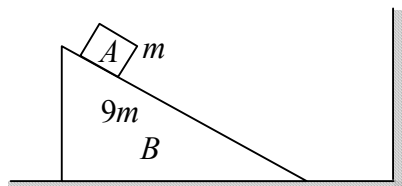
(D) ∞



3. The figure below depicts the paths of two colliding steel balls, A and B. Which of the arrows best represents the impulse applied to ball A during the collision ?

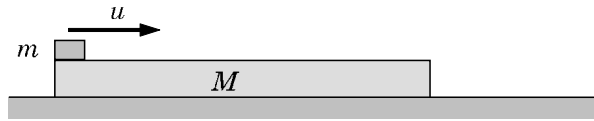


4. As shown in the figure, block A of mass m starts from rest from the top of the wedge of the mass $9m$. Wedge can slide on smooth ground. Block A slides down onto the ground, moves along it with a speed u , has an elastic collision with the wall, and climbs back onto the wedge. All surfaces are frictionless. Choose the wrong alternative.

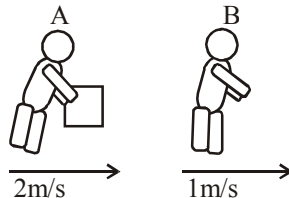


- (A) Just before collision between block A and wall, the center of mass of block A and wedge B remains at rest.
- (B) Just after collision the center of mass of block A and wedge B moves with a velocity $u/5$.
- (C) Finally when block A reaches the maximum height on wedge B, its velocity will be $u/5$.
- (D) Finally block A reaches to the same height, from where it was released.

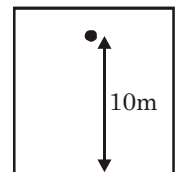
5. A small block of mass m is placed on a long plank of mass M that is placed on a frictionless horizontal floor. The block is abruptly given a velocity u on the plank. If during sliding of the block a distance L relative to the plank, total heat loss is Q , find acceleration of the plank during the sliding of the block?



- (A) $\frac{Q}{mL}$ (B) $\frac{Q}{ML}$ (C) $\frac{Q}{(m+M)L}$ (D) None of these
6. Two astronauts, A and B, both with mass of 60kg, are moving along a straight line in the same direction in a "weightless" spaceship. Relative to the spaceship the speed of A is 2 m/s and that of B is 1 m/s. A is carrying a bag of mass 5 kg with him. To avoid collision with B, A throws the bag with a speed v relative to the spaceship towards B and B catches it. Find the minimum value of v .



- (A) 7.8 m/s (B) 26.0 m/s (C) 14.0 m/s (D) 9.2 m/s
7. A particle of mass 2kg is projected vertically upward in a lift with a velocity 10m/s relative to the lift from a height 10m above the floor of lift. When particle strikes the floor of lift what will be the velocity of the particle just before the strike and the impulse imparted on it by lift respectively.
[At $t = 0$, $V = 10$ m/s & $a = 5\text{m/s}^2$ (Both upwards) for the lift]. Particle does not rebound after it strikes the floor of lift :-

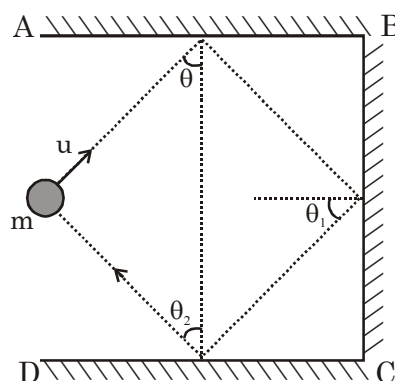


- (A) 0 m/s, 40 N-s (B) 10 m/s upward, No impulse
(C) 10 m/s downward, 60 N-s (D) 30 m/s downward, 20 N-s

Multiple Correct Answer Type

5 Q. [4 M (-1)]

8. Particle of mass ' m ' placed on a horizontal surface surrounded by walls as shown in figure. Coefficients of restitution for collision with walls AB, BC, CD are $e = 1$, $e = 1/2$, $e = 1/4$ respectively. Particle 1st collide with wall AB as shown. After collision from wall BC & CD respectively if again return to same position. Relation between θ_1 , θ_2 & θ will be :-



- (A) $\tan \theta_1 = 2 \tan \theta$ (B) $\tan \theta_2 = 2 \tan \theta$ (C) $\tan \theta_1 = 2 \cot \theta$ (D) $4 \cot \theta_1 = \tan \theta_2$

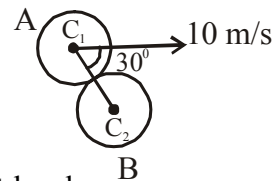
9. A ball A collides elastically with another identical ball B with velocity 10 m/s at an angle of 30° from the line joining their centres C_1 and C_2 . Select the correct alternative(s).

(A) velocity of ball A after collision is 5 m/s

(B) velocity of ball B after collision is $5\sqrt{3}$ m/s

(C) both the balls move at right angles after collision

(D) kinetic energy will not be conserved here, because collision is not head on



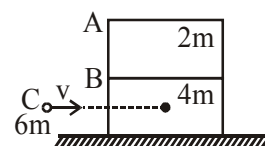
10. A block A of mass $2m$ is placed on other block B of mass $4m$ which is placed on a fixed table as shown in figure. A small object C of mass $6m$ moving horizontally along a line passing through the center of block B and perpendicular to its face with a speed v collides elastically with block B. Friction is absent between ground and block B and surface between blocks is rough. Then :-

(A) Velocity of A just after collision is v

(B) Velocity of B just after collision is $\frac{6}{5}v$

(C) Velocity of centre of mass of system (A + B) is $\frac{4}{5}v$ after collision

(D) Momentum of (C + B) can be conserved during collision.



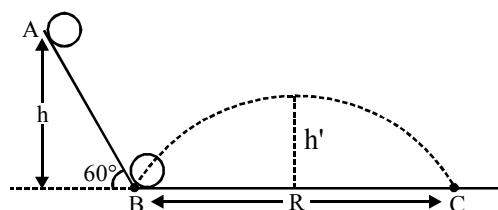
11. In the figure shown, a ball is released from the smooth track. If the ball strikes the horizontal surface of the track and bounces off, it again strikes the horizontal surface at some distance R from B and rises to a height maximum of h' above the surface (If the coefficient of restitution for the collision is e) :-

(A) R is maximum for $e = \frac{1}{\sqrt{3}}$

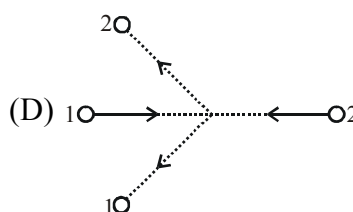
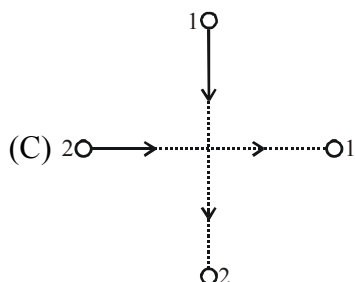
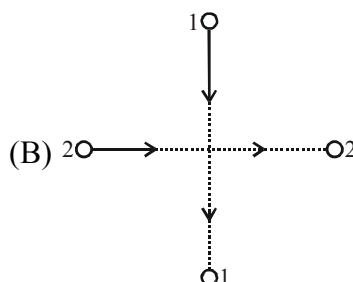
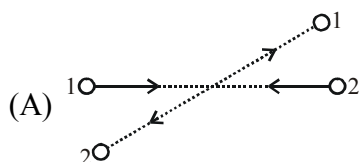
(B) R is maximum for $e = 1$

(C) $h' \leq \frac{3h}{4}$ for all values of e

(D) $h' = h$ for $e = 1$



12. Two identical balls (1 and 2) moving with equal speed collide on a frictionless surface. Collision may or may not be elastic. Which of the following diagrams is/are a physically possible situation ?

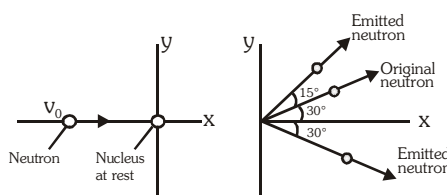


Linked Comprehension Type (Single Correct Answer Type)

(2 Para × 3Q.) [3 M (-1)]

Paragraph for Question 13 to 15

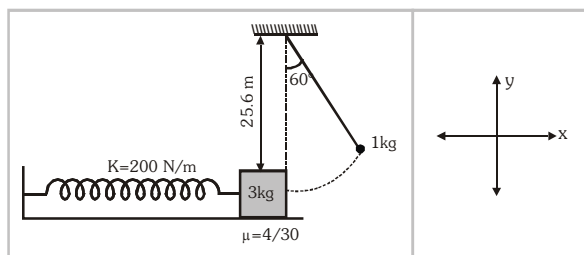
Fission, the process that supplies energy in nuclear power plants, occurs when a heavy nucleus is split into two medium-sized nuclei. One such reaction occurs when a neutron colliding with a ^{235}U (Uranium) nucleus splits that nucleus into a ^{141}Ba (Barium) nucleus and a ^{92}Kr (Krypton) nucleus. In this reaction, two neutrons also are split off from the original ^{235}U . Before the collision, we have the arrangement in figure. After the collision, we have the ^{141}Ba nucleus moving in the +z direction and ^{92}Kr is moving in -z direction. The three neutrons are moving in the xy-plane as shown in Fig. The incoming neutron has an initial velocity of magnitude $3.0 \times 10^3 \text{ m/s}$ and a final velocity of magnitude $\frac{\sqrt{3}}{2} \times 10^3 \text{ m/s}$ in the directions shown.



13. What is the speed of emitted neutron with angle 45° ?
 (A) 750 ms^{-1} (B) $\frac{\sqrt{3}}{2} \times 10^3 \text{ ms}^{-1}$ (C) $3 \times 10^3 \text{ ms}^{-1}$ (D) 1500 m/s
14. What is the speed of emitted neutron with angle 30° ?
 (A) $3 \times 10^3 \text{ ms}^{-1}$ (B) 1960 ms^{-1} (C) 1010 m/s (D) $\frac{\sqrt{3}}{2} \times 10^3 \text{ ms}^{-1}$
15. Which relation is obeyed for velocity of Barium (Ba) and Krypton (Kr).
 (A) $v_{\text{Kr}} = \frac{v_{\text{Ba}}}{2}$ (B) $v_{\text{Kr}} = 4 v_{\text{Ba}}$ (C) $v_{\text{Kr}} = 1.5 v_{\text{Ba}}$ (D) $v_{\text{Kr}} = 6 v_{\text{Ba}}$

Paragraph for Question 16 to 18

As shown in figure there is a small bob having mass 1 kg is attached to the string having negligible mass. Bob is released as shown in figure. Initially when spring is in natural length. ($g = 10 \text{ m/s}^2$)



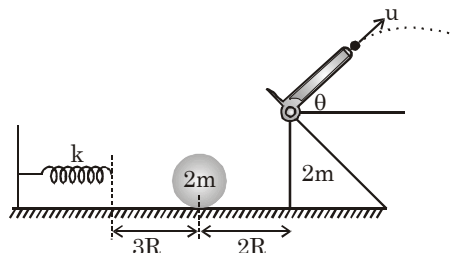
16. If there is no loss in mechanical energy due to collision then find the maximum compression in the spring?
 (A) 106 cm (B) 86 cm (C) 46 cm (D) 96 cm
17. Just after achieving maximum compression in the spring as per previous problem, a bullet of mass 100 g strikes to the block in such a manner that there is maximum loss in mechanical energy due to collision. The velocity of bullet is $\hat{v}_i$. Find the value of v so that elongation of spring is equal to compression of spring? (Assuming there is no further collision with bob)
 (A) 140 m/s (B) 70 m/s (C) 160 m/s (D) 320 m/s
18. Find the total work done by the friction in above two problems?
 (A) -11 J (B) -11.7 J (C) -7.8 J (D) -4 J

SECTION-I

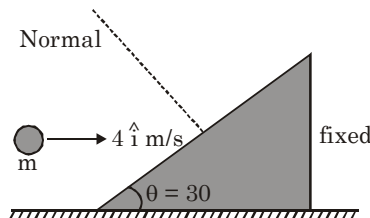
Single Correct Answer Type

5 Q. [3 M (-1)]

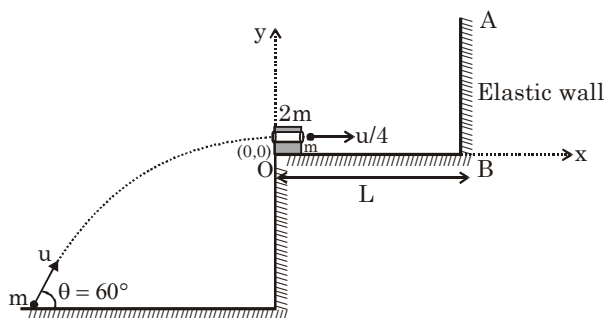
1. In the shown diagram a cannon attached to a moving wedge having total mass (cannon + wedge + bullet = 3m) as shown in figure. Initially system is at rest. Cannon release bullet of mass 'm' with speed u 'w.r.t' wedge. Find out maximum compression in spring in subsequent motion (Collision between wedge and block is perfectly elastic and initially spring is in natural length). Assume all surface are frictionless :-



- (A) $\sqrt{\frac{2m}{k}} \frac{u \cos \theta}{3}$ (B) $\sqrt{\frac{m}{k}} \frac{u \cos \theta}{3}$ (C) $\sqrt{\frac{2m}{k}} \frac{u \sin \theta}{3}$ (D) $\sqrt{\frac{m}{k}} \frac{u \sin \theta}{3}$
2. A ball of mass 'm' is moving with initial velocity $4\hat{i}$ m/sec. It strikes a fixed incline plane such that angle of incidence of the ball with normal = angle of reflection of ball with normal. Velocity vector of ball just after collision will be :- (No friction is present between surfaces of ball and wedge)

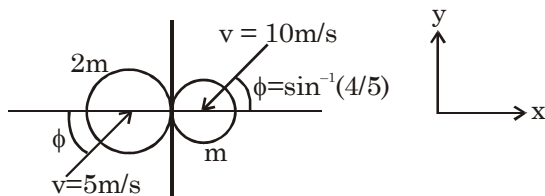


- (A) $(\hat{i} + \sqrt{3}\hat{j})$ m/s (B) $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$ m/s (C) $2(\hat{i} + \sqrt{3}\hat{j})$ m/s (D) $4(\hat{i} + \sqrt{3}\hat{j})$ m/s
3. A bullet of mass m is shot with a velocity u at an angle $\theta = 60^\circ$ with the horizontal which strikes the box of mass $2m$ resting at the edge of a smooth surface 'OB' as shown in figure. Bullet strikes the box horizontally. After that bullet passes through the box and moves with velocity $u/4$ along horizontal direction. If bullet rebound elastically with wall AB. Then find the final position where bullet again strikes mass $2m$.



- (A) $\frac{L}{2}, 0$ (B) $\frac{L}{3}, 0$ (C) $\frac{2L}{3}, 0$ (D) $\frac{L}{4}, 0$

4. Two smooth sphere made of identical material having masses m and $2m$ undergo an oblique impact as shown in figure. The coefficient of restitution $\frac{5}{9}$. The velocity of the mass $2m$ after collision will be :-



- (A) $-\frac{10}{3}\hat{i} - 8\hat{j}$ (B) $-\frac{5}{3}\hat{i} + 4\hat{j}$ (C) $+\frac{5}{3}\hat{i} + 4\hat{j}$ (D) $\frac{10}{3}\hat{i} - 8\hat{j}$
5. Two bodies of masses M and m ($M > m$) are attached to the two ends of a light inextensible string passing over a frictionless pulley. The system is held at rest with the string taut and vertical with both masses at a height 'd' above an inelastic table. The system is now released. Calculate the height to which the larger mass will rise after it has hit the table.

- (A) $\left(\frac{m}{M+m}\right)d$ (B) $\left(\frac{m}{M-m}\right)d$ (C) $\left(\frac{m^2}{M^2+m^2}\right)d$ (D) $\left(\frac{m}{M+m}\right)^2 d$

Multiple Correct Answer Type

7 Q. [4 M (-1)]

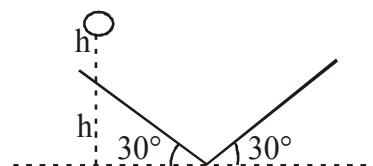
6. A ball is dropped as shown in figure. Both the inclined planes are smooth and ball makes perfect inelastic collision with both planes

(A) maximum height of ball on second plane is $\frac{5}{16}h$

(B) maximum height of ball on second plane is $\frac{5}{4}h$

(C) Total impulse on ball from start to maximum height on second plane is zero.

(D) Total distance covered by ball from start to maximum height on second plane is $\frac{29}{8}h$



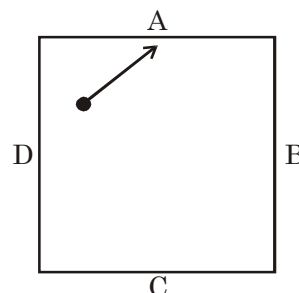
7. A smooth uniform square wall frame of mass m of edge length $2a$ and small height lies in a smooth horizontal plane. At $t = 0$ a particle of mass m hits with speed $v\sqrt{2}$, one of the inner wall of the frame horizontally at an angle 45° from wall normal. Collision is perfectly elastic. The initial collision takes place at the mid point of wall A.

(A) Next collision takes place when $t = \frac{a}{v}$.

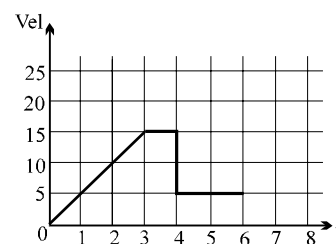
(B) KE of frame when $t = \frac{3a}{2v}$, is mv^2

(C) after collision at B next collision takes place at wall D

(D) particle and frame comes to rest at regular intervals of time



8. The figure shows the velocity as a function of time for an object with mass 10 kg being pushed along a frictionless surface by external force. At $t = 3$ s, the force stops pushing and the object moves freely. It then collides head-on and sticks to another object of mass 25 kg.



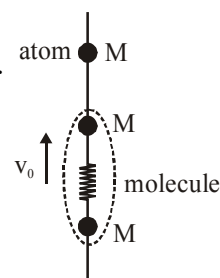
- (A) External force acting on the system is 50 N.
(B) Velocity of the second particle just before the collision is 1 ms^{-1} .
(C) Before collision both bodies are moving in the same direction.
(D) Before collision, bodies are moving in opposite direction.
9. In an elastic collision between spheres A and B of equal mass but unequal radii, A moves along the x-axis and B is stationary before impact. Which of the following is possible after impact?
- (A) A comes to rest
(B) the velocity of B relative to A remains the same in magnitude but reverses in direction
(C) A and B move with equal speeds, making an angle of 45° each with the x-axis.
(D) A and B move with unequal speeds, making angles of 30° and 60° with the x-axis respectively.
10. A molecule capable of vibrations is modeled as two point atoms of mass M connected by a spring. Initially molecule is not vibrating when it collides elastically with a single atom of mass M as shown in figure. Mark the **CORRECT** option (s) :-

(A) Velocity of centre of mass of system consisting of atom and molecules is $\frac{2v_0}{3}$.

(B) Maximum energy stored in spring is $\frac{Mv_0^2}{4}$.

(C) Velocity of centre of mass of molecule after collision is $\frac{v_0}{2}$.

(D) Momentum of molecule after collision is Mv_0 .



11. Two masses ($m < M$) are released from rest inside a smooth fixed hemispherical bowl. They collide at the bottommost point.



- (A) If the collision is elastic, the smaller mass m reaches maximum height of $R \left(\frac{3M - m}{m + M} \right)^2$ from bottom.
- (B) If the collision is elastic, the larger mass M reaches a maximum height of $R \left(\frac{M - 3m}{m + M} \right)^2$ from bottom.
- (C) If the collision is elastic, the smaller mass m reaches maximum height of $R \left(\frac{2M - m}{m + M} \right)^2$ from bottom.
- (D) If the collision is elastic, the larger mass M reaches maximum height of $R \left(\frac{2M - m}{m + M} \right)^2$ from bottom.

12. Two balls of mass m each are connected by a weightless spring (figure) of force constant k and length of L and lie motionlessly on a smooth horizontal table. The third ball of mass m moves with speed of v_0 on the line connecting the centers of first two, and elastically collides with one of them. Assuming that time of impact of balls is very small in comparison to time of spring deformation :-



- (A) The maximum distance between the first two balls during their further movement

$$L_{\max} = L + v_0 \sqrt{\frac{m}{2k}}.$$

- (B) The third ball comes to rest

- (C) The maximum distance between the first two balls during their further movement

$$L_{\max} = L + v_0 \sqrt{\frac{m}{k}}.$$

- (D) The minimum distance between the first two balls during their further movement

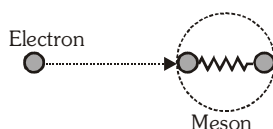
$$L_{\min} = L - v_0 \sqrt{\frac{m}{2k}}.$$

Linked Comprehension Type (1 Para × 3Q.)

[3 M (-1)]

Paragraph for Question 13 to 15

Meson is composed of two quarks and the interaction between the quarks is complicated. Research of meson can be done by studying the inelastic collisions between the meson and high energy electrons. As the collision is quite complicated, scientists invented a simplified model called “parton model” to grasp the main content during collision. In the model, the electron first collides with part of meson (e.g. one of the quarks) elastically. Then the energy and momentum are transferred to the other quark and thus the whole meson during subsequent interaction. This simplified model is described by the following:



An electron of mass M and energy E collides with a quark of mass m_1 in a meson. The other quark in the meson has mass m_2 . The quarks are connected by a massless spring of natural length L which is at equilibrium before collision. All movements are on a straight line and neglect the effect of relativity. Find, after collision,

13. The energy gain of the quark m_1

(A) $\frac{4Mm_1}{(M+m_1)^2}E$ (B) $\frac{2Mm_1}{(M+m_1)^2}E$ (C) $\frac{Mm_1}{(M+m_1)^2}E$ (D) $\frac{3Mm_1}{2(M+m_1)^2}E$

14. The minimum kinetic energy of the meson as a whole system

(A) $\frac{4Mm_1^2}{(M+m_1)^2(m_1+m_2)}E$ (B) $\frac{4Mm_1^2}{(M+m_2)^2(m_1+m_2)}E$
(C) $\frac{2Mm_1^2}{(M+m_2)^2(m_1+m_2)}E$ (D) $\frac{2Mm_1^2}{3(M+m_2)^2(m_1+m_2)}E$

15. The internal energy of the meson which is expressed by the oscillation of the spring.

(A) $\frac{4Mm_1m_2}{(M+m_1)^2(m_1+m_2)}E$

(B) $\frac{4Mm_1m_2}{(M+m_2)^2(m_1+m_2)}E$

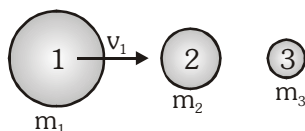
(C) $\frac{2Mm_1m_2}{(M+m_2)^2(m_1+m_2)}E$

(D) $\frac{3Mm_1m_2}{(M+m_2)^2(m_1+m_2)}E$

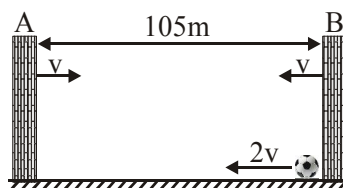
SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 3 Q. [4 M(0)]

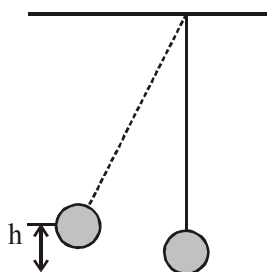
1. The centres of the spheres 1, 2 and 3 lie on a single straight line. Sphere 1 is moving with an (initial) velocity v_1 directed along this line and hits sphere 2. Sphere 2, acquiring after collision a velocity v_2 , hits sphere 3. Both collisions are absolutely elastic. What must be the mass of sphere 2 (in kg) for the sphere 3 to acquire maximum velocity (The masses m_1 and m_3 of spheres 1 and 3 are 9kg & 1kg respectively)?



2. Initially there is a small ball moving with speed $2v$ towards wall A and both walls are moving with constant velocity towards each other as shown in figure. Then find speed v (in m/s) if time taken by the ball in first three collision is 5 sec. (Assume all collision are perfectly elastic and friction is absent).



3. In the arrangement shown, the pendulum A is pulled aside. It is then released and allowed to collide with other pendulum B which is at rest. A perfectly inelastic collision occurs and the system rises to a height $\frac{1}{4}h$. The ratio of the masses of the pendulums $\frac{m_A}{m_B}$ is

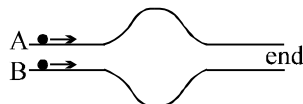


SECTION-I

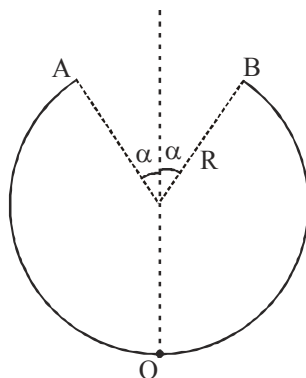
Single Correct Answer Type

5 Q. [3 M (-1)]

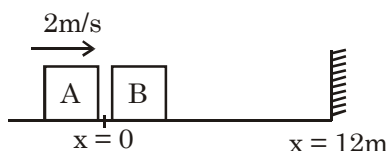
1. Two smooth tracks of equal length have “bumps”-- A up, and B down, both of the same curvature and size. The two balls start simultaneously with the same initial speed = 3 m/s. If the speed of the ball at the bottom of the curve on track B is 4 m/s, then the speed of the ball at the top of the curve on track A is



- (A) = 2 m/s
(B) > 2 m/s
(C) < 2 m/s
(D) not enough information given
2. A wire is bent along an arc with a radius R. A bead is placed on the wire that can move along it without friction. At the initial moment the bead was at point O. The horizontal velocity should be imparted to the bead for it to get on to the wire again at point B after flying a certain distance (AB) through the air, will be :

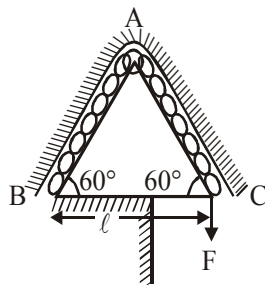


- (A) $\sqrt{gR \left(2 + 2 \cos \alpha + \frac{1}{\cos \alpha} \right)}$
(B) $\sqrt{gR \left(2 + \cos \alpha + \frac{1}{\sin \alpha} \right)}$
(C) $\sqrt{gR \left(\sin \alpha + \frac{1}{2} \cos \alpha \right)}$
(D) $\sqrt{5gR \cos \alpha}$
3. A block A of mass 3kg moving with a speed 2m/s collides elastically with a stationary block B of mass 4kg. The location of block B is at origin at the time of collision. Assume the collision with the wall to be elastic in nature and the size of the blocks is negligible, the x-coordinate of block B when it collides with block A for the second time,



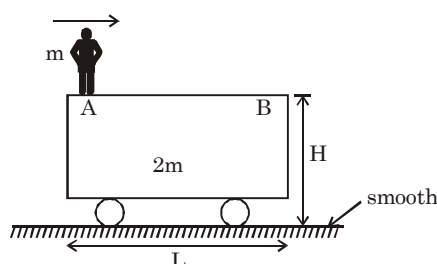
- (A) -3.6 m
(B) -4.8 m
(C) -10.2 m
(D) -15 m

4. A fixed wedge ABC is in the shape of an equilateral triangle of side ℓ . Initially, a chain of length 2ℓ and mass m rests on the wedge inside of a tube as shown. The chain is pulled slightly, as a result it starts sliding out of the tube. Work done by gravity till the time, the chain leaves the wedge will be :-



- (A) $\left(\frac{(\sqrt{3}+1)mg\ell}{2} \right)$ (B) $\left(\frac{(\sqrt{3}+2)mg\ell}{2} \right)$ (C) $\left(\frac{(\sqrt{3}+2)mg\ell}{4} \right)$ (D) $\left(\frac{(\sqrt{3}+4)mg\ell}{4} \right)$

5. A man of mass m is standing on the flat top of a cart of mass $2m$. The length and height of the cart is L and H respectively and it is at rest on a smooth horizontal ground. The man starts running from end A, speeds up and jumps out of the cart at point B with a velocity ' u ' relative to the cart in horizontal direction. The total horizontal distance covered by the man by time he lands on the ground.

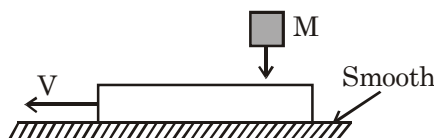


- (A) $L + \frac{2u}{3} \sqrt{\frac{2H}{g}}$ (B) $\frac{2L}{3} + u \sqrt{\frac{2H}{g}}$ (C) $\frac{L}{3} + \frac{2u}{3} \sqrt{\frac{2H}{g}}$ (D) $\frac{2L}{3} + \frac{2u}{3} \sqrt{\frac{2H}{g}}$

Multiple Correct Answer Type

6 Q. [4 M (-1)]

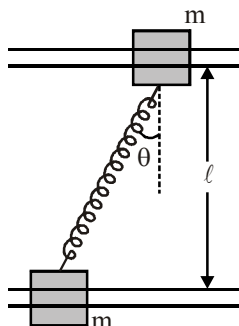
6. A plank is moving along a smooth surface with a constant speed V . A block of mass M is gently placed on it. Initially the block slips and then acquires the constant speed V same as the plank. Through the period, a horizontal force is applied on the plank to keep its speed constant.



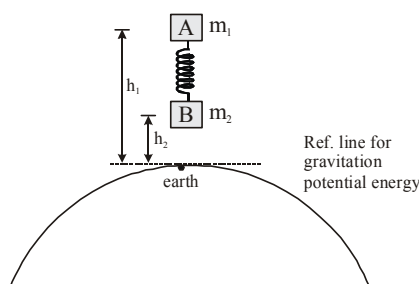
- (A) The work done by frictional force on block will be positive.
(B) Net work done by all the force on plank will be zero.
(C) The work performed by the external force is Mv^2 .

- (D) Heat developed due to friction between the block and the plank is $\frac{1}{2}Mv^2$.

7. The figure below is the TOP-VIEW of two equal masses 'm' that can slide over frictionless rails on a horizontal floor. The rails are separated by distance ℓ as shown. The masses are connected with a massless spring of natural length ℓ and spring constant K. At the initial moment of time the masses are at rest.

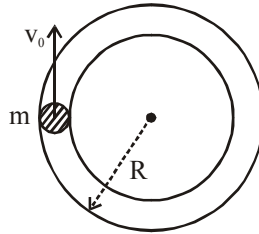


- (A) The maximum magnitude of relative velocity between the two masses is given by $\ell \left(\frac{1 - \cos \theta}{\cos \theta} \right) \sqrt{\frac{2K}{m}}$
- (B) The maximum magnitude of relative velocity between the two masses is given by $\ell \left(\frac{1 - \cos \theta}{\sin \theta} \right) \sqrt{\frac{K}{2m}}$
- (C) As the spring comes to its natural length it is cut suddenly from the center. The speed of lower block as shown in figure just after the spring is cut will be $\frac{\ell}{4} \sqrt{\frac{K}{2m}}$ (Take $\theta = 37^\circ$):
- (D) As the spring comes to its natural length it is cut suddenly from the center. The speed of lower block as shown in figure just after the spring is cut will be $\frac{\ell}{3} \sqrt{\frac{2K}{m}}$ (Take $\theta = 37^\circ$):
8. Figure shows two blocks of masses m_1 and m_2 connected by light spring which at a certain instant, is elongated by 'x'. Heights h_1 and h_2 are very small compared with radius of earth. Take the potential energy of normal spring as zero. Choose the **CORRECT** statement.



- (A) If system = block A, potential energy of system is $m_1 g h_1$
- (B) If system = block A + block B + spring, potential energy of the system is $\frac{1}{2} k x^2$
- (C) If system = block A + block B + spring, potential energy of the system is $\frac{1}{2} k x^2 + m_1 g h_1 + m_2 g h_2$
- (D) If system = block A + block B + spring + earth, potential energy of the system is $\frac{1}{2} k x^2 + m_1 g h_1 + m_2 g h_2$

9. A small ball of mass m is placed in a circular tube of negligible width, mass M and radius R which is kept on a horizontal plane in gravity free space. Friction is absent between tube and ball. Ball is given a velocity v_0 as shown. Then,



- (A) Path of ball from centre of mass of system (ball + tube) will be circular.
 (B) Path of ball from centre of mass of system (ball + tube) will be elliptical
 (C) Radius of curvature of ball at the time of projection of ball is $\frac{MR}{m+M}$
 (D) Normal force between tube and ball if $M = 2m$, at the time of projection of ball is $\frac{2mV_0^2}{3R}$
10. Two bodies of mass m_A & m_B are moving with velocity $\vec{V}_A$ & $\vec{V}_B$ respectively. We know that

$$V_{cm} = \frac{m_A \vec{V}_A + m_B \vec{V}_B}{m_A + m_B}. \text{ If the two bodies collide :-}$$

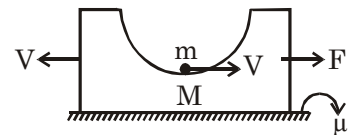
- (A) Maximum possible kinetic energy after collision will be $\frac{1}{2} m_A V_A^2 + \frac{1}{2} m_B V_B^2$.
 (B) Minimum possible kinetic energy after collision will be $\frac{1}{2} (m_A + m_B) V_{cm}^2$.
 (C) Minimum possible kinetic energy after collision may be zero.
 (D) Maximum loss of kinetic energy will be $\frac{1}{2} \frac{m_A m_B}{m_A + m_B} (\vec{V}_A - \vec{V}_B)^2$.
11. A smooth block of mass m slides on a wedge of mass M with a speed V at the lowest position. The wedge is moving towards left with speed V . Coefficient of friction between the wedge & ground is μ . Radius of wedge is R . A constant force F starts acting horizontally on the wedge. At this instant :-

(A) Velocity of centre of mass is $\frac{|m-M|V}{(m+M)}$

(B) Horizontal acceleration of centre of mass is $\frac{F + \mu(M+m)g}{M+m}$

(C) Horizontal acceleration of centre of mass is $\frac{F - \mu \left(Mg + mg + \frac{mV^2}{R} \right)}{(M+m)}$

(D) Horizontal acceleration of centre of mass is $\frac{F + \mu \left(Mg + mg + \frac{4mV^2}{R} \right)}{(M+m)}$

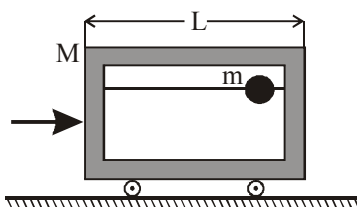


Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for question nos. 12 to 14

A horizontal frictionless rod is threaded through a bead of mass m . The length of the cart is L and the radius of the bead, r , is very small in comparison with L ($r \ll L$). Initially at ($t = 0$) the bead is at the right edge of the cart. The cart is struck and as a result, it moves with velocity v_0 . When the bead collides with the cart's walls, the collisions are always completely elastic. (Given : $m > M$).



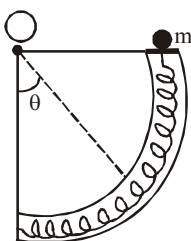
12. What is the velocity of the cart after the first collision, in the center of mass frame?
 (A) $\frac{-mv_0}{m+M}$ (B) $\frac{Mv_0}{m+M}$ (C) $\frac{M-m}{M+m}v_0$ (D) $\frac{2M}{m+M}v_0$
13. The first collision takes place at time t_1 and the second collision takes place at $t = t_2$. Find $t_2 - t_1$.
 (A) $\frac{2L}{v_0}$ (B) $\frac{L}{v_0}$ (C) $\frac{L}{2v_0}$ (D) $\frac{L}{3v_0}$
14. What is the distance the cart travels from $t = 0$ till $t = t_2$ if $m > M$?
 (A) $\frac{2ML}{m+M}$ (B) $\frac{mL}{2(m+M)}$ (C) $\frac{2m}{m+M}L$ (D) $\frac{ML}{2(m+M)}$

SECTION-III

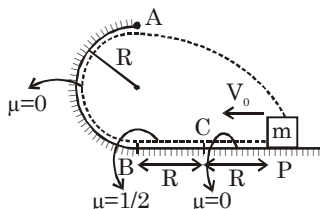
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

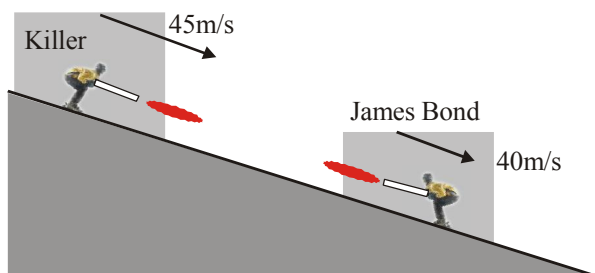
1. A ball of mass m is released in a circular tube of radius R . A spring of spring constant K which is in its natural length is placed in the tube as shown in figure. If angle made by radius vector of the ball with vertical when ball comes at an instantaneous rest is 60° , then the value of K is $\frac{(6mg)\lambda}{\pi^2 R}$. Then find the value of λ :



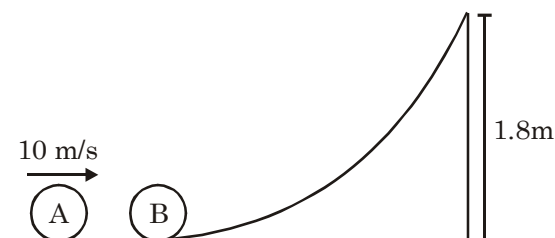
2. A block of mass m is projected with a speed V_0 such that it strikes the point of projection P after describing path as shown by the dotted line. Friction exists from C to B but P to C & curved part is smooth. Fill the value of $\frac{V_0}{6}$. (Take : $R = 15$ m).



3. James skied down a snowy slope (frictionless) in an attempt to escape from the killer. Their speeds are 40 m/s and 45 m/s respectively. James noted that he and the killer carried the same kind of rifle and estimated that their masses M (including body mass, skis, weapon, backpack, etc) were about the same. James realized that each time he fired his weapon back at the killer, his momentum would change; whereas when the killer fired, the killer's momentum would also change. Every time James fired a bullet, the killer would fire back accordingly. Assume that all bullets missed their targets (otherwise this exercise would terminate). Given $M = 100$ kg, mass and the muzzle velocity of a bullet are $m = 0.02$ kg and $v = 500$ m/s (with respect to ground) respectively. If the number of bullets James had to fire in order to assure that the killer couldn't catch up with him is $5x$ then find the value of x .



4. A ball 'A' is moving with velocity 10 m/s collides with an identical ball initially at rest. If ball 'B' just manage to reach the top of inclined plane as shown. Then the value of ' $10e$ ' is (e is coefficient of restitution and all surfaces are smooth).

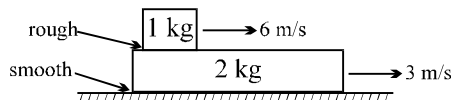


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. In the figure shown, upper block is given a velocity 6 m/s and very long plank, velocity 3 m/s. The following quantities are to be matched when both attain same velocity.



Column I

- (A) Work done by friction on 1 kg block in Joule
- (B) Work done by friction on 2 kg plank in Joule
- (C) Change in momentum in N-s of 2kg plank
- (D) Change in K.E. of system consisting of block and plank in joule

Column II

- (P) Positive
- (Q) Negative
- (R) 3
- (S) 7
- (T) 2

SECTION-I

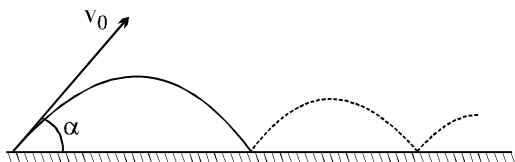
Single Correct Answer Type

6 Q. [3 M (-1)]

1. A particle of mass m and momentum $\vec{P}$ moves on a smooth horizontal table and collides directly and perfectly elastically with a similar particle (of mass m) having momentum $-\vec{2P}$. The loss (-) or gain (+) in the kinetic energy of the first particle in the collision is

(A) $+\frac{p^2}{2m}$ (B) $-\frac{p^2}{4m}$ (C) $+\frac{3p^2}{2m}$ (D) zero

2. A particle of mass ' m ' is projected with velocity v_0 at an angle ' α ' with the horizontal. The coefficient of restitution for any of its impact with the smooth ground is e .

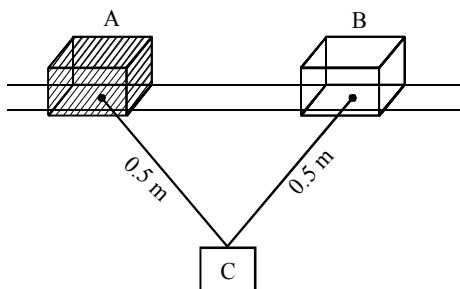


- (A) Total time taken by the particle before it stops moving vertically is $\frac{2v_0 \sin \alpha}{g(1-e)}$
- (B) Total horizontal distance moved in the time before it stops moving vertically is $\frac{v_0^2 \sin 2\alpha}{2g(1-e)}$
- (C) Average force on the particle over the time interval in which it makes first ' n ' impacts with ground equals $\frac{mg}{2}(1-e^n)$ directed upwards.
- (D) Average force on the particle over the time interval in which it makes first ' n ' impact equals $\frac{mg}{2}(1-e^n)$ directed downwards
3. A particle is projected at an angle to the horizontal from a point on a smooth horizontal floor. It is found that ratio of the first maximum height attained to maximum height reached by it after 3rd collision with the floor is $\frac{10^{12}}{2^{36}}$. Find coefficient of restitution between particle and floor.
- (A) 0.72 (B) 0.64 (C) 0.56 (D) None of the above
4. A circus acrobat of mass M leaps straight up with initial velocity V_0 from a trampoline. As he rises up, he takes a trained monkey of mass m hanging from a branch at a height h above the trampoline. What is the maximum height attained by the pair (from the branch) ?

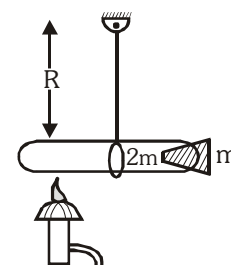
(A) $\frac{M}{M+m} \left(\frac{V_0^2}{2g} - h \right)$ (B) $\left(\frac{M}{M+m} \right)^2 \left(\frac{V_0^2}{2g} - h \right)$

(C) $\left(\frac{m}{M+m} \right) \left(\frac{V_0^2}{2g} - h \right)$ (D) $\left(\frac{m}{M+m} \right)^2 \left(\frac{V_0^2}{2g} - h \right)$

5. In the figure shown two blocks A & B each of mass m having initial separation of 80 cm are placed on smooth horizontal rails. Another block C having mass $2m$ is attached with A & B with the help of string of length 0.5 m each. Block C lies 30 cm vertically below the rails. The coefficient of restitutions is $\frac{1}{\sqrt{2}}$ for any collision. The maximum separation between block A & B on rails after they collide when system is released from the rest position :-



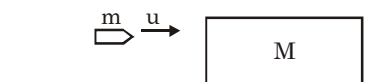
- (A) 30 cm (B) 40 cm (C) 60 cm (D) 80 cm
6. A test tube of mass $2m$ closed with a cork of mass m contains a drop of liquid of negligible mass. When the test tube is heated, the liquid evaporates and the cork flies off under the pressure of the gas. What must be the minimum velocity with which the cork must be ejected such that the test tube describes a full circle of radius R about the pivot? (Assuming test tube as a point object)
- (A) $\sqrt{5Rg}$ (B) $2\sqrt{5Rg}$
(C) $2\sqrt{3Rg}$ (D) $2\sqrt{4Rg}$



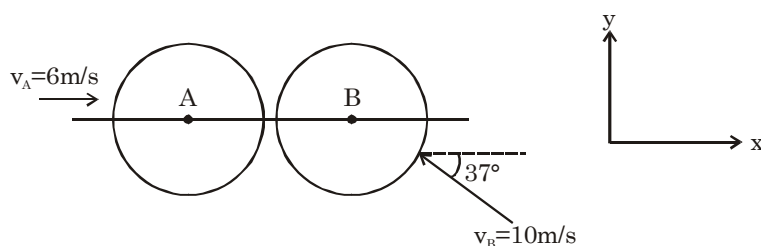
Multiple Correct Answer Type

4 Q. [4 M (-1)]

7. A set of n -identical cubical blocks lies at rest parallel to each other along a line on a smooth horizontal surface. The separation between the near surfaces of any two adjacent blocks is L . The block at one end is given a speed v towards the next one at time $t = 0$. All collisions are completely inelastic, then :
- (A) The last block starts moving at $t = n(n-1) \frac{L}{2v}$
(B) The last block starts moving at $t = (n-1) \frac{L}{v}$
(C) After all collisions the centre of mass of the system will have a speed $\frac{v}{n}$
(D) After all collisions the centre of mass of the system will have a speed v .
8. A bullet of mass m travelling horizontally with speed u strikes a wooden block of mass ' M ' placed on a smooth horizontal plane. The penetration is assumed uniform and the bullet comes to rest after penetrating a distance ' d ' into the block. Then choose the correct statement:
- (A) Ultimately the total loss of kinetic energy is $\frac{mMu^2}{2(M+m)}$
(B) The value of the resistance force (assumed constant) offered by the wood is $F = \frac{mMu^2}{2d(M+m)}$
(C) The distance covered by the bullet w.r.t. the ground before it comes to rest w.r.t. block is $\frac{dm}{M+m}$
(D) The block moves greater distance than the bullet w.r.t. ground.



9. Two particles A & B of masses m and $2m$ respectively are connected through a spring in its natural length. They are projected directly away from each other along spring with the same speed. Mark the correct statement(s) :
- (A) At the moment of maximum distance between them, the particles are moving with same speed in same direction
- (B) At the moment of maximum distance between them, the particles are moving with same speed in opposite direction
- (C) Minimum individual speed of the particles A & B are both zero
- (D) Minimum individual speed of the particles A & B are both non-zero
10. Two identical discs of mass 2kg are moving with initial velocities $\vec{v}_A$ and $\vec{v}_B$ and collide as shown. If the collision is perfectly elastic,



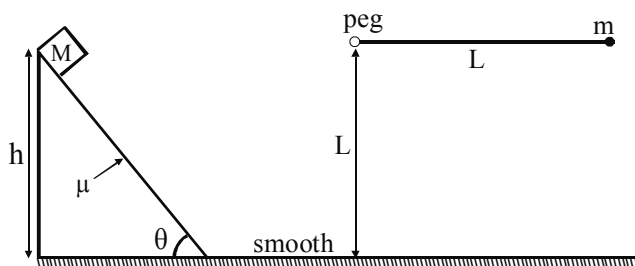
- (A) Velocity of A after collision is $-8\hat{i} \text{ m/s}$.
- (B) Velocity of B after collision is $6\hat{i} \text{ m/s}$.
- (C) Impulse of collision on A is $-28\hat{i} \text{ Ns}$.
- (D) Loss of KE of B during collision is 28 J .

Linked Comprehension Type (Single Correct Answer Type)

(2 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Question 11 to 13

A large mass M is released from rest at the top of an inclined plane of angle θ and friction coefficient μ . At the same time, a pendulum consisting of a small mass m and (massless) string of length L is released at height L (horizontally). The large mass slides down the incline and smoothly onto the table, continues across the frictionless table, and collides with the pendulum's mass just as the pendulum has reached the bottom of its swing and is approaching the large mass. The two masses stick together after the collision.



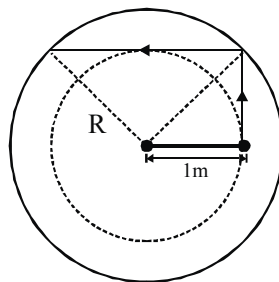
11. Assuming the mass M is released at a height h , what is its velocity when it reaches the flat part of the table?
- (A) $\sqrt{2gh(1 + \mu \tan \theta)}$ (B) $\sqrt{gh(1 + \mu \cot \theta)}$ (C) $\sqrt{2gh(1 - \mu \cot \theta)}$ (D) $\sqrt{gh(1 - \mu \tan \theta)}$
12. For what initial height h of the mass M will the combined mass come to rest immediately after the collision? [Take $\mu=0$]
- (A) $\frac{m^2}{M^2} L$ (B) $\left(\frac{m^2 + M^2}{M^2}\right) L$ (C) $\left(\frac{M^2}{m^2}\right) L$ (D) $\left(\frac{m^2}{m^2 + M^2}\right) L$

13. What must the initial height h have been in order for the pendulum of the combined mass to complete the vertical circle in the counterclockwise direction? [Take $\mu=0$ and $M=2m$]

(A) $h = \frac{L(\sqrt{2} + 3\sqrt{5})^2}{8}$ (B) $h = \frac{L(3\sqrt{2} - \sqrt{5})^2}{4}$ (C) $h = \frac{L(3\sqrt{2} + \sqrt{5})^2}{9}$ (D) none

Paragraph for question 14 to 16

A small ball of mass 1 kg is kept in circular path of radius 1m in fixed concentric smooth horizontal ring of radius R . Angular speed of the ball in the circular motion is 1 rad/s. At a certain moment the string, which kept the ball in the circular path breaks and the ball goes off tangentially to the wall of rigid, fixed ring and bounces off elastically and again hits the ring and bounces off. After completing one round about the centre, the seventh collision takes place exactly at the position of the first collision.

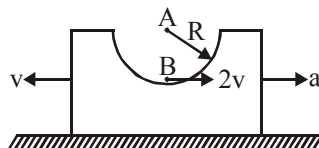


14. Total impulse imparted to the ring by the ball in first three collision will be
(A) $\sqrt{3}$ Ns (B) 2 Ns (C) 1 Ns (D) 3 Ns
15. Following quantities of the ball will remain a constant relative to the centre of the ring during one complete cycle of collisions.
(A) linear momentum (B) kinetic energy
(C) velocity, kinetic energy (D) Acceleration
16. Value of radius of the ring R will be
(A) $\frac{2}{\sqrt{3}}$ m (B) $\sqrt{3}m$ (C) $\sqrt{2}m$ (D) 2m

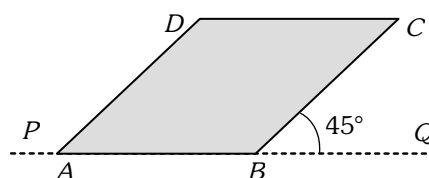
SECTION-III

Numerical Grid Type (Single digit Ranging from 0 to 9) 2 Q. [4 M(0)]

1. A bead kept at the bottom of a wedge moves towards right with a velocity $2v$. If the wedge moves towards left with a velocity v while accelerating towards right with an acceleration a as shown then the magnitude of acceleration of the bead is x units. Find x if $v = 3$ and $R = 9$ units.



2. A thin sheet cut in the shape of a parallelogram of mass 500 g is shown in the figure. Each side of the parallelogram is 6 m. Find the moment of inertia (SI units) of the sheet about the axis PQ that coincides with edge AB of the sheet.



SECTION-I

Single Correct Answer Type

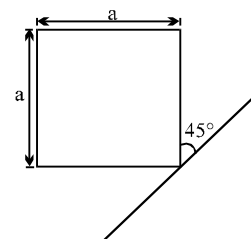
7 Q. [3 M (-1)]

1. A uniform disc of radius R lies in the x - y plane, with its centre at origin, its moment of inertia about z -axis is equal to its moment of inertia about line $y = x + c$. The value of c will be—

(A) $-\frac{R}{2}$ (B) $\pm\frac{R}{\sqrt{2}}$ (C) $\frac{+R}{4}$ (D) $-R$

2. The moment of inertia of a thin sheet of mass M of the given shape about the specified axis is :

(A) $\frac{7}{12} Ma^2$ (B) $\frac{5}{12} Ma^2$
(C) $\frac{1}{3} Ma^2$ (D) $\frac{1}{12} Ma^2$

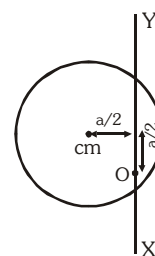


3. Two vertical walls are separated by a distance of 2 metres. Wall 'A' is smooth while wall B is rough with a coefficient of friction $\mu = 0.5$. A uniform rod is adjusted between them such that the rod touches both the walls. The length of the longest rod that can be adjusted between the walls is equal to :-

(A) 2 metres (B) $2\sqrt{2}$ metres (C) $\sqrt{5}$ metres (D) $\frac{\sqrt{17}}{2}$ metres

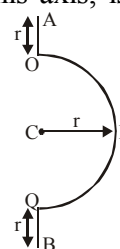
4. The diagram shows a uniform disc of mass M & radius ' a '. If the moment of inertia of the disc about the axis XY is I , its moment of inertia about an axis through O and perpendicular to the plane of the disc is

(A) $\frac{4}{3} Ma^2$ (B) $I + \frac{1}{4} Ma^2$
(C) $2I$ (D) $\frac{I}{2}$

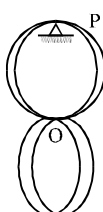


5. A wire frame AOPQB, lying in the horizontal plane, is free to rotate about a vertical axis passing through center C of the same circle and perpendicular to plane of AOPQB. The mass M of the frame is uniformly distributed over its whole length. The moment of inertia of the frame about this axis, is (OA = QB = r and CP = r the radius of semicircular part) :-

(A) $Mr^2 \left(\frac{14 + 3\pi}{3\pi + 6} \right)$ (B) $Mr^2 \left(\frac{14 + 3\pi}{\pi + 2} \right)$
(C) $\frac{Mr^2}{2} \left(\frac{\pi}{2 + \pi} \right)$ (D) $\frac{1}{2} Mr^2$



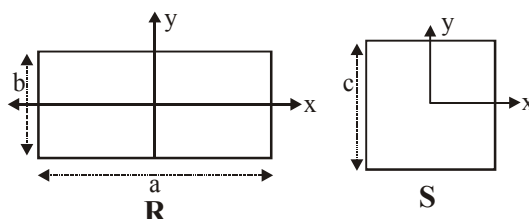
6. Two identical rings each of mass m with their planes mutually perpendicular, radius R are welded at their point of contact O . If the system is free to rotate about an axis passing through the point P perpendicular to the plane of the paper the moment of inertia of the system about this axis is equal to



(A) $6.5 mR^2$ (B) $12 mR^2$ (C) $6 mR^2$ (D) $11.5 mR^2$

CLASS TEST

7. A uniform square plate S (side c) and a uniform rectangular plate R (sides b, a) ($a > b$) have identical areas and masses figure. given (a) $I_{xR}/I_{xS} < 1$; (b) $I_{yR}/I_{yS} > 1$; (c) $I_{zR}/I_{zS} > 1$ which of the following is **CORRECT** :-

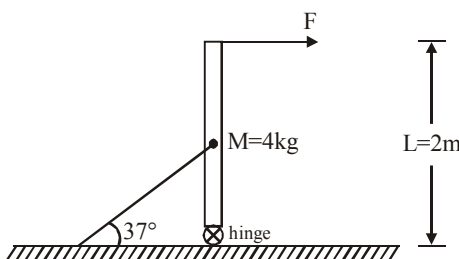


- (A) a, b (B) b, c (C) a, b, c (D) None

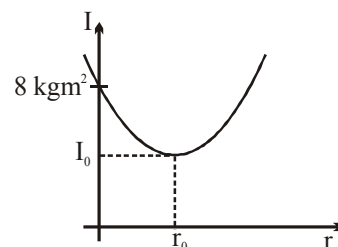
Multiple Correct Answer Type

4 Q. [4 M (-1)]

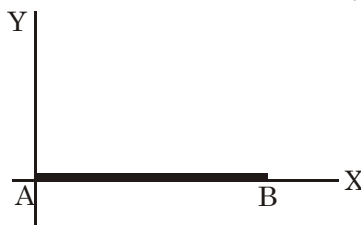
8. A beam with mass 4 kg and length 2 m can pivot freely about a hinge at its lower end as shown in figure. A force $F = 30$ N is applied at the top of the beam and the beam is kept vertical by a cable that makes an angle 37° with the ground and attached to the centre of the beam. (Take $\sin 37^\circ = 0.6$).



- (A) tension in the cable is 75 N.
(B) horizontal component of the force of the hinge on the beam is 30 N.
(C) vertical component of the force of the hinge on the beam is 85 N.
(D) tension in the cable is 100 N.
9. The graph shows the moment of inertia (I) of a uniform rod of mass 1.5 kg about an axis perpendicular to its length with the distance of axis (r) from the end of the rod.

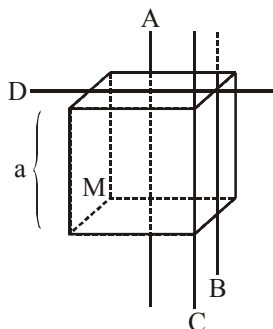


- (A) $I_0 = 2 \text{ kgm}^2$
(B) $r_0 = 2 \text{ m}$
(C) $r_0 = 4 \text{ m}$
(D) $I_0 = 4 \text{ kgm}^2$
10. A wire of mass M and length L is placed along x -axis with end A at origin. The linear mass density λ (of the rod) varies as $\lambda = \lambda_0 x$, where x is distance from end A, λ_0 is a constant.



- (A) The value of λ_0 is $\frac{2M}{L^2}$
- (B) Moment of Inertia of the rod about an axis passing through A and parallel to y -axis is $\frac{1}{2}ML^2$.
- (C) Moment of inertia of the rod about an axis passing through COM of the rod and parallel to y -axis $\frac{ML^2}{18}$.
- (D) Moment of inertia of the rod about an axis passing through B and parallel to y -axis is $\frac{1}{6}ML^2$.

11. Illustrated below is a uniform cubical block of mass M and side a . Mark the correct statement (s) :-



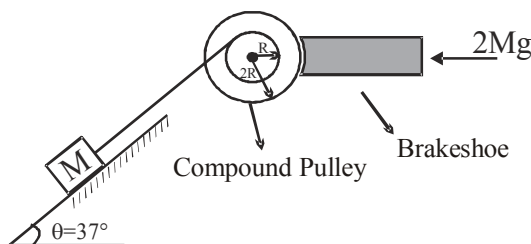
- (A) The moment of inertia about axis A, passing through the centre of mass is $I_A = \frac{1}{6}Ma^2$
- (B) The moment of inertia about axis B, which bisects one of the cube faces is $I_B = \frac{5}{12}Ma^2$
- (C) The moment of inertia about axis C, along one of the cube edges is $I_C = \frac{2}{3}Ma^2$
- (D) The moment of inertia about axis D, which bisects one of the horizontal cube faces is $I_D = \frac{7}{12}Ma^2$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for question nos. 12 to 14

A block of mass M slides on a smooth incline as shown in figure. The compound pulley has moment of inertia $I = 2MR^2$. A Brakeshoe can be pressed against pulley to stop it. The normal force applied on the Brakeshoe has magnitude $2Mg$ and coefficient of friction between pulley and brakeshoe is $\mu = 0.5$. Initially block is at rest and pulley rotates on a frictionless pivot.



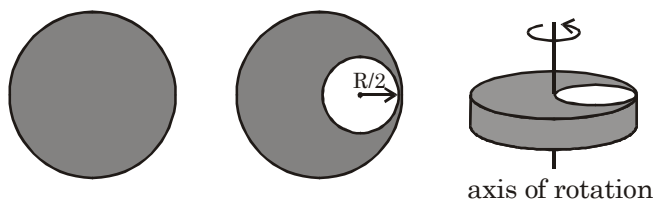
12. Initially at $t = 0$, Brakeshoe is not in contact with the pulley. What is acceleration of block along the incline?
- (A) 4 m/s^2 (B) 3 m/s^2 (C) 2 m/s^2 (D) 1 m/s^2
13. At $t = 7 \text{ sec}$ Brakeshoe is applied on the compound pulley. What is magnitude of acceleration of block thereafter?
- (A) $\frac{13}{15}g$ (B) $\frac{g}{5}$ (C) $\frac{11}{15}g$ (D) $\frac{7}{15}g$
14. Determine time taken by block and pulley system to come to stop after the brake is applied.
- (A) 7 sec (B) 10 sec (C) 3 sec (D) 5 sec

SECTION-III

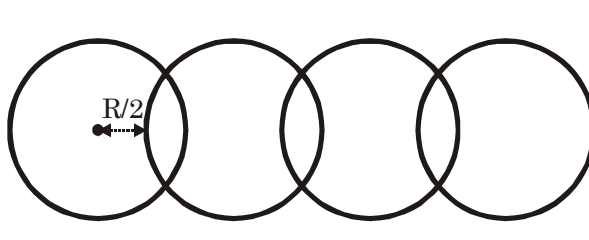
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

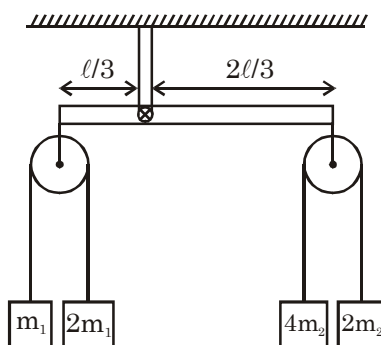
1. A uniform circular disk of radius R is with a mass M . A hole is cut in the disk as shown in the diagram. In terms of the radius R and the mass M of the original disk, What is the moment of inertia of the resulting object about the axis shown? If answer is $\frac{MR^2}{C}$ fill C in OMR sheet after rounding off to nearest integer.



2. The world famous car *AUDI* logo consist of four rings of each mass m and radius R as shown. Then moment of inertia about axis shown will be $\frac{37a}{2}mR^2$ the value of 'a' is



3. The ratio $\left(\frac{m_1}{m_2}\right)$ for which the massless rod remains horizontal is :-



SECTION-IV

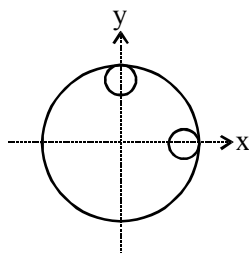
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. **Column-I** : Shows geometrical objects with some modifications.

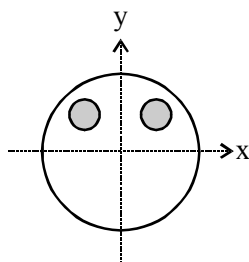
Column-II : Describes corresponding center of mass or moment of inertia about different axis.

(A) A uniform disc centered at origin with two small disc taken out with their center on x & y axis respectively as shown.



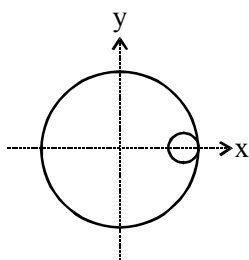
(P) Center of mass of the body have positive x coordinate ($x_{cm} > 0$)

(B) A uniform disc centered at origin with two small disc kept symmetrically with respect to y axis as shown.



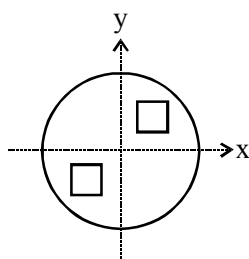
(Q) Center of mass of the body have positive y coordinate ($y_{cm} > 0$)

(C) A uniform solid sphere having center at origin. A uniform spherical cavity is made having center on x-axis.



(R) Moment of inertia about x-axis is equal to moment of inertia about y-axis. ($I_x = I_y$)

(D) A uniform disc centered at origin. Two uniform squares are taken out symmetrically with respect to line $y = -x$ as shown.



(S) Moment of inertia about y axis is equal to moment of inertia about z-axis. ($I_y = I_z$)

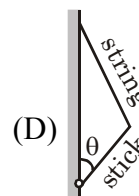
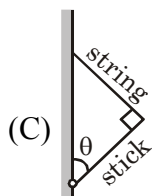
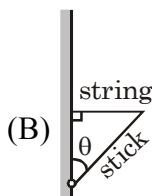
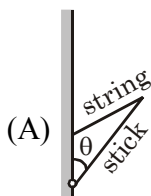
(T) x co-ordinate & y co-ordinate of center of mass is either negative or zero. ($x_{cm} \leq 0, y_{cm} \leq 0$)

SECTION-I

Single Correct Answer Type

10 Q. [3 M (–1)]

1. A stick has its bottom end attached to a wall by a pivot and is held up by a massless string attached to its other end. Which of the following scenarios has the smallest tension in the string ? (Length of stick is same in all scenarios).

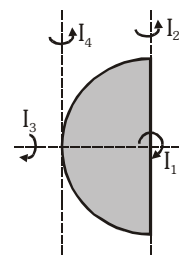


2. Two identical rings A and B are acted upon by torques τ_A and τ_B respectively. A is rotating about an axis passing through the centre of mass and perpendicular to the plane of the ring. B is rotating about a chord at a distance $\frac{1}{\sqrt{2}}$ times the radius from the centre of the ring. If the angular acceleration of the rings is the same, then

- (A) $\tau_A = \tau_B$ (B) $\tau_A > \tau_B$ (C) $\tau_A < \tau_B$
(D) Nothing can be said about τ_A and τ_B as data are insufficient

3. Moment of inertia of a uniform hollow hemisphere of mass m and radius r about various axes are indicated in the figure. Which of the following is correct?

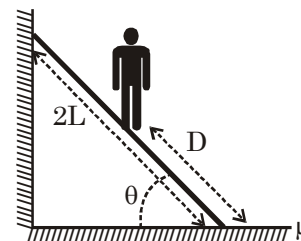
- (A) $I_1 + I_2 = I_3$ (B) $I_2 + I_3 = I_1$
(C) $I_3 + I_4 = I_1$ (D) $I_1 + I_2 = 2I_4$



4. A uniform circular disc has a moment of inertia I and radius of gyration K about an axis perpendicular to the disc and passing through its centre. A smaller concentric disc is taken out from this bigger disc. As a result of which moment of inertia and radius of gyration of the resulting body about the same axis
- (A) Increases and decreases respectively (B) Increases and increases respectively
(C) Decreases and increases respectively (D) Decreases and decreases respectively

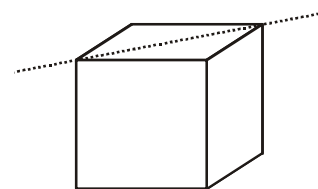
5. A uniform ladder of length $2L$ and mass m leans against a wall in a vertical plane at an angle θ to the horizontal. The floor is rough, having a coefficient of static friction μ . A person of mass M stands on the ladder at a distance D from its base (see figure). If the wall is frictionless, the maximum distance ($D_{\max}$) up the ladder that the person can reach before the ladder slips is :

- (A) $\mu L \tan \theta$ (B) $2\mu L \frac{m}{M} \tan \theta$
(C) $\left\{ 2\mu \left(1 + \frac{m}{M} \right) \tan \theta - \frac{m}{M} \right\} L$ (D) $2\mu L \left(1 + \frac{m}{M} \right) \tan \theta$

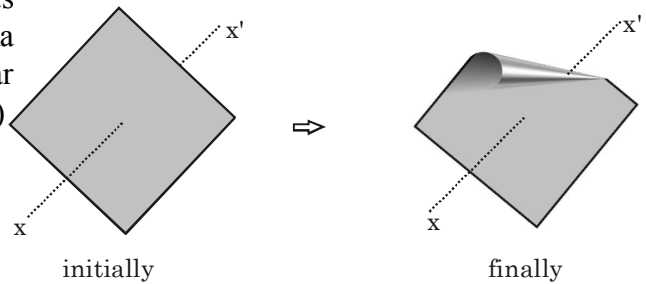


6. The moment of inertia of a cube of mass m and side a about the diagonal of one of its face is :-

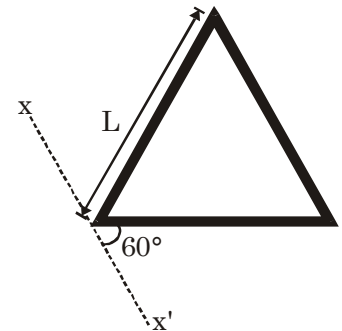
- (A) $\frac{ma^2}{12}$ (B) $\frac{ma^2}{6}$
(C) $\frac{5ma^2}{12}$ (D) $\frac{2ma^2}{3}$



7. A uniform square plate is folded from one of its corner as shown in figure. The moment of inertia about the xx' axis is then (xx' axis is perpendicular to plane of square plate and passes through centre)
- (A) Increased
(B) The same
(C) Decreased
(D) Changed in unpredicted manner

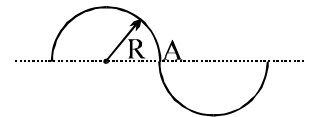


8. Three identical rods of mass m & length L are arranged such that they form equilateral triangle. Moment of inertia about axis xx' will be (xx' axis has same plane that of arrangement shown) :-



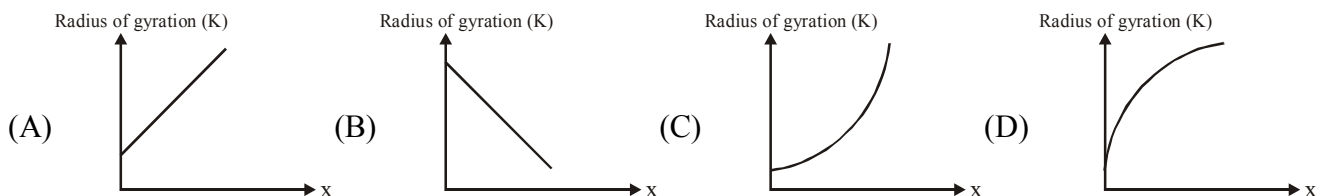
- (A) $\frac{7}{4} mL^2$ (B) $\frac{mL^2}{2}$
(C) $\frac{3}{4} mL^2$ (D) $\frac{5}{4} mL^2$

9. The S-shaped uniform wire shown in figure has a mass M , and the radius of curvature of each half is R . The moment of inertia about an axis through A and perpendicular to the plane of the paper is:



- (A) $\frac{3}{4} MR^2$ (B) MR^2 (C) $\frac{3}{2} MR^2$ (D) $2MR^2$

10. A uniform disc has radius R . If a concentric smaller disc of radius x is removed from the bigger disc then radius of gyration of the remaining disc about an axis passing through centre and perpendicular to plane of disc varies with x , according to the graph :-



Multiple Correct Answer Type

2 Q. [4 M (-1)]

11. A thin uniform rod of mass m and length L is hinged at one end and is free to rotate in the vertical plane. A small ball of mass m is fixed to the other end of the rod. A constant horizontal force $F = \frac{mg}{2}$ is also acting on the rod at the free end when the rod is released from horizontal position as shown in figure.

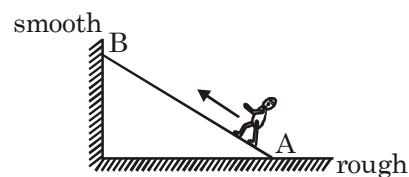
- (A) Acceleration of the ball at the instant of releasing the rod $\frac{9g}{8}$
-

- (B) Acceleration of the ball at the instant of releasing the rod $\frac{7g}{8}$

- (C) The vertical component of reaction force exerted by the hinge when the rod has just been released $\frac{5mg}{16}$

- (D) The vertical component of reaction force exerted by the hinge when the rod has just been released $\frac{9mg}{8}$

12. A ladder AB is supported by a smooth vertical wall and a rough horizontal surface as shown. A boy starts moving very slowly from A to B. The ladder always remains in rest. As the boy moves :-



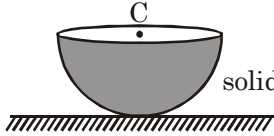
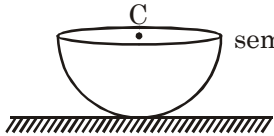
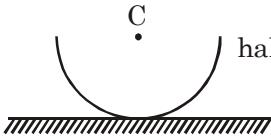
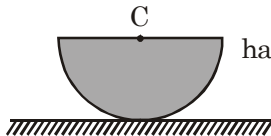
- (A) Magnitude of normal reaction by wall on ladder at point B will increase.
(B) Magnitude of normal reaction by surface on ladder at point A will remain constant only if ladder is uniform.
(C) Magnitude of normal reaction by surface on ladder at point A will remain constant irrespective of the fact that ladder is uniform or nonuniform.
(D) Magnitude of frictional force at A will be equal to magnitude of normal reaction at B.

Matrix List Type ($4 \times 4 \times 4$) (1 Table $\times$ 3Q.) [3(-1)]

(Single options correct)

(Three Columns and Four Rows)

Answer Q.13, Q.14 and Q.15 by appropriately matching the information given in the three columns of the following table. Radius of each object is R.

Column-1 Object of mass m	Column-2 Position of centre of mass from ground	Column-3 Radius of gyration
(I)  solid sphere	(i) $\frac{5R}{8}$	(P) $R\sqrt{\frac{2}{5}}$
(II)  semispherical shell	(ii) $\frac{R}{2}$	(Q) $R\sqrt{\frac{2}{3}}$
(III)  half ring	(iii) $R - \frac{4R}{3\pi}$	(R) $\frac{R}{\sqrt{2}}$
(IV)  half disc	(iv) $R - \frac{2R}{\pi}$	(S) $\frac{R}{2}$

13. Which of the following body has farthest centre of mass from C, what is its distance of centre of mass from ground & its radius of gyration about vertical axis passing through 'C' :-
(A)(III) (iv) (R) (B) (II) (ii) (Q) (C) (I) (iii) (P) (D) (IV) (i) (S)
14. Which of the following body has nearest centre of mass from C. What is its distance of centre of mass from ground and its radius of gyration about vertical line passing through C :-
(A)(II) (iii) (S) (B) (I) (i) (P) (C) (IV) (ii) (R) (D) (III) (iv) (Q)
15. A body having maximum moment of inertia about vertical line passing through 'C', what is its distance of centre of mass from ground and its radius of gyration about vertical line passing through C.
(A) (II) (ii) (Q) (B) (I) (iii) (S) (C) (III) (iv) (R) (D) (IV) (i) (P)

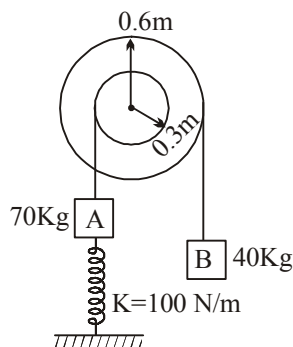
SECTION-II

Numerical Answer Type Question

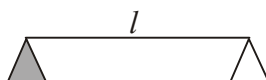
4Q.[3(0)]

(upto second decimal place)

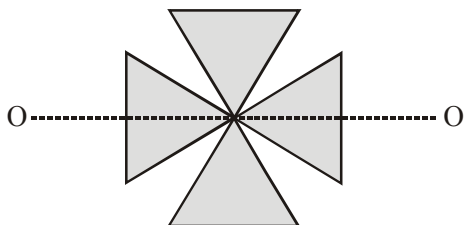
1. The figure shown is in equilibrium. Find out the extension in the spring (in cm).



2. A uniform rod of length l and mass $m = 4$ kg rests on supports at its ends. The right support is quickly removed (see figure). What is the force (in N) on the left support immediately thereafter?

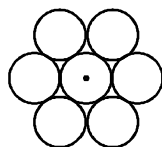


3. Moment of inertia of an equilateral triangular plate of side length a , about an axis passing through one of its vertices perpendicular to its plane is equal to I_0 . If the following symmetric structure is made from the same material, then its moment of inertia about the axis OO' is found to be equal to $N I_0$. Find N .



Each plate is an equilateral triangle of side length $2a$

4. Seven pennies are arranged in a hexagonal, planar pattern so as to touch each neighbor, as shown in the figure below. Each penny is a uniform disk of mass $m = 2$ kg and radius $r = 1$ m. What is the moment of inertia of the system (in kg-m^2) of seven pennies about an axis that passes through the center of the central penny and is normal to the plane of the pennies ?



SECTION-I

Single Correct Answer Type

9 Q. [3 M (-1)]

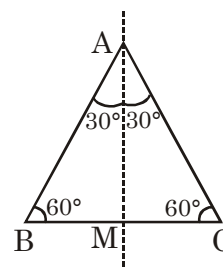
1. A uniform rod of mass 'm' and length '2l' is balanced on triangular prism. Now length $\frac{\ell}{2}$ of rod is cut from one end and placed over the shortened part such that the ends meet. The initial angular acceleration is



- (A) $\frac{3g}{5\ell}$ (B) $\frac{2g}{3\ell}$ (C) $\frac{3g}{7\ell}$ (D) $\frac{5g}{7\ell}$

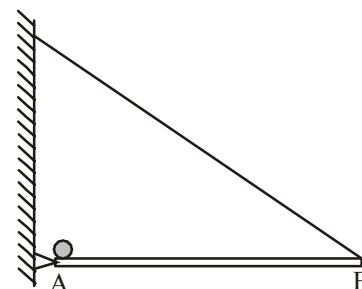
2. A uniform thin wire of length 3L and mass 3M is bent into the form of an equilateral triangle ABC as shown in figure. The moment of inertia of the system about the median AM is :-

- (A) $\frac{ML^2}{4}$ (B) $\frac{ML^2}{12}$
(C) $\frac{ML^2}{6}$ (D) $\frac{2}{3}ML^2$



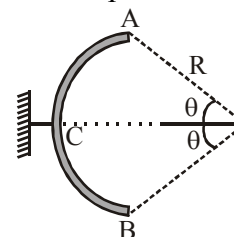
3. Figure shows a uniform rod of $M = 5$ kg and length 8 m supported horizontally by a string of length 10 m and breaking tension 52N. A ball of mass 1 kg starts rolling at $t = 0$ from end A with constant speed of 50 cm/s towards end B. Find the time when the string breaks. ($g = 10$ m/s²)

- (A) 7.92 seconds (B) 8.92 seconds
(C) 9.92 seconds (D) 10.92 seconds

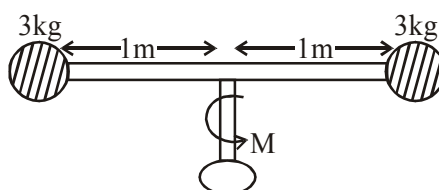


4. A uniform rod AB is bent in the shape of an arc of circle and hinged at point C. Determine the angular acceleration of the rod immediately after it is released from rest in a vertical plane.

- (A) $\frac{g}{R} \sin \theta$ (B) $\frac{2g}{R} \sin 2\theta$
(C) $\frac{g}{2R}$ (D) $\frac{g}{2R} \left(\frac{5 \sin \theta}{\theta} \right)$

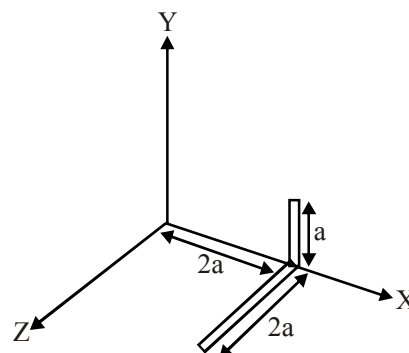


5. The two small spheres each have a mass of 3 kg are attached to the rod of negligible mass. A torque $M = (8t)$ Nm, where t is in seconds, is applied to the rod. The time at which each sphere attains a speed of 3 m/s starting from rest is



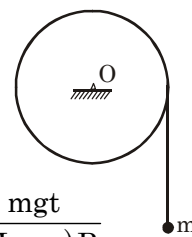
- (A) $\sqrt{2}$ s (B) $1.5\sqrt{2}$ s (C) $2\sqrt{2}$ s (D) $3\sqrt{2}$ s

6. A section of a particular device looks as shown in the figure. It has two thin rods made up of a material of uniform linear mass density λ . The smaller rod is parallel to Y axis and in XOY plane while the larger rod is parallel to Z-axis in XOZ plane. Their dimensions are shown in the figure. The moment of inertia of the structure about Z axis shown is :



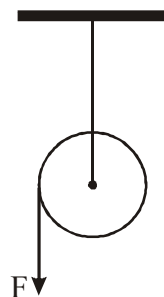
- (A) $\frac{25}{3}\lambda a^3$ (B) $\left(\frac{47}{3}\right)\lambda a^3$
(C) $\frac{37}{3}\lambda a^3$ (D) $\frac{25}{2}\lambda a^3$

7. A uniform disc of mass M and radius R is smoothly pivoted at 'O'. A light inextensible strings wrapped over the disc hangs a particle of mass m. If the system is released from rest, assuming string does not slide. What will be the angular speed of the disc as a function of time.



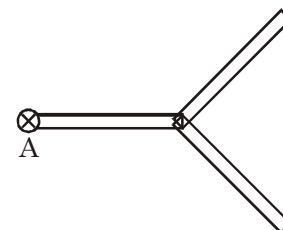
- (A) $\omega = \frac{2mgt}{(M+2m)R}$ (B) $\omega = \frac{4mgt}{(M+2m)R}$ (C) $\omega = \frac{mgt}{(M+2m)R}$ (D) $\omega = \frac{mgt}{(M+m)R}$

8. Suppose the force F in the cord hanging from the disc is given by the relation $F = 3t - 2t^2$ (newtons) where t is in seconds. If the wheel starts from rest, what is the linear speed of a point on its rim 3s later ? Mass of disc = 1kg, Radius of disc = 1m.



- (A) 10 m/s (B) 2.25 m/s
(C) 5 m/s (D) 9 m/s

9. Three identical rods each of mass 'm' and length ' ℓ ' are joined together at the ends at an angle 120° with each other. The moment of inertia about an axis passing through point A and perpendicular to the plane of the rods is :-



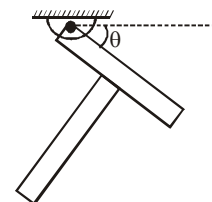
- (A) $m\ell^2$ (B) $2m\ell^2$
(C) $4m\ell^2$ (D) $6m\ell^2$

Multiple Correct Answer Type

5 Q. [4 M (-1)]

10. A T shaped rod is hinged at it's corner as shown and free to rotate in a vertical plane. The rods have mass m each and length ℓ each.

- (A) The angle θ made by upper end in the equilibrium position is $\tan^{-1}(2)$
(B) The angle θ made by upper end in the equilibrium position is 53°
(C) The moment of inertia of the system about the axis passing through

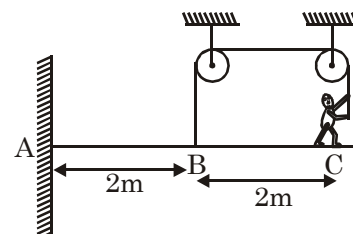


hinge and perpendicular to plane of rod is $\frac{11}{12} m\ell^2$

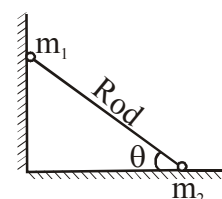
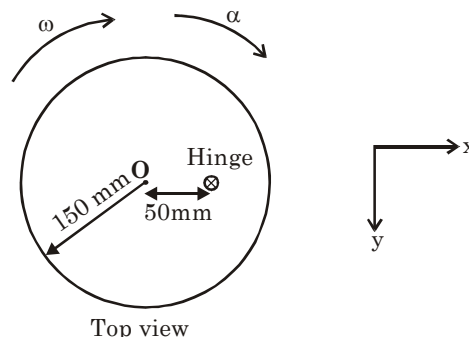
- (D) The moment of inertia of the system about the axis passing through hinge and perpendicular to

plane of rod is $\frac{11}{6} m\ell^2$

11. A man weighing W stands on a horizontal beam (hinged on a wall) of negligible weight at point C and holds a massless rope passing over two smooth pulleys. The rope is attached to point B on the beam as shown. If the system is in equilibrium, then :-



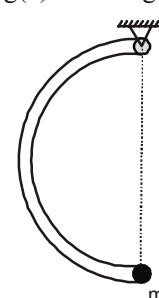
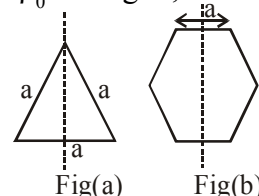
- (A) If $W = 600 \text{ N}$, tension in string is 400 N
 (B) If $W = 600 \text{ N}$ hinge force at A is 200 N
 (C) Hinge force at A is always in vertically downward direction, irrespective of value of W
 (D) Hinge force at A is always in vertically upward direction, irrespective of value of W
12. An eccentric cylinder used in vibrator has a mass of 18 kg and rotates about an axis 50 mm from its geometric centre and perpendicular to the top view as shown in diagram. If the magnitude of ω and α are 10 rad/s and 2 rad/s^2 respectively at the instant shown, then choose the **CORRECT** option(s) :-
 (The whole system is kept in gravity free region)
 (A) Magnitude of x-component of hinge force is 90 N
 (B) Magnitude of y-component of hinge force is 1.8 N
 (C) Net torque required about the hinge is 0.495 N-m
 (D) Net hinge force has positive x & y component.
13. Two beads of mass m_1 and m_2 are connected by a light rigid rod. System is at rest placed between a smooth wall and a rough floor having coefficient of friction μ . Which of the following are correct. (T is the force exerted by the rod, N_1 is the normal force exerted by the wall, N_2 is the normal force exerted by the ground and f is the friction force exerted by the ground.)
- (A) Minimum value of θ so that system does not slip is $\cot^{-1} \left[\mu \left(1 + \frac{m_2}{m_1} \right) \right]$
 (B) $N_1 = T \cos \theta$
 (C) $N_2 = (m_1 + m_2)g$
 (D) $f = T \cos \theta$
14. A rigid body is observed in equilibrium in a particular non rotating, non inertial frame. What can you conclude, if the body is observed from an inertial frame.
 (A) The body is in rotational equilibrium but not in translational equilibrium
 (B) Net torque of all the forces on body about its centre of mass is a null vector.
 (C) Net torque of all the forces on the body about any point that is collinear with line of acceleration of mass centre is a null vector.
 (D) Net torque of all the forces on the body about all point on a line that is parallel to the line of acceleration of mass centre is a null vector.



SECTION-III

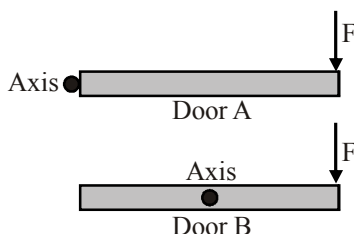
Numerical Grid Type (Single digit Ranging from 0 to 9) 4 Q. [4 M(0)]

1. A thin rod of total length l and of mass m has variable linear density given by $\rho = \rho_0 [1 + (x/l)]$, where x is the distance from one end. Calculate the moment of inertia (in kg m^2) about an axis perpendicular to the length of the rod and passing midway, at $x = l/2$. (Take $\rho_0 = 1 \text{ kg/m}$, $l = 2\text{m}$)
2. The moment of inertia of the plate in fig(a) about the axis shown is I . The moment of inertia of the hexagonal plate of same material shown in figure(b) about the axis given in figure(b) is $10\alpha I$, find value of α .
3. Half ring of mass m and radius R is released from the position shown in diagram. A small point mass of same mass is also fixed at the end as shown in figure. If the initial acceleration of point mass m is $\frac{2g}{\beta\pi}$ then



find the value of β .

4. The drawing shows the top view of two doors. The doors are uniform and identical. Door A rotates about an axis through its left edge, while door B rotates about an axis through the center. The same force F is applied perpendicular to each door at its right edge and the force remains perpendicular as the door turns. Starting from rest, door A rotates through a certain angle in 3 s. If door B takes $\frac{3n}{\sqrt{2}}$ sec to rotate through the same angle. Fill the value of 'n' in OMR sheet.



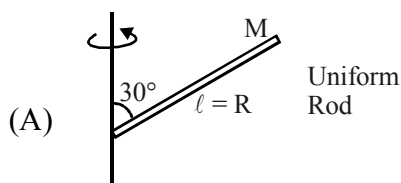
SECTION-IV

Matrix Match Type (4×4)

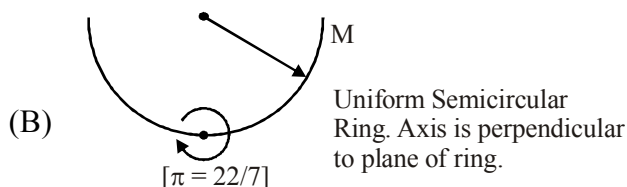
1 Q. [8 M (for each entry +2(0))]

1. Column-I

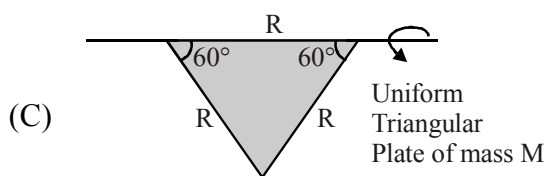
Column-II



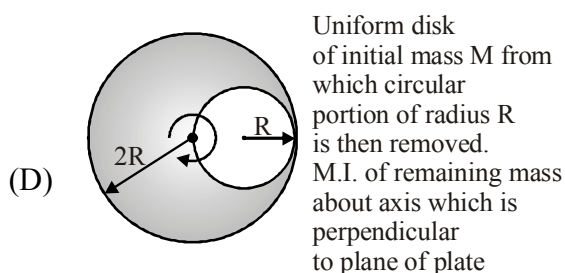
(P) $\frac{8MR^2}{11}$



(Q) $\frac{MR^2}{12}$



(R) $\frac{13MR^2}{8}$



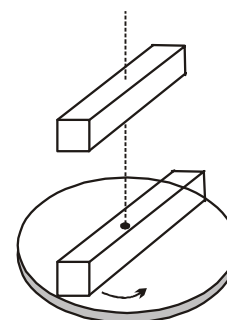
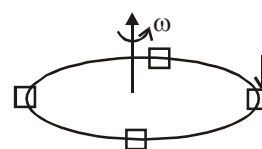
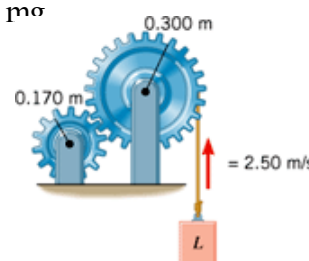
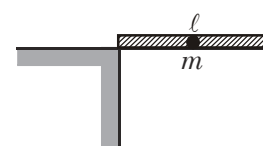
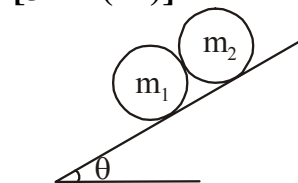
(S) $\frac{MR^2}{8}$

SECTION-I

Single Correct Answer Type

8 Q. [3 M (-1)]

- Two spheres having masses m_1 and m_2 are kept on a on a inclined plane as shown. Both of them are in equilibrium. What is a necessary condition for this? (μ_1 is coefficient of friction between m_1 and inclined plane and μ_2 is coefficient of friction between m_2 and inclined plane).
(A) $m_1 > m_2$ (B) $m_1 < m_2$ (C) $\mu_1 > \mu_2$ (D) $\mu_1 < \mu_2$
- The left end of a massless stick with length ℓ is placed on the corner of a table, as shown in Fig. A point mass m is attached to the center of the stick, which is initially held horizontal. It is then released. Immediately afterward, what normal force does the table exert on the stick?
(A) 0 (B) $mg/6$ (C) $mg/2$ (D) $m\sigma$
- The two-gear combination shown in the drawing is being used to lift the load L with a constant upward speed of 2.50 m/s. The rope that is attached to the load is being wound onto a cylinder behind the big gear. The depth of the teeth of gears is negligible compared to the radii. Determine the angular velocity (magnitude) of the smaller gear.
(A) $\frac{17}{250}$ rad/s (B) $25 \times \frac{30}{17}$ rad/s (C) $\frac{250}{17}$ rad/s (D) $25 \times \frac{17}{30}$ rad/s
- A particle of mass 1 kg moves along the curve $y = x^2$. Magnitude of angular momentum of particle about origin when particle has x co-ordinate as $\frac{1}{2}$ m and x component of velocity as 4m/s is :-
(A) 4 J – S (B) 3 J – S (C) 2 J – S (D) 1 J – S
- A horizontal wheel with buckets on its circumference revolves about a frictionless axis. Water falls into the buckets at a uniform rate of mass m per unit time. Treating the buckets as small compared with the wheel, find the angle turned through by the wheel in time t . The moment of inertia of the wheel and buckets about the vertical axis is I and r is the radius of the circumference on which the buckets are placed. Initial angular velocity of the wheel system is Ω .
(A) $\frac{I\Omega}{mr^2}$ (B) $\frac{mr^2}{I\Omega} \ln\left(1 + \frac{mrt^2}{I}\right)$
(C) $\frac{I\Omega}{mr^2} \log_e\left(1 + \frac{mr^2t}{I}\right)$ (D) $\frac{I\Omega}{mr^2} \ln\left(\frac{mr^2t}{I}\right)$
- A uniform disk turns at 2.4 rev/s around a frictionless axis. A nonrotating rod, of the same mass as the disk and length equal to the disk's diameter is dropped onto the freely spinning disk figure. They then both turn around the axis with their centres superposed. What as the angular frequency in rev/s of the combination.
(A) 1.2 rev/s (B) 2.0 rev/s
(C) 1.44 rev/s (D) 0.96 rev/s



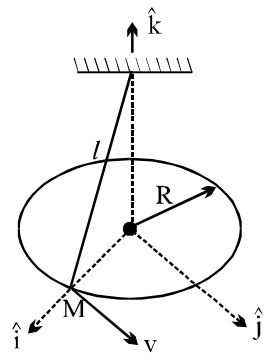
7. A conical pendulum consists of a mass M suspended from a string of length l . The mass executes a circle of radius R in a horizontal plane with speed v . At time t , the mass is at position $R\hat{i}$ and has velocity $v\hat{j}$. At time t , the angular momentum vector of the mass M about the point from which the string is suspended is :

(A) $MvR\hat{k}$

(B) $Mv/l\hat{k}$

(C) $Mvl \left[\frac{\sqrt{l^2 - R^2}}{l} \hat{i} + \frac{R}{l} \hat{k} \right]$

(D) $-Mvl \left[\frac{\sqrt{l^2 - R^2}}{l} \hat{i} + \frac{R}{l} \hat{k} \right]$



8. A solid body of mass m in the form of a double cylinder is kept on a rough surface and friction is sufficient for pure rolling for this body. If forces are acting on it as shown in figure, and moment of inertia of this body

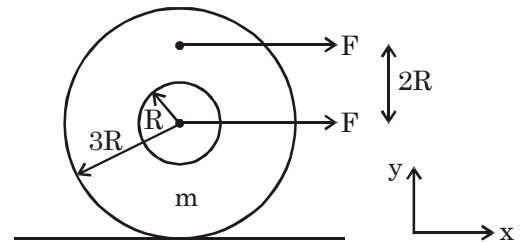
about centre of mass is $I = \frac{5}{2}mR^2$; the direction of friction on the body is :-

(A) in forward (+x) direction

(B) in backward (-x) direction

(C) no friction

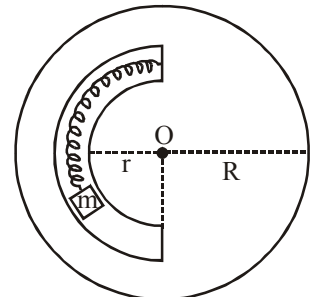
(D) cannot be predicted



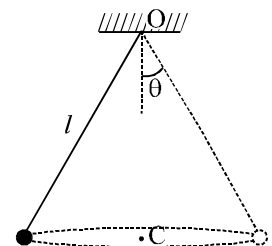
Multiple Correct Answer Type

3 Q. [4 M (-1)]

9. A disk of radius R mass m has a thin groove of radius r in it. It is free to rotate along a vertical axis, passing through its centre O . The spring shown is initially compressed and centre of system O is at rest. Consider the duration, when spring is released and block (also of mass m) is moving inside smooth groove :-
- (A) Block and disk both have equal K.E.
(B) Block and disk both have angular momentum of same magnitude, about O .
(C) Block and disk have equal magnitude of force acting on them.
(D) Block and disk have equal magnitude of torques acting on them, about O .



10. A particle of mass m is suspended from point O and undergoes circular motion in horizontal plane as conical pendulum as shown in figure.
- (A) Angular momentum of particle about point of suspension does not remain constant.
(B) Angular momentum of particle about centre of circle remains constant.



(C) Average force during half rotation is $\frac{2mg \tan \theta}{\pi}$.

(D) Average torque about axis OC during half rotation is zero

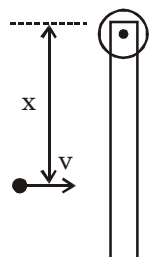
11. A rod of mass M and length L is placed on a smooth horizontal table and hinged about vertical axis and free to rotate in horizontal plane. A particle hit the rod perpendicular to the length at a distance x ($x > 0$) from the hinge, then

(A) For all values of x , angular momentum of 'rod + particle' is conserved about atleast one point

(B) For exactly one value of x , angular momentum of 'rod + particle' system is conserved about infinitely many point.

(C) For all value of x , linear momentum of 'rod + particle' system is conserved.

(D) For exactly one value of x , linear momentum of 'rod + particle' system is conserved.



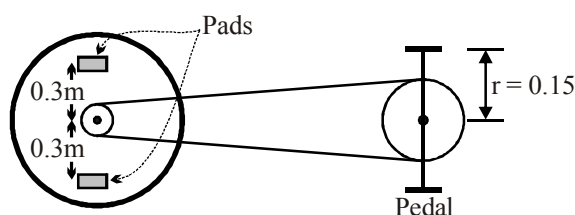
Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.)

[3 M (-1)]

Paragraph for question nos. 12 to 14

An exercise bike has the basic construction of a bicycle with a single heavy disc wheel. In addition to friction in the bearings and the transmission system, resistance to pedaling is provided by two narrow friction pads that push with equal force on each side of the wheel. The coefficient of kinetic friction between the pads and the wheel is 0.4, and the pads provide a total retarding force of 20 N tangential to the wheel. The pads are located at a position 0.3 m from the center of the wheel. The distance, recorded on the odometer, is considered to be the distance that a point on the wheel 0.3 m from the center moves. The pedals move in a circle of 0.15 m in radius and complete one revolution, while a transmission system allows the wheel to rotate twice.



12. If the wheel has a kinetic energy of 30 J when the cyclist stops pedaling, how many rotations will it make before coming to rest ?
(A) Less than 1 (B) Between 1 and 2 (C) Between 2 and 3 (D) Between 3 and 4
13. During a second workout, the cyclist reduces the force on the friction pads by 50%, then pedals for two times the previous distance in $\frac{1}{2}$ the previous time. How does the amount of energy dissipated by the pads in the second workout compare with energy dissipated in the first workout ?
(A) One-eighth as much (B) One-half as much
(C) Equal (D) Two times as much
14. A cyclist's energy consumption rate during the workout is 500W. If the cyclist wishes to expend at least 300 kcal of energy, how long must the cyclist exercise at this rate ? (Take 1 calorie = 4.2 J)
(A) 0.6 min (B) 3.6 min (C) 36 min (D) 42 min

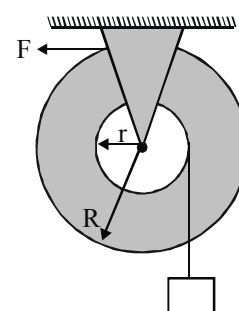
SECTION-II

Numerical Answer Type Question (upto second decimal place)

1. A bicyclist traveling with speed $v = 4$ m/s on flat road is making a turn with a radius $r = 64$ m as seen in figure. Calculate angle θ made by him with vertical (in radians) for the values given. If your answer is $A \times 10^{-3}$ rad, fill A in the OMR sheet.



2. A double pulley shaped device mounted on a horizontal frictionless axis is used to lift a 30 kg box as shown in figure. The outer radius R of the device is 0.50 m, and the radius r of the hub is 0.20 m. When a constant horizontal force of magnitude 152 N is applied in the left direction to a rope wrapped around the outer circumference of the device, the box, which is suspended from a rope wrapped around the hub, has an upward acceleration of magnitude 0.80 m/s². The rotational inertia (in kg-m²) of the device about its axis of rotation is I . Fill $10I$ in the OMR sheet.

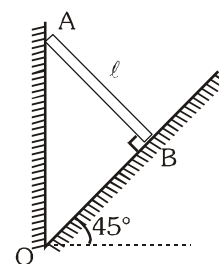


SECTION-III

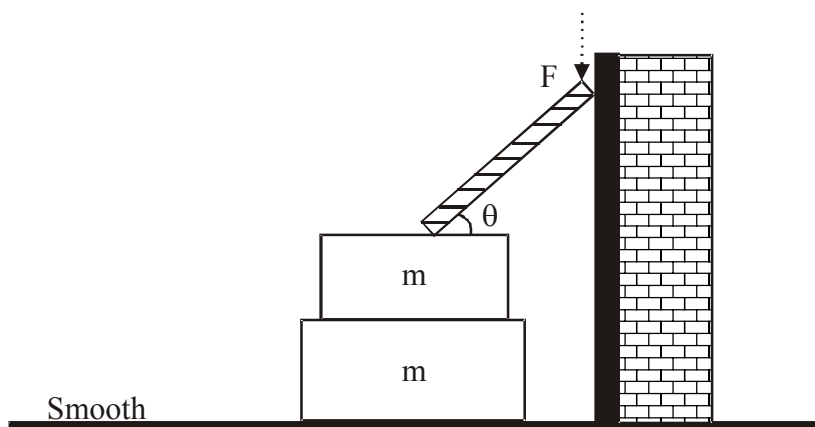
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

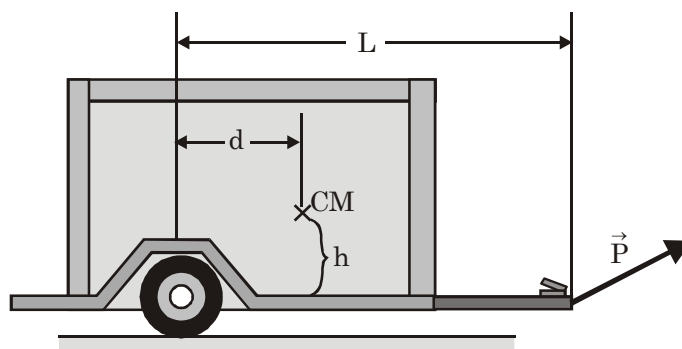
1. At the bottom edge of a smooth vertical wall, an inclined plane is kept at an angle of 45° from the horizontal. A uniform ladder of length ℓ and mass M rests on the inclined plane against the wall such that it is perpendicular to the inclined, as shown in the figure. The minimum coefficient of friction necessary so that the ladder does not slip on the incline is found to be $1/\alpha$. Find the value of α .



2. In the diagram shown the rod is rigid and massless. A constant force $F = mg$ has been applied vertically downwards on the rod. The wall and the horizontal surface are perfectly smooth. Assume no relative motion to be present between the lower contact point of the rod and the upper surface of the block. If the minimum value of co-efficient of friction between the blocks such that initially no relative motion occurs between the blocks, is equal to $\frac{1}{\alpha}$ then find the value of α . The initial value of the angle θ is equal to 45° .



3. A trailer with loaded weight $\vec{F}_g$ is being pulled by a vehicle with a force $\vec{P}$ as shown in figure. The trailer is loaded such that its center of mass (CM) is located as shown. Ignore the force of rolling friction and let 'a' represent the x component of the acceleration of the trailer. Assume $a = 4.00 \text{ m/s}^2$ and $h = 2.50 \text{ m}$. What must be the value of "d" so that vertical component of force $\vec{P}$ (P_y) is equal to zero.



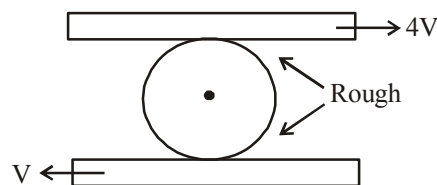
SECTION-I

Single Correct Answer Type

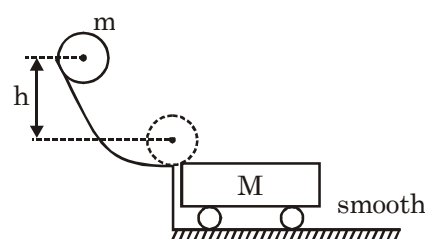
5 Q. [3 M (–1)]

1. The arrangement shows a solid cylinder of radius R placed between two planks. There is no slipping at rough surfaces. The planks are given a horizontal constant velocity as shown. The angular velocity of the cylinder is:

- (A) $\frac{5V}{4R}$ (B) $\frac{2V}{3R}$
(C) $\frac{3V}{4R}$ (D) $\frac{5V}{2R}$

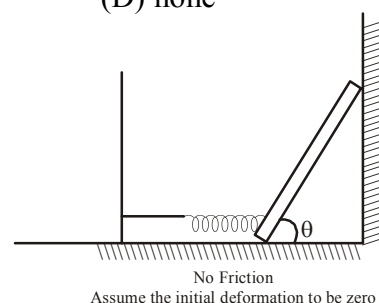


2. A uniform disc of mass $m = 12$ kg slides down along smooth, frictionless hill, which ends in a horizontal plane without break. The disc is released from rest at a height of $h = 1.25$ m (it has no initial speed and it does not rotate), and lands on the top of a cart of mass $M = 6$ kg, which can move on a frictionless surface. The coefficient of kinetic friction between the cart and the disc is $\mu = 0.4$. Find minimum length of the cart (in m) so that the disc begins to roll without slipping before losing contact with the cart. ($g = 10$ m/s²)



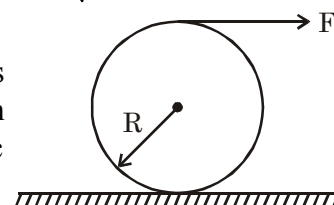
- (A) $7/8$ (B) $7/4$ (C) 5.25
3. In the diagram shown below, the rod is uniform having mass M and length L . When it is making an angle $\theta = 60^\circ$, the rod is released from the state of rest (and spring in its natural length). The rod comes to state of rest when the angle θ reduces to 30° . What should be the initial compression in the spring (when $\theta = 60^\circ$) so that the whole system can remain in the state of rest? (there is no friction between any surfaces)

(D) none



- (A) $\frac{(\sqrt{3}-1)}{2\sqrt{3}}\ell$ (B) $\frac{3(\sqrt{3}-1)}{2}\ell$ (C) $\frac{(\sqrt{3}-1)}{4\sqrt{3}}\ell$ (D) $\frac{2(\sqrt{3}-1)}{\sqrt{3}}\ell$

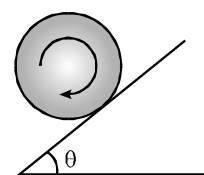
4. Light thread is slightly wound on a uniform solid cylinder of radius R . The cylinder is placed on a smooth horizontal table and the thread is pulled horizontally as shown by applying a constant force F . How much length of the thread is unwound from the cylinder by the time its kinetic energy becomes equal to k .



- (A) $\frac{3k}{2F}$ (B) $\frac{2k}{3F}$ (C) $\frac{k}{2F}$ (D) $\frac{k}{F}$

5. A hollow cylinder of mass m and radius R is spinned to a clockwise angular velocity ω_0 and then gently placed on an inclined plane for which coefficient of friction $\mu = \tan\theta$, θ is the angle of inclined plane with horizontal. The centre of mass of the cylinder will remain stationary for time :-

- (A) $\omega_0 R / g \sin \theta$ (B) $2\omega_0 R / 3g \sin \theta$
(C) $2\omega_0 R / 5g \sin \theta$ (D) $\omega_0 R / 2g \sin \theta$



Multiple Correct Answer Type

6 Q. [4 M (-1)]

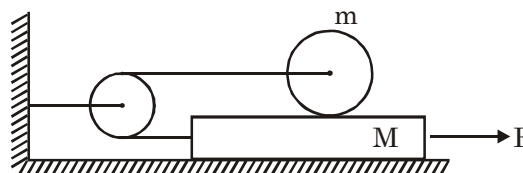
6. A solid cylinder of mass m and radius ' R ' and a plank of mass M placed on smooth surface are pulled with a force ' F ' as shown in figure. Given that sufficient friction is present between cylinder and plank surface to prevent sliding of cylinder. Select the **CORRECT** alternatives:-

(A) Acceleration of centre of mass of cylinder is $\frac{F}{M+3m}$

(B) Angular acceleration of cylinder is $\frac{2F}{R(M+3m)}$

(C) Tension in the string is $\frac{2mF}{M+3m}$

(D) Friction force on cylinder is $\frac{mF}{(M+3m)}$



7. Figure shows two identical particles 1 and 2, each of mass m , moving in opposite directions with same speed $\vec{v}$ along parallel lines. At a particular instant, $\vec{r}_1$ and $\vec{r}_2$ are their respective position vectors drawn from point A which is in the plane of the parallel lines. Choose the correct option(s).

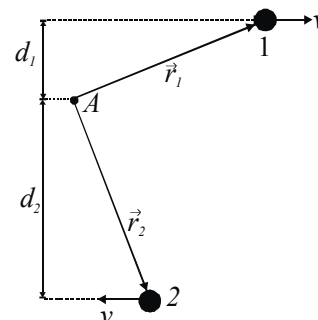
[$\hat{e}$ represents a unit vector coming out of the page]

(A) Angular momentum of particle 1 about A is $mv d_1 (-\hat{e})$

(B) Angular momentum of particle 2 about A is $mv d_2 (\hat{e})$

(C) Total angular momentum of the system about A is $mv(d_2 - d_1)\hat{e}$

(D) Total angular momentum of the system about A is $mv(d_1 + d_2)(-\hat{e})$



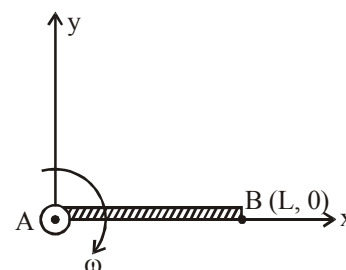
8. A rod of mass m , length L is rotating with uniform angular velocity ' ω ', about one of its ends, and axis along z -axis. At any instant, the rod is found along x -axis, as shown in figure. At this instant, the angular momentum of rod about A (0, 0) is $\vec{L}_1$ and B (L, 0) is $\vec{L}_2$. (Both A and B are fixed points and gravity is absent) Then :

(A) $\vec{L}_1$ and $\vec{L}_2$ are equal in magnitude and have same direction.

(B) $\vec{L}_1$ and $\vec{L}_2$ have opposite direction.

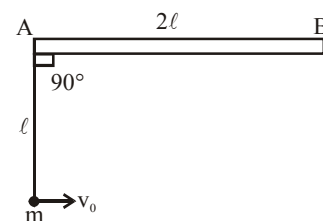
(C) $\frac{|\vec{L}_1|}{|\vec{L}_2|} = 1$ and torque appear to act when B is taken as origin at this instant

(D) $\frac{|\vec{L}_1|}{|\vec{L}_2|} = 2$ & no torque acts at this instant either from A, or B.

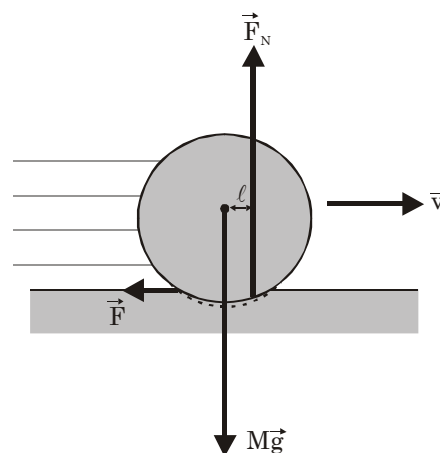


9. A uniform rod of mass m and length 2ℓ lies on smooth horizontal surface. A particle of same mass m is connected to string of length ℓ , whose other end is connected to rod. Initially string is taut and both rod and string lies in same horizontal plane with 90° angle between them. If particle is given initially velocity v_0 perpendicular to string, then just after giving velocity v_0 to particle :-

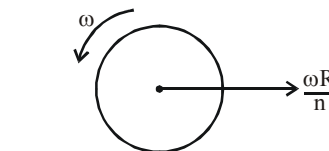
- (A) linear acceleration of centre of mass of rod will be $\frac{v_0^2}{4\ell}$
- (B) angular acceleration of rod will be $\frac{6v_0^2}{5\ell^2}$
- (C) tension in string will be $\frac{mv_0^2}{5\ell}$
- (D) angular acceleration of rod will be $\frac{3v_0^2}{5\ell^2}$



10. A rolling ball slows down because the normal force does not pass exactly through the CM of the ball, but passes in front of the CM. This is rolling friction. v & ω both decrease such that $v = r\omega$ is maintained. Assume that deformation is small so that friction acts at r_0 from centre. (r_0 is radius of ball).
- (A) The friction force opposes translation as well as rotation.
- (B) The normal force produces a torque which is greater in magnitude than torque of friction.
- (C) Friction force opposes translation but aids rotation.
- (D) $\ell = \frac{5}{7}\mu r_0$ where μ is the coefficient of static friction.



11. A disc of radius R is given a forward velocity $\frac{\omega R}{n}$ and an angular velocity ω as shown and is left over a rough surface :-
- (A) For $n = 1.5$ disc will come back to its initial point
- (B) For $n = 3$ disc will come back to its initial point
- (C) For $n = 1.5$ kinetic friction on disc is backward
- (D) For $n = 3$ kinetic friction on disc is backward

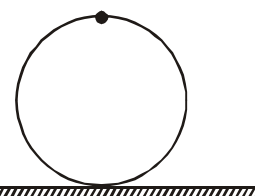


Linked Comprehension Type (1 Para $\times$ 2 Q.)
(Multiple Correct Answer Type)

[4 M (-1)]

Paragraph for question no. 12 and 13

A small particle of mass m is fixed to the perimeter of a ring of same mass and radius r . The system comprising of particle and ring is placed on a horizontal plane. Friction is negligible on horizontal plane. Initially particle is at top most point, then the system is released from rest. Answer next two questions when the particle is at the same height as the centre of ring after being released from topmost point. Assume that the ring stays in vertical plane during its motion under consideration.



12. Mark the **CORRECT** option(s) :

- (A) Speed of centre of mass of system is $\sqrt{\frac{gr}{4}}$
- (B) Angular velocity of system is $\omega = \sqrt{\frac{g}{r}}$
- (C) Contact point of ring with ground is instantaneously at rest.
- (D) Centre of ring is instantaneously at rest.

13. Mark the **CORRECT** option(s) :-

- (A) Instantaneous acceleration of the centre of ring is $\frac{g}{2}$
 (B) Acceleration of centre of ring initially points towards left and then towards right.
 (C) The force exerted by the ring on the ground at the instant given is $\frac{3}{2}mg$
 (D) The force exerted by the ring on the ground at the instant given is $2mg$

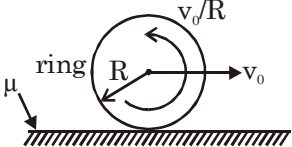
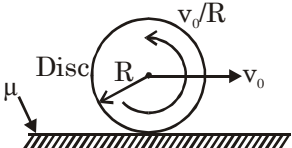
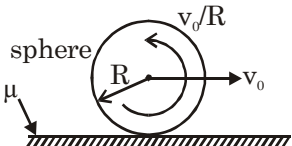
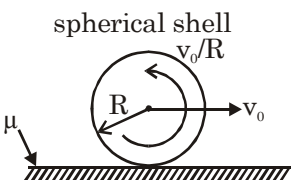
Matching list based comprehension Type ($4 \times 4 \times 4$)

1 Table $\times$ 3 Q. [3(-1)]

Single option correct

(Three Columns and Four Rows)

Answer Q.14, Q.15 and Q.16 by appropriately matching the information given in the three columns of the following table.

Column-1	Column-2	Column-3
Mass of the Object is 'm' situation is given at $t = 0$	Kinetic energy at the time when slipping ceases.	Time till slipping ceases.
(I) 	(i) Zero	(P) $\frac{v_0}{\mu g}$
(II) 	(ii) $\frac{1}{12}mv_0^2$	(Q) $\frac{2v_0}{3\mu g}$
(III) 	(iii) $\frac{9}{70}mv_0^2$	(R) $\frac{4v_0}{7\mu g}$
(IV) 	(iv) $\frac{mv_0^2}{30}$	(S) $\frac{4v_0}{5\mu g}$

14. Name the object that has maximum loss in kinetic energy, its kinetic energy after slipping ceases and the time till slipping ceases.

- (A)(II) (iii) (Q) (B) (I) (i) (P) (C) (III) (iii) (R) (D) (IV) (iv) (S)

15. Name the object has minimum loss in kinetic energy during slipping, its kinetic energy after slipping ceases and the time till slipping ceases :-

- (A)(II) (ii) (P) (B) (I) (i) (S) (C) (III) (iii) (R) (D) (IV) (iv) (Q)

16. Name the object, which stops when slipping ceases, its kinetic energy at the time of slipping ceases and time till slipping ceases :-

- (A)(II) (ii) (R) (B) (III) (iii) (S) (C) (IV) (iv) (Q) (D) (I) (i) (P)

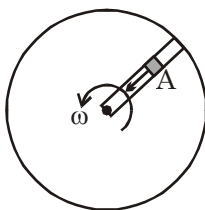
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. A disk is rotating in a horizontal plane with constant $\omega = \frac{\pi}{2}$ rad/s. A small particle A of mass m is moved with in a frictionless radial groove, towards centre with uniform radial speed of 1 m/s relative to platform. If particle goes from $r = 2$ m to $r = 1$ m, the net impulse on particle A is

expressed as $I_{\text{net}} = \alpha m \sqrt{\left(\frac{\pi}{2} - 1\right)^2 + (\pi + 1)^2}$. Find value of $(3\alpha + 2)$.

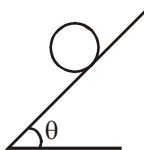


SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. A sphere is given an initial angular velocity ω_0 and kept on a long rough inclined plane of inclination θ and coefficient of friction μ . In each option is column-I, match appropriate description with column-II.



Column-I

- (A) $\theta = 37^\circ, \mu = 1, \omega_0$ clockwise
(B) $\theta = 45^\circ, \mu = 1, \omega_0$ anticlockwise
(C) $\theta = 53^\circ, \mu = \frac{1}{3}, \omega_0$ clockwise
(D) $\theta = 30^\circ, \mu = \frac{1}{2}, \omega_0$ anticlockwise

Column-II

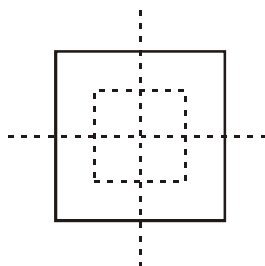
- (P) Acceleration of sphere just after $t = 0$ is zero.
(Q) Acceleration of sphere just after $t = 0$ is upwards.
(R) The sphere will start rolling purely while ascending
(S) The sphere will start rolling purely while descending
(T) Sphere will never roll purely

SECTION-I

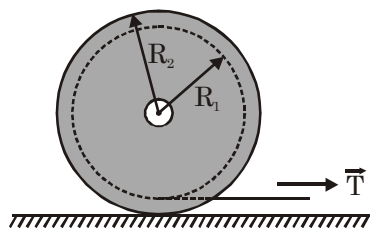
Single Correct Answer Type

7 Q. [3 M (-1)]

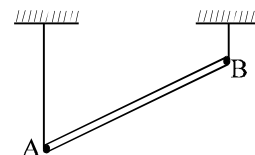
1. A concentric square hole is made in a square uniform plate (as shown in figure). About an axis passing through the center of the plate and perpendicular to its plane



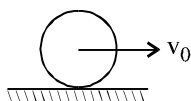
- (A) Moment of inertia increases but radius of gyration decreases.
(B) moment of inertia decreases but radius of gyration increases.
(C) both moment of inertia and radius of gyration increase.
(D) moment of inertia and radius of gyration both decrease.
2. A spool consists of a cylinder of radius R_1 wrapped with thread and its end caps have radius R_2 as shown in the end view illustrated in figure. The mass of the spool, including the thread, is m , and its moment of inertia about an axis through its center is I . The spool is placed on a rough horizontal surface so that it rolls without slipping. When a force $\vec{T}$ acting to the right is applied to the free end of the thread. Friction force exerted by the surface on the spool is given by :



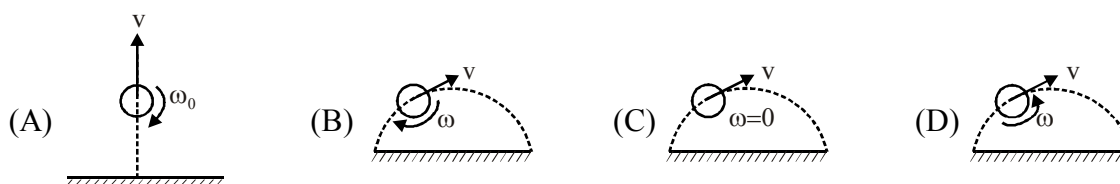
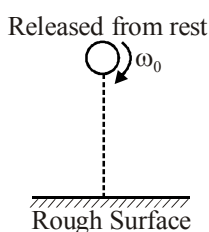
- (A) $f = \left(\frac{I + mR_1R_2}{I + mR_2^2} \right) T$, towards right
(B) $f = \left(\frac{I + mR_1R_2}{I + mR_2^2} \right) T$, towards left
(C) $f = \left(\frac{I - mR_1R_2}{I + mR_2^2} \right) T$, towards right
(D) $f = \left(\frac{I - mR_1R_2}{I + mR_2^2} \right) T$, towards left
3. Two strings support a uniform rod as shown. String at end B is cut. Which of the following is true just after cut
- [I] initial acceleration of A is vertical
[II] initial acceleration of A is horizontal
[III] initial acceleration of centre of mass of rod is vertical
[IV] initial acceleration of centre of mass of rod is horizontal
- (A) [I] & [II] (B) [II] & [III] (C) [III] & [IV] (D) [I] & [IV]



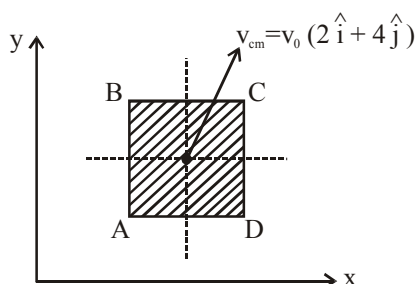
4. A sphere of mass m is projected on a rough ground with a velocity of v_0 without any spin. This is observed from ground and by an observer moving with constant velocity v_0 . For both the frames, origin is on ground and direction of motion is positive x -axis. Choose the **incorrect** statement.



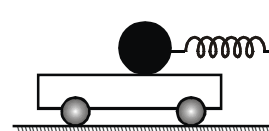
- (A) Change in angular momentum of the ball about origin in any time interval is same from both frames.
 (B) Work done by friction on the ball in any time interval is same from both frames
 (C) Total heat dissipated in any time interval is same from both frames
 (D) Change in momentum of ball in any time interval is same from both frames.
5. A disc is given an angular speed ω_0 and released from a certain height (as shown in figure). Motion of disc is observed after collision with the rough surface. Velocity of centre of mass of ball and direction of ω is shown in figure after the collision. Mark possible path, disc **CAN** follow after the collision.



6. A uniform solid square plate ABCD of mass m and side a is moving in x - y horizontal smooth plane. The velocity of centre of mass is $v_0(2\hat{i} + 4\hat{j})$ m/s. The end A of square plate is suddenly fixed by a pin, find the new velocity of centre of mass of square :



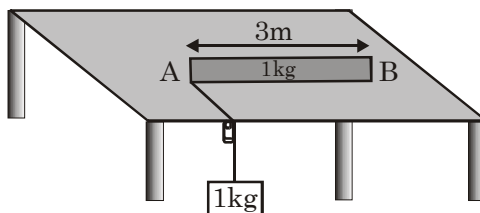
- (A) $v_0(2\hat{i} + 4\hat{j})$ (B) $\frac{3v_0}{4}(-\hat{i}) + \frac{3v_0}{4}\hat{j}$ (C) $\frac{3v_0}{4}\hat{i} - \frac{3v_0}{2}\hat{j}$ (D) $3v_0\hat{i} - 3v_0\hat{j}$
7. In the figure shown, the mass of the disc as well as that of the trolley is M . The spring is ideal and has stiffness k . The trolley can move horizontally on smooth floor and the disc can roll on the trolley surface without slipping. The spring is compressed and the system released so that oscillations begin. The
- (A) acceleration of centre of disc = twice of that of trolley
 (B) acceleration of centre of disc = thrice of that of trolley
 (C) acceleration of centre of disc = half of that of trolley
 (D) acceleration of centre of disc = that of trolley



Multiple Correct Answer Type

4 Q. [4 M (-1)]

8. There is a rod of length "3m" & mass "1kg" lying on a smooth horizontal table. A cord is led through a pulley, and its vertical part is attached to block of mass "1kg". The mass of the pulley and the friction are negligible. Select the **CORRECT** statement :-



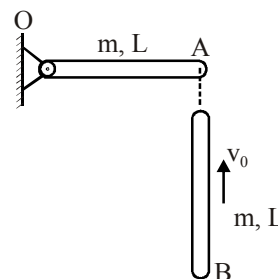
- (A) Acceleration of the centre of mass of the rod is 5 m/s^2
 (B) Acceleration of the centre of mass of the rod is 2 m/s^2
 (C) Acceleration of the point on rod at distance of 2m from end A is zero.
 (D) Angular acceleration of rod is 4 rad/sec^2
9. Rod B sticks to rod A on collision. Collision takes place on horizontal plane. Rod-A is hinged at O. Friction is absent every where.

(A) Angular velocity of system just after collision is $\frac{2v_0}{5L}$

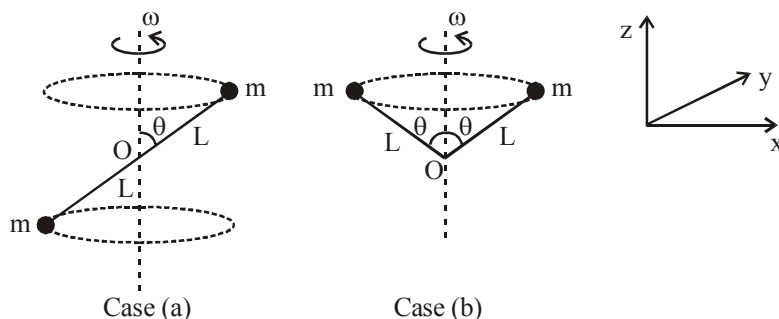
(B) Velocity of centre of mass of system just after collision is $\sqrt{\frac{9}{40}} v_0$

(C) Centre of mass of system is at a distance of $\frac{\sqrt{10} L}{4}$ from O

(D) Kinetic energy of system just after collision is $\frac{9}{40} mv^2$.

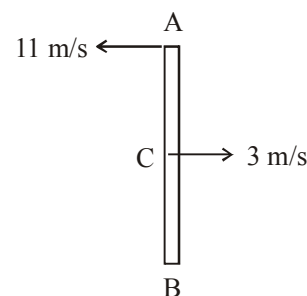


10. Two particles connected with massless rod are rotating with angular velocity ω about axis as shown in figure.



- (A) Total angular momentum about O in case (a) and (b) is $2m L^2 \sin^2\theta \omega \hat{k}$.
 (B) Total angular momentum about O is $2m L^2 \sin^2\theta \omega \hat{k}$ in case (a) only.
 (C) Total angular momentum about O is $2m L^2 \sin^2\theta \omega \hat{k}$ in case (b).
 (D) In case (a) a horizontal torque must act to rotate system with constant ω .

11. A uniform rod AB of length 7m is undergoing combined rotational and translational motion on smooth horizontal surface such that, at some instant of time, velocities of its end point A and centre C are both perpendicular to the rod and opposite in direction, having magnitude 11 m/s and 3 m/s respectively as shown in the fig. Now consider that no external force and torque is acting on the rod. Then select correct options.



- (A) acceleration of point A is 56 m/s^2
(B) acceleration of point B is 56 m/s^2
(C) at the instant shown in the figure acceleration of point C is zero.
(D) angular velocity of the rod is 4 rad/s

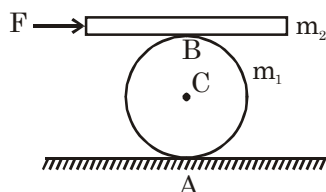
Linked Comprehension Type (1 Para $\times$ 2Q.)

[3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question no. 12 and 13

A man pushes a cylinder of mass m_1 with the help of a plank of mass m_2 as shown. There is no slipping at any contact. The horizontal component of the force applied is F .



12. The acceleration of plank (a_p) will be :

- (A) $\frac{F}{3m_1 + 8m_2}$ (B) $\frac{F}{4m_1 + 8m_2}$ (C) $\frac{4F}{3m_1 + 8m_2}$ (D) $\frac{8F}{3m_1 + 8m_2}$

13. The frictional force at contact point A is :

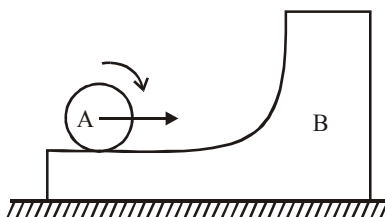
- (A) $\frac{3m_1 F}{3m_1 + 8m_2}$ (B) $\frac{3m_2 F}{3m_1 + 8m_2}$ (C) $\frac{m_1 F}{3m_1 + 8m_2}$ (D) $\frac{m_2 F}{8m_1 + 3m_2}$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

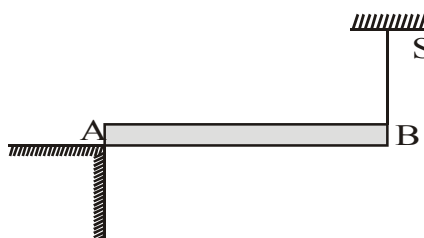
4 Q. [4 M (0)]

1. In the figure shown, a uniform solid cylinder A is initially rolling with velocity 14 m/s on the horizontal surface of wedge B of identical mass. All surfaces are smooth and B has no initial velocity. Find maximum height (in m) reached by cylinder. ($g = 9.8 \text{ m/s}^2$)

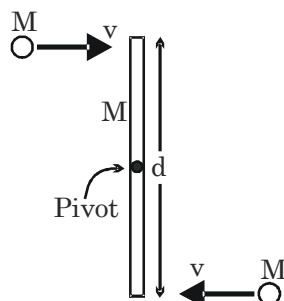


2. A linear rod of mass m , length ℓ is placed as shown in figure such that its one end rests on a rough table of friction coefficient μ . When string BS is cut, rod AB starts falling from rest. After rotating an angle

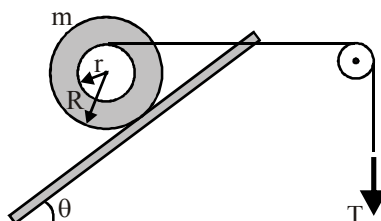
$\theta = \tan^{-1}\left(\frac{\mu}{2x}\right)$, rod starts slipping on the table, find x ?



3. Two particles, each of mass M and speed v , move as shown. They simultaneously strike the ends of a uniform rod at rest, of mass M and length d which is pivoted at its center. The particles stick to the ends of the rod. Find the ratio of total initial kinetic energy of the two particles plus rod system with the total loss in kinetic energy during collision ?



4. A spool, initially at rest, is kept on a frictionless incline making an angle $\theta = 37^\circ$ with the horizontal. The mass of the spool is 2 kg and it is pulled by a string as shown with a force of $T = 10\text{ N}$. The string connecting the spool and the pulley is initially horizontal. Find the initial acceleration of the spool on the incline. Express your answer in m/s^2 .

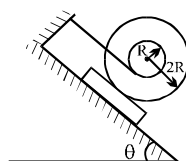


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. In the arrangement shown here friction between the spool of mass m , moment of inertia I and the plank of mass M is sufficient to prevent sliding and incline makes angle θ with horizontal. There is no slipping of rope. Assume that the system starts from rest rolling.



Column-I

- (A) When $m > M$ and there is no friction between the plank's and the inclined surface
(B) When $M > m$ and there is no friction between the plank's and the inclined surface
(C) When $m = M$ and there is no friction between the plank's and the inclined surface
(D) When $m = M$ and friction coefficient between the plank's and the inclined surface is $\tan \theta$

Column-II

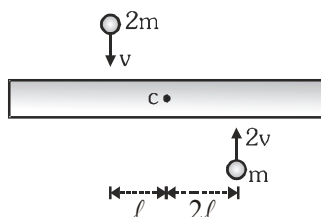
- (P) Plank will ascend
(Q) Plank will descend
(R) Plank will remain stationary
(S) Spool may rotate clockwise
(T) Spool may rotate anticlockwise

SECTION-I

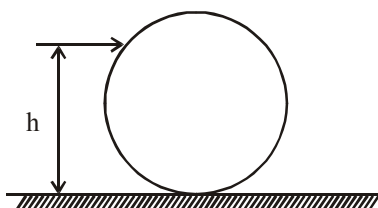
Single Correct Answer Type

8 Q. [3 M (–1)]

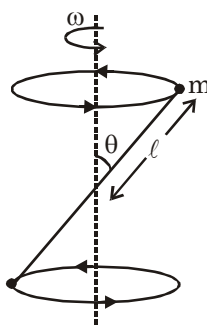
1. A uniform bar of length ' 6ℓ ' and mass ' $8m$ ' lies on a smooth horizontal table. Two point masses m and $2m$ moving in the same horizontal plane with speed $2v$ and v respectively, strike the bar (as shown in the fig.) and stick to the bar after collision. Total energy (about the center of mass, c) will be :-



- (A) $\frac{2mv^2}{5}$ (B) $\frac{mv^2}{5}$ (C) $\frac{3mv^2}{5}$ (D) mv^2
2. Two particles of equal mass m at A and B are connected by a rigid light rod AB lying on a smooth horizontal table. An impulse J is applied at A in the plane of the table and perpendicular at AB. Then the velocity of particle at A is:
- (A) $\frac{J}{2m}$ (B) $\frac{J}{m}$ (C) $\frac{2J}{m}$ (D) zero
3. A solid spherical ball of mass 1 kg and radius 1 m is hit horizontally at a height h above ground such that it starts rolling purely on the smooth ground. What is value of h ?

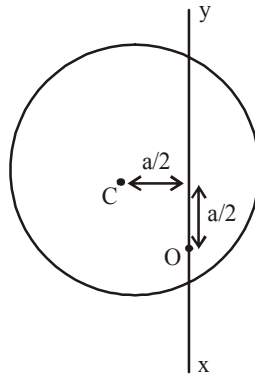


- (A) $\frac{7}{5}m$ (B) $\frac{5}{3}m$ (C) $\frac{3}{2}m$ (D) $\frac{4}{3}m$
4. A thin massless rod of length 2ℓ has equal point masses m attached at its ends (see figure). The rod is rotating about an axis passing through its centre and making angle θ with it. The magnitude of the rate of change of its angular momentum $\left| \frac{d\vec{L}}{dt} \right|$ about centre of rod is :

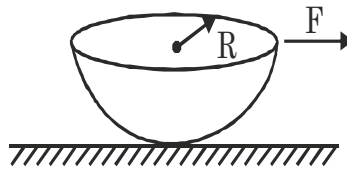


- (A) $2m\ell^2\omega^2 \sin\theta\cos\theta$ (B) $2m\ell^2\omega^2 \sin\theta$ (C) $2m\ell^2\omega^2 \sin^2\theta$ (D) $2m\ell^2\omega^2 \cos^2\theta$

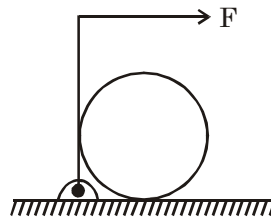
5. The figure shows a uniform disc of mass m and radius a . The moment of inertia about an axis xy is I . The moment of inertia about an axis passing through O and perpendicular to the plane of the disc will be:-



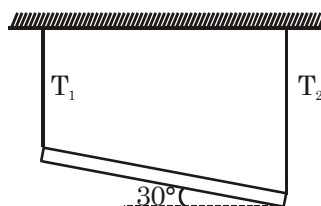
- (A) $I + \frac{1}{3}ma^2$ (B) $I + \frac{1}{4}ma^2$ (C) $2I$ (D) $I/2$
6. A thin uniform hemispherical bowl of mass m and radius R is lying on a smooth horizontal surface. A horizontal force F is now applied perpendicular to the rim of the bowl (see figure). The instantaneous angular acceleration of the bowl will be :-



- (A) $\frac{20}{3} \frac{F}{MR}$ (B) $\frac{10}{3} \frac{F}{MR}$ (C) $\frac{40}{3} \frac{F}{MR}$ (D) $\frac{6}{5} \frac{F}{MR}$
7. A uniform rod of mass M and length L is hinged at its lower end. The rod can rotate freely in vertical plane and there is no friction at the hinge or at the ground. A ball of mass M and radius $R = \frac{L}{3}$ is placed and a constant force F is applied as shown. The acceleration of centre of mass of the rod initially is :



- (A) $\frac{3F}{4M}$ (B) $\frac{F}{M}$ (C) $\frac{9F}{8M}$ (D) $\frac{3F}{M}$
8. A uniform rod of mass m and length ℓ is connected with two ideal strings from ends of rod as shown in figure, if tensions in strings are T_1 & T_2 respectively then :-



- (A) $T_1 > \frac{mg}{2}$ (B) $T_2 > \frac{mg}{2}$ (C) $T_1 < \frac{mg}{2}$ (D) $T_1 = T_2 = \frac{mg}{2}$

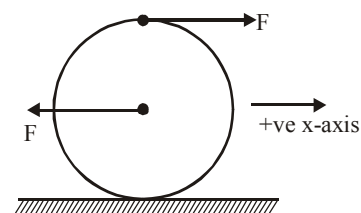
Multiple Correct Answer Type

2 Q. [4 M (-1)]

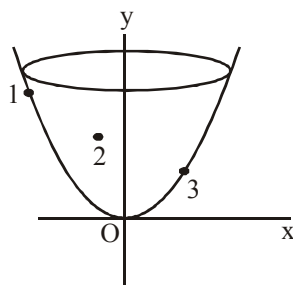
9. A disc is kept on a perfect rough surface at rest and now two forces of equal magnitude are applied in opposite direction as shown in the figure.

Mark the **CORRECT** statement(s) :-

- (A) Disc will remain at rest
- (B) Disc will move towards the x-direction
- (C) Frictional force on the disc will be zero
- (D) Frictional force will act towards the x-direction



10. Three particles, each of mass m are placed at the points (x_1, y_1, z_1) , (x_2, y_2, z_2) and (x_3, y_3, z_3) on the inner surface of a paraboloid of revolution obtained by rotating the parabola, $x^2 = 4ay$ about the y -axis. Neglect the mass of the paraboloid. (y -axis is along the vertical)



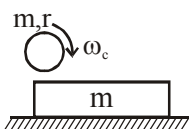
- (A) The moment of inertia of the system about the axis of the paraboloid is $I = 4ma(y_1 + y_2 + y_3)$.
- (B) If potential energy at O is taken to be zero, the potential energy of the system is $mg(y_1 + y_2 + y_3)$.
- (C) If the particle at (x_1, y_1, z_1) slides down the smooth surface, its speed at O is $\sqrt{2gy_1}$
- (D) If the paraboloid spins about OY with an angular speed ω , the kinetic energy of the system will be $ma(y_1 + y_2 + y_3)\omega^2$.

Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 3Q. & 1 Para $\times$ 2Q.) [3 M (-1)]

Paragraph for Question no. 11 to 13

In the figure shown a plank of mass m is lying at rest on a smooth horizontal surface. A disc of same mass m and radius r is rotated to an angular speed ω_0 and then gently placed on the plank. If we consider the plank and the disc as a system then frictional force between them is an internal force. Momentum of the system changes due to external force only. It is found that finally slipping ceases, and 50% of total kinetic energy of the system is lost. Assume that plank is long enough. μ is coefficient of friction between disc and plank.



11. Final velocity of the plank is :-

- (A) $\frac{r\omega_0}{4}$
- (B) $\frac{r\omega_0}{\sqrt{10}}$
- (C) $\frac{r\omega_0}{2}$
- (D) $\frac{r\omega_0}{2\sqrt{10}}$

12. Magnitude of the change in angular momentum of the disc about center of mass of the disc :-

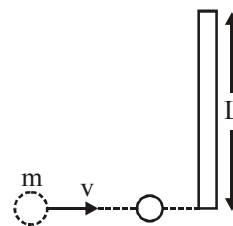
- (A) $\frac{3}{4} mr^2\omega_0$
- (B) $\frac{1}{4} mr^2\omega_0$
- (C) zero
- (D) $\frac{1}{2} mr^2\omega_0$

13. Distance moved by the plank from the placing of disc on the plank till the slipping ceases between disc and plank :-

- (A) $\frac{r^2\omega_0^2}{16\mu g}$
- (B) $\frac{r^2\omega_0^2}{8\mu g}$
- (C) $\frac{r^2\omega_0^2}{32\mu g}$
- (D) $\frac{r^2\omega_0^2}{200\mu g}$

Paragraph for Questions No. 14 and 15

A small disc and a thin uniform rod of length L , whose mass is η times greater than the mass of the disc, lie on a smooth horizontal plane. The disc is set in motion, in horizontal direction and perpendicular to the rod, with velocity v , after which it elastically collides with the end of the rod.



14. The velocity of the disc after collision is

- (A) $\frac{(4+\eta)v}{4-\eta}$ (B) ηv (C) $\left(\frac{4-\eta}{4+\eta}\right)v$ (D) None of these

15. Angular velocity of the rod after the collision is :-

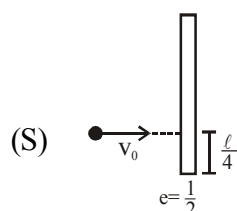
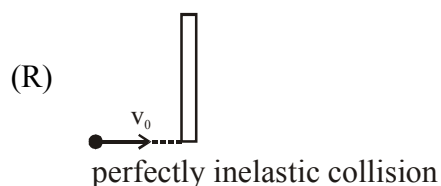
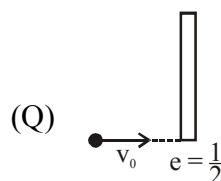
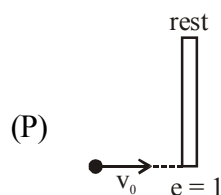
- (A) $\frac{2v}{L}$ (B) $\frac{\eta v}{L}$ (C) $\left(\frac{4+v}{4-\eta}\right)$ (D) $\frac{12v}{(4+\eta)L}$

Matching List Type (4×4)

1 Q. [3 M (-1)]

16. List-I, shows four system of rod + particle undergoing collision on a smooth horizontal floor. Mass of particle & rod in each case is m , & M respectively. In List-II relationships between velocity (v) of centre of rod and angular velocity (ω) of the rod just after the collision are given. Choose appropriate option (length of rod is ℓ)

List-I



List-II

(1) $\frac{\omega \ell}{3}$

(2) $\frac{\omega \ell}{2}$

(3) $\frac{\omega \ell}{6}$

(4) $\frac{2\omega \ell}{3}$

Codes :

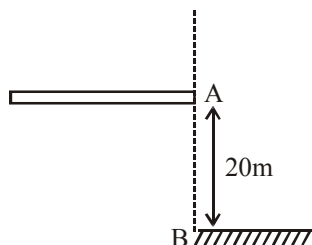
	P	Q	R	S
(A)	4	3	1	2
(B)	2	1	3	2
(C)	3	3	3	1
(D)	1	2	3	4

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. A 1 m long rod is dropped (keeping it horizontal) from a height of 20 m. The end A of rod collides with end B of a stationary horizontal table. The impact is perfectly elastic. The post impact angular velocity of rod is 10α (in rad/sec.), find value of α .

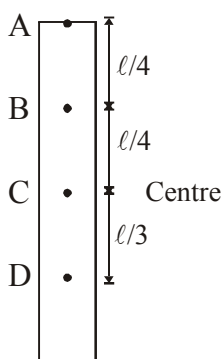


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Figure shows a uniform thin rod of length l , A, B, C, D are four points on the rod with point C, marked on the centre of mass of the rod. Positions of A, B, C, D are shown in the figure.



Column I

- (A) Moment of inertia of the rod is maximum about the axis perpendicular to the rod and passing through...
- (B) Moment of inertia will be least if the rod is hinged about
- (C) If the rod is kept on a smooth horizontal surface, and a ball having same mass collides elastically then the angular velocity of the rod will be maximum, if the ball strikes at.....
- (D) If the rod is kept on a smooth horizontal surface, and a ball having same mass collides elastically then the speed of centre of mass of the rod will be maximum, if the ball strikes at.....

Column II

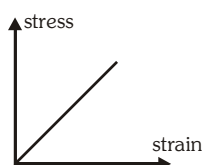
- (P) Points A
- (Q) Point B
- (R) Point C
- (S) Point D
- (T) Point A & D

SECTION-I

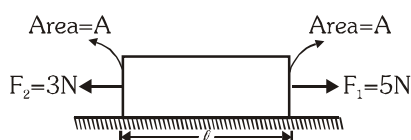
Single Correct Answer Type

10 Q. [3 M (-1)]

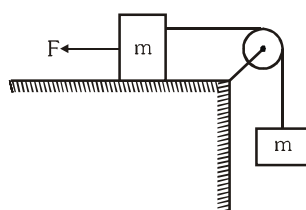
1. A graph is plotted between stress in the wire and corresponding strain produced in it within elastic limit. The area under stress-strain curve gives



- (A) elastic potential energy (B) elastic potential energy per unit length
(C) elastic potential energy per unit volume (D) $\frac{1}{2}$ elastic potential energy per unit volume
2. A block of mass M area of cross-section A and length ℓ is placed on smooth horizontal floor. Two forces $F_1 = 5\text{ N}$ and $F_2 = 3\text{ N}$ are applied on the block as shown in the figure. If Y is Young's modulus of elasticity of material, then total extension in the length of the block will be



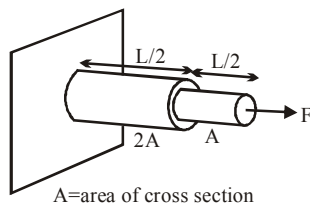
- (A) $\frac{8\ell}{AY}$ (B) $\frac{2\ell}{AY}$ (C) $\frac{4\ell}{AY}$ (D) $\frac{5\ell}{2AY}$
3. Consider the situation shown in figure. The force F is equal to mg . If area of cross-section of the string is A and its Young's modulus is Y , the strain developed in the wire is (the string is light and there is no friction anywhere)



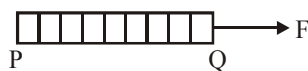
- (A) $\frac{mg}{AY}$ (B) $\frac{2mg}{AY}$ (C) $\frac{mg}{2AY}$ (D) $\frac{3mg}{2AY}$
4. A thin steel ring of inner radius r and cross-sectional area A is fitted on to a wooden disc of radius R ($R > r$). If Young's modulus be Y , then the tension in the steel ring is
- (A) $AY\left(\frac{R}{r}\right)$ (B) $AY\left(\frac{R-r}{r}\right)$ (C) $\frac{Y}{A}\left(\frac{R-r}{r}\right)$ (D) $\frac{Yr}{AR}$
5. Water freezes inside a pipe, normally it expands by about 9% due to freezing. What would be the pressure increase inside the pipe? The bulk modulus of ice is $2.00 \times 10^9 \text{ N/m}^2$.
- (A) $1.80 \times 10^8 \text{ N/m}^2$ (B) $3.60 \times 10^8 \text{ N/m}^2$
(C) $9 \times 10^7 \text{ N/m}^2$ (D) $7.2 \times 10^8 \text{ N/m}^2$

CLASS TEST

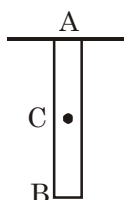
6. The bar shown in the figure is made of a single piece of material. It is fixed at one end and consists of two segments of equal length $\frac{L}{2}$ but different cross-sectional area A and $2A$. What is the change in length of the entire system under the action of an axial force F . [Consider the shape of joint to remain circular, Y is young's modulus].



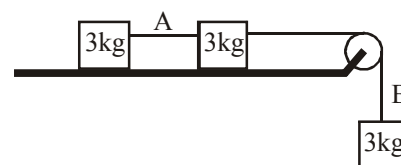
- (A) $\frac{3FL}{4AY}$ (B) $\frac{3FL}{8AY}$ (C) $\frac{3FL}{2AY}$ (D) $\frac{2FL}{3AY}$
7. A wire of cross-section A is stretched horizontally between two clamps located $2l$ meters apart. A weight W kg is suspended from the mid point of the wire. If the mid point sags vertically through a distance $x \ll l$ the strain produced is
- (A) $\frac{2x^2}{l^2}$ (B) $\frac{x^2}{l^2}$ (C) $\frac{x^2}{2l^2}$ (D) none of these
8. The given diagram shows a uniform massive strip of rubber marked with equal divisions kept in a gravity free space. The rubber strip is pulled by a force F applied at end Q until it has extended to twice its original length. Which of the following diagrams best shows the separation of the divisions on the extended strip? (Assuming rubber has achieved constant length finally)



- (A) (B) (C) (D)
9. A wire of uniform cross-section is hanging vertically and due to its own weight its length changes. There is a point C on the wire such that change in length AC is equal to the change in length BC . Points A , B & C are shown in figure. Find $\frac{AC}{BC}$.



- (A) 1 (B) $\sqrt{2}$ (C) $\sqrt{2} - 1$ (D) $\frac{1}{\sqrt{2}}$
10. Three equal masses of 3 kg are connected by two massless strings of cross sectional area 0.005 cm^2 and Young modulus is $2 \times 10^{11} \text{ N/m}^2$ each. The longitudinal strain in the wires
- (A) are equal
(B) cannot be different
(C) must be different
(D) may or may not be different



Multiple Correct Answer Type

4 Q. [4 M (-1)]

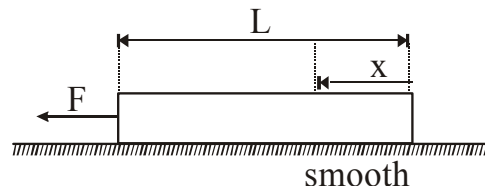
11. A rod of mass m , uniform cross sectional area A and length L is accelerated by applying force F as shown in figure on a smooth surface. If Young's modulus of elasticity of the material of rod is Y . (Consider x as measured from the right end)

(A) Tension in rod as a function of distance x is $\frac{Fx}{2L}$

(B) Strain in rod is $\frac{F}{2AY}$

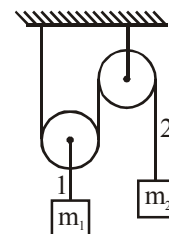
(C) Elastic potential energy stored in the rod is $\frac{F^2 L}{8AY}$

(D) There is no stress in rod.

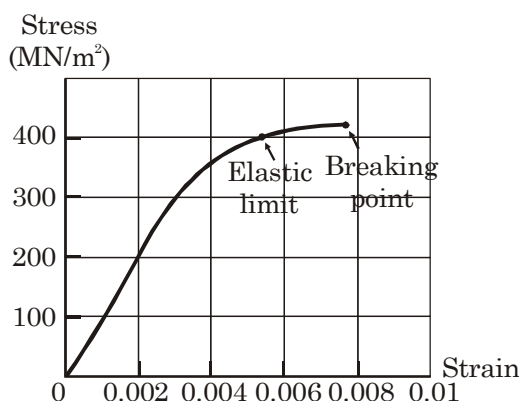


12. In the figure shown the system is in equilibrium. In this system we have used two wires 1 and 2. The Young's modulus and radius of the wire 1 are two times and $2^{1/4}$ times respectively that of wire 2, then.

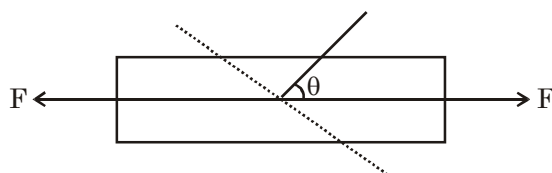
- (A) Elastic energy density in wire 1 is equal to that of in wire 2
(B) Elastic energy density in wire 1 is greater than that of in wire 2
(C) Elastic energy density in wire 1 is less than that of in wire 2
(D) Stress in wire 1 is greater than stress in wire 2.



13. Consider a solid whose stress strain curve is shown here in the figure :-



- (A) The Young modulus for the solid is 10^{11} N/m^2 .
(B) If the solid is subjected to any stress greater than 350 MN/m^2 , it will permanently deform.
(C) If the solid is cast into a wire of cross section 1 mm^2 , it will break under a weight of 50 kg .
(D) If the solid is cast into a wire of cross section 1 mm^2 and subjected to a force of 150 N , it will have a strain of 0.0015
14. Consider a bar of uniform cross-section, subjected to equal and opposite tensile forces at its ends. A plane section is shown in the diagram by dotted line. Normal of this section makes an angle θ with the axis of bar. With change in θ , orientation of this plane changes.



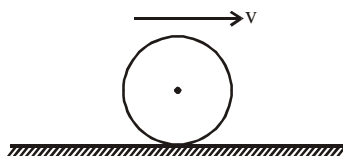
- (A) For $\theta = 90^\circ$, tensile stress on this plane is maximum.
(B) For $\theta = 0^\circ$, tensile stress on this plane is maximum.
(C) For $\theta = 45^\circ$, shearing stress on this plane is maximum.
(D) For $\theta = 90^\circ$, shearing stress on this plane is maximum.

Linked Comprehension Type (Single Correct Answer Type)

(2 Para × 2Q.) [3 M (-1)]

Paragraph for Question 15 & 16

A ring of mass m and radius R is rolling with constant velocity v on a horizontal surface. Young's modulus of material of ring is y and crosssectional area of the wire by which ring is made is A .



15. Tensile force applied by one half of the ring on the other half, due to its motion is :-

(A) $\frac{mv^2}{R}$ (B) $\frac{mv^2}{2R}$ (C) $\frac{mv^2}{\pi R}$ (D) $\frac{mv^2}{2\pi R}$

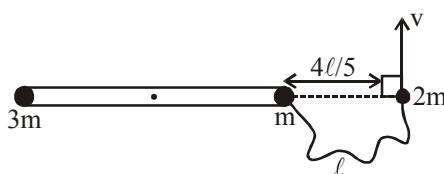
16. Change in radius of the ring due to its motion is :-

(A) $\frac{mv^2}{4\pi Ay}$ (B) $\frac{mv^2}{2\pi Ay}$ (C) $\frac{2mv^2}{\pi Ay}$ (D) $\frac{mv^2}{\pi Ay}$

Paragraph for Question Nos. 17 & 18

A light and thin rod of length ℓ is lying on horizontal table. Two small balls of masses $3m$ and m are fixed on two ends of a massless rod. The rod is free to rotate about fixed vertical axis passing through midpoint of rod. A third ball of mass $2m$ is fastened to ball of mass m through a string of length ℓ as shown in figure. The ball is projected horizontally with speed V as shown.

17. The angular velocity of rod, just after the string becomes taut is



(A) $\frac{18v}{59\ell}$ (B) $\frac{9v}{59\ell}$ (C) $\frac{59v}{18\ell}$ (D) $\frac{3v}{5\ell}$

18. The impulse imparted by string is

(A) $\frac{mv}{10}$ (B) $\frac{5mv}{59}$ (C) $\frac{59mv}{50}$ (D) $\frac{60mv}{59}$

SECTION-I

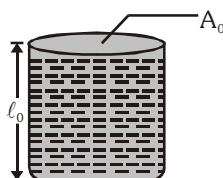
Single Correct Answer Type

15 Q. [3 M (-1)]

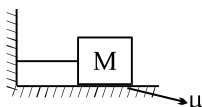
1. The table gives the initial length ℓ_0 , change in temperature ΔT and change in length $\Delta \ell$ of four rods. Which rod has greatest coefficient of linear expansion

Rod	$\ell_0(\text{m})$	$\Delta T(^{\circ}\text{C})$	$\Delta \ell(\text{mm})$
A1	1	100	1
A2	1	100	2
A3	1.5	50	3
A4	2.5	20	4

- (A) A_1 (B) A_2 (C) A_3 (D) A_4
2. Density of substance at 0°C is 10 g/cc and at 100°C , its density is 9.7 g/cc. The coefficient of linear expansion of the substance will be—
(A) $10^2\text{ }^{\circ}\text{C}^{-1}$ (B) 10^{-2} K^{-1} (C) 10^{-3} K^{-1} (D) $10^{-4}\text{ }^{\circ}\text{C}^{-1}$
3. When a copper ball is heated, the largest percentage increase will occur in its
(A) diameter (B) area (C) volume (D) density
4. If a thermometer reads freezing point of water as 20°C and boiling point as 150°C , how much thermometer will read when the actual temperature is 60°C
(A) 98°C (B) 110°C (C) 40°C (D) 60°C
5. The figure shows a glass tube (linear co-efficient of expansion is α) completely filled with a liquid of volume expansion co-efficient γ . On heating length of the liquid column does not change. Choose the correct relation between γ and α .



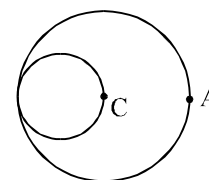
- (A) $\gamma = \alpha$ (B) $\gamma = 2\alpha$ (C) $\gamma = 3\alpha$ (D) $\gamma = \frac{\alpha}{3}$
6. The density of a uniform rod with cross-section A is d, its specific heat capacity is c and the coefficient of its linear expansion is α . Calculate the amount of heat that should be added in order to increase the length of the rod by l .
(A) $\frac{dAc l}{\alpha}$ (B) $\frac{\alpha A c l}{d}$ (C) $\frac{dc l}{A \alpha}$ (D) $\frac{c l \alpha d}{A}$
7. A wire having cross-sectional area S is attached to wall on one side and a block of mass M on the other side which placed on a horizontal surface having coefficient of friction μ as shown. Material of wire has coefficient of thermal expansion α and Young's modulus Y. At initial temperature there is no stress in the wire. Now the wire is cooled. At what decrease in temperature the block will begin to move?



- (A) $\frac{\mu M g}{\alpha Y S}$ (B) $\frac{2 \mu M g}{\alpha Y S}$ (C) $\frac{\mu M g}{2 \alpha Y S}$ (D) none

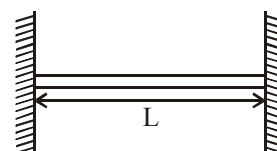
8. A liquid of volumetric thermal expansion coefficient $= \gamma$ and bulk modulus B is filled in a spherical tank of negligible heat expansion coefficient. Its radius is R and wall thickness is ' t ' ($t \ll R$). When the temperature of the liquid is raised by θ , the tensile stress developed in the walls of the tank is :
- (A) $B\gamma\theta R/2t$ (B) $B\gamma\theta R/t$ (C) $2B\gamma\theta R/t$ (D) $B\gamma\theta R/4t$

9. A uniform circular disc with the hole lies over a smooth horizontal surface, C is centre of the disc and A is a point on the rim of the disc as shown. As the temperature is increased.



- (A) C remains at rest & A to right
(B) Both A & C move to right
(C) Both remain at rest
(D) C moves to left & A to right
10. A uniform rod having length L , is made of material having density ρ , coefficient of thermal expansion α , and Young's modulus Y . If it placed between two vertical walls having separation L and coefficient of friction for one wall is 2μ and other wall is μ , what is the minimum rise in temperature so that the rod is in equilibrium.

- (A) $\frac{L\rho g}{\alpha Y \mu}$ (B) $\frac{2L\rho g}{\alpha Y \mu}$
(C) $\frac{L\rho g}{2\alpha Y \mu}$ (D) None



11. What must be the lengths of steel and copper rods at 0°C for the difference in their lengths to be 10 cm at any common temperature?

($\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ }^\circ\text{K}^{-1}$ and $\alpha_{\text{copper}} = 1.8 \times 10^{-5} \text{ }^\circ\text{K}^{-1}$)

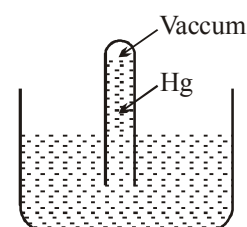
- (A) 30 cm for steel and 20 cm for copper (B) 20 cm for steel and 30 cm for copper
(C) 40 cm for steel and 30 cm for copper (D) 30 cm for steel and 40 cm for copper
12. A uniform metal rod fixed at its ends of 2 mm^2 cross-section is cooled from 40°C to 20°C . The coefficient of the linear expansion of the rod is 12×10^{-6} per degree & it's young modulus of elasticity is 10^{11} N/m^2 . The energy stored per unit volume of the rod is :

- (A) 2880 J/m^3 (B) 1500 J/m^3 (C) 5760 J/m^3 (D) 1440 J/m^3

13. Two rods one of aluminium of length l_1 having coefficient of linear expansion α_a , and other steel of length l_2 having coefficient of linear expansion α_s are joined end to end. The expansion in both the rods is same on variation of temperature. Then the value of $\frac{l_1}{l_1 + l_2}$ is

- (A) $\frac{\alpha_s}{\alpha_a + \alpha_s}$ (B) $\frac{\alpha_s}{\alpha_a - \alpha_s}$ (C) $\frac{\alpha_a + \alpha_s}{\alpha_s}$ (D) None of these

14. A barometer made of a very narrow tube (see fig.) is placed at normal temperature and pressure. The coefficient of volume expansion of mercury is $0.00018/^\circ\text{C}$ and that of the tube is negligible. The temperature of mercury in the barometer is now raised by 1°C but the temperature of the atmosphere does not change. Then



Statement-1 : The mercury height in the tube remains unchanged.

Statement-2 : The atmospheric pressure remains same.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true

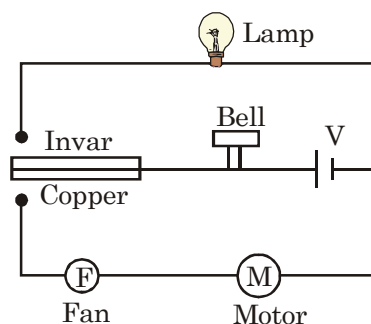
15. A liquid whose coefficient of volume expansion is γ_r completely fills a sealed metal tank, at negligible pressure. The coefficient of linear expansion of the metal is α , and the compressibility of the liquid is k . If the temperature of the system is increased by t , the new pressure of the liquid will be

(A) $\gamma_r \frac{t}{k}$ (B) $\frac{t}{k} (\gamma_r - 3\alpha)$ (C) $tk(\gamma_r - 3\alpha)$ (D) $\frac{t}{k} (\gamma_r - \alpha)$

Multiple Correct Answer Type

2 Q. [4 M (-1)]

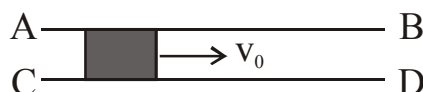
16. A fire alarm circuit is as shown. Due to fire the exposed bimetallic strip made of invar and copper ($\alpha_{\text{invar}} < \alpha_{\text{copper}}$) is heated up. Which of the following will switch on.



- (A) Bell (B) Motor (C) Lamp (D) Fan
17. In the figure, AB and CD are two vertical walls of a horizontal rectangular groove. A rectangular block of mass m just fits in the groove. Base of the block is smooth and coefficient of friction between walls and block is μ . Contact area of block with each wall is A . Linear coefficient of thermal expansion and young's modulus of material of block are α and Y respectively. Temperature θ of block increases uniformly as $\Delta\theta = t$ (t = time). Initial velocity of block is v_0 then

(A) Retardation of block at time t is $\frac{\mu YA \alpha t}{m}$

(B) Velocity at time t is $\left(v_0 - \frac{\mu YA \alpha t^2}{m} \right)$



(C) Time taken by block to stop is $\sqrt{\frac{mv_0}{\mu YA \alpha}}$

(D) work done by friction is $\frac{1}{2} mv_0^2$

SECTION-II

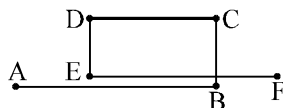
Numerical Answer Type Question

2Q.[3M(0)]

(upto second decimal place)

1. A thin ring of an alloy of inner diameter 20 cm and cross-sectional area 1 mm², is heated until it easily slides on a rigid cylinder of diameter 20.025 cm. [For the alloy : $\alpha = 10^{-5}/^\circ\text{C}$, $Y = 400 \text{ GPa}$] when the ring cools down, what is the tension (in N) in it?

2. The length of sections AB and EF is l_1 in the figure, and their coefficient of thermal expansion is same ($\alpha_1 = 2 \times 10^{-5} / ^\circ\text{C}$). The length of section CD is l_2 and its expansion coefficient is $\alpha_2 = 1 \times 10^{-5} / ^\circ\text{C}$. What must be the ratio l_1/l_2 to ensure that distance AF remains the same at all temperatures? (Points A, B, E and F reside on the same line.)

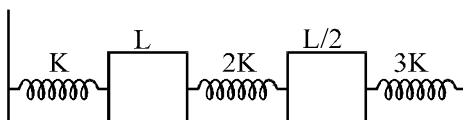


SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. The system shown consists of 3 springs & two rods. If the temperature of the rods is increased by ΔT , if the force exerted by springs on wall is $\frac{x}{11}KL\alpha\Delta T$, then find value of x. Neglect friction & thermal stress & take the coefficient of linear expansion of the material of rods equal to α .



SECTION-I

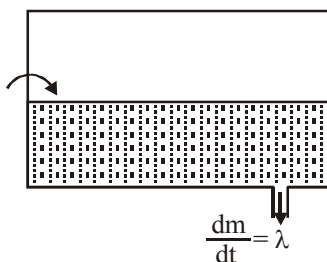
Single Correct Answer Type

6 Q. [3 M (–1)]

- Earth receives 1400 Wm^{-2} of solar power. If all the solar energy falling on a lens of area 0.2 m^2 is focused on to a block of ice of mass 280 g (at 0°C), the time taken to melt the ice will be (Take latent heat of fusion of ice = $3.3 \times 10^5 \text{ Jkg}^{-1}$)
(A) 303 s (B) 330 s (C) 33 s (D) 5.5 s
- A substance of mass $M \text{ kg}$ requires a power input of P watts to remain in the molten state at its melting point. When the power source is turned off. The sample completely solidifies in time t seconds. The specific latent heat of fusion of the substance is :
(A) Pt (B) Pt/M (C) PtM (D) PM/t
- At NTP water boils at 100°C . Deep down the mine, water will boil at a temperature
(A) 100°C (B) $> 100^\circ\text{C}$ (C) $< 100^\circ\text{C}$ (D) Will not boil at all
- In two experiments with a continuous flow calorimeter to determine the specific heat capacity of a liquid, an input power of 60 W produced rise of 10 K in the liquid. When the power was doubled, the same temperature rise was achieved by making the rate of flow of liquid three times faster. If power loss in each case is same then find power lost to the surroundings :-
(A) 20 W (B) 30 W (C) 40 W (D) 120 W
- A well insulated box contains water (specific heat = C and very large conductivity) of mass m_0 and temperature T_0 at time $t = 0$. If heat is being added to it uniformly at a constant rate $dQ/dt = R$ and

water is leaking from it at a constant rate $\frac{dm}{dt} = \lambda$ then the temperature of the water at time

$t \left(t < \frac{m_0}{\lambda} \right)$ will be:-



- (A) $T = \frac{R}{2\lambda C} \ln \left(\frac{m_0}{m_0 - \lambda t} \right) + T_0$ (B) $T = \frac{R}{\lambda C} \ln \left(\frac{m_0}{m_0 - \lambda t} \right) + T_0$
- (C) $T = \frac{R}{C\lambda} e^{(m_0 - \lambda t)}$ (D) $T = \frac{R}{2C\lambda} e^{(m_0 - \lambda t)}$

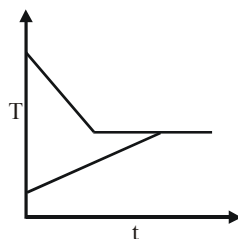
- How many minimum ice cubes each of mass 10 gm and temperature 0°C should be added to a glass of water containing 100 gm water at 20°C to bring the temperature of mixture below 5°C . The heat capacity of glass is $50 \text{ cal/}^\circ\text{C}$. Neglect heat losses.
(A) 3 (B) 4 (C) 6 (D) 12

Multiple Correct Answer Type

7 Q. [4 M (–1)]

7. A sample A of liquid water and a sample B of ice of identical mass are kept in two neighbouring chambers in an otherwise insulated container. The chambers can exchange heat with each other.

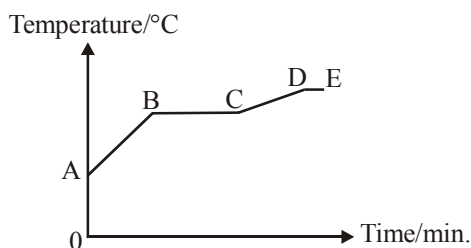
The graph of temperatures of the two chambers is plotted with time. $S_{\text{ice}} = \frac{S_{\text{water}}}{2}$:-



- (A) Finally the contents in sample A is water.
(B) Equilibrium temperature is freezing point of water
(C) Ice melts partly
(D) Finally the contents in sample B is ice only.
8. In three experiments, a material A at a particular low temperature T_C and a material B at a particular high temperature T_H are placed in an isolated and insulated container. When they reach thermal equilibrium with each other (No phase change occurs), their final temperature T_f is measured. The masses m_A & m_B and specific heats C_A & C_B of the material are given in the table. Assume that heat transferred is Q in the experiment. Then which of the following is/are correct :-

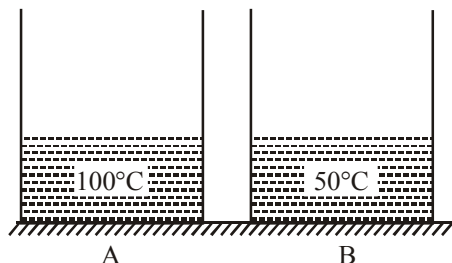
Experiment	m_A	C_A	m_B	C_B
1	m	c	m	c
2	m	c	$2m$	c
3	m	$2c$	m	c

- (A) $(T_f)_1 > (T_f)_2 > (T_f)_3$ (B) $Q_3 > Q_1, Q_2 > Q_1$ (C) $Q_2 > Q_1 > Q_3$ (D) $(T_f)_2 > (T_f)_1 > (T_f)_3$
9. A well insulated substance in solid state is heated at a constant rate until it vaporizes completely. The temperature-time graph of the substance is shown below. Which of the following statements is/are true?



- (A) The specific heat capacity of the substance in solid state is greater than that of the substance in liquid state.
(B) The specific latent heat of fusion of the substance is greater than the specific latent heat of vaporization of the substance.
(C) If the rate of heating increases, the slope of AB will be increased but the length of BC remains unchanged.
(D) If the mass of the substance is doubled, the length of DE is also doubled but the slope of CD is halved.

10. 2 kg water at 100°C and 2.5 kg water at 50°C is kept in two identical containers A and B respectively of water equivalent 0.5 kg. If water of container A is poured into container B the final temperature of mixture is T_1 and if water of container B is poured into container A the final temperature is T_2 . (heat loss is negligible)



- (A) $15T_1 = 14T_2$ (B) $T_1 = T_2$ (C) $T_1 = 70^{\circ}\text{C}$ (D) $T_2 = 70^{\circ}\text{C}$
11. 50 gm ice at -10°C is mixed with 10gm steam at 100°C . When the mixture finally reaches its steady state inside a calorimeter of water equivalent 1.5 gm then : [Assume calorimeter was initially at 0°C , Take latent heat of vaporization of water = 540 cal/gm, Latent heat of fusion of water = 80 cal/gm and specific heat capacity of water = 1 cal/gm- $^{\circ}\text{C}$, specific heat of ice = 0.5 cal/gm- $^{\circ}\text{C}$]
- (A) Mass of water remaining is 60 gm (B) Mass of ice remaining is 3 gm
(C) Mass of steam remaining is 1.20 gm (D) Final temperature is between 0°C and 100°C
12. A piece of ice having temperature 0°C is thrown into a thermally insulated container, filled with water of volume 5 dm^3 at a temperature of 50°C . Match the final amount of water in the container if after the ice melts the temperature of the system is $T^{\circ}\text{C}$?
- (A) $T = 15, V = \frac{35}{19} \text{ dm}^3$ (B) $T = 0, V = \frac{65}{8} \text{ dm}^3$
(C) $T = 5, V = \frac{130}{17} \text{ dm}^3$ (D) $T = 10, V = \frac{20}{9} \text{ dm}^3$
13. A calorimeter of specific heat $0.42 \text{ J/g}^{\circ}\text{C}$ weighing 40gm contains 50gm of water mixed with 50gm of ice at 0°C and system is in thermal equilibrium. Now steam at 100°C is passed into the mixture until the temperature rises to 20°C . Take $L_F = 80 \text{ cal/gm}$, $L_V = 540 \text{ cal/gm}$. Select the **CORRECT** alternatives.
- (A) Water equivalent of calorimeter is 4gm. (B) Water equivalent of calorimeter is 40 gm.
(C) Mass of steam condensed is 9.8 gm. (D) Mass of steam condensed is 11.26gm.

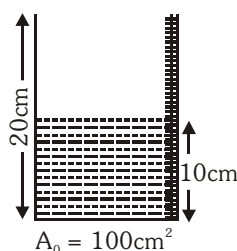
Linked Comprehension Type (1 Para $\times$ 3Q.)

[3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question 14 to 16

At 20°C a liquid is filled upto 10 cm height in a container of glass of length 20 cm and cross-sectional area 100 cm^2 . Scale is marked on the surface of container. This scale gives correct reading at 20°C . Given $\gamma_L = 5 \times 10^{-5} \text{ K}^{-1}$, $\alpha_g = 1 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$



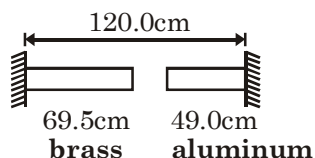
14. The volume of liquid at 40°C is :-
(A) 1002 cc (B) 1001 cc (C) 1003 cc (D) 1000.5 cc
15. The actual height of liquid at 40°C is-
(A) 10.01 cm (B) 10.006 cm (C) 10.6 cm (D) 10.1 cm
16. The reading of scale at 40°C is-
(A) 10.01 cm (B) 10.004 cm (C) 10.006 cm (D) 10.04 cm

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. A brass rod is 69.5 cm long and an aluminum rod is 49.0 cm long when both rods are at an initial temperature of 0°C. The rods are placed in line with a gap of 1.5 cm between them. The distance between the far ends of the rods is maintained at 120.0 cm throughout. The temperature is raised until the two rods are barely in contact. The coefficients of linear expansion of brass and aluminum are $2.0 \times 10^{-5} \text{ K}^{-1}$ and $2.4 \times 10^{-5} \text{ K}^{-1}$ respectively. In figure, the temperature (in °C) at which contact of the rods just occurs is x. x/585 (to the nearest integer) is



2. A thin copper wire has a length L at 0°C. When heated to t°C, its length increases by 0.1 %. If a thin copper plate of dimension L × 3L is heated through the same range of temperature, its surface area gets increased by x %. Fill value of 10x.

SECTION-IV

Matrix Match Type (4 × 5)

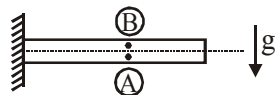
1 Q. [8 M (for each entry +2(0))]

1. Column-I shows various arrangements of rods having same cross sectional area and same length. In each figure two points A & B are indicated by dots (•). Young's modulus and thermal coefficient of expansion of the rod having point (A) are (Y_A, α_A) & having point (B) are (Y_B, α_B) if (A) and (B) are in different rods. Each arrangement undergoes various processes also mentioned in column-I. In column-II is written statements about the rods.

Match them correctly

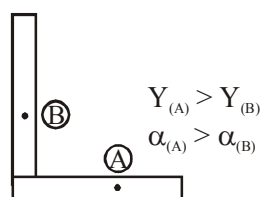
Column-I

- (A) A single horizontal heavy rod is free at one end & fixed at other.



[dotted line passes through the middle of the rod and (A) is below the line and (B) above it]

- (B) Two rods are joined together and temperature is increased

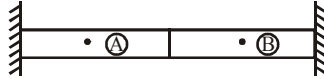


[rod is kept on horizontal smooth ground]

Column-II

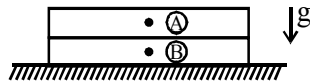
- (P) Tensile stress is present at (A)
- (Q) Tensile stress is present at (B)

- (C) Two rods are joined in series and fixed at both ends, in gravity free space. Initially rods are at natural length, when temperature is increased



$$Y_{(A)} > Y_{(B)}$$

- (D) Two rods are fixed to each other and placed on a smooth surface. Temperature is increased but the rods are not allowed to bend.



$$\alpha_{(A)} > \alpha_{(B)}, Y_{(A)} < Y_{(B)}$$

- (S) Compressive stress is present at (B)

- (T) There is no elastic potential energy in rod having

point (A)

SECTION-I

Single Correct Answer Type

7 Q. [3 M (-1)]

- Which of the following steps would make the pipe more capable of withstanding the increased pressure?
(A) Increase the radius of pipe and decrease thickness of it's walls.
(B) Increase the radius of pipe and thickness of it's walls.
(C) decrease the radius of pipe and thickness of it's walls.
(D) decrease the radius of the pipe and increase the thickness of its walls.
- A glass bowl of water equivalent of 10 g is at room temperature of 20°C when 40 g water at 100°C is poured in it. Also 5g of ice at 0°C is put in the water. If no heat is lost to the surroundings, the final temperature of the mixture is
(A) 74°C (B) 55°C (C) 80°C (D) 69°C
- A solid metal sphere of radius 'r' and specific heat 's' is rotated about an axis passing through its centre at a speed of 'n' rotations per second. It is suddenly stopped and 25% of its energy is used in increasing its temperature, the rise in temperature of sphere is
(A) $\left(\frac{\pi^2 n^2 r^2}{5s}\right)$ (B) $\left(\frac{\pi^2 n^2 r^2}{10s}\right)$ (C) $\left(\frac{2\pi^2 n^2 r^2}{5s}\right)$ (D) $\left(\frac{4\pi^2 n^2 r^2}{5s}\right)$
- A storage type geyser contains 10 kg of water at 10°C. The geyser has a power rating = 1000 W and efficiency of conversion of electric power to heat = 75%. If the geyser remains on for 20 minutes, what will be the final temperature of water in the geyser?
(A) 37°C (B) 31.4°C (C) 28.2°C (D) 18.6°C
- A thermally insulated vessel contains two liquids A & B with initial temperatures T_1 & T_2 and specific heats s_1 & s_2 respectively. They are separated by a non-conducting piston. When the partition is removed the final equilibrium temperature is average of the initial temperatures of the liquids. The ratio of masses of two liquids m_1/m_2 is :
(A) $\frac{s_2}{s_1}$ (B) $\frac{s_1}{s_2}$ (C) $\sqrt{\frac{s_1}{s_2}}$ (D) $\sqrt{\frac{s_2}{s_1}}$
- If a 5 kg body falls to the ground from a height of 30 m and if all its mechanical energy is converted into heat, then heat produced will be (approx)
(A) 350 cal (B) 1500 cal (C) 60 cal (D) 6 cal
- A rod of length L_0 (fixed at both ends) is supplied heat to raise its temperature by T. If γ is the coefficient of volume expansion of the rod and Y is Young's modulus of the rod then the energy density stored in the rod is:
(A) $\frac{1}{2} \gamma^2 T^2 Y$ (B) $\frac{1}{3} \gamma^2 T^2 Y^3$ (C) $\frac{1}{18} \frac{\gamma^2 T^2}{Y}$ (D) $\frac{1}{18} \gamma^2 T^2 Y$

Multiple Correct Answer Type

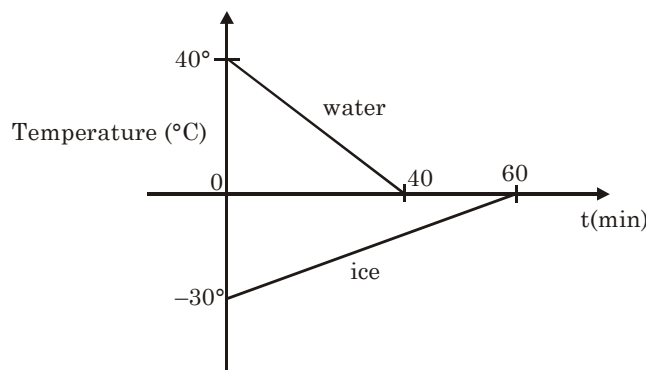
2 Q. [4 M (-1)]

8. A 0.6 kg sample of water and a sample of ice are placed in two compartments A and B that are separated by a conducting wall, in a thermally insulated container. The rate of heat transfer from the water to ice through the conducting wall is constant P , until thermal equilibrium is reached. The temperature T of the liquid water and ice are shown in graph as a function of time t .

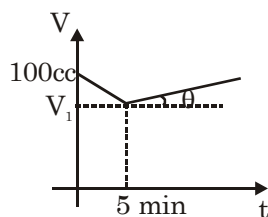
Given specific heat of ice = 2100 J/kg-K

specific heat of water = 4200 J/kg-k

latent heat of fusion = $3.3 \times 10^5 \text{ J/kg}$



- (A) The value of rate P is 42 W .
 (B) The initial mass of ice in container is equal to 3.6 kg .
 (C) The initial mass of ice in container is equal to 2.4 kg .
 (D) The mass of ice formed due to conversion from water till thermal equilibrium is reached, is equal to 0.12 kg .
9. A block of ice at 0°C was being heated at a constant rate. The volume of sample as a function of time is shown in the graph below. Take $\rho_{\text{ice}} = 0.9 \text{ gm/cc}$, $\rho_{\text{water}} = 1 \text{ gm/cc}$, $\gamma_{\text{water}} = 10^{-3} / ^\circ\text{C}$. Specific heat capacity of water = $1 \text{ cal/gm } ^\circ\text{C}$, latent heat of fusion of ice = 80 cal/gm :-



- (A) $V_1 = 90 \text{ cc}$
 (B) $\frac{dH}{dt} = 24 \text{ cal/sec}$
 (C) $\tan \theta = 1.44 \text{ cc/min}$
 (D) Water will start boiling at 8.24 min .

Linked Comprehension Type (Single Correct Answer Type)

(2 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Question 10 to 12

Given a liquid of initial temperature -20°C is warmed up in a microwave oven, using 2520 watts . The mass of the liquid is $M = 1 \text{ kg}$. The specific heat of liquid is $s = 2(T - 253) \text{ cal/gm-K}$, where T is in Kelvin. (Given $T = \theta^\circ + 273$) (Here θ is in $^\circ\text{C}$)

10. Determine the temperature vs. time function

- (A) $6\theta^2 + 20\theta + 1000 = 5t$
 (B) $5\theta^2 + 200\theta + 2000 = 3t$
 (C) $6\theta^2 + 30\theta + 3000 = 5t$
 (D) $5\theta^2 + 20\theta + 1000 = 3t$

11. The time after which temperature of liquid becomes 20°C (in sec)
(A) 1500 (B) 1233.33 (C) 1000 (D) 2666.67
12. If 1 kg water at temperature 50°C is added with a liquid of mass 1 kg at 20°C , with spicific heat $2 \text{ cal/gm } ^{\circ}\text{C}$, then final temperature of the mixture is (in $^{\circ}\text{C}$) ?
(A) 33 (B) 32 (C) 25 (D) 30

Paragraph for Question 13 to 15

You are program director in ISRO (Indian Space Research Organisation) to conduct a feasibility study into the SDI (Strategic Defence Initiative) missile defence system. You are to consider the viability of a ground based laser defence against ICBM's, which will use powerful CO_2 lasers to bore holes in the incoming missiles, thus destroying them. The lasers have an output power of 50W in a beam diameter of 1 mm. The laser beam is fired at the missile when it is 10 km away and the beam loses 3% of its intensity for every kilometre travelled. The outer skin of the missile is aluminium that is 3 cm thick. When the laser fires the skin temperatures of the missile is -50°C and must be heated to its boiling point at 2600°C . The density of aluminium is 2.4 g/cm^3 and heat capacity is $0.9 \text{ J/g}^{\circ}\text{C}$.

13. The amount of heat required to reach the outer skin of aluminium to its boiling point is
(A) 129.7 J (B) 140.7 J (C) 196.4 J (D) 554.7 J
14. The power of laser beam when it strikes missile surface is (Take $(0.97)^{10} = 0.738$)
(A) 50W (B) 24.8 W (C) 36.9 W (D) None of these
15. How long will it take the laser to burn through the outer skin of the missile, thus destroying it? Assume that all of the laser power that reaches the missile goes into heating the 1 mm diameter spot.
(A) 3.5 s (B) 3.2 s (C) 2.4 s (D) 7.1 s

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1Q. [3M(0)]

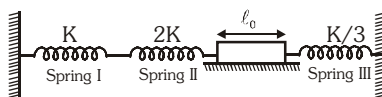
1. The elastic limit of a steel cable is $2.40 \times 10^8 \text{ Pa}$ and the cross-section area is 4.00 cm^2 . Find the maximum upward acceleration (in m/s^2) that can be given to a 800 kg elevator supported by the cable if the stress should not exceed one third of the elastic limit.

SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

1. A rod of length ℓ_0 having co-efficient of linear expansion α is joined as in figure. Initially all strings are in natural length and friction is absent. Temperature of rod is increased by ΔT .



Column I

- (A) Compression in spring I.
- (B) Compression in spring II.
- (C) Compression in spring III.
- (D) Elongation in rod.

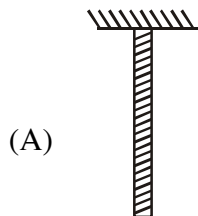
Column II

- (P) $\ell_0 \alpha \Delta T$
- (Q) $\frac{2\ell_0}{3} \alpha \Delta T$
- (R) $\frac{\ell_0}{9} \alpha \Delta T$
- (S) $\frac{2\ell_0}{9} \alpha \Delta T$
- (T) $\frac{3\ell_0}{10} \alpha \Delta T$

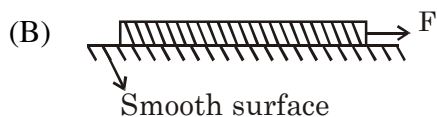
2. In Column-I different configuration of uniform rod of mass m , length ' L ', cross-sectional area ' A ' and Young's modulus of elasticity ' Y ' are as shown. Match the column-I with column-II.

Column-I

Column-II



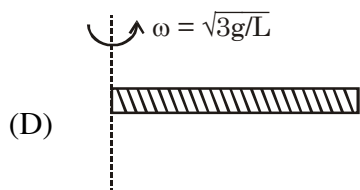
A rod is hanging vertically with fixed support



A constant force $F = mg$ is applied at one end of the rod.



Equal and opposite forces ($F = mg$) are applied at both end of the rod



A rod is rotating with constant angular velocity about end point in gravity free space.

(P) Total elongation in the rod is $\frac{mgL}{2AY}$

(Q) Total elongation in the rod is $\frac{mgL}{AY}$

(R) Uniform stresses developed in the rod.

(S) Non-uniform stress developed in the rod.

(T) Elastic potential energy stored in the

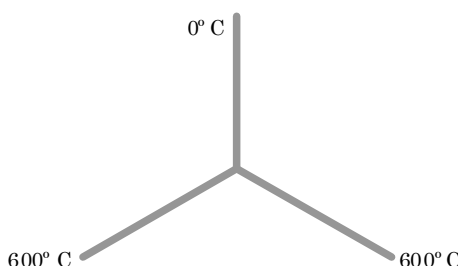
rod is $\frac{m^2 g^2 L}{6AY}$

SECTION-I

Single Correct Answer Type

9 Q. [3 M (-1)]

1. Three identical rods made of the same material have been joined as shown in the figure. The free ends of the rods are maintained at temperatures 0°C , 600°C and 600°C as shown in the figure. The temperature at the junction of the three rods is



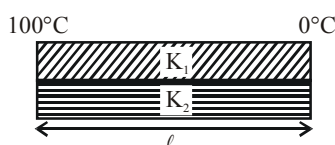
- (A) 200°C (B) 300°C (C) 400°C (D) 500°C
2. One mole of oxygen is contained in a rigid container with walls of inner surface area A , thickness d and thermal conductivity K . The gas is at initial temperature T_0 and the surrounding temperature of air is $2T_0$. The temperature of the gas at time t is (R is gas constant) :-

(A) $T = T_0 \left(2 - e^{\frac{-2KA}{5dR}t} \right)$ (B) $T = T_0 \left(2 - e^{\frac{2KA}{5dR}t} \right)$ (C) $T = T_0 e^{\frac{-2KA}{5dR}t}$ (D) $T = \frac{2}{5} T_0 e^{\frac{KA}{dR}t}$

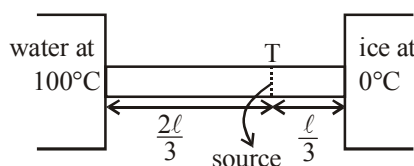
3. Two rods of identical dimensions (Area = A , length ℓ each) are kept as shown in figure. One end of both the rods is kept at 100°C & another end at 0°C . The rods have different thermal conductivity given by

$K_1 = K_0 \left(1 + \frac{x}{\ell} \right)$ and $K_2 = \frac{K_0}{2 \ln 2}$ where x is the distance from the end having 100°C temperature & K_0

is a +ve constant. Find the rate of heat flow through this combination in steady state :-



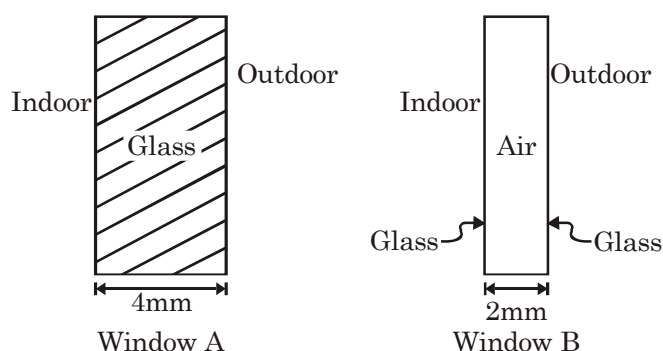
- (A) $\frac{150 K_0 A}{\ln 2 \ell}$ (B) $150 \frac{K_0 A}{\ell}$ (C) $\frac{100 K_0 A}{\ln 2 \ell}$ (D) $\frac{3 K_0 A}{4 \ln 2 \ell}$
4. The rod connecting two reservoir and connected to a source is as shown in diagram. The rod is in steady state. Find the temperature of source, so that rate of melting of ice is 16 times that rate of vaporization. (Latent heat of vaporization is 540 Cal/gm and latent heat of fusion is 80 Cal/gm)



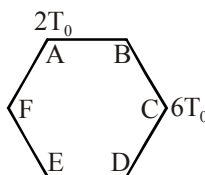
- (A) 540°C (B) 580°C (C) 640°C (D) 740°C

5. A manufacturer is able to offer two models of heat-conserving windows, as described below. Window A is a simple pane of glass, 4 mm thick. Window B, on the other hand, consists of two extremely thin panes of glass, separated by an air gap of 2mm, as shown in the figure below. If the thermal conductivity of glass is known to be $0.8 \text{ Wm}^{-1}\text{K}^{-1}$ and that of air is $0.025 \text{ Wm}^{-1}\text{K}^{-1}$, then the ratio of heat flow Q_A

through Window A to the heat flow Q_B through Window B is given by $\frac{Q_A}{Q_B} =$



- (A) 4 (B) 16 (C) $\frac{1}{4}$ (D) $\frac{1}{16}$
6. A rod of length L with sides fully insulated is of a material whose thermal conductivity varies with temperature as $K = \frac{\alpha}{T}$, where α is a constant. The ends of the rod are kept at temperature T_1 and T_2 . The temperature T at x , where x is the distance from the end whose temperature is T_1 is
- (A) $T_1 \left(\frac{T_2}{T_1} \right)^{\frac{x}{L}}$ (B) $\frac{x}{L} \ln \frac{T_2}{T_1}$ (C) $T_1 e^{\frac{T_2 x}{T_1 L}}$ (D) $T_1 + \frac{T_2 - T_1}{L} x$
7. Six identical conducting rods having uniform cross-section area are fixed to make a regular hexagon as shown. The ends A and C are maintained at constant temperature $2T_0$ and $6T_0$. After the steady state is reached, pick up the correct relation between temperatures T_B , T_D , T_E and T_F of ends B, D, E and F respectively : (Neglect convection and radiation losses)



- (A) $T_D > T_E = T_B > T_F$ (B) $T_D = T_B > T_E > T_F$
(C) $T_D > T_E > T_F = T_B$ (D) $T_B > T_F = T_E > T_D$

8. In the figure-II temperature of inner surface of cylinder is 0°C and temperature of outer surface is 100°C . In figure-I temperature of one end of cylinder is 100°C while another end is at 0°C

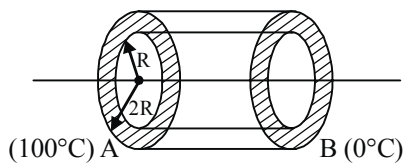


Figure-I

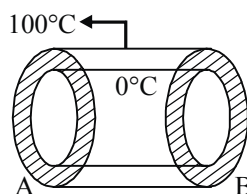


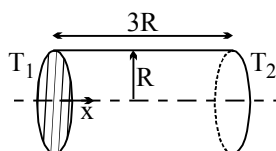
Figure-II

$\frac{dT}{dr}$ = temperature gradient in radial direction at steady state in figure-II

$\frac{dT}{dl}$ = temperature gradient in linear direction at steady state in figure-I.

Both cylinders are made of uniform material.

- (A) $\frac{dT}{dr} = \text{constant}$ $\frac{dT}{dl} \neq \text{constant}$ (B) $\frac{dT}{dr} \neq \text{constant}$ $\frac{dT}{dl} = \text{constant}$
(C) $\frac{dT}{dr} = \text{constant}$ $\frac{dT}{dl} = \text{constant}$ (D) $\frac{dT}{dr} \neq \text{constant}$ $\frac{dT}{dl} \neq \text{constant}$
9. For a conducting cylinder with variable thermal conductivity along the axis the magnitude of rate of heat flow if temperature difference across ends is $(T_1 - T_2)$,



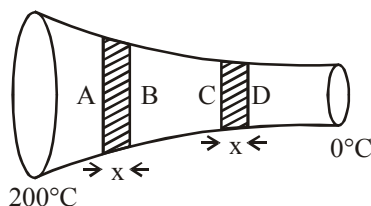
solid cylinder, flow along axis,
variable k as $k = k_0(1 + x/(3R))$
 x is distance from left end

- (A) $6\pi k_0 R(T_1 - T_2)$ (B) $\frac{\pi k_0 R}{3 \ln 2}(T_1 - T_2)$ (C) $\pi k_0 R(T_1 - T_2)$ (D) $\frac{4\pi k_0 R}{\ln 2}(T_1 - T_2)$

Multiple Correct Answer Type

5 Q. [4 M (-1)]

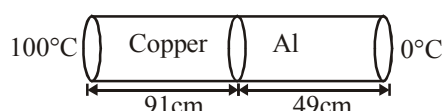
10. Two ends of conducting rod of varying cross-section are maintained at 200°C and 0°C . In steady state:



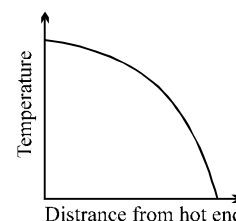
- (A) Temperature difference across AB and CD are equal
(B) Temperature difference across AB is greater than that across CD.
(C) Temperature difference across AB is less than that across CD.
(D) Temperature gradient increases as we move from end A to B.
11. A cubical box is used to keep cold drinks in a bus. Area of each face of cubical box is 0.80 m^2 and each wall has thickness 2.00 cm . It is filled with ice and water which remains at 0°C . Temperature outside each wall is 30°C then: (Thermal conductivity of material of box = $1 \times 10^{-2}\text{ W/m-k}$)
(Latent heat of fusion of ice = $3.2 \times 10^5\text{ J/kg}$)
(A) Total heat current in the box is 60 J/s
(B) Total heat current in the box is 72 J/s
(C) Mass of ice melted in 1 hour is 810 gm
(D) Mass of ice melted in 1 hour is 690 gm
12. A bar of Copper length 91 cm and Aluminium of length 49 cm are joined together end to end. Both rod

have circular cross-section with diameter $\sqrt{\frac{1}{50\pi}}\text{ m}$. Free end of Copper & Aluminium are maintain at

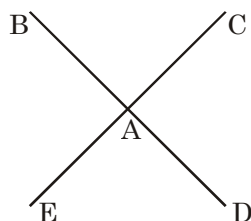
100°C & 0°C respectively. For maintaining temperature steam at 100°C is continuously flowing at the end of copper & ice is continuously melting at the end of Aluminium. After reaching steady state, choose correct statement(s) [Thermal conductivity of copper is $91\text{ cal m}^{-1}\text{ }^{\circ}\text{C}^{-1}\text{ s}^{-1}$ & Thermal conductivity of Aluminium is $49\text{ cal m}^{-1}\text{ }^{\circ}\text{C}^{-1}\text{ s}^{-1}$, $L_f = 80\text{ cal/g}$ & $L_v = 540\text{ cal/g}$]. Neglect any thermal resistance at the junction.



- (A) Heat flow per unit time across the junction is 25 cal/sec .
(B) The rate of melting ice is $\frac{5}{16}\text{ g/sec}$.
(C) Rate of condensation of steam is $\frac{5}{108}\text{ g/sec}$.
(D) Temperature of junction is 50°C
13. The ends of a long homogenous bar are maintained at different temperatures and there is no loss of heat from the sides of the bar due to conduction or radiation. The graph of temperature against distance of the bar when it has attained steady state is shown here. The graph shows
(A) the temperature gradient is not constant
(B) the bar has uniform cross-sectional area
(C) the cross-sectional area of the bar increases as the distance from the hot end increases
(D) the cross-sectional area of the bar decreases as the distance from the hot end increases.



14. Four identical rods which have thermally insulated lateral surfaces are joined at point A. Points B, C, D and E are connected to large reservoirs. If heat flows into the junction from point B at a rate of 1W and from point C at 3W towards A, flows out from D at 5W, which relation(s) is/are correct for temperature of these points ?



- (A) $T_A < T_E$ (B) $T_B = T_C$ (C) $T_C > T_D$ (D) $T_B = T_E$

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Question no. 15 and 16

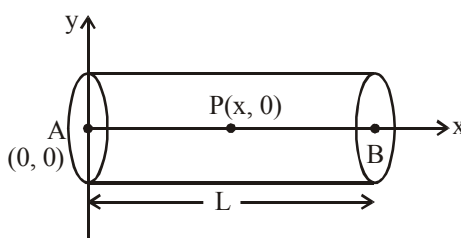
The rate of flow of heat through any cross-section is given by,

$$\frac{dQ}{dt} = -KS \left(\frac{dT}{dx} \right).$$

Where K is thermal conductivity, S is cross sectional area and $\frac{dT}{dx}$ is temperature gradient. For given

rod, temperature at point P is given by, $T = 100 - 2x^2 - 4x$, where T is in kelvin and x is in metre.

$L = 2$ m, $K = 100$ Watt/m-K, $S = 0.1$ m², $M = 5$ kg, $\alpha = 10^{-4}$ /K.



15. Temperature gradient at A and B would be :

- (A) $-4 \left(\frac{K}{m} \right), -2 \left(\frac{K}{m} \right)$ (B) $-4 \left(\frac{K}{m} \right), -12 \left(\frac{K}{m} \right)$
(C) $100 \left(\frac{K}{m} \right), 84 \left(\frac{K}{m} \right)$ (D) $0, -2 \left(\frac{K}{m} \right)$

16. The rate of heat absorption in the rod would be :

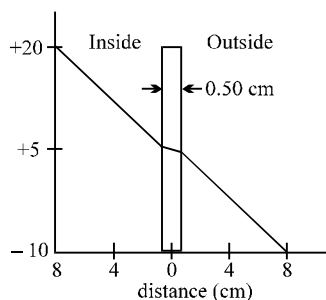
- (A) 160 Watt (B) 60 Watt (C) 0 (D) 80 Watt

SECTION-III

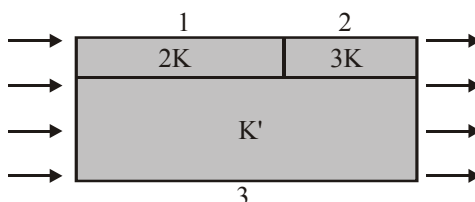
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

1. An idealized representation of the air temperature as a function of distance from a single-pane window on a calm, winter day is shown in figure. The window dimensions are $60 \text{ cm} \times 60 \text{ cm} \times 0.50 \text{ cm}$. $k_{\text{air}} = 0.026 \text{ W/m-K}$, $k_{\text{glass}} = 1.0 \text{ W/m-K}$. If rate of heat flow out through the window is $0.6x$ then find the value of x .



2. Two rods 1 and 2 of length 2ℓ and ℓ , thermal conductivities $2k$ and $3k$ are joined end to end. The area of cross-sectional of each rod be A . This composite rod is connected with another rod 3 of length 3ℓ and conductivity k' . The area of cross-section of the rod 3 is $2A$. If the rod 3 conducts the heat two times that through the other rods, find the value of $\frac{4k'}{k}$.



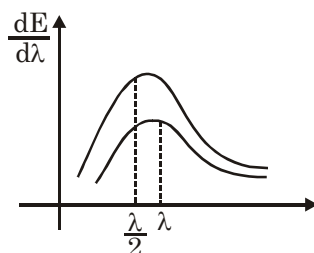
3. Three identical rods are connected between two containers. One of them is at 100°C and another is at 0°C . If rods are connected in parallel then the rate of melting of ice is $q_1 \text{ g/s}$. If they are connected in series, then the rate is $\frac{q_1}{x}$. Then value of x will be.

SECTION-I

Single Correct Answer Type

5 Q. [3 M (–1)]

- Assuming Newton's law of cooling to be valid. The temperature of body changes from 60°C to 40°C in 7 minutes. Temperature of surroundings being 10°C , its temperature after next 7 minutes, is :-
(A) 7°C (B) 14°C (C) 21°C (D) 28°C
- There are two thin spheres A and B of the same material and same thickness. They emit like black bodies. Radius of A is double that of B. A and B have same temperature T . When A and B are kept in a room of temperature T_0 ($< T$), the ratio of their rates of cooling (rate of fall of temperature) is : [assume negligible heat exchange between A and B]
(A) 2 : 1 (B) 1 : 1 (C) 4 : 1 (D) 8 : 1
- A body radiates energy 5W at a temperature of 127°C . If the temperature is increased to 927°C , then it radiates energy at the rate of—
(A) 410 W (B) 81 W (C) 405 W (D) 20 W
- A solid sphere of radius R and a hollow sphere of inner radius r and outer radius R made of copper are heated to the same temperature and are allowed to cool in the same environment. Then, choose the **CORRECT** statement :-
(A) Hollow sphere cools faster
(B) Solid sphere cools faster
(C) Both the spheres attain room temperature at the same time
(D) The rate of loss of heat of the solid sphere is twice that of the hollow sphere
- The variation of spectral emissive power for two spherical objects having radius R and $2R$ respectively are shown below. The ratio of their emissive powers are :-

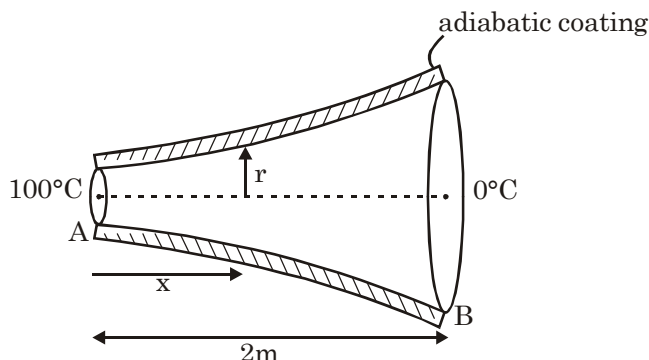


- (A) 16 : 1 (B) 4 : 1 (C) 32 : 1 (D) 8 : 1

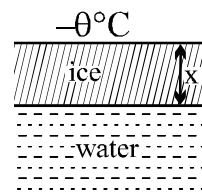
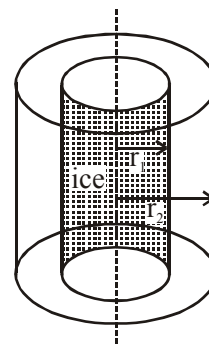
Multiple Correct Answer Type

4 Q. [4 M (–1)]

6. Radius of a metallic rod varies with x (distance from end A) as $r = (x + 1)^{3/2}$ m. End A and B of rod are maintain at 100°C and 0°C respectively. Thermal conductivity and length of rod are $\frac{3}{\pi}$ W/m-K and 2m respectively. Assume steady state flow of heat in rod and rod is covered with adiabatic coating :-



- (A) Rate of heat flow through the rod is 600 W.
(B) Value of x at which temperature is 50°C , is 70 cm
(C) Temperature is approximately 15.6°C at $x = 1\text{m}$
(D) Ratio of thermal resistance of rod between ($x = 0$ & $x = 1$) and ($x = 1$ & $x = 2$) is $\frac{27}{5}$
7. Two cylindrical copper vessels A and B have circular bases of thickness ' d ' and ' $2d$ ' respectively. Their diameters are R and $2R$ respectively. The heights of the standing water in vessels are $2h$ and h respectively. Both A and B are placed on hot plate having constant temperature $T_0 > 100^\circ\text{C}$. When the water in both vessels starts to boil,
(A) The rate of vaporisation of water in vessels A and B are in ratio of 1 : 2
(B) The ratio of time taken for the whole water in vessels to vaporise is 1 : 2
(C) The ratio of time taken for the whole water in vessels to vaporise is same for A & B
(D) The rates of vaporisation of water in vessels A & B are the same
8. A 100 cm long cylindrical flask with inner and outer diameter 2 cm and 4 cm respectively is completely filled with ice at 0°C as shown in the figure. The constant temperature outside the flask is 40°C . (Thermal conductivity of the flask is $0.693 \text{ W/m}^\circ\text{C}$, $L_{\text{ice}} = 80 \text{ cal/gm}$ & $\ln 2 = 0.693$). (Flat faces are thermally insulated)
(A) Initial rate of heat flow from outside to the flask is $80 \pi \text{ J/s}$
(B) Initial rate at which ice melts is $\frac{\pi}{4200} \text{ kg/s}$
(C) Initial rate at which ice melts is $100 \pi \text{ kg/s}$
(D) Initial rate of heat flow from outside to flask is $40 \pi \text{ J/s}$
9. Consider the shown case of a freezing lake due to negative environmental temperature ($-\theta^\circ\text{C}$). Thickness (x) of ice layer is small in comparison to depth of lake. Rate of increase in x will be greater
(A) if environmental temperature increases
(B) for larger thickness of ice layer
(C) if environmental temperature decreases
(D) for smaller thickness of ice layer



Linked Comprehension Type (Single Correct Answer Type)

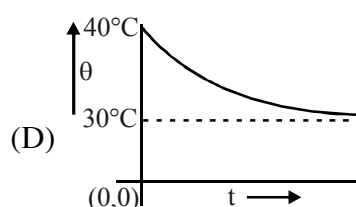
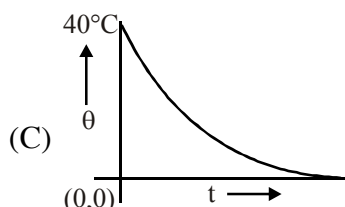
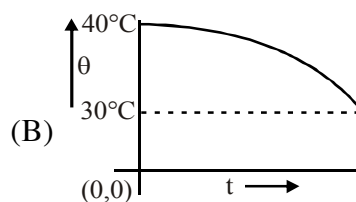
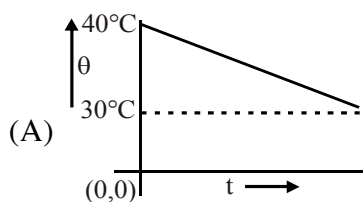
(2 Para × 3Q.)

[3 M (-1)]

Paragraph for Question No. 10 to 12

A body cools in a surrounding of constant temperature 30°C . Its heat capacity is $2\text{J}/^{\circ}\text{C}$. Initial temperature of the body is 40°C . Assume Newton's law of cooling is valid. The body cools to 36°C in 10 minutes.

10. In further 10 minutes it will cool from 36°C to :
(A) 34.8°C (B) 32.1°C (C) 32.8°C (D) 33.6°C
11. The temperature of the body in $^{\circ}\text{C}$ denoted by θ the variation of θ versus time t is best denoted as



12. When the body temperature has reached 36°C , it is heated again so that it reaches to 40°C in 10 minutes. Assume that the rate of loss of heat at 38°C is the average rate of loss for the given time. The total heat required from a heater by the body is :
(A) 7.2 J (B) 0.728 J (C) 16 J (D) 32 J

Paragraph for Question 13 to 15

A metal block is placed in a room which is at 10°C for long time. Now it is heated by an electric heater of power 500 W till its temperature becomes 50°C . Its initial rate of rise of temperature is $2.5^{\circ}\text{C}/\text{sec}$. The heater is switched off and now a heater of 100W is required to maintain the temperature of the block at 50°C . (Assume Newtons Law of cooling to be valid)

13. What is the heat capacity of the block?
(A) $50\text{ J}/^{\circ}\text{C}$ (B) $100\text{ J}/^{\circ}\text{C}$ (C) $150\text{ J}/^{\circ}\text{C}$ (D) $200\text{ J}/^{\circ}\text{C}$
14. What is the rate of cooling of block at 50°C if the 100W heater is also switched off?
(A) $5^{\circ}\text{C}/\text{s}$ (B) $0.5^{\circ}\text{C}/\text{s}$ (C) $1^{\circ}\text{C}/\text{s}$ (D) $0.1^{\circ}\text{C}/\text{s}$
15. What is the heat radiated per second when the block was 30°C ?
(A) 100 W (B) 80 W (C) 50 W (D) 30 W

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1Q.[3 M(0)]

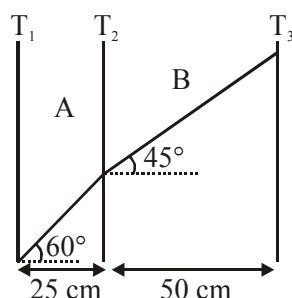
1. A metallic rod of cross-sectional area 9.0 cm^2 and length 0.54 m , with the surface insulated to prevent heat loss, has one end immersed in boiling water and the other in ice-water mixture. The heat conducted through the rod melts the ice at the rate of 1 gm for every 35 sec . Find the thermal conductivity of the rod in SI unit. [Latent heat of fusion of ice = 80 cal/g & $J = 4.2\text{ J/cal}$]

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

- A hot body placed in a surrounding of temperature θ_0 obeys Newton's law of cooling $\frac{d\theta}{dt} = -k(\theta - \theta_0)$, where $k = \frac{\ln 10}{5} \text{ s}^{-1}$. If the maximum heat that the body can loose is Q_0 . Then find the time (in seconds) starting from $t = 0$ at which the body will loose 90% of this maximum heat Q_0 .
- Two conductors A and B each of cross section area 5 cm^2 are connected in series. Variation of temperature (in $^\circ\text{C}$) along the length (in cm) is as shown in the figure. If thermal conductivity of A is $200 \text{ J/m-sec}^\circ\text{C}$ and thermal conductivity of B (in $\text{J/m-sec}^\circ\text{C}$ upto one decimal place.) is k then find $\sqrt{\frac{k}{8\sqrt{3}}}$.



SECTION-IV

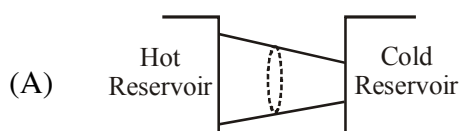
Matrix Match Type (4 × 4)

1 Q. [8 M (for each entry +2(0))]

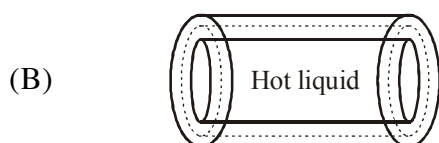
- Column I shows some heat conductors of various shapes. Column II shows certain parameters related with heat transfer. Match the column I with column II. Note that comparison is to be made in direction of heat flow and in steady state. In all the figures the surfaces shown in dotted are isothermal surfaces.

Column I

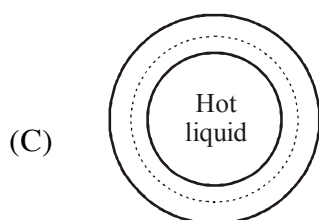
Column II



Tapering rod is well insulated



The flat faces are thermally insulating surrounding at low temperature



Hollow sphere surrounding at lower temperature

(P) Modulus of temperature gradient is increasing as we move from higher temperature to lower temperature

(Q) The rate of heat transfer across the dotted surfaces shown, is constant

(R) Modulus of Temperature gradient is decreasing as we move from higher temperature to lower temperature

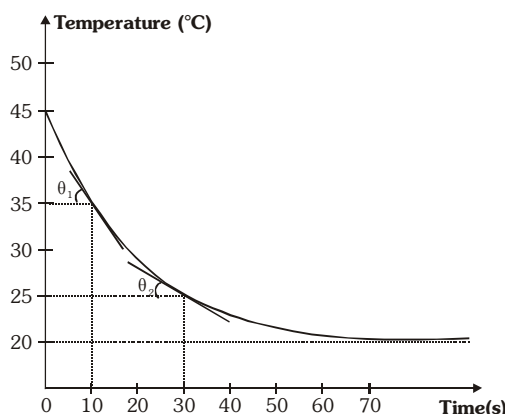
(S) Rate of heat transfer across dotted surface is variable

SECTION-I

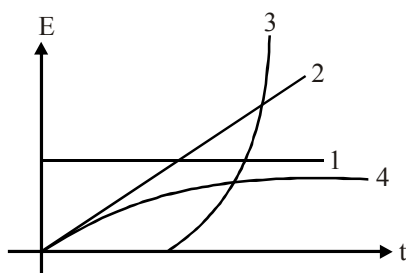
Single Correct Answer Type

6 Q. [3 M (-1)]

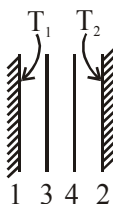
1. The temperature of a well stirred liquid kept open to a cold surrounding is plotted against time. The value of $\sec^2\theta_1$ is :-



- (A) $1 + 9 \tan^2\theta_2$ (B) $1 + \tan^2\theta_2$ (C) $1 + 3 \tan^2\theta_2$ (D) $3 + \tan^2\theta_2$
2. When a body is placed in surroundings at a constant temperature of 20°C , and heated by a 10-W heater, its temperature remains constant at 40°C . If the temperature of the body is now raised from 20°C to 80°C in 5 minutes at a uniform rate, the total heat it will lose to the surroundings will be- (Assume rate of heat loss is directly proportional to excess temperature over surrounding).
- (A) 3000 J (B) 3600 J (C) 4500 J (D) 5400 J
3. A ball of surface temperature T is in thermal equilibrium with its environment. Which of the curve gives the energy E emitted by the sphere as a function of time ' t '?



- (A) 1 (B) 2 (C) 3 (D) 4
4. Two large black plane surfaces are maintained at constant temperature T_1 and T_2 ($T_1 > T_2$). Two thin black plates are placed between the two surfaces and in parallel to these. After some time, steady conditions are obtained. What is the ratio of heat transfer rate between plate-1 & plate-3 to the ratio of original (when plate-3 & plate-4 was not present) heat transfer rate between plate-1 & plate-2 (η) in steady state?



- (A) $\eta = \frac{1}{2}$ (B) $\eta = \frac{1}{3}$ (C) $\eta = 1$ (D) $\eta = 0$

5. A copper sphere is suspended in an evacuated chamber maintained at 300 K. The sphere is maintained at a constant temperature of 500 K by heating it electrically. A total of 300 W of electric power is needed to do it. When **half** of the surface of the copper sphere is completely blackened, 600W is needed to maintain the same temperature of the sphere. Calculate the emissivity of copper.

(A) $e = \frac{1}{3}$ (B) $e = \frac{2}{3}$ (C) $e = \frac{1}{2}$ (D) $e = \frac{1}{6}$

6. The solar constant for a planet is S. The surface temperature of the sun is T K. The sun subtends an small angle θ at the planet. Find S. [Note: Solar constant is rate at which radiations are received from sun per unit surface area]

(A) $\frac{\sigma T^4 \theta^2}{4}$ (B) $\frac{\sigma T^4 \theta^2}{2}$ (C) $\frac{\sigma T^4 \theta}{4}$ (D) $\frac{\sigma T^4 \theta^2}{5}$

Multiple Correct Answer Type

5 Q. [4 M (–1)]

7. Two spherical black-bodies A and B, having radii r_A and r_B , where $r_B = 2r_A$ emit radiations with peak intensities at wavelengths 400 nm and 800 nm respectively. If their temperature are T_A and T_B respectively in Kelvin scale, their emissive powers are E_A and E_B and energies emitted per second are P_A and P_B then:

(A) $T_A / T_B = 2$ (B) $P_A / P_B = 4$ (C) $E_A / E_B = 8$ (D) $E_A / E_B = 4$

8. If a solid sphere and a hollow sphere of the same material and radius are heated to the same temperature:

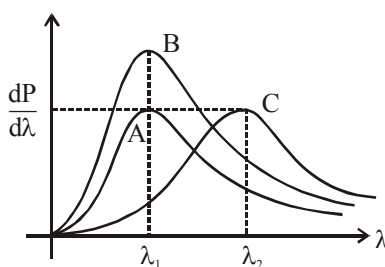
- (A) Both will emit equal amount of radiation per sec in the beginning.
(B) Both will absorb equal amount of radiation per sec in the beginning.
(C) The initial rate of cooling will be same for both the spheres.
(D) The two spheres will have same temperature at any instant.

9. Two identical objects A and B (emissivity e_A and e_B ; $e_A \neq e_B$) are placed in an enclosure. Temperature of body A, B and enclosure are same and constant :-

- (A) Heat emitted by body A is not equal to heat emitted by body B
(B) Heat absorbed by body A is not equal to heat absorbed by body B
(C) If $e_A > e_B$, body A absorbs more heat than body B
(D) If $e_A < e_B$, body A emits less heat than body B

10. Three stars radiate energy such that their radiated power depends on wavelength of radiation emitted as shown below. They can be considered as black bodies. The area under graph for A & C are equal. Which of the following statement(s) are **CORRECT**? dP is power emitted by star from wavelength λ

to $\lambda + d\lambda$. $\frac{dP}{d\lambda}$ is plotted against λ .



- (A) Emissive power of A = Emissive power of B < Emissive power of C.
(B) Temperature of A < Temperature of B = Temperature of C.
(C) Temperature of A = Temperature of B > Temperature of C.
(D) Radius of A < Radius of B; Radius of C > Radius of A.

11. A metal cylinder of mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C. The cylinder temperature rises uniformly to 25°C in 5 min and finally becomes constant at 45°C. Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings,
- (A) the rate of loss of heat of the cylinder to surrounding at 20°C is 2W
 (B) the rate of loss of heat of the cylinder to surrounding at 45°C is 12W
 (C) the rate of loss of heat of the cylinder to surrounding at 20°C is 5W.
 (D) the rate of loss of heat of the cylinder to surrounding at 45°C is 30W.

Linked Comprehension Type

(1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for question nos. 12 to 14

By varying the voltage applied to the kettle, you can change power consumption P. Depending on the P, kettle with water can be heated to different maximum temperatures. This dependence is shown in

Table-1

Power P (in Watt)	0	100	200	300
Temperature T(in °C)	20	40	60	80

Table-2 shows the variation of the temperature with time when the kettle of power 300W is switched off. The heat capacity of the empty kettle $C_0 = 100 \text{ J/K}$, specific heat of water $S_w = 4200 \text{ J/kg} \cdot \text{K}$. The density of water $\rho = 1000 \text{ kg/m}^3$. (Assume newton's law of cooling is valid)

Table-2

Time t (in sec)	0	60	300	600	1200	2400
Temperature T (in °C)	80	75	60	45	30	20

12. If the power consumption is 400W
- (A) The water will come to the boiling point and start boiling
 (B) The water will reach boiling point but will not boil
 (C) The water will not reach boiling point but start boiling
 (D) The water will neither reach boiling point nor start boiling.
13. The rate of heat loss at $t = 600 \text{ sec}$ after the heater is switched off (as in table-2) is.
- (A) 100 W (B) 125 W (C) 150 W (D) 200 W
14. How many litres of water is present in the kettle?
- (A) 1 litre (B) 0.629 litre (C) 0.415 litre (D) 0.798 litre

Paragraph for Question Nos. 15 & 16

According to Newton's law of cooling, rate of cooling of a body is directly proportional to the difference of temperature of body & temperature of surrounding, we should be very clear about the proportionality constant, which does not depend on the temperature of body.

15. A body is kept in room temperature of the room is 30°C. Initial temperature of the body is 40°C & becomes 36°C in 2 minutes, then find the time taken by the body to decrease the temperature further by 4°C.
- (A) 2 minute (B) 4 minute (C) 6 minute (D) 8 minute
16. A body is kept in room, at temperature 30°C. Initial temperature of the body is 40°C & becomes 36°C in 2 minutes. If the same body is taken to the other room with room temperature 14°, the find the time taken by the body to decrease the temperature from 36°C to 32°C will be
- (A) 4/5 minutes (B) 6 minutes (C) 4 minutes (D) None of these

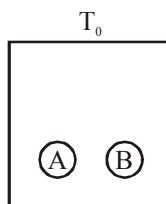
Matching List Type (4 × 4)

1Q.[3 M (–1)]

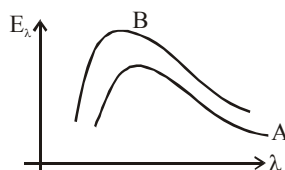
17. Four different situations are shown in list-I. Corresponding to the given situations relation between heat energy & temperature are shown in list-II. Match the correct list.

List-I

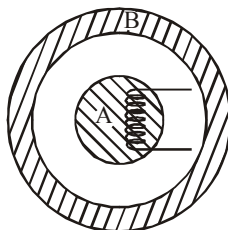
(P) A & B are 2 spheres of same size kept in an isothermal chamber at equilibrium is kept at constant temperature T_0 . A is black body & B is gray body with emissivity 0.8. (consider steady state)



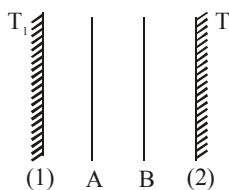
(Q) A & B are the graphs of spectral emissive power E_λ vs. wave length of two black bodies A and B of same surface area



(R) A is a solid sphere & B is a thick spherical shell. A is connected to a power source. Surfaces of A and B are perfect black bodies (consider steady state)



(S) (1) and (2) are large black body surfaces maintained at temperature T_1 and T_2 respectively. A and B are large plates with emissivity $e = 1$. (Consider steady state only, $T_1 > T_2$)



List-II

(1) $T_A > T_B$ or $T_A = T_B$

(2) $T_A < T_B$

(3) Heat energy radiated by A is greater than B

(4) Heat energy radiated by B is greater than A

Codes :

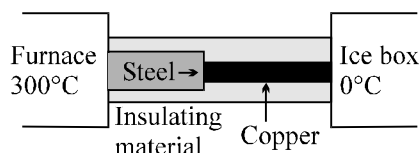
	P	Q	R	S
(A)	1	4, 2	3	2
(B)	3, 1	2, 4	1	1
(C)	3, 1	2	1	1
(D)	3, 1	2	3, 1	1

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. What is the temperature of the steel-copper junction in the steady state of the system shown in the figure. Length of the steel rod = 25 cm, length of the copper rod = 50 cm, temperature of the furnace = 300°C , temperature of the other end = 0°C . The area of cross section of the steel rod is twice that of the copper rod. If your answer is N (in $^\circ\text{C}$) fill value of $N/20$. (Thermal conductivity of steel = $50 \text{ J s}^{-1} \text{ m}^{-1} \text{ K}^{-1}$ and of copper = $400 \text{ J s}^{-1} \text{ m}^{-1} \text{ K}^{-1}$)

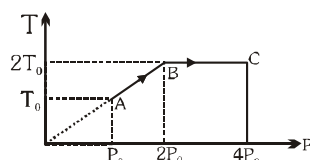


SECTION-I

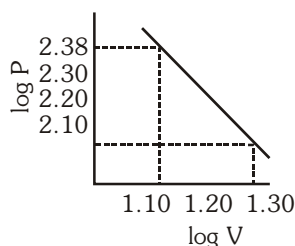
Single Correct Answer Type

12 Q. [3 M (–1)]

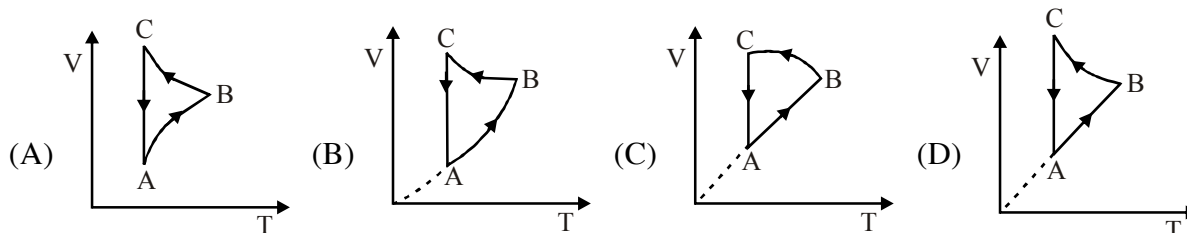
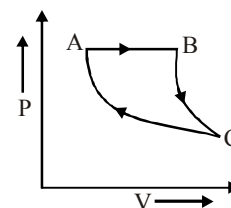
1. One mole of an ideal gas is taken through the process ABC as shown in the figure. The total work done on the gas is:



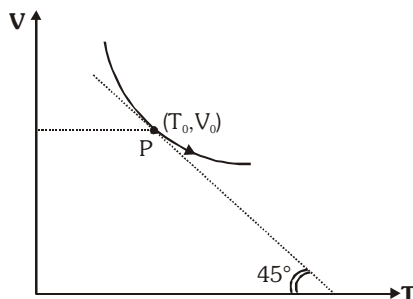
- (A) zero (B) $2RT_0 \ln 2$ (C) $-2RT_0 \ln 2$ (D) $4RT_0 \ln 2$
2. Two moles of helium gas ($\gamma = 5/3$), assumed ideal, are initially at 27°C and occupy a volume of 20 litres. The gas is first expanded at constant pressure till its volume is doubled. It then undergoes an adiabatic change until the temperature returns to its initial value. [$R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$]
- (A) Final volume of the gas is $75\sqrt{2}$ litre
(B) Final pressure of the gas is $0.44 \times 10^5 \text{ N/m}^2$
(C) Work done under isobaric process is 4765 J
(D) Work done under adiabatic process is 7506 J
3. 2 moles of an ideal monoatomic gas occupying volume V is adiabatically expanded from temperature 300K to a volume of $2\sqrt{2} V$. Then the final temperature & change in internal energy are respectively ($R = 8.3$)
- (A) 150 K, -3735 J (B) 140 K, -3735 J (C) 150 K, -3537 J (D) 140 K, -3537 J
4. Logarithm of readings of pressure and volume for an ideal monoatomic gas were plotted on a graph as shown in figure. By measuring the gradient, we can conclude that process is :



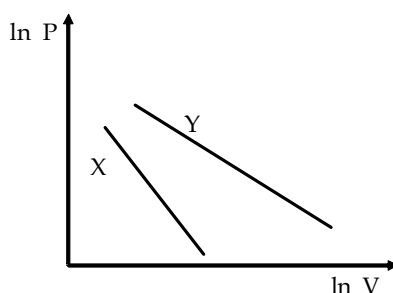
- (A) adiabatic (B) isothermal (C) isobaric (D) none of these
5. For a given thermodynamic process, the $P - V$ diagram is as shown below:
Which of the following is the $V - T$ diagram for the process?
- A $\rightarrow$ B : isobaric
B $\rightarrow$ C : adiabatic
C $\rightarrow$ A : isothermal



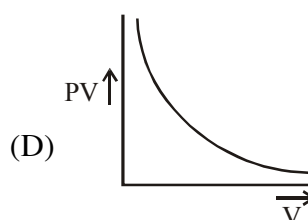
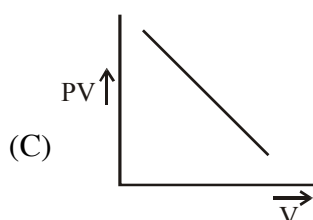
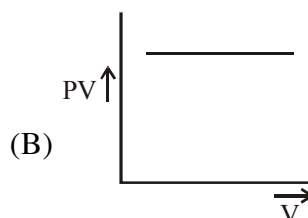
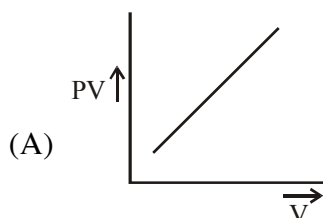
6. Figure shows the adiabatic curve for 2 moles of an ideal gas. The Bulk modulus $\left(\text{i.e. } B = \frac{dP}{-dV/V} \right)$ at the point P will be :-



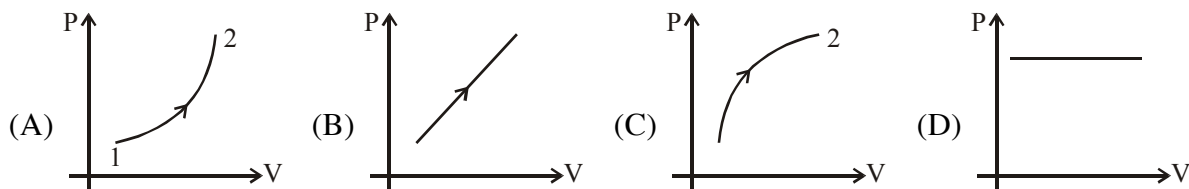
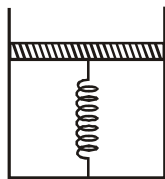
- (A) $R \left(1 + \frac{T_0}{V_0} \right)$ (B) $2R \left(1 + \frac{T_0}{V_0} \right)$ (C) $\frac{2RT_0}{V_0}$ (D) None of these
7. For two different gases X and Y, having degrees of freedom f_1 and f_2 and molar heat capacities at constant volume C_{v_1} and C_{v_2} respectively, for adiabatic process, the $\ln P$ versus $\ln V$ graph is plotted as shown :-



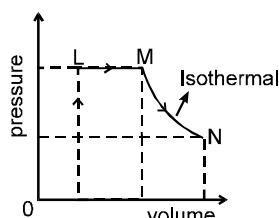
- (A) $f_1 > f_2$ (B) $f_2 > f_1$ (C) $C_{v_2} = C_{v_1}$ (D) $C_{v_1} > C_{v_2}$
8. Temperature of mixture of 2 moles of an ideal diatomic gas and 3 moles of an ideal monoatomic gas is increased by 3°C at constant pressure. What is the work done by the gas mixture? (R is universal gas constant)
- (A) $18 R$ (B) $24 R$ (C) $15 R$ (D) $27 R$
9. Which one of the following graphs represents the behaviour of an ideal gas for isothermal process.



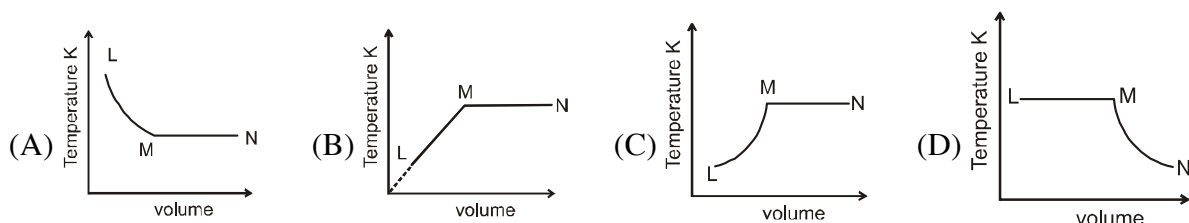
10. The piston is massless and the spring is ideal and initially stretched. The piston cylinder arrangement encloses an ideal gas. If the gas is heated quasistatically, the PV graph is :-



11. A fixed mass of an ideal gas undergoes changes of pressure and volume starting at L, as shown in Figure.



Which of the following is correct :



12. Consider an ideal diatomic gas undergoing a process $PT^{\frac{3}{2}} = \text{constant}$. The molar heat capacity for the process (R is universal gas constant) is :

- (A) $\frac{11R}{2}$ (B) $5R$ (C) $\frac{9}{4}R$ (D) $\frac{12}{5}R$

Multiple Correct Answer Type

2 Q. [4 M (-1)]

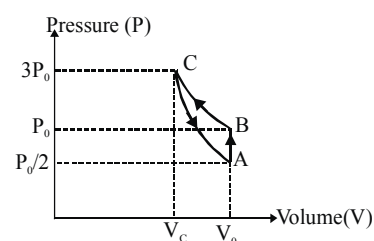
13. An ideal gas can be expanded from an initial state to a certain volume through two different processes : (i) $PV^2 = K$ and (ii) $P = KV^2$, where K is a positive constant. Then, choose the correct option from the following.

- (A) Final temperature in (i) will be greater than in (ii).
(B) Final temperature in (ii) will be greater than in (i).
(C) Work done by the gas in both the processes would be equal.
(D) Total heat given to the gas in (i) is greater than in (ii).

14. One mole of an ideal gas is carried through a thermodynamic cycle as shown in the figure. The cycle consists of an isochoric, an isothermal and an adiabatic processes. The adiabatic exponent of the gas is γ . Choose the correct option(s).

- (A) $\gamma = \frac{\ln 6}{\ln 3}$
(C) BC is adiabatic

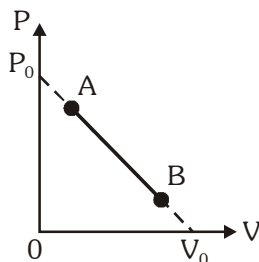
- (B) $\gamma = \frac{\ln 5}{\ln 3}$
(D) AC is adiabatic



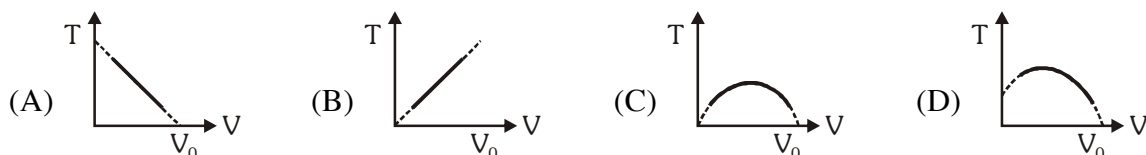
Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Questions no 15 to 17

One mole of an ideal monatomic gas undergoes a linear process from A to B, in which its pressure P and its volume V change as shown in figure.



15. The absolute temperature T versus volume V for the given process is



16. The maximum temperature of the gas during this process is

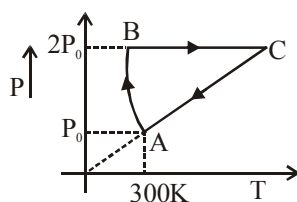
(A) $\frac{P_0 V_0}{2R}$ (B) $\frac{P_0 V_0}{4R}$ (C) $\frac{3P_0 V_0}{4R}$ (D) $\frac{3P_0 V_0}{2R}$

17. As the volume of the gas is increased, in some range of volume the gas expands with absorbing the heat (the endothermic process); in the other range the gas emits the heat (the exothermic process). Then the volume after which if the volume of gas is further increased the given process switches from endothermic to exothermic is

(A) $\frac{2V_0}{8}$ (B) $\frac{3V_0}{8}$ (C) $\frac{5V_0}{8}$ (D) None of these

Paragraph for Questions no 18 and 19

One mole of ideal monoatomic gas is taken through cyclic process ABC as shown in the graph with pressure temperature as coordinate axes. Process AB is defined as $PT = \text{constant}$. Take universal gas constant to be R then :



18. Work done on gas in process AB is :-

(A) $675 R$ (B) $975 R$ (C) $300 R$ (D) None of these

19. Decrease in internal energy of gas in process CA is :-

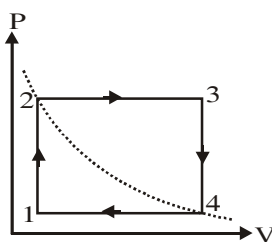
(A) $-350 R$ (B) $-450 R$ (C) $-550 R$ (D) None of these

SECTION-I

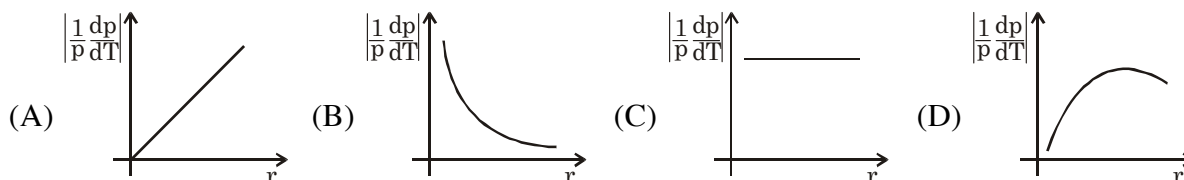
Single Correct Answer Type

6 Q. [3 M (-1)]

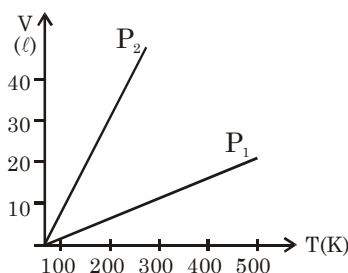
1. One mole of an ideal gas undergoes a cyclic process, consisting of two isochores and two isobars. Temperature at 1 and 3 equal to T_1 and T_3 respectively. The work done by the gas over the cycle, if the point 2 and 4 lie on the same isotherm :-



- (A) $\frac{R(T_1 + T_3)}{2}$ (B) $R(\sqrt{T_3} - \sqrt{T_1})^2$ (C) $\frac{R}{2}(\sqrt{T_1} + \sqrt{T_3})^2$ (D) $R\sqrt{T_1 T_3}$
2. The radius of a spherical enclosure containing a monatomic gas increases linearly with temperature such that $r = r_0 + kT$. If the walls of the spherical enclosure is adiabatic. Choose the correct graph of variation of $\left| \frac{1}{p} \frac{dp}{dT} \right|$ with radius (r) :-

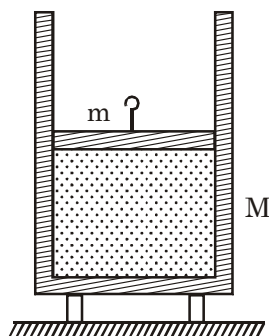


3. Volume versus temperature graphs for a given mass of an ideal gas are shown in figure at two different values of constant pressure. What can be inferred about relation between P_1 & P_2 ?

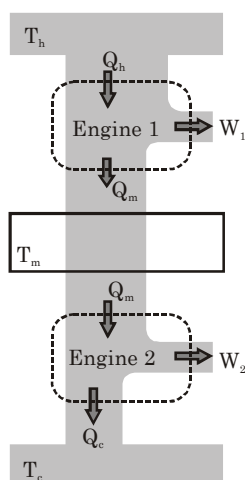


- (A) $P_1 > P_2$ (B) $P_1 = P_2$ (C) $P_1 < P_2$ (D) data is insufficient

4. As shown, a piston chamber of cross section area A is filled with an ideal gas. A sealed piston of mass m is at the middle height of the cylinder at equilibrium. The friction force between the chamber wall and the piston can be ignored. The mass of the rest of the chamber is M . The atmosphere pressure is P_0 . If the piston is pulled slowly up to full height of chamber and temperature remains unchanged. The maximum value of M such that the chamber can be lifted off the ground is :



- (A) $M = \frac{P_0 A - 2mg}{2g}$ (B) $M = \frac{P_0 A - mg}{2g}$ (C) $M = \frac{P_0 A - mg}{g}$ (D) $\frac{2(P_0 A - mg)}{g}$
5. Vessel A contains an ideal gas at a pressure 5×10^5 Pa and is connected with a heat source which maintains its temperature at 300 K. Another vessel B which has four times greater inner volume contains the same gas at a pressure 10^5 Pa and is connected to a heat source which maintains its temperature at 400 K. What will be the pressure of entire system if two vessels are connected by a narrow tube tap :-
- (A) 10^5 Pa (B) 2×10^5 Pa (C) 4×10^5 Pa (D) 5×10^5 Pa
6. Suppose that two heat engines are connected in series, such that the heat released by the first engine is used as the heat absorbed by the second engine, as shown in figure. The efficiencies of the engines are ϵ_1 and ϵ_2 , respectively. The net efficiency of the combination is given by

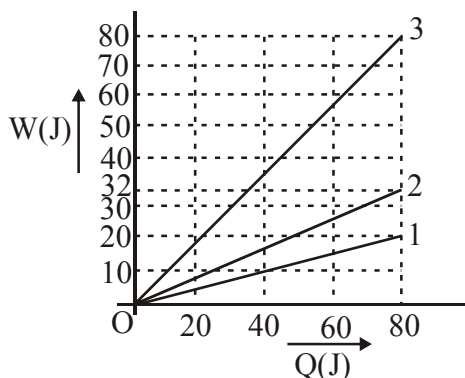


- (A) $\epsilon_{\text{net}} = \epsilon_1 + \epsilon_2$ (B) $\epsilon_{\text{net}} = \epsilon_1 + \epsilon_2 - \epsilon_1 \epsilon_2$ (C) $\epsilon_1 + \epsilon_2 - \sqrt{\epsilon_1 \epsilon_2}$ (D) $\epsilon_1 + \epsilon_2 - 2\epsilon_1 \epsilon_2$

Multiple Correct Answer Type

6 Q. [4 M (-1)]

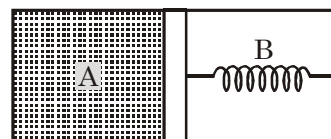
7. In the figure shown, the amount of heat supplied to one mole of an ideal gas is plotted on the horizontal axis and the amount of work performed by the gas is drawn on the vertical axis. One of the straight lines in the figure is an isotherm and the other two are isobars of two gases. The initial states of both gases are same. Mark the **CORRECT** statement(s).



- (A) Curve 3 corresponds to isothermal process
(B) Curve 1 corresponds to a polyatomic gas
(C) Curve 2 corresponds to a monatomic gas
(D) Process 1 and 2 are isobaric process.
8. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless and insulated piston of cross section area S into two equal parts A and B. Part A has an ideal diatomic gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant K is connected with the piston and the wall of the container as shown. Initially the spring is unstretched. The ideal gas in chamber A is allowed to expand slowly with the help of an external agent. After achieving equilibrium state, external agent is removed. Let in equilibrium the spring be compressed by a length x_0 . then

(A) Final pressure of the gas is Kx_0/S

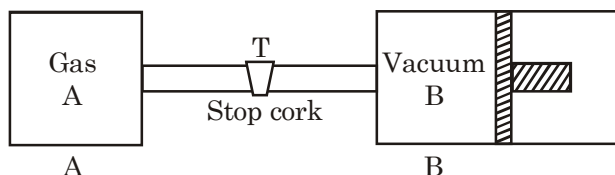
(B) Work done by the gas is $\frac{5}{2} \left[P_0 V_0 - Kx_0^2 - \frac{Kx_0 V_0}{S} \right]$



(C) Magnitude of change in internal energy of the gas is $\frac{1}{2} Kx_0^2$

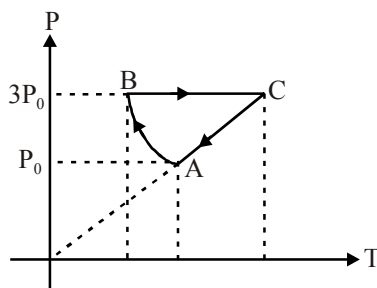
(D) Work done by an external agent is $\frac{5V_0}{2} \left[\frac{Kx_0}{S} - P_0 \right] + 3Kx_0^2$

9. An ideal diatomic gas is confined in a cylinder A of volume V_0 . Then, cylinder is connected to another cylinder B with the help of tube of negligible volume. The cylinder B is fitted with movable piston, which can be adjusted from outside. Initially, piston is adjusted so that the volume of B is same as volume of A, i.e., V_0 . B is evacuated and stop cork is opened so that gas expands and occupies the volume $2V_0$.



- (A) During this free expansion, the internal energy of this system remains constant
(B) With open stop cork, now the piston is slowly moved to compress the gas back to cylinder A at constant temperature T . Then for n number of moles of gas, work done on the gas is $nRT \ln 2$
(C) The heat absorbed by the gas in the above process is $-nRT \ln 2$
(D) With open stop cork, now the piston is slowly moved to compress the gas back to cylinder A at constant temperature T . Then for n moles gas, work done on the gas is $\frac{1}{2} (nRT \ln 2)$

10. A process for monoatomic gas is defined by $V = aT^2$, where a is a constant. In this process temperature of 1 mole of a gas is increased by 100 K; then (V = Volume, T = Temperature)
- (A) Molar heat capacity of this process is negative.
 (B) The work done by the gas in the process is 250 R.
 (C) The work done by the gas in the process is 200 R.
 (D) The work done is given by $\int_{v_1}^{v_2} PdV$ in which P is a function of volume.
11. One mole of ideal monoatomic gas expands according to the law $P \propto \frac{1}{V^2}$. Initial temperature of gas is T_0 , Select the **CORRECT** statement :-
- (A) Workdone by the gas is $\frac{RT_0}{2}$ when its volume is doubled.
 (B) Temperature is increasing in the process.
 (C) Molar specific heat capacity of gas in the process is $\frac{R}{2}$
 (D) Heat is absorbed by the gas in the process.
12. One mole of monoatomic gas is taken through cyclic process shown below. $T_A = 300$ K. Process AB is defined as $PT = \text{constant}$.



- (A) Work done in process AB is $-400 R$.
 (B) Change in internal energy in process CA is $900 R$.
 (C) Heat transferred in the process BC is $2000 R$.
 (D) Change in internal energy in process CA is $-900 R$.

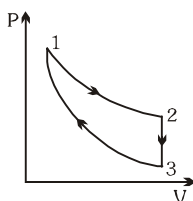
Linked Comprehension Type (2 Para $\times$ 3Q.)

[3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question 13 to 15

Three processes compose a thermodynamics cycle shown in the PV diagram. Process 1 $\rightarrow$ 2 takes place at constant temperature. Process 2 $\rightarrow$ 3 takes place at constant volume, and process 3 $\rightarrow$ 1 is adiabatic. During the complete cycle, the total amount of work done is 10 J. During process 2 $\rightarrow$ 3, the internal energy decrease by 20J and during process 3 $\rightarrow$ 1, 20 J of work is done on the system.



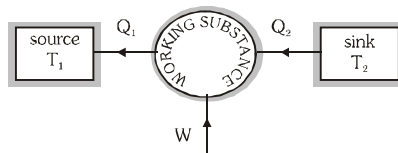
13. How much heat is added to the system during process 1 $\rightarrow$ 2 is
- (A) 0 (B) 10 J (C) 20 J (D) 30 J

14. Change in temperature in $2 \rightarrow 3$ if sample consists of one mole of monoatomic gas is
(A) $10/3R$ (B) $20/3R$ (C) $30/3R$ (D) $40/3R$
15. Average molar specific heat for process $1 \rightarrow 2 \rightarrow 3$ is
(A) $R/4$ (B) $R/2$ (C) $3R/4$ (D) R

Paragraph for Question 16 to 18

Refrigerator is an apparatus which takes heat from a cold body, work is done on it and the work done together with the heat absorbed is rejected to the source. An ideal refrigerator can be regarded as Carnot's ideal heat engine working in the reverse direction.

The coefficient of performance of refrigerator is defined as



$$\beta = \frac{\text{Heat extracted from cold reservoir}}{\text{work done on working substance}} = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

A Carnot's refrigerator takes heat from water at 0°C and discards it to a room temperature at 27°C . 1kg of water at 0°C is to be changed into ice at 0°C . ($L_{\text{ice}} = 80 \text{ kcal/kg}$)

16. How many calories of heat are discarded to the room ?
(A) 72.8 kcal (B) 87.9 kcal (C) 80 kcal (D) 7.9 kcal
17. What is the work done by the refrigerator in this process ($1 \text{ cal} = 4.2 \text{ joule}$)
(A) 7.9 kJ (B) 33.18 kJ (C) 43.18 kJ (D) 23.18 kJ
18. What is the coefficient of performance of the machine ?
(A) 11.1 (B) 10.1 (C) 9.1 (D) 8.1

SECTION-I

Single Correct Answer Type

13 Q. [3 M (-1)]

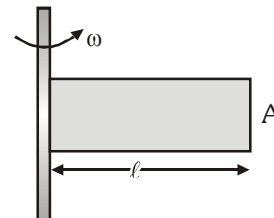
1. A closed cylinder of length ' ℓ ' containing a liquid of variable density $\rho(x) = \rho_0 (1 + \alpha x)$. Find the net force exerted by the liquid on the axis of rotation. (Take the cylinder to be massless and A = cross sectional area of cylinder)

(A) $\rho_0 A \omega^2 \ell^2 \left[\frac{1}{2} + \frac{1}{3} \alpha \ell \right]$

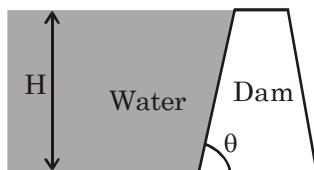
(B) $\rho_0 A \omega^2 \ell^2 \left[\frac{1}{2} + \frac{2}{3} \alpha \ell \right]$

(C) $\rho_0 A \omega^2 \ell^2 \left[\frac{1}{2} + \alpha \ell \right]$

(D) $\rho_0 A \omega^2 \ell^2 \left[\frac{1}{2} + \frac{4}{3} \alpha \ell \right]$



2. An open cubical tank was fully filled with water. When the tank was accelerated on a horizontal plane along one of its side was found that one third of volume of water spilled out. The acceleration was :-
(A) $g/3$ (B) $2g/3$ (C) $3g/2$ (D) None of these
3. A dam (height H , width W , one wall inclined at angle θ) is fully filled with water (density ρ). If g is the acceleration due to gravity, the resultant force exerted by the water on the dam wall is given by (Neglect atmospheric pressure) :-



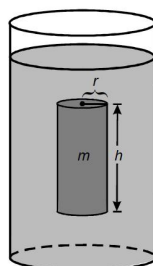
(A) $\frac{(\rho g H^2 W)}{2}$

(B) $\frac{(\rho g H^2 W)}{(2 \sin \theta)}$

(C) $\frac{(\rho g H W)}{(2 \cos \theta)}$

(D) $\frac{(\rho g H^2 W)}{(2 \cos \theta)}$

4. A right circular cylinder has a mass m , radius r , and a height h . The cylinder is completely submerged in a fluid of density ρ , as shown in the diagram and is released. What is the magnitude of the net force on the cylinder initially?



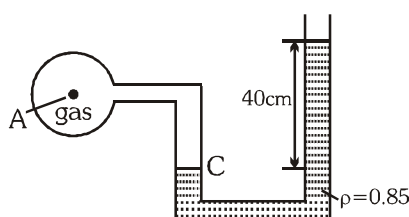
(A) $r \rho g h + m g$

(B) $|r \rho g h - m g|$

(C) $\pi r^2 \rho g h + m g$

(D) $|\pi r^2 \rho g h - m g|$

5. The manometer shown in the fig. contains oil of relative density or specific gravity $s = 0.85$. Determine the absolute pressure in the bulb if the atmospheric pressure is 100 kPa. ($g = 10 \text{ m/s}^2$)



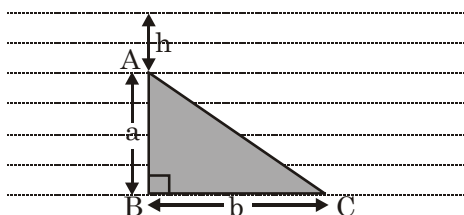
(A) 100.4 kPa

(B) 109 kPa

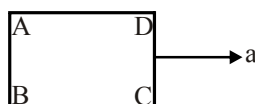
(C) 103.4 kPa

(D) 120.4 kPa

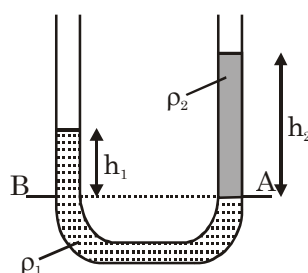
6. A vertical triangular plate ABC is placed inside water with side BC parallel to water surface as shown. The force on one surface of plate by water is (density of water is ρ and atmospheric pressure P_0) :-



- (A) $(P_0 + h\rho g)\frac{1}{2}ab$ (B) $(P_0 + h\rho g)\frac{1}{2}ab + \frac{a^2b}{3}\rho g$
(C) $[P_0 + \rho g(h + a)]\frac{1}{2}ab$ (D) 0
7. In a cylindrical container open to the atmosphere from the top a liquid is filled upto 10 m depth. Density of the liquid varies with depth from the surface as $\rho(h) = 100 + 6h^2$ where h is in meter and ρ is in kg/m^3 . The pressure at the bottom of the container will be : (atmosphere pressure = 10^5 Pa , $g = 10 \text{ m/sec}^2$)
(A) $1.7 \times 10^5 \text{ Pa}$ (B) $1.4 \times 10^5 \text{ Pa}$ (C) $1.6 \times 10^5 \text{ Pa}$ (D) $1.3 \times 10^5 \text{ Pa}$
8. A liquid is kept in a cylindrical vessel. When the vessel is rotated about its axis, the liquid rises at its sides. If the radius of the vessel is 0.05 m and the speed of rotation is 2 revolutions per second, the difference in the heights of the liquid at the centre and at the sides of the vessels will be (take $g = 10 \text{ ms}^{-2}$ and $\pi^2 = 10$)
(A) 2 cm (B) 4 cm (C) 1 cm (D) 8 cm
9. A closed rectangular tank is completely filled with water and is accelerated horizontally with an acceleration towards right, side view of container is given as shown. Pressure is (i) maximum at, and (ii) minimum at

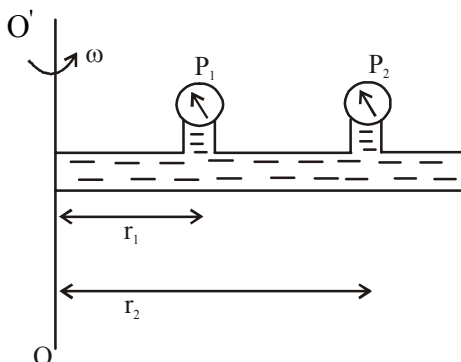


- (A) (i) B (ii) D (B) (i) C (ii) D (C) (i) B (ii) C (D) (i) B (ii) A
10. Two immiscible liquids are poured in a U-tube having densities ρ_1 and ρ_2 . The ratio of height of the liquids above their interface $\left(\frac{h_1}{h_2}\right)$ is :



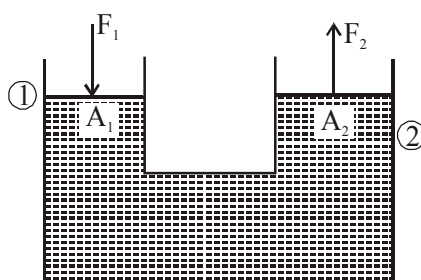
- (A) directly proportional to their densities
(B) inversely proportional to their densities
(C) directly proportional to square of their densities
(D) equal

11. A tube filled with water and closed at both ends rotates uniformly in a horizontal plane about the axis OO' . The manometers fixed in the tube wall at distances r_1 and r_2 from the rotational axis indicate pressures P_1 and P_2 respectively. If is the ρ_w density of water then find the angular velocity of rotation of tube.

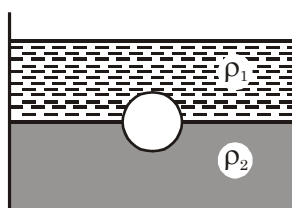


- (A) $\sqrt{\frac{2(P_2 - P_1)}{(r_2 - r_1)^2 \rho_w}}$ (B) $\sqrt{\frac{2(P_2 - P_1)}{(r_2^2 - r_1^2) \rho_w}}$ (C) $\sqrt{\frac{(P_2 - P_1)}{(r_2^2 - r_1^2) \rho_w}}$ (D) $\sqrt{\frac{(P_2 - P_1)}{(r_2 - r_1)^2 \rho_w}}$

12. The diagram below shows a hydraulic lift. A force is applied at side 1 and an output force is generated at side 2. Which of the following is true? (Given : $A_2 > A_1$)



- (A) The force at side 1 is greater than the force at side 2
(B) The force at side 1 is less than the force at side 2
(C) The pressure at side 1 is greater than the pressure at side 2
(D) The pressure at side 1 is less than the pressure at side 2
13. A solid uniform ball of volume V floats on the interface of two immiscible liquids (figure). The specific gravity of the upper liquid is ρ_1 and that of lower one is ρ_2 and the specific gravity of ball is ρ ($\rho_1 < \rho < \rho_2$). The fraction of the volume of the ball in the upper liquid is :

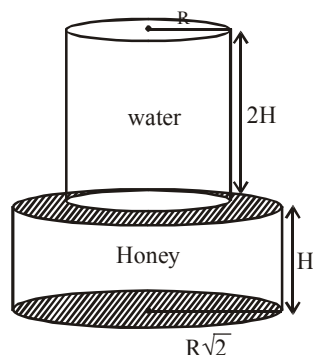


- (A) $\frac{\rho_2}{\rho_1}$ (B) $\frac{\rho_2 - \rho}{\rho_2 - \rho_1}$ (C) $\frac{\rho - \rho_1}{\rho_2 - \rho_1}$ (D) $\frac{\rho_1}{\rho_2}$

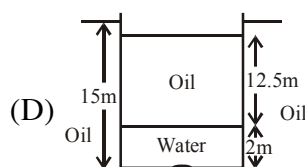
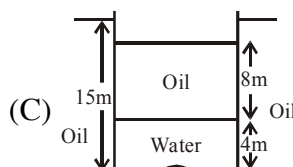
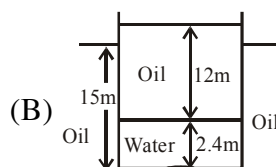
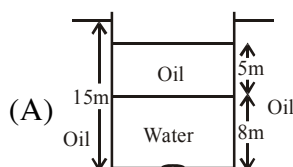
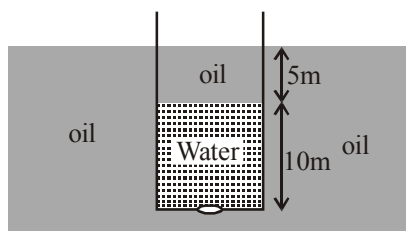
Multiple Correct Answer Type

2 Q. [4 M (-1)]

14. A bottle is kept on the ground as shown in the figure. The bottle can be modelled as having two cylindrical zones. The lower zone of the bottle has a cross-sectional radius of $R\sqrt{2}$ and is filled with honey of density 2ρ . The upper zone of the bottle is filled with the water of density ρ and has a cross-sectional radius R . The height of the lower zone is H while that of the upper zone is $2H$. If now the honey and the water parts are mixed together to form a homogeneous solution. (Assume that total volume does not change)



- (A) The pressure inside the bottle at the base will remain unaltered
(B) The normal reaction on the bottle from the ground will remain unaltered
(C) The pressure inside the bottle at the base will increase by an amount $\left(\frac{1}{2}\right)\rho gH$
(D) The pressure inside the bottle at the base will decrease by an amount $\left(\frac{1}{4}\right)\rho gH$
15. A vessel of uniform cross-section open at the top with an orifice at its bottom contains oil (relative density 0.8) on top of water. It is immersed vertically in a large open tank of same oil as shown in figure. In which of the following configuration will liquid level remain same. (Water does not come out or oil does not enter vessel) :- (Take : area of orifice = 1 mm^2 and area of vessel as 100 cm^2)



Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

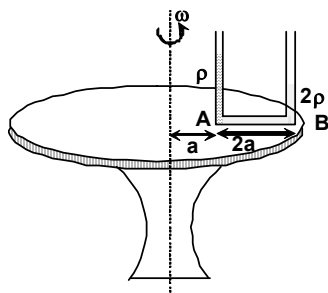
Paragraph for Questions 16 to 18

A U tube containing two different liquid of density ρ and 2ρ is fixed vertically on a rotating table as shown in the figure. Initially table is at rest and table can rotate about a vertical axis passing through the centre of the table. The interface of two liquids of densities ρ and 2ρ respectively lies at the point A in a

U tube at rest. The height of liquid column above A is $\frac{8}{3}a$, where $AB = 2a$. The cross-sectional area of

the tube is S. Now the table is whirled with angular velocity ω about a vertical axis as shown in figure

such that the interface of the liquids shifts towards B by $\frac{2}{3}a$, then at this instant ;



16. Find $P_B - P_A$.

- (A) ρga (B) $2\rho ga$ (C) $3\rho ga$ (D) $4\rho ga$

17. Find the force exerted by liquid of ρ on liquid of 2ρ at the interface.

- (A) $\rho S a g \left(\frac{9}{4} \right)$ (B) $\rho S a g \left(\frac{29}{3} \right)$ (C) $\rho S a g \left(\frac{9}{10} \right)$ (D) $\rho S a g \left(\frac{29}{10} \right)$

18. Find the value of ω :-

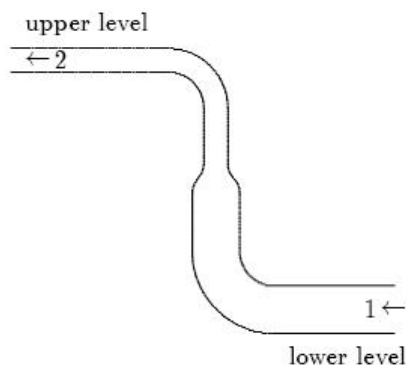
- (A) $\sqrt{\frac{9g}{4a}}$ (B) $\sqrt{\frac{9g}{32a}}$ (C) $\sqrt{\frac{9g}{8a}}$ (D) $\sqrt{\frac{9g}{16a}}$

SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

1. A concrete sphere of radius R has a cavity of radius r which is packed with sawdust. The specific gravities of concrete and sawdust are respectively 2.4 and 0.3. For this sphere to float with its entire volume submerged under water, ratio of mass of concrete to mass of sawdust will be :-
(A) 8 (B) 4 (C) 3 (D) None of these
2. A piece of copper having an internal cavity weighs 264 g in air and 221 g in water. If the density of copper is 8.8 g/cm^3 , find the volume of the cavity :-
(A) 33 cm^3 (B) 23 cm^3 (C) 13 cm^3 (D) 9 cm^3
3. A balloon of total mass ' M ' and a fixed size starts coming down with an acceleration f ($f < g$). The fraction of the total mass of the balloon which must be dropped from it so that it starts going up with an acceleration of ' f ' (assuming negligible air resistance) is :-
(A) $\frac{f}{g+f}$ (B) $\frac{2f}{(g+f)}$ (C) $\frac{g+f}{f}$ (D) $\frac{2f}{g+2f}$
4. The reading of a spring balance when a mass is weighed on it in air is 40gm. When the mass is immersed in water, its reading is 20gm. The specific gravity of the mass is :
(A) 3 (B) 2 (C) 1.5 (D) 2.5
5. An ice cube floats in water at 0°C . The cube has a mass 900 gm and density of 0.9 gm/cc . Water has density of 1 gm/cc . Find the change in gravitational potential energy of the system when entire ice melts. The temperature of water remains at 0°C .
(A) Zero (B) Increases of 45 mJ
(C) Decreases by 45 mJ (D) Decreases by 36 mJ
6. Water is pumped through the hose shown below, from a lower level to an upper level. Compared to the water at point 1, the water at point 2 :-

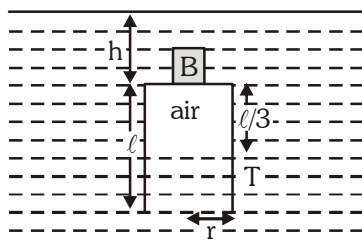


- | | |
|--|---|
| (A) has greater speed and greater pressure | (B) has greater speed and less pressure |
| (C) has less speed and less pressure | (D) has less speed and greater pressure |

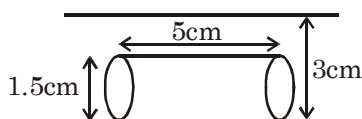
Multiple Correct Answer Type

4 Q. [4 M (–1)]

7. A light cylindrical tube 'T' of length ℓ and radius 'r' containing air is inverted in water (density d). One end of the tube is open and the other is closed. A block 'B' of density $2d$ is kept on the tube as shown in the figure. the tube stays in equilibrium in the position shown. (Assume the atmospheric pressure is to be P_0). Pick up the correct statement(s). (Neglect mass of tube and mass of air enclosed).



- (A) the volume of block B is $\frac{\pi r^2 \ell}{3}$
- (B) the volume of block B is $\frac{2\pi r^2 \ell}{3}$
- (C) the pressure of air trapped in the tube is $P_0 + dg \left(h + \frac{\ell}{3} \right)$
- (D) the pressure of air trapped in the tube is $P_0 + dg \left(h + \frac{2\ell}{3} \right)$
8. A cylindrical glass of height 5cm and diameter 1.5 cm is lying horizontally in a large tank filled with water to height 3cm. The wall have negligible thickness. Now the glass is slowly raised from open end as shown.

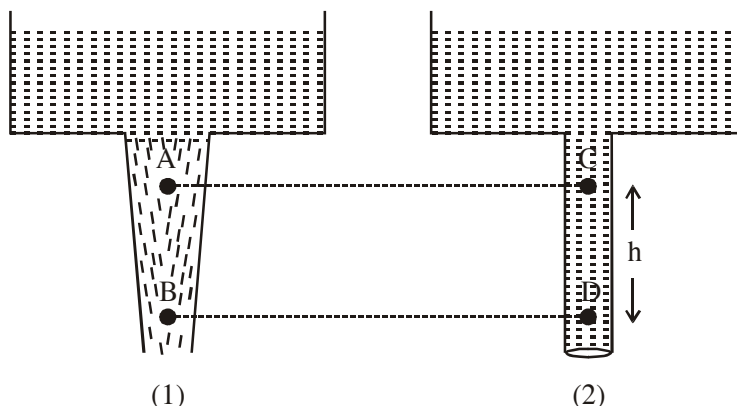


It is finally made to stand on it's base. The mass of glass is negligible. $\rho_{\text{water}} = 1000 \text{ kg/m}^3$

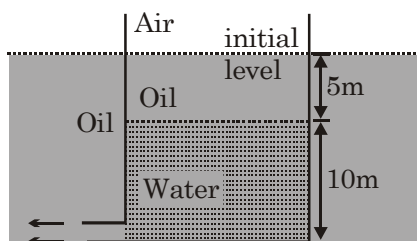


- (A) Finally, the glass is 40 % empty.
- (B) The normal force on glass when it is standing is $\frac{9\pi}{1600} \text{ N}$
- (C) Finally, the glass is 20% empty.
- (D) The normal force on glass when it is standing is zero.

9. Consider two set up shown in diagram, liquid is ideal (surface tension and viscosity can be neglected). Choose correct option(s).



- (A) $P_B - P_A = P_D - P_C$ i.e. pressure difference between corresponding points in two cases is same
 (B) $P_B - P_A < P_D - P_C$
 (C) $P_B - P_A > P_D - P_C$
 (D) Bernaulli's theorem is applicable in both vessels.
10. A tank with a small orifice contains oil on top of water. It is immersed in a large tank of the same oil. Water flows through the hole. Density of oil is 800 kg/m^3 .



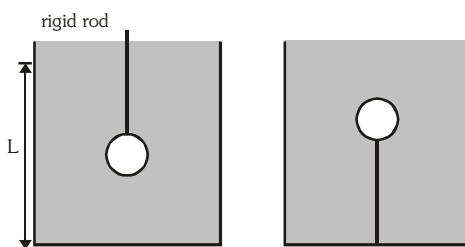
- (A) The velocity of this flow initially is $\sqrt{40} \text{ m/s}$
 (B) When the flow stops, the position of the oil-water interface in the tank is 8m from orifice.
 (C) The velocity of this flow decreases and finally becomes zero.
 (D) The velocity of this flow initially is $\sqrt{280} \text{ m/s}$.

Linked Comprehension Type (Single Correct Answer Type)

(2 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Question no. 11 to 13

A cylindrical container of length L is full to the brim with a liquid which has mass density ρ . It is placed on a weight-scale; the scale reading is w . A light ball which would float on the liquid if allowed to do so, of volume V and mass m is pushed gently down and held beneath the surface of the liquid with a rigid rod of negligible volume as shown on the left.

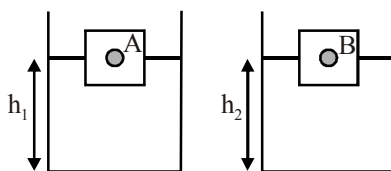


11. What is the mass M of liquid which overflowed while the ball was being pushed into the liquid?
 (A) ρV (B) m (C) $m - \rho V$ (D) none of these

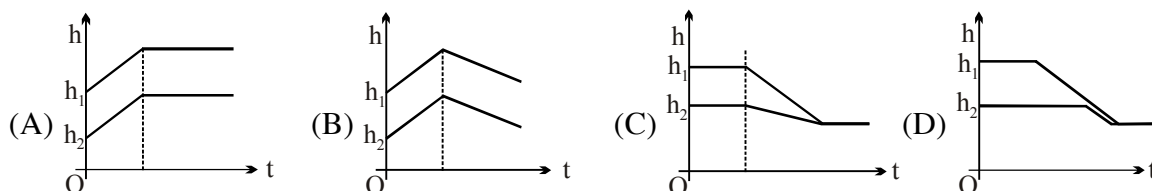
12. What is the reading of the scale when the ball is fully immersed
(A) $w - \rho Vg$ (B) w (C) $w + mg - \rho Vg$ (D) none of these
13. If instead of being pushed down by a rod, the ball is held in place by a thin string attached to the bottom of the container as shown on the right. What is the tension T in the string?
(A) $(\rho V - m)g$ (B) ρVg (C) mg (D) none of these

Paragraph for Question no. 14 to 16

Two ice cubes of side 10 cm, having cavity of volume 20 cm^3 at centre of cube but filled with different materials A and B respectively. The specific gravity of material A is 4.9 and specific gravity of material B is 1.9. Now these cubes are placed in two different vessels of same base area as shown in figure. The water level before putting blocks in vessels are same. Assume that ice melts uniformly from all sides and with same constant rate in both the vessels. (specific gravity of ice = 0.9)



14. Find the ratio of initial submerged volumes of the blocks containing A and B respectively
(A) $\frac{49}{46}$ (B) 1 : 1 (C) $\frac{49}{19}$ (D) 4 : 1
15. Choose the correct statement :
(A) Both cubes sink simultaneously
(B) A and B sink only after complete melting of ice.
(C) A sinks prior to B
(D) B sinks prior to A
16. Choose the correct graph showing the variation of heights of water-level in two vessels with time



SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. A vertical glass tube of length $\ell = 2\text{m}$ is open both the ends. It is immersed in water in vertical position. After lifting it out while we close the upper end with a finger, we find that half the length of the tube is filled with water. What was the length of air column (in dm) before lifting out? (Take $P_{\text{atm}} = 10^5 \text{ N/m}^2$ and neglect surface tension).

SECTION-IV

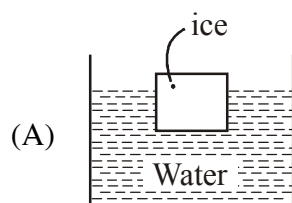
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

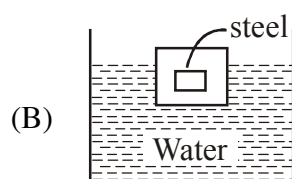
1. When ice melts completely, level of liquid in which ice is submerged

Column-I

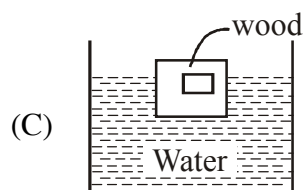
Column-II



(P) Increases



(Q) Decreases



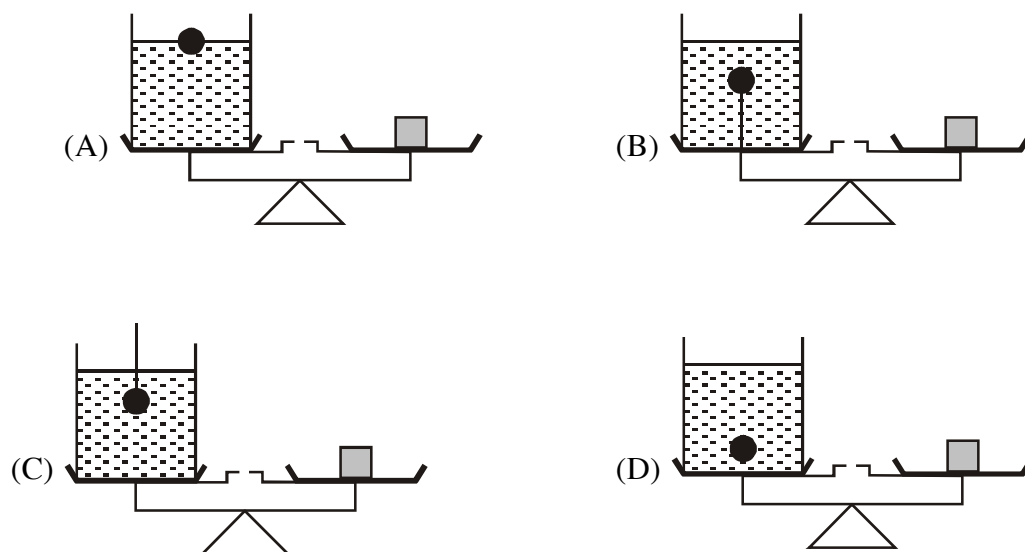
(R) Remains same

SECTION-I

Single Correct Answer Type

7 Q. [3 M (-1)]

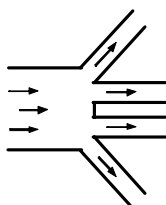
1. With a wooden ball and a high water tank we hold four experiments. In the first experiment we have a floating ball weighed in a vessel. In the second experiment, the ball is weighted while tied to the bottom of the vessel, the third time the ball held under water by using the thin rod, finally, during the fourth experiment the ball is taken deep inside and released. In all the four cases weight is recorded. In which case the weight recorded is more than the weight of ball + water + tank?



2. A cylindrical vessel filled with water upto a height h weighs w and is resting on a horizontal plane. The side wall of the vessel has a plugged circular hole touching the bottom. The coefficient of friction between the bottom of the vessel and the plane is μ . The minimum diameter of the hole so that the vessel just begins to move on the floor after the removal of the plug is

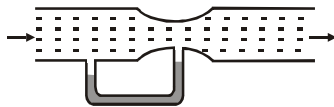
(A) $\sqrt{\frac{\pi\mu w}{\rho hg}}$ (B) $\sqrt{\frac{\mu w}{2\pi\rho hg}}$ (C) $\sqrt{\frac{\mu w}{\pi\rho hg}}$ (D) $\sqrt{\frac{2\mu w}{\pi\rho hg}}$

3. Water is flowing through a channel that is 12 m wide with a speed of 0.75 m/s. The water then flows into four identical channels that have a width of 4.0 m each. The depth of the water does not change as it flows into the four channels. What is the speed of the water in one of the smaller channels?

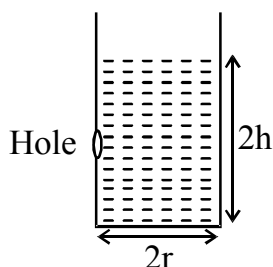


(A) 0.56 m/s (B) 2.3 m/s (C) 0.25 m/s (D) 0.75 m/s

4. Water flows through the tube shown. Area of cross-section of wide and narrow part are 5 cm^2 & 2 cm^2 . The rate of flow is $500 \text{ cm}^3/\text{sec}$. Find difference in mercury level of U-tube :-



- (A) 2.9 cm (B) 1.9 cm (C) 0.9 cm (D) None of these
5. A light cylindrical vessel of radius r is kept on a rough horizontal surface so that it cannot slide but can topple. It is filled with water upto a height ' $2h$ ' and a very small hole of area ' a ' is punched in it so that the water that will come out of it falls at the maximum distance from its wall along horizontal surface. Water comes out horizontally from the hole. The maximum value of h for which the cylinder does not topple is :



- (A) $\frac{\pi r^3}{a}$ (B) $\frac{2\pi r^3}{a}$ (C) $\frac{3\pi r^3}{2a}$ (D) $\frac{4\pi r^3}{3a}$
6. Water is being poured in a vessel at a constant rate $\alpha \text{ m}^3/\text{s}$. There is a small hole of area a at the bottom of the tank. The maximum level of water in the vessel is proportional to
- (A) α (B) α^2 (C) α^{-1} (D) α^{-2}
7. A small body with relative density d_1 falls in air from a height ' h ' on to the surface of a liquid of relative density d_2 where $d_2 > d_1$. The time elapsed after entering the liquid to the instant when it comes to instantaneous rest inside liquid :

- (A) $\sqrt{\frac{2h}{g}} \frac{d_2}{d_1}$ (B) $\sqrt{\frac{2h}{g}} \frac{d_1}{d_2 - d_1}$ (C) $\sqrt{\frac{2h}{g}} \frac{d_1}{d_2}$ (D) $\sqrt{\frac{2h}{g}} \frac{d_2 - d_1}{d_1}$

Multiple Correct Answer Type

2 Q. [4 M (-1)]

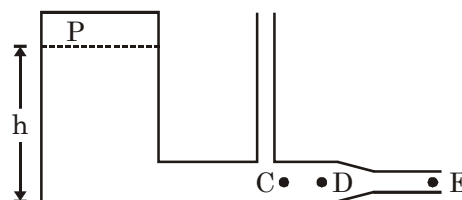
8. As shown in figure, a liquid of density ρ is standing in a sealed container to a height h . The container contains compressed air at a gauge pressure of p . The horizontal outlet pipe has a cross-sectional area ' A ' at C and D. The cross-sectional area is ' $A/2$ ' at E. Find correct options.

- (A) The velocity of liquid at C will be $\left[\frac{(p + \rho gh)}{2\rho} \right]^{1/2}$

- (B) The discharge rate is given by $\frac{A}{2\rho} (p + \rho gh)^{1/2}$

- (C) The discharge rate is given by $\frac{A}{\sqrt{\rho}} (p + \rho gh)^{1/2}$

- (D) The discharge rate is given by $\frac{A}{\sqrt{2\rho}} (p + \rho gh)^{1/2}$



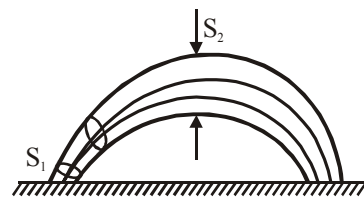
9. Water jet is projected at an angle to the horizontal. At the point of projection, the area of the jet is S_1 and at the highest point, the area of the jet is S_2 . The initial velocity of projection is u .

(A) The angle of projection is $\cos^{-1}\left(\frac{S_1}{S_2}\right)$

(B) The range on the level ground is $\frac{2u^2}{g} \frac{S_1}{S_2} \sqrt{1 - \frac{S_1^2}{S_2^2}}$

(C) The maximum height reached from the ground is $\frac{2u^2}{g} \left(1 - \frac{S_1^2}{S_2^2}\right)$

(D) The rate of volume flow is $S_2 u$



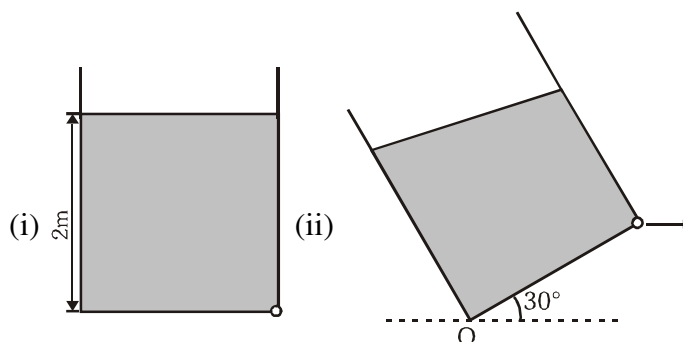
Linked Comprehension Type

(1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question 10 to 12

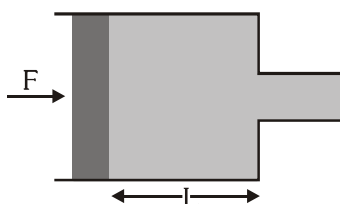
Velocity of efflux in Torricelli's theorem is given by $v = \sqrt{2gh}$, here h is the height of hole from the top surface, after that, motion of liquid can be treated as projectile motion.



10. Liquid is filled in a vessel of square base ($2\text{m} \times 2\text{m}$) upto a height of 2m as shown in figure (i). In figure (ii) the vessel is tilted from horizontal at 30° . What is the velocity of efflux in this case. Liquid does not spill out?
- (A) 3.29 m/s (B) 4.96 m/s (C) 5.67 cm (D) 2.68 m/s
11. What is its time of fall of liquid on the ground?
- (A) $\frac{1}{\sqrt{2}} \text{ s}$ (B) $\frac{1}{\sqrt{3}} \text{ s}$ (C) $\frac{1}{\sqrt{5}} \text{ s}$ (D) $\sqrt{2} \text{ s}$
12. At what distance from point O, will be liquid strike on the ground?
- (A) 5.24 m (B) 6.27 m (C) 4.93 m (D) 3.95 m

Paragraph for Question No. 13 & 14

A container is filled with water as shown. A constant force is applied on the piston slowly to remove the water as shown. The volume of the water inside the container is V . The cross-sectional area of the orifice is ' a ' and take the density of water as ρ . Cross-sectional area of the cylinder is A .



13. What is the velocity of water which is coming out considering $A \gg a$?

- (A) $\frac{\sqrt{F}}{A\rho}$ (B) $\frac{\sqrt{2F}}{A\rho}$ (C) $\frac{F}{\sqrt{2} A\rho}$ (D) $\sqrt{\frac{2F}{A\rho}}$

14. What is the time taken to completely squeeze the water?

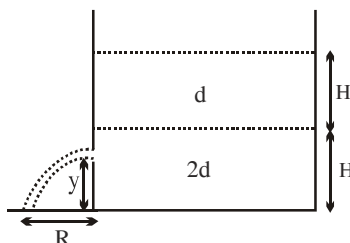
- (A) $\frac{V}{a} \sqrt{\frac{A\rho}{2F}}$ (B) $\frac{V}{a} \sqrt{\frac{A\rho}{F}}$ (C) $\frac{V}{a} \sqrt{\frac{2A\rho}{F}}$ (D) $\frac{2V}{a} \sqrt{\frac{a\rho}{F}}$

SECTION-III

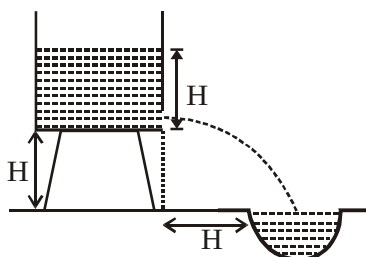
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

- A cube (density 0.5 gm/cc) of side 10 cm is floating in water kept in a cylindrical beaker of base area 1500 cm². When a unknown mass is kept on wooden block the level of water rises in the beaker by 2mm. If the unknown mass is 75α (in gm) then the value of α .
- Two immiscible liquids are filled in a container of large cross-section as shown in the diagram. In the given situation y can be varied. Horizontal range of liquid stream coming out of small hole will depend upon y . Densities of ideal liquids are $2d$ and d , respectively as shown in diagram. Let maximum range of liquid stream is $R_{\max}$ and corresponding value of y is y_0 . Find $(R_{\max}/R_{\max}-y_0)$



- A cylindrical vessel of height $H = 4$ m & area of cross section 1 m² filled with water rests on a stand of same height H . It has a small plugged hole near its bottom. When plug is removed the liquid starts to come out.



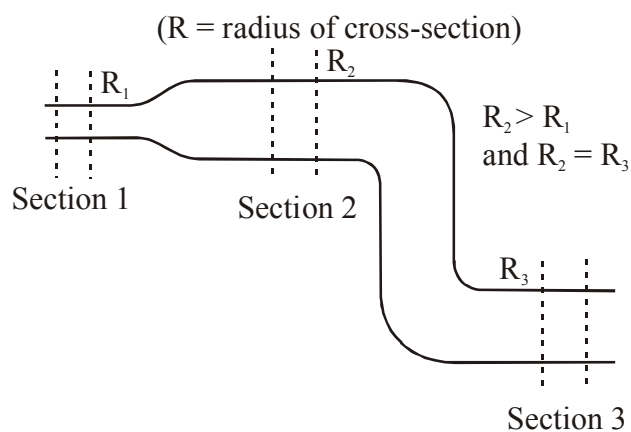
Find the volume (in m³) of liquid collected in a large initially empty vessel lying on floor at a distance H from the stand. Assume that water falling on to the floor does not flow into the vessel.

SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

- An arrangement of the pipes of circular cross-section is shown in the figure. The flow of water (incompressible and nonviscous) through the pipes is steady in nature. Three sections of the pipe are marked in which section 1 and section 2 are at same horizontal level, while being at a greater height than section 3. Correctly match order of the different physical parameter with the options given. In column-I certain statements are given and numbers given in column-II represent the section shown in figure. Match the statements in column-I with corresponding ranking in column-II



Column-I

- (A) Order of volume flow rate in section
- (B) Order of total energy of unit weight while flowing through sections.
- (C) Order of pressure in the sections.
- (D) Order of flow speed in sections

Column-II

- (P) $1 > 2 > 3$
- (Q) $3 > 2 > 1$
- (R) $1 > 2 = 3$
- (S) $1 = 2 = 3$
- (T) $1 = 2 > 3$

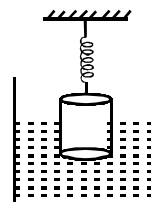
SECTION-I

Single Correct Answer Type

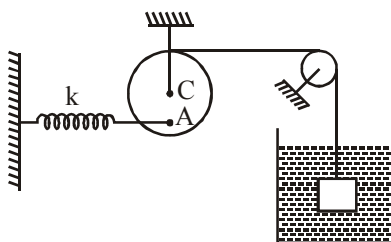
5 Q. [3 M (-1)]

1. A solid cylinder of mass m and volume v is suspended from ceiling by a spring of spring constant k . It has cross-section area A . It is submerged in a liquid of density ρ upto half its length. If a small block of mass M_0 is kept at the centre of the top, the amplitude of small oscillation will be :-

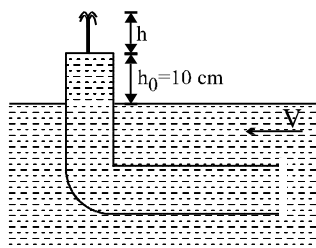
- (A) $\frac{M_0 g}{K + v\rho g}$ (B) $\frac{M_0 g}{K + A\rho g}$
(C) $\frac{(M_0 + m)g}{K + A\rho g}$ (D) $\frac{M_0 g}{K}$



2. A cubical block of volume v and density 3ρ is placed inside a liquid of density ρ and attached to a spring of spring constant k as shown in the figure. Assuming ideal spring and pulley and spring is attached at A which is at $R/2$ from centre. The compression in the spring at equilibrium is :

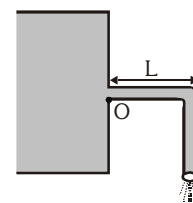


- (A) $\frac{4\rho v g}{k}$ (B) $\frac{2\rho v g}{k}$ (C) $\frac{3\rho v g}{k}$ (D) $\frac{\rho v g}{k}$
3. A bent tube is lowered into water stream as shown in the figure. The velocity of the stream relative to the tube is equal to $V = 2$ m/s. The closed upper end of the tube located at height $h_0 = 10$ cm above free surface of water has a small orifice. To what height h will the water get spurt?

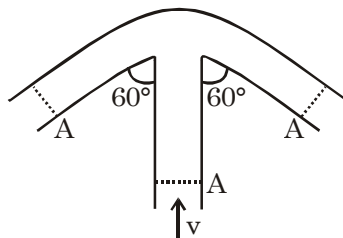


- (A) 5 cm (B) 10 cm (C) 20 cm (D) 40 cm
4. Water flows out of a big tank along a big tube bent at right angles, the inner radius of tube is r . The length of horizontal section of the tube is equal to ' L '. The water flow rate is Q litres/sec. Moment of reaction forces due to water flow acting on tube's wall, w.r.t. point O . Where ρ is the density of water

- (A) $\frac{Q^2 \rho L}{\pi r^2}$ clockwise (B) $\frac{Q^2 \rho L}{\pi r^2}$ anticlockwise
(C) $\frac{Q^2 \rho L}{2\pi r^2}$ clockwise (D) $\frac{Q^2 \rho L}{2\pi r^2}$ anticlockwise



5. An ideal liquid of density ρ is pushed with velocity v through the central limb of the tube shown in the figure. What force does the liquid exert on the tube? The cross-sectional areas of the three limbs are equal to A each. Assume stream-line flow.

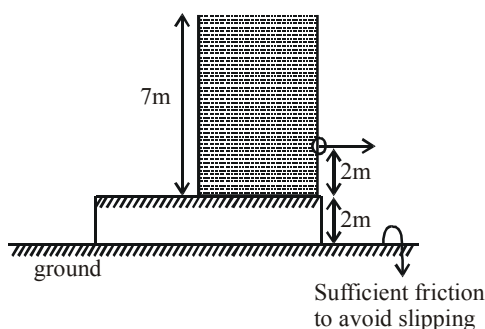


- (A) $\frac{9}{8} \rho A v^2$ (B) $\frac{5}{4} \rho A v^2$ (C) $\frac{3}{2} \rho A v^2$ (D) $\rho A v^2$

Multiple Correct Answer Type

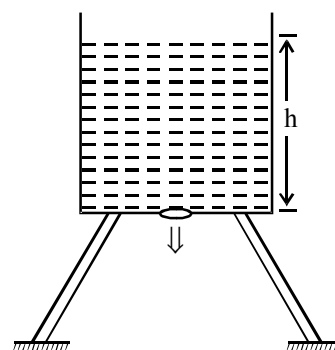
3 Q. [4 M (-1)]

6. A cylindrical vessel is full of water (density $\rho = 1 \text{ gm/cc}$) is placed on horizontal ground as shown. A small hole of area ($= 1 \text{ cm}^2$) is made on a curved wall at a height of 2m from the bottom of container (container does not move) choose **CORRECT** statement(s) ($g = 10 \text{ m/s}^2$) :-

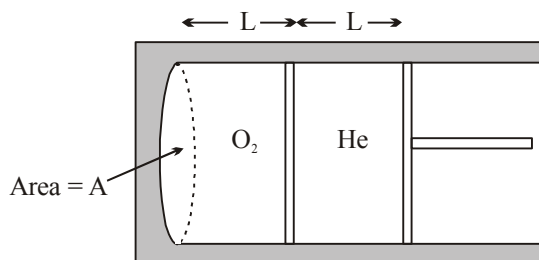


- (A) Horizontal range of water jet is maximum when water level is at height of 8 m from ground
(B) Maximum horizontal force on container by table is 10 Newton
(C) Horizontal range of water jet on ground is $4\sqrt{5}$ meter
(D) Horizontal range of water jet will decrease then increase with decrease of water level.
7. A small hole of area a is at the bottom of a container of area A . The liquid is filled up to height h from base. As liquid comes out then $\left[a = \frac{A}{3} \right]$.

- (A) level of liquid in container falls at rate of $\sqrt{\frac{gh}{4}} \text{ m/s}$.
(B) magnitude of acceleration of top surface of liquid is $\frac{8g}{8} \text{ m/s}^2$.
(C) magnitude of acceleration of top surface of liquid is $\frac{8g}{3} \text{ m/s}^2$.
(D) None of these.



8. An insulating cylinder contains equal volumes of He and O₂ separated by a massless freely moving adiabatic piston as shown. The gas is compressed by moving the insulating piston so that volume of He becomes half. Select the correct alternative(s).



- (A) Pressure in He chamber will be equal to pressure in O₂ chamber
(B) Pressure in He chamber will be less than pressure in O₂ chamber
(C) Volume of He chamber will be equal to volume of O₂ chamber

(D) Volume of O₂ chamber will be $\frac{(LA)}{(2)^{25/21}}$.

Linked Comprehension Type (Single Correct Answer Type)

(3 Para × 2Q.) [3 M (-1)]

Paragraph for Question Nos. 9 and 10

An engineer is instructed to design a tank to measure the heat capacity of liquids. The tank is designed and it consists of insulating walls of negligible heat capacity, with a hole in the bottom, a drain plug fitted with a light rod to cover the hole, and 10 electric immersion heaters each of 5 KW. The tank has been shown in the adjacent figure. For testing, water is filled in the tank at 20° C and heaters are switched on. After 15 hour of switching on the heaters water starts boiling at normal atmospheric pressure. Now, water is drained by lifting the drain plug and again filled with oil having specific gravity 0.7. The engineer reported that the oil took 3.15 hour to raise its temperature from 20° C to 60° C at normal atmospheric pressure.

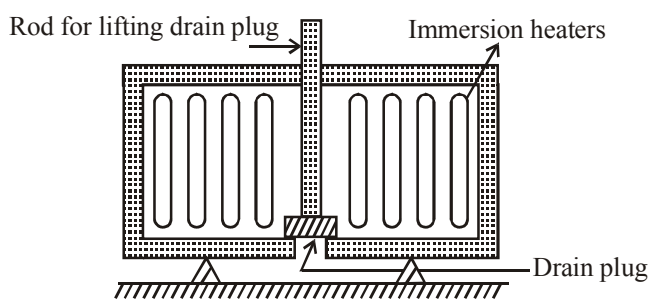
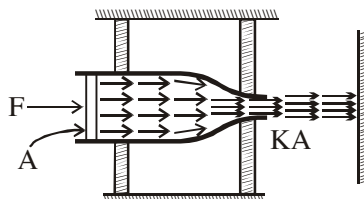


Figure : Synthetic lubricating oil tank

9. From the report of the engineer the specific heat capacity of the oil is approximately :-
(A) 60 % of specific heat of water (B) 80% of specific heat of water
(C) 90% of specific heat of water (D) equal to specific heat of water
10. It is suggested that the air in the tank above the oil can be pressurized at 4 atm above normal air pressure, which of the following is **INCORRECT** if pressure is increased to the suggested value.
(A) The time required to heat the oil would be greatly extended.
(B) The drain plug would be more difficult to lift.
(C) Fluid velocity would be increased when the tank is drained.
(D) The time required to drain the tank would decrease.

Paragraph for Questions 11 and 12

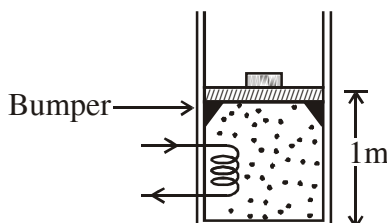
The area of the inner cross section of a horizontal tube (fixed) is A , while the area of cross section of its part on the right hand side is KA , where $K < 1$. The two parts are attached smoothly with out any sharp turns as given. In the tube there is some liquid of density ρ and negligible viscosity with the help of a piston at left hand side, the liquid can be pushed out of this part of tube. The water which leaves the tube hits a vertical wall and spreads out, creating a tapering film of fluid



11. At what constant speed does the piston moves if external force has constant value F
- (A) $\sqrt{\frac{2F}{\rho A} \left(\frac{K}{1-K} \right)}$ (B) $\sqrt{\frac{2F}{\rho A} \left(\frac{K^2}{1-K^2} \right)}$ (C) $\sqrt{\frac{2F}{\rho A} \left(\frac{K}{1-K^2} \right)}$ (D) $\sqrt{\frac{2F}{\rho A} \left(\frac{1-K^2}{K^2} \right)}$
12. Value of horizontal force which is exerted by tube on support
- (A) $\frac{KF}{1-K^2}$ (B) $\frac{K^2}{1-K^2} F$ (C) $\frac{2K}{1-K^2} F$ (D) $-\frac{K-1}{K+1} F$

Paragraph for Questions 13 and 14

Thermal insulated loaded piston of total mass 1 ton (with the load) in a thermally insulated cylinder of cross section area 0.5 m^2 . The piston is supported by a bumper at a height of 1 m measured from the bottom of cylinder. Air in closed chamber is heated by an heater so that load & piston can be raised by 0.5 m . Initially both the inside pressure and ambient air pressure is 10^5 pa and the temperature is 300 K



(Treat air as diatomic gas)

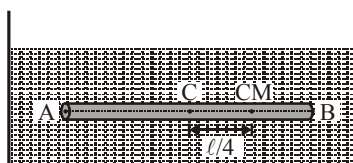
13. Total heat supplied by heater during complete process
- (A) 25 kJ (B) 105 kJ (C) 130 kJ (D) 150 kJ
14. The percentage of added heat is used for raising the piston is
- (A) 33.33% (B) 50% (C) 19.23% (D) 23.07%

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1 Q. [3(0)]

1. A non uniform cylindrical rod of mass 5 kg , length 2 m and radius 0.1 m is having its center of mass at a distance $\frac{1}{2}\text{ m}$ from the centre and lying on the axis of the rod. The rod is kept in a liquid of uniform density 1400 kg/m^3 . The moment of inertia of the rod about the centre of mass is 4 kg-m^2 . Find the angular acceleration (in SI unit) of the rod just after the rod is released from the position shown in figure. If your answer is $n\pi$, fill the value of n .



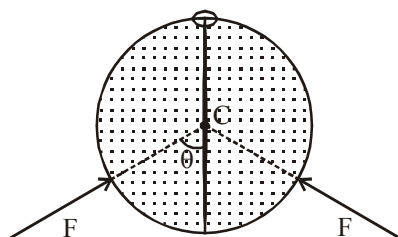
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

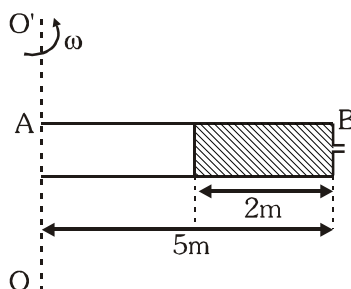
3 Q. [4 M (0)]

1. Two hemispheres of radius R combined together by two minimum forces ' F ' to form a sphere (completely filled with water) as shown in figure. A small pin hole of radius a at top of the sphere as shown in figure.

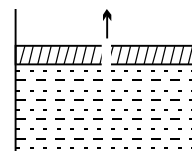
If the system is in equilibrium at $\tan \theta = \frac{x}{2}$. Find the value of x . (gravity is present and no other support is given to hemispheres) Neglect atmospheric pressure.



2. A horizontal oriented tube AB of length 5 m rotates with a constant angular velocity 0.5 rad/s about a stationary vertical axis OO' passing through the end A. The tube is filled with ideal fluid. The end A of the tube is open, the closed end B has a very small orifice. The velocity with which the liquid comes out from the hole in m/s.



3. In the figure shown water is filled in the cylindrical tank of cross-sectional area 100 cm^2 up to a height of 10 cm. The mass of the thin, tight fitting and frictionless piston is 100 kg. The small hole in the piston has area $2\sqrt{2} \text{ mm}^2$. Due to weight of the piston water spurts upwards in a jet from the hole. Find the time (in sec) in which the tank becomes empty. If your answer is $n/5$. (Assume that spurting water does not fall back on piston)

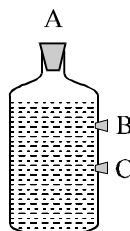


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. A bottle is filled with water, above which a little air at atmospheric pressure is present. Plugs are blocking the three small holes (A, B & C). (Hole B is just below the upper most layer of water)



Column-I

- (A) A & C are opened
- (B) B & C are opened
- (C) Only C is opened
- (D) All the holes are opened

Column-II

- (P) Air comes into the bottle from upper most open hole
- (Q) Water flows out of holes B & C
- (R) Very little water comes out and the flow stops
- (S) Pressure of air inside bottle above the water remains unchanged.

SECTION-I

Single Correct Answer Type

17 Q. [3 M (–1)]

- If a drop of liquid breaks into smaller droplets, it results in lowering of temperature of the droplets. Let a drop of radius R , break into N small droplets each of radius r . Estimate the decrease in temperature. (Given : T = Surface tension, s = Specific Heat Capacity, ρ = Density of liquid)

(A) $\frac{2T}{\rho s} \left(\frac{1}{R} + \frac{1}{r} \right)$ (B) $\frac{4T}{\rho s} \left(\frac{1}{R} - \frac{1}{r} \right)$ (C) $\frac{3T}{\rho s} \left(\frac{1}{R} - \frac{1}{r} \right)$ (D) $\frac{2T}{\rho s} \left(\frac{1}{R} - \frac{1}{r} \right)$
- A sphere of radius 10 cm and density 500 kg/m^3 is under water of density 1000 kg/m^3 . The acceleration of the sphere is 9.80 m/s^2 upward. Viscosity of water is 1.0 centipoise. If $g = 9.81 \text{ m/s}^2$, the velocity of the sphere is nearly

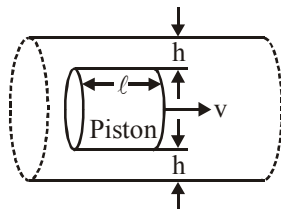
(A) 9 m/s (B) 10 m/s (C) 11 m/s (D) 12 m/s
- Two solid spheres of same radius and materials of specific gravities 4 and 8 are released in water. The terminal velocity attained by the heavier sphere is:

(A) double than that of lighter sphere
(B) more than double than that of lighter sphere
(C) less than double than that of lighter sphere
(D) data is not sufficient
- Two large vertical parallel plates separated by a gap of d have a highly viscous liquid of density ρ and viscosity coefficient η , flowing steadily under gravity in the gap. The velocity gradient of flow near plates surface is

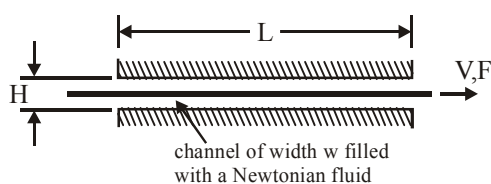
(A) $\frac{2\rho dg}{\eta}$ (B) $\frac{3\rho dg}{\eta}$ (C) $\frac{\rho dg}{3\eta}$ (D) $\frac{\rho dg}{2\eta}$
- Assertion** : A helium filled balloon does not rise indefinitely in air but halts after a certain height.
Reason : Viscosity opposes the motion of balloon.
Choose any one of the following four responses:
(A) both (A) and (R) are true and (R) is the correct explanation of (A)
(B) both (A) and (R) are true but (R) is not correct explanation of (A)
(C) if (A) is true but (R) is false
(D) if (A) is false and (R) is true
- A spherical solid ball of volume V is made of a material of density ρ_0 . It is falling through a liquid of density ρ' ($\rho' < \rho_0$). Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v . i.e., $F_{\text{viscous}} = -kv^2$, $k > 0$. The terminal speed of the ball is -

(A) $\sqrt{\frac{Vg(\rho_0 - \rho')}{k}}$ (B) $\frac{Vg\rho_0}{k}$ (C) $\sqrt{\frac{Vg\rho_0}{k}}$ (D) $\frac{V(\rho_0 - \rho')}{k}$

7. A cylindrical piston of length ℓ and radius R is moved with constant velocity v in a cylindrical tube filled with oil. If the clearance between the piston and tube is h ($\ll R$), assuming η as the coefficient of viscosity of the oil, find the external force required to derive the piston with constant velocity.



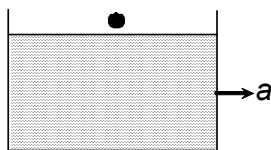
- (A) $F = \pi R \eta \frac{v \ell}{h}$ (B) $F = 2 \pi R \eta \frac{v \ell}{h}$
- (C) $F = \pi R^2 \eta \frac{v \ell}{h}$ (D) $F = 2 \pi R^2 \eta \frac{v \ell}{h}$
8. The graph between terminal velocity (along y-axis) and square of radius (along x-axis) of spherical body of density ρ allowed to fall through a fluid of density σ is a :-
- (A) Straight line with positive slope
(B) Straight line with negative slope
(C) Straight line perpendicular to x-axis
(D) Straight line perpendicular to y-axis
9. During a coating process, a thin, flat tape of width w is pulled through a channel of length L containing a Newtonian fluid of density ρ and dynamic viscosity μ . The fluid is in contact with both sides of the tape. Estimate the force required to pull the tape through the channel if the tape has velocity V and the channel has height H . You may assume that the tape is much thinner than H and tape is in middle of channel.



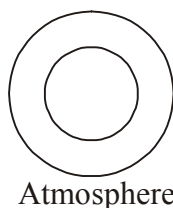
- (A) $\frac{4\mu VLw}{H}$ (B) $\frac{2\mu VLw}{H}$ (C) $\frac{\mu VLw}{H}$ (D) $\frac{\mu VLw}{2H}$
10. Eight equal drops of water each of radius $r = 2$ mm are falling through air with a terminal velocity of 16 cm/s. The eight drops combine to form a big drop. The terminal velocity of drop will be
- (A) 16 cm/s (B) 32 cm/s (C) 64 cm/s (D) 30 cm/s
11. A solid iron ball and an aluminium ball of same diameter are released together in a lake. Which ball will reach the bottom first. (neglect viscosity and $\rho_{Al} < \rho_{Fe}$)
- (A) Aluminium Ball
(B) Iron Ball
(C) Both will reach at the same time
(D) Data insufficient

CLASS TEST

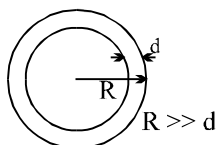
12. A very small ball of radius r and mass m is released in a large tank having a liquid of density ρ and moving with an acceleration a as shown in figure. If coefficient of viscosity the liquid is η , then terminal speed of the ball with respect to container will be (neglect buoyant force)



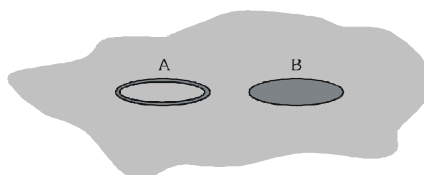
- (A) $\frac{mg}{6\pi\eta r}$ (B) $\frac{ma}{6\pi\eta r}$ (C) $\frac{m}{6\pi\eta r} \sqrt{g^2 + a^2}$ (D) $\frac{m}{6\pi\eta r} \sqrt{g + a}$
13. A ball of mass m and radius r is gently released in a viscous liquid. The mass of the liquid displaced by it is m' such that $m > m'$. The terminal velocity is proportional to
- (A) $\frac{m - m'}{r}$ (B) $\frac{m + m'}{r}$ (C) $\frac{(m + m')}{r^2}$ (D) $(m - m') r^2$
14. A soap bubble of radius R is surrounded by another soap bubble of radius $2R$, as shown. Take surface tension = S . Then, the pressure inside the smaller soap bubble, in excess of the atmospheric pressure, will be



- (A) $4S/R$ (B) $3S/R$ (C) $6S/R$ (D) None of these
15. A soap bubble has radius R and thickness d ($d \ll R$) as shown. It collapses into a spherical drop. The ratio of excess pressure in the drop to the excess pressure inside the bubble is :

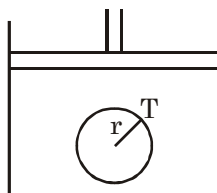


- (A) $\left(\frac{R}{3d}\right)^{\frac{1}{3}}$ (B) $\left(\frac{R}{6d}\right)^{\frac{1}{3}}$ (C) $\left(\frac{R}{24d}\right)^{\frac{1}{3}}$ (D) None
16. A rigid ring A and a rigid thin disk B both made of same material, when gently placed on water, just manage to float due to surface tension as shown in the figure. Both the ring and the disk have same radius. What can you conclude about their masses (thickness may be different)?



- (A) Both have the same mass.
(B) Mass of the ring is half of that of the disk.
(C) Mass of the ring is double to that of the disk.
(D) More information is needed to decide.

17. In a cylinder-piston arrangement, air is under a pressure P_1 . A soap bubble of radius r lies inside the cylinder, soap bubble has surface tension T . The radius of soap bubble is to be reduced to half. The new pressure P_2 to which air should be compressed isothermally is : (Assume r is very small as compared to height of cylinder)



- (A) $P_1 + \frac{4T}{r}$ (B) $4P_1 + \frac{12T}{r}$ (C) $8P_1 + \frac{24T}{r}$ (D) $P_1 + \frac{8T}{r}$

Multiple Correct Answer Type

1 Q. [4 M (-1)]

18. Consider a spherical bubble of radius R at a depth of h from the surface of water. At the instant when force become zero (atmospheric pressure is P_0 , surface tension, viscosity & density of the water is T , η & ρ). Then (neglect density of air in compare to water) :

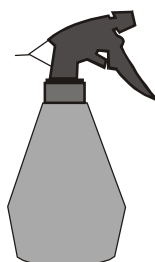
- (A) Pressure inside the bubble is $P_0 + \rho gh + \frac{2T}{R}$
(B) Velocity of the sphere can not remain constant
(C) Velocity of the sphere is given by $\frac{2}{9} \frac{\rho g r^2}{\eta}$
(D) As sphere moves up its radius increases.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. If you squeeze the handle of a spray bottle (figure), you cause air to flow horizontally across the opening of a tube that extends down into the liquid almost to the bottom of the bottle. If the air is moving at 50.0 m/s, what is the pressure difference between the top of the tube and the atmosphere? Assume that the density of air is $\rho = 1.20 \text{ kg/m}^3$. If your answer is 250X (in Pa) fill value of X.

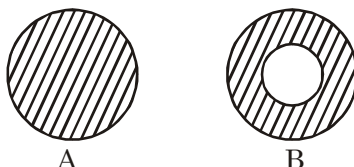


SECTION-I

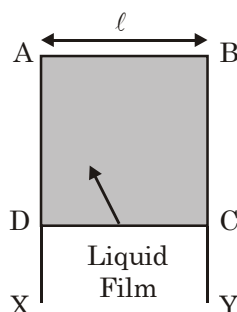
Single Correct Answer Type

10 Q. [3 M (-1)]

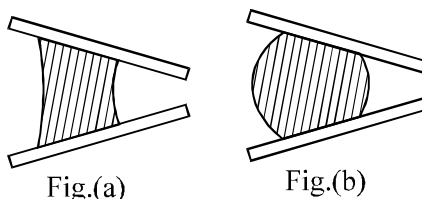
1. Two balls A and B of same outer radius $2R$ are released from rest in water of sufficient depth. Both balls are made from same material. Ball A is completely solid and ball B is hollow upto radius R . Find ratio of terminal velocity of A and terminal velocity of B (specific gravity of balls is 2)



- (A) 1 (B) $\frac{7}{6}$ (C) $\frac{4}{5}$ (D) $\frac{4}{3}$
2. A mercury droplet of radius R and surface tension α is broken into 8 smaller droplets of equal size. The work done by the external agency is :
- (A) $\frac{4}{3} \pi R^3 \alpha$ (B) $\pi R^2 \alpha$ (C) $8 \pi R^2 \alpha$ (D) $4 \pi R^2 \alpha$
3. A ball of radius R falls through oil with terminal velocity 5 cm/sec. Another ball of same material, has radius $3R$. Its terminal velocity in the same oil will be [Use Stoke's law but neglect buoyant force]
- (A) 4.5 cm/sec (B) 0.45 m/s (C) 4.5 m/sec (D) 15 cm/sec
4. A liquid film is formed over a frame ABCD as shown in figure. A massless wire CD can slide without friction. The mass to be hung from CD to keep it in equilibrium is (Surface tension of liquid is T)

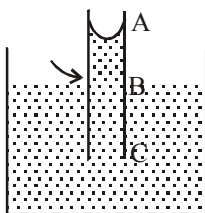


- (A) Tl/g (B) $2Tl/g$ (C) $g/2Tl$ (D) $Tl/2g$
5. Two soap bubbles with radii r_1 and r_2 ($r_1 > r_2$) come in contact. Their common surface has a radius of curvature r .
- (A) $r = \frac{r_1 + r_2}{2}$ (B) $r = \frac{r_1 - r_2}{r_1 - r_2}$ (C) $r = \frac{r_1 - r_2}{r_1 + r_2}$ (D) $r = \sqrt{r_1 r_2}$
6. Figure (a) and (b) show water drop and mercury drop in two identical conical glass pipes, then



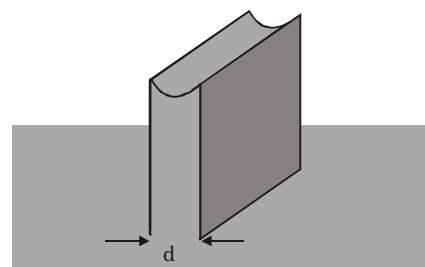
- (A) The water drop tends to move towards narrow end and mercury drop towards wide end.
(B) The water drop tends to move towards wide end and mercury drop towards narrow end.
(C) both tend to move towards wide end.
(D) both tend to move towards narrow end.

7. Water rises to a height of 16.3 cm in a capillary of height 18 cm above the water level. If the tube is cut at a height of 12 cm :-
 (A) water will come as a fountain from the capillary tube
 (B) water will stay at a height of 12 cm in the capillary tube
 (C) the height of water in the capillary will be 10.3 cm
 (D) water will flow down the sides of the capillary tube
8. A glass capillary tube open at both ends is dipped in water, we can see that the water first rises in the tube and finally remains at rest at certain height as shown in figure. If this capillary tube is cut from B then the diagram of part AB is :- (B is the point where water level intersects the tube)

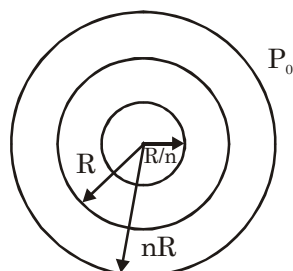


9. Two long parallel glass plates has water between them. Contact angle between glass and water is zero. If separation between the plates is 'd' (d is small). Surface tension of water is 'T'. Atmospheric pressure = P_0 . Then pressure inside water just below the air water interface is :-

- (A) $P_0 + \frac{2T}{d}$ (B) $P_0 + \frac{2T}{(d/2)}$
 (C) $P_0 + \frac{8T}{d}$ (D) $P_0 - \frac{2T}{d}$



10. Consider three concentric soap bubbles of radius $\frac{R}{n}$; R and nR as shown. Surface tension for each bubble is T and atmospheric pressure is P_0 . Value of n for which pressure (P) of gas inside smallest bubble is minimum (n is integer greater than 1) :-



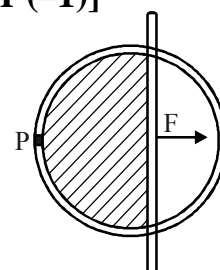
- (A) 2 (B) 3 (C) $n \rightarrow \infty$
 (D) for every value of n, value of P remain same

Multiple Correct Answer Type

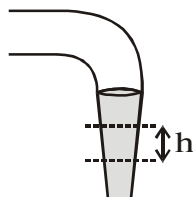
5 Q. [4 M (-1)]

11. A circular wire, 10 cm in diameter, with a slider wire on it, is in a horizontal plane. A liquid film is formed, bounded by the wires, on the left side of the slider, as shown. The surface tension of the liquid is 0.1 N/m. An applied force 16 mN, perpendicular to the slider, maintains the film in equilibrium. Ignore the sag in the film. What can be the distance between point P and slider.

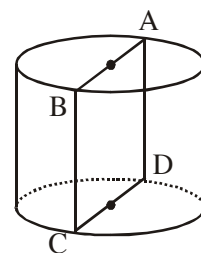
- (A) 8 cm (B) 2 cm
 (C) 5 cm (D) slider cannot be in equilibrium



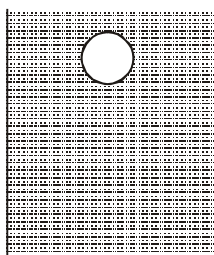
12. From a water tap, liquid comes out in form of cylinder. As water moves down, radius of that cylinder decreases. Consider, a very small element of height h and radius R (due to small height radius is assumed to be constant)



This element is given in the diagram. An imaginary cross-section ABCD divides this element in two equal semi cylinders. S is surface tension of liquid. Now choose correct statement(s)



- (A) Surface energy of this elementary cylinder is $S[2\pi Rh + 2\pi R^2]$
- (B) Pressure inside the cylinder is more than atmospheric pressure by $\frac{2S}{R}$
- (C) Force exerted by one half cylinder on another half cylinder due to surface tension is $2Sh$.
- (D) Surface energy of this elementary cylinder is $S \times 2\pi Rh$
13. A sphere is dropped into a viscous liquid of viscosity η from some height. If the density of material and liquid are ρ and σ respectively ($\rho > \sigma$) then which of the following is incorrect.
- (A) the acceleration of the sphere just after entering the liquid is $g\left(\frac{\rho - \sigma}{\rho}\right)$
- (B) Time taken to attain terminal speed $t \propto \rho^0$
- (C) At terminal speed, the viscous force is maximum
- (D) At terminal speed, the net force acting on the sphere is zero
14. A solid sphere moves at a terminal velocity of 20 m/s in air at a place where $g = 10 \text{ m/s}^2$ downwards. The sphere is taken in a hall, where $g = 5 \text{ m/s}^2$ downwards having air of same viscosity and sphere is pushed down at a speed of 40 m/s. [Neglect buoyancy of air]
- (A) Its initial acceleration will be 15 m/s^2 downward.
- (B) Its initial acceleration will be 15 m/s^2 upward
- (C) The magnitude of acceleration will decrease as time passes and becomes zero after some time
- (D) New terminal velocity of the solid sphere will be 10 m/s downwards
15. A ball of mass "1 kg" and specific gravity $\left(\frac{\rho_m}{\rho_w}\right) = 2$ is released inside the tank filled with water as shown. Resistance offered by water to ball is $2.5 V$ where V is velocity of the ball. Select the **CORRECT** statement (ρ_m & ρ_w are density of ball and water respectively):-



- (A) Momentum of water is zero when ball reaches steady velocity.
- (B) Momentum of water is 1 kg m/s in upward direction when ball reaches steady state.
- (C) Momentum of ball is 2 kg m/s in downward direction when ball reaches steady state.
- (D) Momentum of [Ball + water] system is 2 kg m/s in downward direction when ball reaches steady state.

Linked Comprehension Type (Single Correct Answer Type)

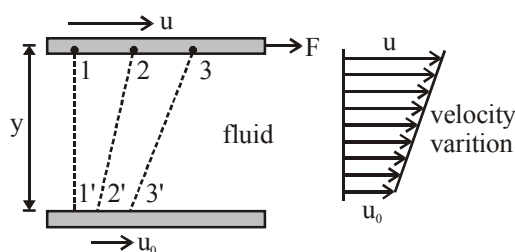
(1 Para × 2Q.) [3 M (-1)]

Paragraph for Questions no. 16 and 17

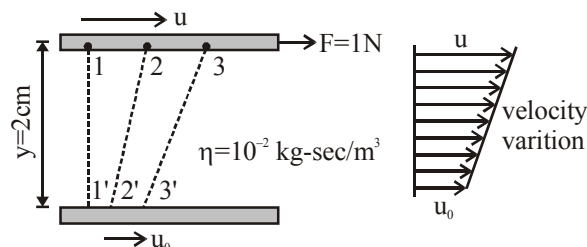
In the shown figure, both the plates are moving. The fluid between the plates moves such that fluid particles in contact of the plates move with same velocity as the plates. The fluid particles in the gap moves in a linear velocity distribution function. As per Newton law of viscosity the tangential force F is

given by $F = \eta A \frac{u - u_0}{y}$, where η = coefficient of viscosity = $10^{-2} \text{ kg -sec/m}^3$, A = area of the plate =

1 m^2 , y = distance between both the plates = 2 cm . The velocity gradient is given by $\frac{du}{dy}$.



16. If the tangential force $F = 1 \text{ Newton}$ and $u_0 = 1 \text{ m/s}$, then the velocity profile for the given figure is :-



- (A) $u = 100 y$ (B) $u = 200 y$ (C) $u = 200 y + 1$ (D) $u = 100 y + 1$
17. In the previous question, the constant velocity of the upper plate would be :
- (A) 1 m/s (B) 2 m/s (C) 3 m/s (D) 4 m/s

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

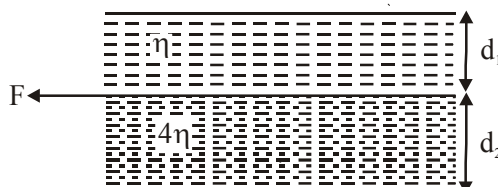
1. A glass rod of diameter $d_1 = 1.5 \text{ mm}$ is inserted symmetrically into a glass capillary with inside diameter $d_2 = 2.0 \text{ mm}$. Then the whole arrangement is vertically oriented and brought in contact with water. To what height (in cm) will the water rise in the capillary? Given surface tension of water = 75 mN , density of water = 10^3 kg/m^3 , $g = 10 \text{ m/s}^2$.

SECTION-I

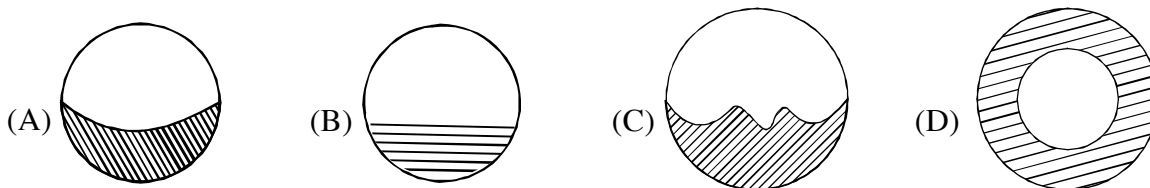
Single Correct Answer Type

8 Q. [3 M (–1)]

1. A thin plate separates two liquids of coefficients of viscosity η and 4η kept between two fixed plates as shown. Value of d_1 and d_2 is adjusted such that force required to pull the plate is minimum. If total distance between fixed plate is constant then find d_2/d_1 .



- (A) 1 (B) 2 (C) 3 (D) 4
2. Consider a spherical body, one third of its volume is filled with a liquid which has contact angle zero with the surface of sphere. If this sphere (along with liquid) is taken to gravity free space then shape of liquid inside sphere will look like.



3. A drop of water volume 0.05 cm^3 is pressed between two glass-plates, as a consequence of which, it spreads between the plates. The area of contact with each plate is 40 cm^2 . If the surface tension of water is 70 dyne/cm , the minimum normal force required to separate out the two glass plates in newton is approximately (assuming angle of contact is zero) :

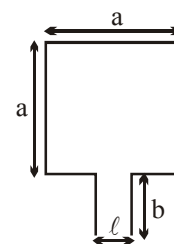
(A) 45 N (B) 100 N (C) 90 N (D) None of these

4. An open capillary tube is lowered in a vessel with mercury. The difference between the levels of the mercury in the vessel and in the capillary tube $\Delta h = 4.6 \text{ mm}$. What is the radius of curvature of the mercury meniscus in the capillary tube? Surface tension of mercury is 0.46 N/m , density of mercury is 13.6 gm/cc .

(A) $\frac{1}{340} \text{ m}$ (B) $\frac{1}{680} \text{ m}$ (C) $\frac{1}{1020} \text{ m}$ (D) Information insufficient

5. A light open rigid wire frame floats on the surface of water as shown in figure. What force will act on the frame, immediately after some soap solution is dropped inside it? α_1 and α_2 are the surface tensions of water and soap respectively ($\alpha_1 > \alpha_2$)

(A) zero
(B) $(\alpha_1 - \alpha_2) \ell$
(C) $(\alpha_1 + \alpha_2) (4a + 2b + \ell)$
(D) $(\alpha_1 - \alpha_2) (4a + 2b - \ell)$



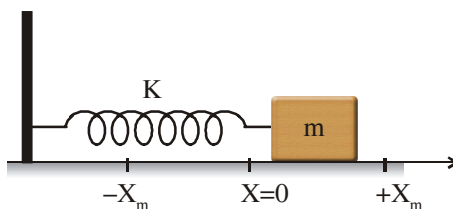
6. Water rises to a height of 10 cm in a certain capillary tube. An another identical tube when dipped in mercury the level of mercury is depressed by 3.42 cm . Density of mercury is 13.6 g/cc . The angle of contact for water in contact with glass is 0° and mercury in contact with glass is 135° . The ratio of surface tension of water to that of Hg is :-

(A) 1 : 3 (B) 1 : 4 (C) 1 : 5.5 (D) 1 : 6.5

7. A simple pendulum is suspended in a car. The car starts moving on a horizontal road according to equation $x = \frac{g}{2}\sqrt{3}t^2$. Find the time period of oscillation of the pendulum.

(A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $\pi\sqrt{\frac{2\ell}{g}}$ (C) $2\pi\sqrt{\frac{\ell}{8g}}$ (D) $2\pi\sqrt{\frac{\ell}{g\sqrt{3}}}$

8. At $t = 0$ the displacement of the block in a linear oscillator as shown is -0.08m . At the same moment $t = 0$, its velocity is -1.6 m/s and acceleration is 32 m/s^2 . Choose the **INCORRECT** statement.

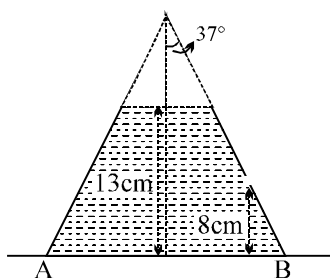


- (A) Angular frequency of the motion is 20 rad/s .
(B) Amplitude of the motion is 11.3 cm .
(C) Phase constant ϕ of the motion if the equation of motion is expressed as $x = A \sin(\omega t + \phi)$ is $\frac{\pi}{4}$
(D) Phase constant ϕ of the motion if the equation of motion is expressed as $x = A \sin(\omega t + \phi)$ is $\frac{5\pi}{4}$

Multiple Correct Answer Type

2 Q. [4 M (-1)]

9. Curved surface of a vessel has shape of a truncated cone having semi vertex angle 37° . Vessel is full of water (density $\rho = 1000\text{ kg/m}^3$) upto a height of 13 cm and is placed on a smooth horizontal plane. Upper surface is opened to atmosphere. A hole of 1.5 cm^2 is made on curved wall at a height of 8 cm from bottom as shown in figure. Area of water surface in the vessel is large as compared to the area of hole.



- (A) Initial velocity of efflux is 1 m/sec
(B) Initial velocity of efflux is 5 m/sec
(C) Horizontal force required to keep the vessel in static equilibrium is 0.15 N
(D) Horizontal force required to keep the vessel in static equilibrium is 0.12 N .
10. A soap film (surface tension $= T$) is formed on a rectangular wire frame as shown in figure. A small, light, inextensible loop of thread of length " ℓ " is gently placed over the soap film. The loop stays on the film in the irregular fashion in which it was placed. Now, a hole is pricked in the film inside the loop with a needle and the thread finally comes to equilibrium position. Choose the correct option(s)
- (A) The thread will take a square shape.
(B) The thread will take a circular shape.
(C) Tension in the thread will be $2T\ell$.
(D) Tension in the thread will be $\frac{T\ell}{\pi}$.



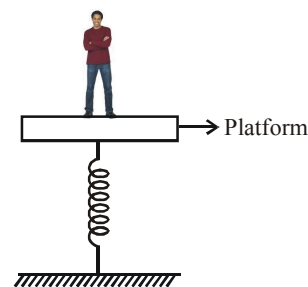
Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.)

[3 M (-1)]

Paragraph for Question 11 to 13

Mr. Saransh having mass 50kg is standing on a platform of mass 10 kg which oscillates up & down (doing SHM) of frequency 0.5 sec^{-1} & amplitude 0.4 m. Platform has a weighing machine fitted in it (which is massless) on which person is standing. (Take $\pi^2 = 10$ & $g = 10 \text{ m/s}^2$). Assume that at $t = 0$ (platform + man) is at their highest point of oscillation.



11. Maximum reading of weighing machine is :-
 (A) 60 kg (B) 70 kg (C) 84 kg (D) 50 kg
12. Find compression in spring when weighing machine reads 55 kg :-
 (A) 1m (B) 1.1 m (C) 1.4 m (D) 0.1 m
13. Find minimum time after which speed of person is $0.2 \pi \text{ m/s}$:-
 (A) $\frac{1}{6} \text{ sec}$ (B) $\frac{1}{3} \text{ sec}$ (C) 1 sec (D) $\frac{1}{4} \text{ sec}$

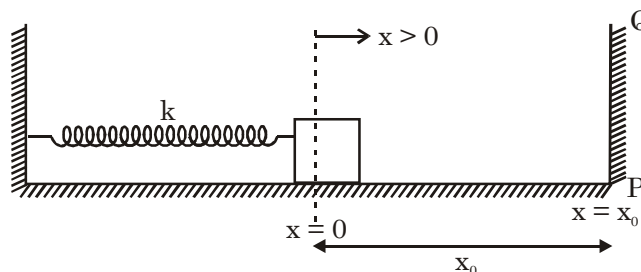
Linked Comprehension Type (Multiple Correct Answer Type)

(1 Para × 2 Q.)

[4 M (-1)]

Paragraph for Question no. 14 and 15

A block is attached to a spring and is placed on a horizontal smooth surface as shown, in which spring is unstretched. Now the spring is given an initial compression $2x_0$ and block is released from rest. Collision with the wall PQ are elastic.



14. Find the time period of motion of the block :-
 (A) $\frac{2\pi}{3} \sqrt{\frac{m}{k}}$ (B) $\frac{4\pi}{3} \sqrt{\frac{m}{k}}$ (C) $\frac{3\pi}{2} \sqrt{\frac{m}{k}}$ (D) $\frac{\pi}{2} \sqrt{\frac{m}{k}}$
15. If coefficient of resitution $e = 1/2$ in the collision between block and the wall then select the **CORRECT** statement(s).
 (A) New maximum compression in the spring is x_0 .
 (B) New maximum compression in the spring is $x_0 \sqrt{\frac{7}{4}}$.
 (C) New angular frequency after collision is $\sqrt{\frac{k}{m}}$.
 (D) New angular frequency after collision is $\sqrt{\frac{k}{2m}}$.

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1Q.[3(0)]

1. A syringe is filled with water upto volume 20 cm^3 . The area of cross-section of the cylinder is 5 cm^2 . The syringe is held vertically and its massless piston is pushed upward by external agent with constant speed. A water beam coming out of the small nozzle (hole area 1 mm^2) has speed 2 m/s . Neglecting friction and viscous nature of water, find the work done by the agent [in 10^{-2} J] in fully emptying the syringe. (Take $g = 10 \text{ m/s}^2$)

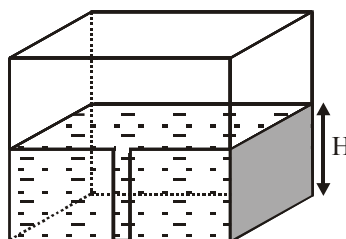


SECTION-III

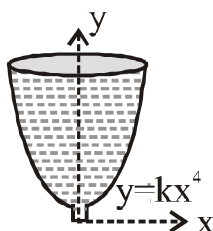
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

1. Figure shows a large container containing water to height ' $H = 8 \text{ m}$ '. The front portion of the container has a rectangular portion cut out from it. Because of that water comes out of it causing the water level to reduce at a certain rate. How many times (let's call it ' η ') will this rate decrease when its level changes from $H = 8 \text{ m}$ to $h = 2 \text{ m}$. Fill $\eta/4$ in OMR sheet.

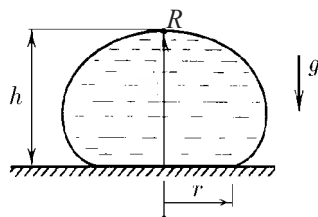


2. Figure shows container filled with water. The equation for the shape of container is $y = kx^4$ in two dimension. There is a small hole at the bottom of the container. The ratio of time taken for the height of water level to reduce to three-fourth from initial level to the time taken for the height of water level to reduce to half from three-fourth level is x . Value of x is. Assume cross section of container is circular.

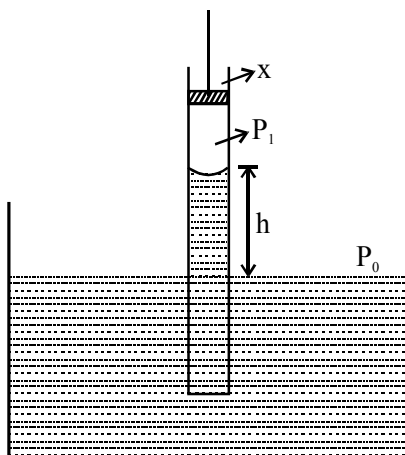


3. The figure below shows a drop of mercury on a glass surface. The radius of curvature of the drop at its upper point is $R = 1.7$ mm. What is the mass 'm' (in mg) of the drop, if its height $h = 2$ mm, the contact radius of the drop with the horizontal plane on which it is located is equal to $r = 1$ mm? The density of the mercury is $\rho = 13500$ kg/m³, the surface tension of mercury is $\sigma = 425$ mN/m.

Take $g = 10$ m/s². Write $\frac{m}{11\pi}$ to its nearest integer in OMR sheet. Take contact angle as 180°



4. A capillary of radius $r = 0.2$ mm is dipped vertically in a liquid of density, ($\rho = 1$ gm / cc). A small piston (x) is inserted inside the capillary which maintains the constant pressure of air ($P_1 = 200$ dynes/cm²) above the hemispherical meniscus. If ambient pressure is $P_0 = 1000$ dynes/cm² and surface tension of liquid is $T = 72$ dynes/cm, then find the height h of liquid in cm. (weight of curved part of liquid is ignored)

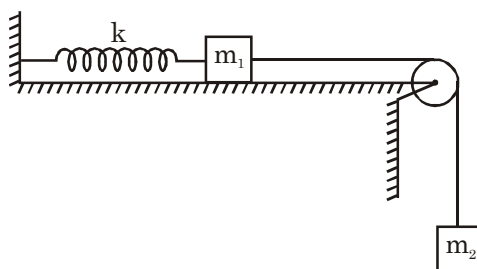


SECTION-I

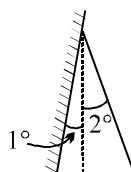
Single Correct Answer Type

11 Q. [3 M (-1)]

1. m_1 & m_2 are connected with a light inextensible string with m_1 lying on smooth table and m_2 hanging as shown in figure. m_1 is also connected to a light spring which is initially unstretched and the system is released from rest :



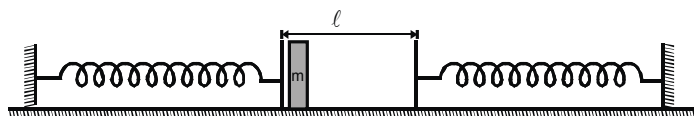
- (A) System performs SHM with angular frequency given by $\sqrt{\frac{k(m_1 + m_2)}{m_1 m_2}}$
- (B) System performs SHM with angular frequency given by $\sqrt{\frac{k}{m_1 + m_2}}$
- (C) Tension in the string will be zero when system is released.
- (D) Maximum displacement of m_1 will be $\frac{m_2 g}{k}$.
2. A simple pendulum of length 1 m is allowed to oscillate with amplitude 2° . It collides elastically with a wall inclined at 1° to the vertical. Its time period will be (use $g = \pi^2$)



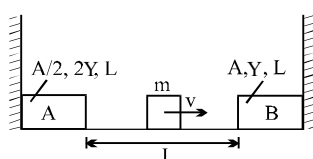
- (A) $2/3$ sec (B) $4/3$ sec (C) 2 sec (D) none
3. A particle moving about its equilibrium position with equation $y = -ax - bt$. Interpret the condition. Where a and b are non zero constant :-
- (A) It will always perform the SHM (B) It can never perform the SHM
- (C) It can perform SHM only when $t \geq \frac{bx}{a}$ (D) It can perform SHM only when $t \leq \frac{bx}{a}$
4. A thin spherical shell full of water is attached to the end of a rope and allowed to swing back and forth as a pendulum from a fixed support. It has a hole in its bottom that allows water to leak out. How does the period of motion change with the loss of water?
- (A) The period does not change.
- (B) The period continuously decreases.
- (C) The period continuously increases.
- (D) The period increases to some maximum and then decreases again.
5. The period of oscillation of simple pendulum of length L suspended from the roof of a vehicle which moves without friction down an inclined plane of inclination α is given by

- (A) $2\pi\sqrt{\frac{L}{g \cos \alpha}}$ (B) $2\pi\sqrt{\frac{L}{g \sin \alpha}}$ (C) $2\pi\sqrt{\frac{L}{g}}$ (D) $2\pi\sqrt{\frac{L}{g \tan \alpha}}$

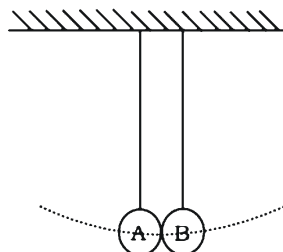
6. A block of mass m is oscillating on smooth surface between two light springs of spring constant k separated by a distance ℓ colliding elastically with the springs. If the velocity of the block is increased by an external impulse when it is not touching either of the spring then time period



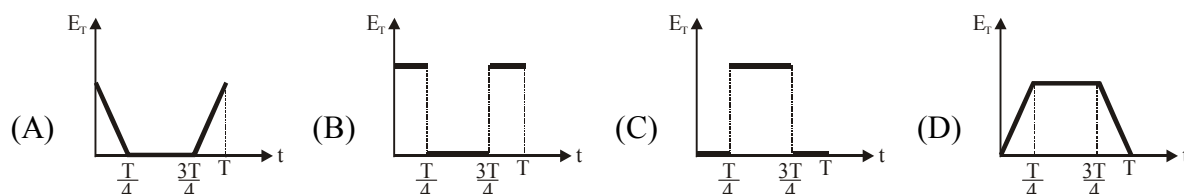
- (A) Increases. (B) Decreases (C) remains same. (D) Time period is independent of ℓ
7. In the given figure, two elastic rods A & B are rigidly joined to end supports. A small mass 'm' is moving with velocity v between the rods. All collisions are assumed to be elastic & the surface is given to be frictionless. The time period of small mass 'm' will be : [A=area of cross section, Y = Young's modulus, L=length of each rod ; here, an elastic rod may be treated as a spring of spring constant $\frac{YA}{L}$]



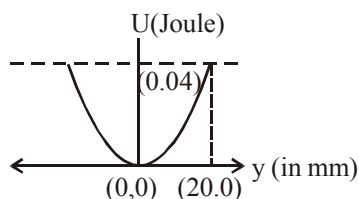
- (A) $\frac{2L}{v} + 2\pi\sqrt{\frac{mL}{AY}}$ (B) $\frac{2L}{v} + 2\pi\sqrt{\frac{2mL}{AY}}$ (C) $\frac{2L}{v} + \pi\sqrt{\frac{mL}{AY}}$ (D) $\frac{2L}{v}$
8. A particle at the end of a spring executes simple harmonic motion with a period t_1 , while the corresponding period for another spring is t_2 . If the period of oscillation with the two springs in series is T , then
- (A) $T^2 = t_1^2 + t_2^2$ (B) $T = t_1 + t_2$ (C) $T^{-1} = t_1^{-1} + t_2^{-1}$ (D) $T^{-2} = t_1^{-2} + t_2^{-2}$
9. The potential energy of a particle of mass 'm' situated in a unidimensional potential field varies as $U(x) = U_0 [1 - \cos ax]$, where U_0 and a are constants. The time period of small oscillations of the particle about the mean position is (Given $|a| < 1$)
- (A) $2\pi\sqrt{\frac{m}{aU_0}}$ (B) $2\pi\sqrt{\frac{am}{U_0}}$ (C) $2\pi\sqrt{\frac{m}{a^2U_0}}$ (D) $2\pi\sqrt{\frac{a^2m}{U_0}}$
10. Two pendulums with identical bobs and lengths are suspended from a common support such that in rest position the two bobs are in contact (figure). After being displaced by 5° the bob A is released from rest, at $t = 0$ subsequently it collides elastically head-on with the other bob.



The graph showing variation in energy of pendulum A with time, for $0 \leq t \leq T$ (where T is the period of either pendulum).



11. The variation in potential energy of a harmonic oscillator is as shown in fig. The spring constant is :-



- (A) $1 \times 10^2 \text{ Nm}^{-1}$ (B) $2 \times 10^2 \text{ Nm}^{-1}$ (C) $0.667 \times 10^2 \text{ Nm}^{-1}$ (D) $3 \times 10^2 \text{ Nm}^{-1}$

Multiple Correct Answer Type

4 Q. [4 M (-1)]

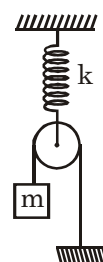
12. In following diagram a block of mass m is released from rest at $t = 0$ when spring was in its natural length then which of the following statement(s) is correct ?

(A) Maximum elongation in the spring is $\frac{4mg}{k}$.

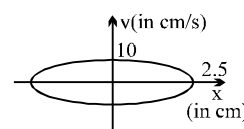
(B) Maximum speed of the block is $g\sqrt{\frac{4m}{k}}$.

(C) If block is displaced from mean position then time period of SHM is $2\pi\sqrt{\frac{4m}{k}}$.

(D) If block is displaced from mean position then time period of SHM is $\pi\sqrt{\frac{2m}{k}}$.



13. The figure shows a graph between velocity and displacement (from mean position) of a particle performing SHM:



(A) the time period of the particle is 1.57s

(B) the maximum acceleration will be 40cm/s^2

(C) the velocity of particle is $2\sqrt{21} \text{ cm/s}$ when it is at a distance 1 cm from the mean position.

(D) none of these

14. A particle moves such that its position from origin is given by $x = 2 + 3 \sin^2 \omega t$. Choose **CORRECT** statement(s) about its motion :-

(A) Particle performs S.H.M. of amplitude $\frac{3}{2} \text{ m}$ and frequency $\frac{\omega}{\pi}$

(B) Particle performs oscillations but no S.H.M.

(C) Particle performs S.H.M. about mean position at $x = \frac{7}{2} \text{ m}$

(D) Particle performs oscillations of amplitude 3m.

15. A particle of mass m moves rectilinearly under the action of a force $F = \alpha - \beta x$. Here x is distance from the point where it was at rest ($x = 0$). (α, β are positive constants) :-

(A) Motion is simple harmonic

(B) Amplitude of particle is $\frac{\alpha}{\beta}$

(C) Mean position of particle may be at $\frac{\alpha}{\beta}$

(D) Maximum acceleration of particle is $\frac{\beta}{m}$.

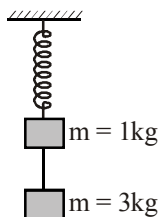
SECTION-II

Numerical Answer Type Question

1Q.[3(0)]

(upto second decimal place)

- Two masses are executing SHM under influence of a vertical spring as shown with an amplitude of 2cm and angular frequency of 5 rad/s. When the extension of spring is maximum, suddenly the string breaks and 3kg mass falls off. What is the maximum velocity of the 1kg mass (in cm/s) during subsequent motion ?

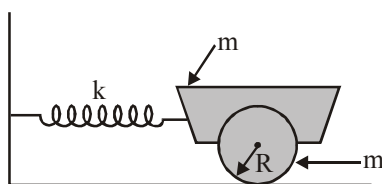


SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

- A linear harmonic oscillator of force constant 1.2×10^6 N/m and amplitude 0.01m has a total mechanical energy of 160 joule. Its maximum kinetic energy is equal to $(\alpha \times 10^\beta)$ J ; $0 < (\alpha, \beta) < 9$. Find the value of $\alpha + \beta$.
- A cart consists of a body of mass m and two wheels, each of mass m and radius R. The cart is attached to a spring of constant k. The other end of the spring is fixed to a wall as shown in figure. If time period of oscillation is $\beta \times 2\pi \sqrt{\frac{m}{k}}$ then find the value of β

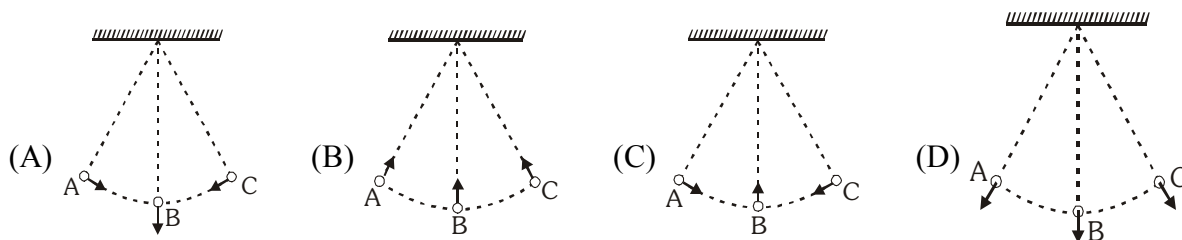


SECTION-I

Single Correct Answer Type

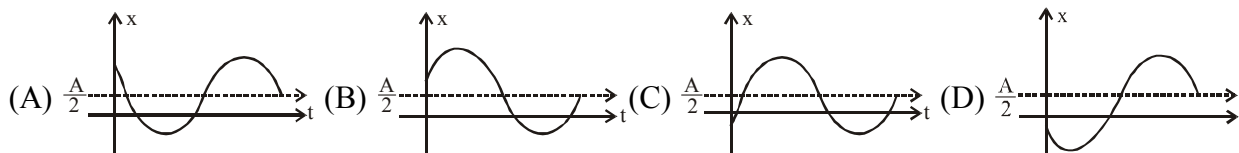
10 Q. [3 M (-1)]

1. A simple pendulum of mass m swings about point B between extreme positions A and C. Net force acting on the bob at these three points is correctly shown by

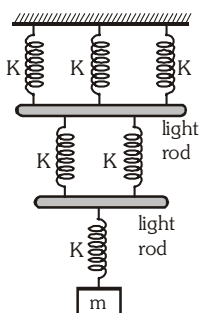


2. A particle oscillating in simple harmonic motion is :
 (A) never in equilibrium because it is in motion
 (B) never in equilibrium because there is always a force
 (C) in equilibrium at the ends of its path because its velocity is zero there
 (D) in equilibrium at the centre of its path because the acceleration is zero there
3. On the superposition of two harmonic oscillations represented by
 $x_1 = a \sin(\omega t + \phi_1)$ and $x_2 = a \sin(\omega t + \phi_2)$
 a resulting oscillation with the same time period and amplitude is obtained. The value of $\phi_1 - \phi_2$ is :-
 (A) 120° (B) 90° (C) 60° (D) 15°
4. Two pendulums of time periods 3 s and 7 s respectively start oscillating simultaneously from two opposite extreme positions. After how much time they will be in phase
 (A) $\frac{21}{8}$ s (B) $\frac{21}{4}$ s (C) $\frac{21}{2}$ s (D) $\frac{21}{10}$ s
5. A particle executes SHM with amplitude of oscillations A and time period T . Find the magnitude of average acceleration for the period of time in which it moves from mean position by a distance $\frac{A}{2}$.
 (A) $\frac{3\pi A}{T^2}$ (B) $\frac{12\pi A}{T^2}$ (C) $\frac{12\pi A(2-\sqrt{3})}{T^2}$ (D) None of these
6. Two particles A and B perform SHM along the same straight line with the same amplitude 'a', same frequency 'f' and same equilibrium position 'O'. The greatest distance between them is found to be $3a/2$. At some instant of time they have the same displacement from mean position. What is the displacement?
 (A) $a/2$ (B) $a\sqrt{7}/4$ (C) $\sqrt{3}a/2$ (D) $3a/4$
7. Two particles are in SHM with same angular frequency and amplitudes A and $2A$ respectively along same straight line with same mean position. They cross each other at position $A/2$ distance from mean position in opposite direction. The phase between them is :
 (A) $\frac{5\pi}{6} - \sin^{-1}\left(\frac{1}{4}\right)$ (B) $\frac{\pi}{6} - \sin^{-1}\left(\frac{1}{4}\right)$ (C) $\frac{5\pi}{6} - \cos^{-1}\left(\frac{1}{4}\right)$ (D) $\frac{\pi}{6} - \cos^{-1}\left(\frac{1}{4}\right)$

8. A particle performing S.H.M. about mean position $x = \frac{A}{2}$ and at $t = 0$. It has displacement $\frac{-3A}{4}$ and moving away from the origin. Then which of the following is its possible graph between position (x) and time (t)



9. The time period of small oscillations of mass m :-



- (A) $2\pi\sqrt{\frac{m}{6K}}$ (B) $2\pi\sqrt{\frac{11m}{6K}}$ (C) $2\pi\sqrt{\frac{6m}{11K}}$ (D) $2\pi\sqrt{\frac{m}{K}}$

10. A system is shown in the figure. The time period for small oscillations of the two blocks will be :-

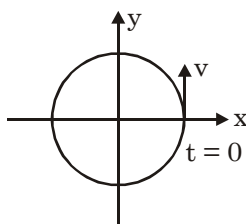


- (A) $2\pi\sqrt{\frac{3m}{k}}$ (B) $2\pi\sqrt{\frac{3m}{2k}}$ (C) $2\pi\sqrt{\frac{3m}{8k}}$ (D) $2\pi\sqrt{\frac{3m}{4k}}$

Multiple Correct Answer Type

3 Q. [4 M (-1)]

11. A particle is moving in uniform circular motion with its initial position, velocity & coordinate system is as shown. Identify the correct graphs :-



- (A) x component of velocity (B) y component of force (C) Angular velocity (D) Position x-t graph

12. A particle of mass m executing S.H.M. with amplitude a and frequency ω . At a distance x from the equilibrium point particle receive a impulse J in the direction of motion. Then choose **CORRECT** statement(s) :

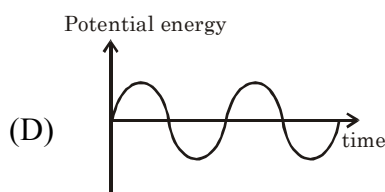
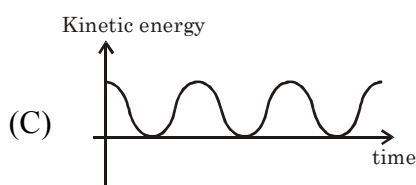
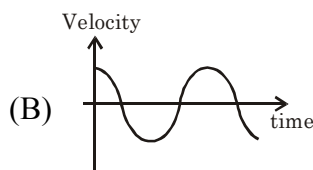
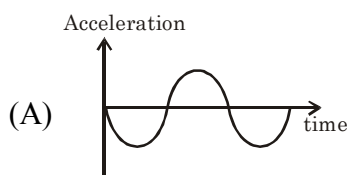
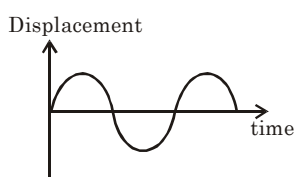
(A) Amplitude of new S.H.M. will increase.

(B) Time period of particle of new S.H.M. remain constant.

(C) Amplitude of new S.H.M. is $\sqrt{\left(\frac{J}{m\omega} + (\sqrt{a^2 - x^2})\right)^2 + x^2}$.

(D) Frequency of new S.H.M. of particle may increase.

13. The graph besides shows how the displacement of a particle executing simple harmonic motion varies with time t . The options show the variation of different parameters with time. Select the **CORRECT** alternative(s) :

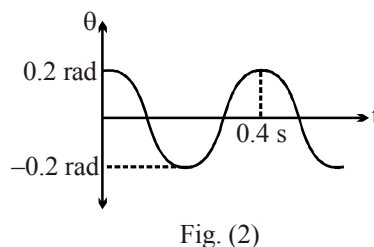
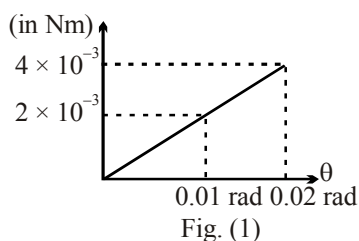


Linked Comprehension Type (Single Correct Answer Type)

(2 Para $\times$ 2Q.) [3 M (-1)]

Paragraph for Question no. 14 and 15

A torsional pendulum is made by attaching a disc to a wire. Figure-1 gives the magnitude of torque needed to rotate the disc about its center v/s rotation angle θ .

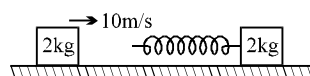


The disc is rotated and released. Figure-2 shows the angular position (θ) with time (t). Take $\pi^2 = 10$.

14. What is the moment of inertia of the disc ?
 (A) $4 \times 10^{-5} \text{ kgm}^2$ (B) $8 \times 10^{-4} \text{ kgm}^2$ (C) $8 \times 10^{-5} \text{ kgm}^2$ (D) $8 \times 10^{-6} \text{ kgm}^2$
15. What is the maximum angular speed of the disc ?
 (A) $\pi \text{ rad/sec}$ (B) $5\pi \text{ rad/sec}$ (C) $2\pi \text{ rad/sec}$ (D) $3\pi \text{ rad/sec}$

Paragraph for Question no. 16 and 17

A 2 Kg block moving with 10 m/s strikes a spring of constant π^2 N/m attached to 2 Kg block at rest kept on a smooth floor as shown in figure.



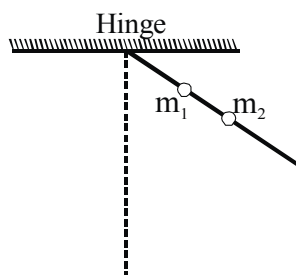
16. The time for which rear moving block remain in contact with spring will be
- (A) $\sqrt{2}$ sec (B) $\frac{1}{\sqrt{2}}$ sec (C) 1 sec (D) $\frac{1}{2}$ sec
17. The velocity of the rear 2 kg block after it separates from the spring will be :
- (A) 0 m/s (B) 5 m/s (C) 10 m/s (D) 7.5 m/s

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. Two point masses m_1 and m_2 are fixed to a light rod hinged at one end. The masses are at distances ℓ_1 and ℓ_2 respectively from the hinge. Find the time period of oscillation (small amplitude) of this system in seconds if $m_1 = 2m_2$, $\ell_1 = 15$ cm, $\ell_2 = 30$ cm. Fill the answer of x if time period is $x \left(\frac{\pi}{10} \right)$ seconds.



SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. The speed (v) of a particle moving along a straight line, when it is at a position (x) from mean position on the line is given by : $v^2 = 144 - 9x^2$

Column-I

Column-II

- | | |
|--|------------------------------------|
| (A) Displacement of the particle for $t > \frac{2\pi}{3}$ | (P) $T = 2\pi/3$ units |
| (B) Time period of simple harmonic motion | (Q) Less than distance moved by it |
| (C) Maximum displacement from the fixed point is | (R) 27 units |
| (D) Magnitude of acceleration at a distance 3 cm from the fixed point is | (S) 4 units |
| | (T) 3 units |